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RSICC DATA LIBRARY COLLECTION

MCNPXDATA

Standard Neutron, Photon, and Electron Data Libraries for MCNP4C
for MCNPX 2.3.0 or 2.4.0

Contributed by:

Los Alamos National Laboratory
Los Alamos, New Mexico



RADIATION SAFETY INFORMATION COMPUTATIONAL CENTER

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Documentation for DLC-205/MCNPXDATA Data Package

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(September 2002)

RSICC DATA LIBRARY DLC-205

1. NAME AND TITLE OF DATA LIBRARY

MCNPXDATA: Standard Neutron, Photon, Electron, and Proton Data Libraries for MCNPX 2.3.0 or 2.4.0.

2. NAME AND TITLE OF DATA RETRIEVAL PROGRAMS

MAKXS: Prepares MCNP Cross-Section Libraries for MCNPX.

3. CONTRIBUTOR

Los Alamos National Laboratory, Los Alamos, New Mexico.

4. HISTORICAL BACKGROUND AND INFORMATION

These cross-section libraries are released by the Diagnostics Applications Group, X-5, at Los Alamos National Laboratory for use with the MCNPX Monte Carlo code package. This release includes all of the X-5 distributed neutron data libraries, the photon libraries, and electron libraries that have been released as of September 2002. An updated XSDIR file and information and report files are also included. These data were previously released in DLC-105, DLC-181, DLC-189, and DLC-200.

Additions to the June 2002 **D00205ALLCP02** release included the LA150U photonuclear data and LA150H proton data tables. A modification was made to correct a small error in the original release of LA150N. The ZAID for Pb-208 was changed from 82208.24c to 82208.25c to differentiate from the earlier version.

The September 2002 **D00205ALLCP03** release includes all files in D00205ALLCP02 plus the EL03 electron data library. **EL03 requires new functionality in MCNP4C or MCNPX 2.4.0.** EL03 cannot be used with MCNP4B, MCNPX2.3.0 or earlier versions.

Several README files provide information regarding each data library of this release. Additional documentation for some of the individual libraries and example SPECS files for use with MAKXS are also provided. The XSDIR file is specific to this release and may not work with other packages. Currently LA150N is the default library for continuous-energy neutron transport. Additionally, MCPLIB02 is the default library for photon transport. The default electron library is EL in D00205ALLCP02 and is EL03 in D00205ALLCP03.

5. APPLICATION OF THE DATA

DLC-205/MCNPXDATA is for use with MCNPX and can be used with Version 4B, 4C and later of the MCNP transport code. Note the warning above regarding EL03 and earlier releases of MCNP. This library provides a comprehensive set of cross sections for a wide range of radiation transport applications using MCNPX.

6. SOURCE AND SCOPE OF DATA

The DLC-205/MCNPXDATA package is comprised of DLC-200/MCNPDATA, which was released for use with MCNP4C; plus the LA150N library of 42 high energy neutron data tables, LA150U photonuclear data for 12 isotopes, and LA150H proton data tables for 41 isotopes. The EL03 library is omitted from D00205ALLCP02 for MCNPX2.3.0, but it is included in D00205ALLCP03 for MCNPX 2.4.0. Electron tables with zaid numbers ending in .01e will work in MCNPX 2.3.0; others will not. In LA150N, the neutron energy is extended to 150 MeV except for Be-9, which only goes to 100 MeV. This library typically extends ENDF/B-VI data from 20 MeV to 150 MeV; therefore, charged particle and recoil nuclei data will sometimes not be available below 20 MeV. Exceptions are noted in the MCNPX User's Manual. All standard neutron libraries

used with MCNP4C (originally distributed in DLC-200 and now included in DLC-205) can be used with MCNPX 2.3.0; however, they will not contain emission data for charged particles or recoil nuclei; therefore, these products will not be tracked upon production. The new neutron libraries developed for use with MCNP4C will work with MCNPX 2.4.0, but some features specific to MCNP4C will not function in MCNPX 2.3.0. In particular, features related to probability tables and delayed neutrons will be ignored in MCNPX 2.3.0.

A wide variety of continuous-energy, discrete, multigroup, thermal and dosimetry neutron data libraries are available in this release. The continuous-energy neutron data libraries available include: LA150N, ENDF60, RMCCS, RMCCSA, ENDF5U, ENDF5P, NEWXS, ENDF5MT*, MISC5XS**, ENDF85, KIDMAN, 100XS, URESA***, ENDF6DN***, ENDF62MT***, and ENDF92***. The discrete neutron data libraries include: NEWXSD, DRMCCS, and DRE5. The multigroup neutron data library is MGXSNP, and the thermal S(alpha, beta) libraries are TMCCS and THERXS. The neutron dosimetry libraries are 531DOS, 532DOS, and LLLDOS. The photon transport libraries are MCPLIB and MCPLIB02, and the electron libraries are EL and EL03. The photon and electron data libraries contain data for elements having $Z < 95$. The data libraries, as distributed, are in ASCII or type 1, format.

* The data library ENDF5MT contains data previously available in the library EPRIXS, along with the U600K data library.

** The data library MISC5XS contains corrected data for ENDF/B-V based Zr as described below, and the libraries previously known as IRNAT, MISCXS, ARKRC, TM169, GDT2GP, and T2DDC. The ENDF/B-V Zr data was corrected for five ZAIID's from the libraries RMCCS, DRMCCS, ENDF5P, DRE5, and EPRIXS. Below is a summary of the changes that have been made for Zr:

	(Previous)		(Corrected)
RMCCS	40000.51c	?	40000.57c MISC5XS 300K
DRMCCS	40000.51d	?	40000.57d MISC5XS 300K
ENDF5P	40000.50c	?	40000.56c MISC5XS 300K
DRE5	40000.50d	?	40000.56d MISC5XS 300K
EPRIXS	40000.53c	?	40000.58c MISC5XS 600K

*** These four data libraries (URES, ENDF6DN, ENDF62MT, and ENDF92) were released in DLC-200. In the February 2001 update, URES was replaced with URESA. Please consult the README files and the more detailed documentation provided for descriptions of these libraries.

The LA150N neutron data library includes 42 isotopes: 1H, 2H, 9Be (100 MeV), natC, 14N, 16O, 27Al, 28Si, 29Si, 30Si, 31P, natCa, 50Cr, 52Cr, 53Cr, 54Cr, 54Fe, 56Fe, 57Fe, 58Ni, 60Ni, 61Ni, 62Ni, 64Ni, 63Cu, 65Cu, 93Nb, 182W, 183W, 184W, 186W, 196Hg, 198Hg, 199Hg, 200Hg, 201Hg, 202Hg, 204Hg, 206Pb, 207Pb, 208Pb, 209Bi.

The LA150U photonuclear library includes 12 isotopes: 12C, 16O, 27Al, 28Si, 40Ca, 56Fe, 63Cu, 181Ta, 184W, 206Pb, 207Pb, 208Pb.

LA150H Proton Data Library (updated 11/20/01) includes 41 isotopes: 1H, 2H, 12C, 14N, 16O, 27Al, 28Si, 29Si, 30Si, 31P, 40Ca, 50Cr, 52Cr, 53Cr, 54Cr, 54Fe, 56Fe, 57Fe, 58Ni, 60Ni, 61Ni, 62Ni, 64Ni, 63Cu, 65Cu, 93Nb, 182W, 183W, 184W, 186W, 196Hg, 198Hg, 199Hg, 200Hg, 201Hg, 202Hg, 204Hg, 206Pb, 207Pb, 208Pb, 209Bi.

7. DISCUSSION OF THE DATA RETRIEVAL PROGRAM

The MAKXS program is required if the user wants to translate the data into binary form. It is distributed with the MCNPX code. This translation saves disk space and cuts down on running time of MCNPX. Input to the executable MAKXS includes the SPECS file, the XSDIR file (nuclide data file directory) and the data libraries. The user should study the SPECS and XSDIR files and read the comments in the MAKXS source for SPECS input specifications. Some platforms have a restriction on the length of filenames, and the SPECS file may need to be appropriately edited.

8. DATA FORMAT AND COMPUTER

ASCII (Type 1); D00205/ALLCP/02 for use with MCNPX2.3.0 (June 2002).

ASCII (Type 1); D00205/ALLCP/03 for use with MCNPX2.4.0 (September 2002).

All data libraries are distributed in a GNU compressed Unix tar file. The data are in ASCII format and can be used with MCNPX on all computer platforms supported by the code. Note that the trailing "1" is no longer appended to file names for Type 1 (ASCII) libraries.

9. TYPICAL RUNNING TIME

Run times vary according to problems and computer speed.

10. REFERENCES

When referring to the LA150 data, please cite the following publication: M. B. Chadwick, P. G. Young, S. Chiba, S. C. Frankle, G. M. Hale, H. G. Hughes, A. J. Koning, R. C. Little, R. E. MacFarlane, R. E. Prael, and L. S. Waters, "Cross Section Evaluations to 150 MeV for Accelerator-Driven Systems and Implementation in MCNPX," Nuclear Science and Engineering 131, Number 3 (March 1999) 293.

a.) The following documents are included in electronic (PDF) form with the package on the distribution CD in the DOC/D205DOC.PDF file:

Revised Appendix G for the MCNP4C manual (December 18, 2000).

R. C. Little, "Readme_LA150N," (July 2000, revised May 2002).

R. C. Little, "Readme_LA150H," (May 2002).

M. C. White, "Readme_LA150U," (June 2001).

S. C. Frankle and R. C. Little, "README_4C.TXT," informal overview of the DLC-200/MCNPDATA package (April 2000).

S. C. Frankle, "README_ENDF60," informal overview of the ENDF60 neutron data library (April 2000).

J. S. Hendricks, S. C. Frankle, J. D. Court, "ENDF/B-VI Data for MCNP," and errata, Los Alamos National Laboratory report LA-12891 (1994).

H. G. Hughes, "Information on the MCPLIB02 Photon Library," Los Alamos National Laboratory memorandum X-6:HG-93-77 (revised 1996).

R. C. Little, "Summary Documentation for the 100XS Neutron Cross Section Library," Los Alamos National Laboratory memorandum, XTM:95-259 and LA-UR-96- 24 (1995).

R. C. Little and R. E. Seamon, "Dosimetry/Activation Cross Sections for MCNP," Los Alamos National Laboratory memorandum (March 13, 1984).

R. C. Little, "Neutron and Photon Multigroup Data Tables for MCNP3B," Los Alamos National Laboratory memorandum X-6:RCL-87-225 (1987).

R. C. Little and R. E. MacFarlane, "ENDF/B-VI Neutron Library for MCNP with Probability Tables," Los Alamos National Laboratory Research Note XCI-RN(U)98-041, LA-UR-98-5718 (December 1998).

S. C. Frankle, "ENDF62MT, A Multi-temperature Neutron Library for MCNP (Rev.0), Los Alamos National Laboratory memorandum XTM:96-153 (April 1996).

C. Werner, "New Data Library for MCNP Delayed Neutron Capability," Los Alamos National Laboratory memorandum XCI:CJW-99-25 (April 1999).

R. C. Little, "ReadURES.txt" (February 2001).

K. J. Adams, "Electron Upgrade for MCNP4B," X-5-RN(U)-00-14 (May 25, 2000).

b.) Additional documentation is distributed in electronic (PDF) format in the distribution tar file and will be extracted to the user's mcnpdata/DOC subdirectory.

M. C. White, "Class 'u' ACE format Photonuclear Data," LANL Memo, file mcw-00-89.pdf (July 26, 2000).

M. C. White, "Release of the LA150U Photonuclear Data Library," LANL Memo, file mcw-00-87.pdf (Revised March 21, 2001).

M. C. White, "User Interface for Photonuclear Physics in MCNP(X)," LANL Memo, file mcw-00-88.pdf (Revised March 21, 2001).

M. C. White, "A Brief Primer for Simulating Photonuclear Interactions with MCNP(X)," LANL Memo, file mcw-00-89.pdf (Revised March 21, 2001).

A. A. Adams, S. C. Frankle, and R. C. Little, "Revised Criticality Benchmarks for MCNP," Los Alamos National Laboratory memorandum (1996).

J. D. Court, R. C. Brockhoff, and J. S. Hendricks, "Lawrence Livermore Pulsed Sphere Benchmark Analysis of MCNP ENDF/B-VI," Los Alamos National Laboratory report LA-12885 (1994).

J. D. Court, J. S. Hendricks, and S. C. Frankle, "MCNP ENDF/B-VI Validation: Infinite Media Comparisons of ENDF/B-VI and ENDF/B-V," and errata, Los Alamos National Laboratory report LA-12887 (1994).

J. D. Court and J. S. Hendricks, "Benchmark Analysis of MCNP ENDF/B-VI Iron," Los Alamos National Laboratory report LA-12884 (1994).

S. C. Frankle, "Photon Production Assessment for the MCNP ENDF/B-VI Data Library," Los Alamos National Laboratory report LA-13093 (1995).

S. C. Frankle, "Cross-section and Reaction Nomenclature for MCNP Continuous- energy Libraries and DANTSYS Multigroup Libraries," Los Alamos National Laboratory memorandum XTM:SCF-96-313 (1996).

S. C. Frankle, "Experience with ENDF60 and QA of a Continuous-Energy Library," Los Alamos National Laboratory memorandum XTM:SCF-96-363 (1996).

11. CONTENTS OF LIBRARY

Included are the documents referenced in 10.a and files described in Section 6. The ASCII libraries are distributed on CD-ROM in a GNU compressed Unix tar file. The data can be distributed on the same CD as the MCNPX 2.3.0 or MCNPX 2.4.0 code packages. WinZIP 8.0 or later is required to expand the files under Windows operating systems.

12. DATE OF ABSTRACT

August 2000, revised February 2001, May 2002, September 2002.

KEYWORDS: NEUTRON CROSS SECTIONS; GAMMA-RAY CROSS SECTIONS; ELECTRON CROSS SECTIONS; MULTIGROUP CROSS SECTIONS BASED ON ENDF/B; MCNP FORMAT; WORKSTATION

S E C T I O N 1

APPENDIX G

NEUTRON CROSS-SECTION LIBRARIES

This appendix is divided into five sections. Section I lists some of the more frequently used ENDF/B reaction types that can be used with the FMn input card. TABLE G-1 in Section II lists the currently available $S(\alpha, \beta)$ data available for use with the MTm card. Section III provides a brief description of the available continuous-energy and discrete neutron data libraries. TABLE G-2 in Section III is a list of the continuous-energy and discrete neutron data libraries maintained by X-5. Section IV describes the multigroup data library MGXSNP (TABLE G-3), and Section V describes the dosimetry data libraries (TABLE G-4).

I. ENDF/B REACTION TYPES

The following partial list includes some of the more useful reactions for use with the FMn input card and with the cross-section plotter (see pages 3-87 and B-10.) The complete ENDF/B list can be found in the ENDF/B manual.¹ The MT column lists the ENDF/B reaction number. The FM column lists special MCNP reaction numbers that can be used with the FM card and cross-section plotter.

Generally only a subset of reactions are available for a particular nuclide. Some reaction data are eliminated by MCNP in cross-section processing if they are not required by the problem. Examples are photon production in a MODE N problem, or certain reaction cross sections not requested on an FM card. FM numbers should be used when available, rather than MT numbers. If an MT number is requested, the equivalent FM number will be displayed on the legend of cross-section plots.

Neutron Continuous-energy and Discrete:

<u>MT</u>	<u>FM</u>	<u>Microscopic Cross-Section Description</u>
1	-1	Total (see note 1 following)
2	-3	Elastic (see note 1 following)
16		(n,2n)
17		(n,3n)
18		Total fission (n,fx) if and only if MT=18 is used to specify fission in the original evaluation.
	-6	Total fission cross section. (equal to MT=18 if MT=18 exists; otherwise equal to the sum of MTs 19, 20, 21, and 38.)
19		(n,f)
20		(n,n'f)
21		(n,2nf)
22		(n,n'α)

APPENDIX G ENDF/B REACTION TYPES

28		(n,n'p)
32		(n,n'd)
33		(n,n't)
38		(n,3nf)
51		(n,n') to 1 st excited state
52		(n,n') to 2 nd excited state
90		(n,n') to 40 th excited state
91		(n,n') to continuum
101	-2	Absorption: sum of MT=102-117 (neutron disappearance; does not include fission)
102		(n, γ)
103		(n,p)
104		(n,d)
105		(n,t)
106		(n, ³ He)
107		(n, α)

In addition, the following special reactions are available for many nuclides:

202	-5	total photon production
203		total proton production (see note 3 following)
204		total deuterium production (see note 3 following)
205		total tritium production (see note 3 following)
206		total ³ He production (see note 3 following)
207		total alpha production (see note 3 following)
301	-4	average heating numbers (MeV/collision)
	-7	nubar (prompt or total)
	-8	fission Q (in print table 98, but not plots)

S(α,β):

<u>MT</u>	<u>FM</u>	<u>Microscopic Cross-Section Description</u>
1		Total cross section
2		Elastic scattering cross-section
4		Inelastic scattering cross-section

Neutron and Photon Multigroup:

<u>MT</u>	<u>FM</u>	<u>Microscopic Cross-Section Description</u>
1	-1	Total cross section
18	-2	Fission cross section

APPENDIX G

ENDF/B REACTION TYPES

	-3	Nubar data
	-4	Fission chi data
101	-5	Absorption cross section
	-6	Stopping powers
	-7	Momentum transfers
n		Edit reaction n
202		Photon production
301		Heating number
318		Fission Q
401		Heating number times total cross section

Photons (see note 4 following):

<u>MT</u>	<u>FM</u>	<u>Microscopic Cross-Section Description</u>
501	-5	Total
504	-1	Incoherent (Compton + Form Factor)
502	-2	Coherent (Thomson + Form Factor)
522	-3	Photoelectric with fluorescence
516	-4	Pair production
301	-6	Heating number

Electrons (see note 5 following):

<u>MT</u>	<u>FM</u>	<u>Microscopic Cross-Section Description</u>
	1	de/dx electron collision stopping power
	2	de/dx electron radiative stopping power
	3	de/dx total electron stopping power
	4	electron range
	5	electron radiation yield
	6	relativistic β^2
	7	stopping power density correction
	8	ratio of rad/col stopping powers
	9	drange
	10	dyield
	11	rng array values
	12	qav array values
	13	ear array values

APPENDIX G

ENDF/B REACTION TYPES

Notes:

1. At the time they are loaded, the total and elastic cross sections from the data library are thermally adjusted by MCNP to the temperature of the problem, if that temperature is different from the temperature at which the cross-section set was processed (see page 2-29.) If different cells have different temperatures, the cross sections first are adjusted to zero degrees and adjusted again to the appropriate cell temperatures during transport. The cross-section plot will *never* display the *transport* adjustment. Therefore, for plotting, reactions 1 and -1 are equivalent and reactions 2 and -3 are equivalent. But for the FM card, reactions -1 and -3 will use the zero degree data and reactions 1 and 2 will use the transport-adjusted data.
For example, if a library evaluated at 300° is used in a problem with cells at 400° and 500°, the cross-section plotter and MT=-1 and MT=-3 options on the FM card will use 0° data. The MT=1 and MT=2 options on the FM card will use 400° and 500° data.
2. The nomenclature between MCNP and ENDF/B is sometimes inconsistent in that MCNP often refers to the number of the reaction type as R whereas ENDF/B uses MT. They are one and the same, however. The problem arises because MCNP has an MT input card used for the $S(\alpha,\beta)$ thermal treatment.
3. The user looking for total production of p, d, t, ^3He and ^4He should be warned that in some evaluations, such processes are represented using reactions with MT (or R) numbers other than the standard ones given in the above list. This is of particular importance with the so-called "pseudolevel" representation of certain reactions which take place in light isotopes. For example, the ENDF/B-V evaluation of carbon includes cross sections for the $(n,n'\alpha)$ reaction in MT = 52 to 58. The user interested in particle production from light isotopes should check for the existence of pseudolevels and thus possible deviations from the above standard reaction list.
4. There are two photon transport libraries maintained by X-5, MCPLIB and MCPLIB02.^{2,3} The photon library MCPLIB provides data for transporting photons with energies from 1 keV to 100 MeV. The default photon library MCPLIB02 provides data up to 100 GeV. Photon transport data are not provided for $Z > 94$, and coupled neutron-photon problems cannot be run for these nuclides.
5. X-5 maintains one electron transport library, EL. The MT numbers used for xs plotting are taken from Print Table 85 columns and are not from ENDF.

II. S(α,β) DATA FOR USE WITH THE MT_m CARD

TABLE G-1
Thermal S(α,β) Cross-Section Libraries

ZAID	Date of Processing	Material Description	Nuclides*	Temp (°K)
THERXS1 (Source: LANL)				
smeth.01t	04/10/88	Solid methane	1001	22
lmeth.01t	04/10/88	Liquid methane	1001	100
hpara.01t	03/03/89	Para H	1001	20
hortho.01t	03/03/89	Ortho H	1001	20
dpara.01t	05/30/89	Para D	1002	20
dortho.01t	05/30/89	Ortho D	1002	20
TMCCS1 (Source: ENDF)				
lwtr.01t	10/22/85	H in light water	1001	300
lwtr.02t	10/22/85	H in light water	1001	400
lwtr.03t	10/22/85	H in light water	1001	500
lwtr.04t	10/22/85	H in light water	1001	600
lwtr.05t	10/22/85	H in light water	1001	800
poly.01t	10/22/85	H in polyethylene	1001	300
h/zr.01t	10/22/85	H in Zr-hydride	1001	300
h/zr.02t	10/22/85	H in Zr-hydride	1001	400
h/zr.04t	10/22/85	H in Zr-hydride	1001	600
h/zr.05t	10/22/85	H in Zr-hydride	1001	800
h/zr.06t	10/22/85	H in Zr-hydride	1001	1200
benz.01t	09/08/86	Benzene	1001, 6000, 6012	300
benz.02t	09/08/86	Benzene	1001, 6000, 6012	400
benz.03t	09/08/86	Benzene	1001, 6000, 6012	500
benz.04t	09/08/86	Benzene	1001, 6000, 6012	600
benz.05t	09/08/86	Benzene	1001, 6000, 6012	800
hwtr.01t	10/22/85	D in heavy water	1002	300
hwtr.02t	10/22/85	D in heavy water	1002	400
hwtr.03t	10/22/85	D in heavy water	1002	500
hwtr.04t	10/22/85	D in heavy water	1002	600
hwtr.05t	10/22/85	D in heavy water	1002	800

APPENDIX G
MCNP NEUTRON CROSS-SECTION LIBRARIES

TABLE G-1 (Cont.)
Thermal $S(\alpha,\beta)$ Cross-Section Libraries

ZAID	Date of Processing	Material Description	Nuclides*	Temp (°K)
be.01t	10/24/85	Be metal	4009	300
be.04t	10/24/85	Be metal	4009	600
be.05t	10/24/85	Be metal	4009	800
be.06t	10/24/85	Be metal	4009	1200
beo.01t	09/08/86	Be oxide	4009, 8016	300
beo.04t	09/08/86	Be oxide	4009, 8016	600
beo.05t	09/08/86	Be oxide	4009, 8016	800
beo.06t	09/08/86	Be oxide	4009, 8016	1200
grph.01t	09/08/86	Graphite	6000, 6012	300
grph.04t	09/08/86	Graphite	6000, 6012	600
grph.05t	09/08/86	Graphite	6000, 6012	800
grph.06t	09/08/86	Graphite	6000, 6012	1200
grph.07t	09/08/86	Graphite	6000, 6012	1600
grph.08t	09/08/86	Graphite	6000, 6012	2000
zr/h.01t	09/08/86	Zr in Zr-hydride	40000	300
zr/h.02t	09/08/86	Zr in Zr-hydride	40000	400
zr/h.04t	09/08/86	Zr in Zr-hydride	40000	600
zr/h.05t	09/08/86	Zr in Zr-hydride	40000	800
zr/h.06t	09/08/86	Zr in Zr-hydride	40000	1200

*Nuclides for which the $S(\alpha,\beta)$ data are valid. For example, lwtr.01t provides scattering data only for ^1H ; ^{16}O would still be represented by the default free-gas treatment.

III. MCNP NEUTRON CROSS-SECTION LIBRARIES

TABLE G-2 lists all the continuous-energy and discrete neutron data libraries maintained by X-5. The entries in each of the columns of TABLE G-2 are described as follows:

ZAID – The nuclide identification number with the form ZZZAAA.nnX
 where ZZZ is the atomic number,
 AAA is the mass number (000 for naturally occurring elements),
 nn is the neutron cross-section identifier
 X=C for continuous-energy neutron tables
 X=D for discrete-reaction tables

ATOMIC – The atomic weight ratio (AWR) is the ratio of the atomic mass of the

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MCNP NEUTRON CROSS-SECTION LIBRARIES

- WEIGHT - nuclide to a neutron. This is the AWR that is contained in the original
RATIO - evaluation and that was used in the NJOY processing of the evaluation.
- LIBRARY - Name of the library that contains the data file for that ZAID. The number
in brackets following a file name refers to one of the special notes at the
end of TABLE G-2.
- SOURCE - Indicates the originating evaluation for that data file.
- ENDF/B-V.# or ENDF/B-VI.# (such as B-V.0 and B-VI.1) are the
Evaluated Nuclear Data Files, a US effort coordinated by the National
Nuclear Data Center at Brookhaven National Laboratory. The
evaluations are updated periodically by evaluators from all over the
country, and the release number of the evaluation is given. This is not
necessarily the same as the ENDF revision number for that evaluation.
For example, Pu-242 is noted as ENDF/B-VI.2 as it is from release 2 of
ENDF/B-VI, but it is revision 1 of that evaluation.
- LLNL - evaluated nuclear data libraries compiled by the Nuclear
Data Group at Lawrence Livermore National Laboratory. The number in
the library name indicates the year the library was produced or received.
- T-2 - evaluations from the Nuclear Theory and Applications group
T-2 at Los Alamos National Laboratory.
- :T-2 or —:X-5 - indicates the original evaluation has been
modified by the Los Alamos National Laboratory groups T-2 or X-5.
- DATE of - Denotes the year that the evaluation was completed or accepted. In
EVALUATION - cases where this information is not known, the date that data library was
produced is given. If minor corrections were made to an evaluation, the
original evaluation date was kept. The notation "<1985" means "before"
1985.
- TEMP - Indicates the temperature (°K) at which the data were processed. The
temperature enters into the processing of the evaluation into a data file
only through the Doppler broadening of cross sections. The user must
be aware that without the proper use of the TMP card, MCNP will
attempt to correct the data libraries to the default 300°K by modifying
the elastic and total cross sections only.

Doppler broadening refers to a change in cross section resulting
from thermal motion (translation, rotation and vibration) of nuclei in a
target material. Doppler broadening is done on all cross sections for
incident neutrons (nonrelativistic energies) on a target at some
temperature (TEMP) in which the free-atom approximation is valid. In

APPENDIX G

MCNP NEUTRON CROSS-SECTION LIBRARIES

general an increase in the temperature of the material containing neutron-absorbing nuclei in a homogeneous system results in Doppler broadening of resonances and an increase in resonance absorption. Furthermore, a constant cross section at zero °K goes to $1/v$ behavior as the temperature increases. You should not only use the best evaluations but also use evaluations that are at temperatures approximating the temperatures in your application.

- LENGTH – The total length of a particular cross-section file in words. It is understood that the actual storage requirement in an MCNP problem will often be less because certain data that are not needed for a problem may be expunged.
- NUMBER of – The number of energy points on the grid used for the neutron cross
ENERGIES – section for that data file. In general, a finer energy grid (or greater number of points) indicates a more accurate representation of the cross sections, particularly through the resonance region.
- E_{max} – The maximum incident neutron energy for that data file. For all incident neutron energies greater than E_{max} , MCNP assumes the last cross section value given.
- GPD – “yes” means that photon-production data are included;
“no” means that such data are not included.
- $\bar{u}$ – for fissionable material, $\bar{u}$ indicates the type of fission nu data available.
“pr” means that only prompt nu data are given;
“tot” means that only total nu data are given;
“both” means that prompt and total nu are given.
- CP “yes” means that secondary charged-particles data are present;
“no” means that such data are not present.
- DN “yes” means that delayed neutron data are present;
“no” means that such data are not present.
- UR “yes” means that unresolved resonance data are present;
“no” means that such data are not present.

TABLE G-2 contains no indication of a “recommended” library for each isotope. Because of the wide variety of applications, no one set is “best.” The default cross-section set for each isotope is determined by the XSDIR file being used (see page 2-21.)

Finally, you can introduce a cross-section library of your own by using the XS input card.

APPENDIX G MCNP NEUTRON CROSS-SECTION LIBRARIES

TABLE G-2
Continuous-Energy and Discrete Neutron Data Libraries Maintained by X-5

ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	U	CP	DN	UR
Z = 1 ***** Hydrogen *****													
** H-1 **													
1001.35c	0.9992	endl85	LLNL	<1985	0.0	3506	330	20.0	yes	no	no	no	no
1001.42c	0.9992	endl92	LLNL	<1992	300.0	1968	121	30.0	yes	no	no	no	no
1001.50c	0.9992	rmccs	B-V.0	1977	293.6	2766	244	20.0	yes	no	no	no	no
1001.50d	0.9992	drmccs	B-V.0	1977	293.6	3175	263	20.0	yes	no	no	no	no
1001.53c	0.9992	endf5mt[1]	B-V.0	1977	587.2	4001	394	20.0	yes	no	no	no	no
1001.60c	0.9992	endf60	B-VI.1	1989	293.6	3484	357	100.0	yes	no	no	no	no
** H-2 **													
1002.35c	1.9968	endl85	LLNL	<1985	0.0	2507	135	20.0	yes	no	no	no	no
1002.50c	1.9968	endf5p	B-V.0	1967	293.6	3987	214	20.0	yes	no	no	no	no
1002.50d	1.9968	dre5	B-V.0	1967	293.6	4686	263	20.0	yes	no	no	no	no
1002.55c	1.9968	rmccs	T-2	1982	293.6	5981	285	20.0	yes	no	no	no	no
1002.55d	1.9968	drmccs	T-2	1982	293.6	5343	263	20.0	yes	no	no	no	no
1002.60c	1.9968	endf60	B-VI.0	1967[2]	293.6	2704	178	20.0	yes	no	no	no	no
** H-3 **													
1003.35c	2.9901	endl85	LLNL	<1985	0.0	1269	76	20.0	no	no	no	no	no
1003.42c	2.9901	endl92	LLNL	<1992	300.0	2308	52	30.0	no	no	no	no	no
1003.50c	2.9901	rmccs	B-V.0	1965	293.6	2428	184	20.0	no	no	no	no	no
1003.50d	2.9901	drmccs	B-V.0	1965	293.6	2807	263	20.0	no	no	no	no	no
1003.60c	2.9901	endf60	B-VI.0	1965	293.6	3338	180	20.0	no	no	no	no	no
Z = 2 ***** Helium *****													
** He-3 **													
2003.35c	2.9901	endl85	LLNL	<1985	0.0	2481	182	20.0	yes	no	no	no	no
2003.42c	2.9901	endl92	LLNL	<1992	300.0	1477	151	30.0	yes	no	no	no	no
2003.50c	2.9901	rmccs	B-V.0	1971	293.6	2320	229	20.0	no	no	no	no	no
2003.50d	2.9901	drmccs	B-V.0	1971	293.6	2612	263	20.0	no	no	no	no	no
2003.60c	2.9890	endf60	B-VI.1	1990	293.6	2834	342	20.0	no	no	no	no	no
** He-4 **													
2004.35c	3.9682	endl85	LLNL	<1985	0.0	1442	78	20.0	no	no	no	no	no
2004.42c	3.9682	endl92	LLNL	<1992	300.0	1332	49	30.0	no	no	no	no	no
2004.50c	4.0015	rmccs	B-V.0	1973	293.6	3061	345	20.0	no	no	no	no	no
2004.50d	4.0015	drmccs	B-V.0	1973	293.6	2651	263	20.0	no	no	no	no	no
2004.60c	4.0015	endf60	B-VI.0	1973	293.6	2971	327	20.0	no	no	no	no	no
Z = 3 ***** Lithium *****													
** Li-6 **													
3006.42c	5.9635	endl92	LLNL	<1992	300.0	7805	294	30.0	yes	no	no	no	no
3006.50c	5.9634	rmccs	B-V.0	1977	293.6	9932	373	20.0	yes	no	no	no	no
3006.50d	5.9634	drmccs	B-V.0	1977	293.6	8716	263	20.0	yes	no	no	no	no
3006.60c	5.9634	endf60	B-VI.1	1989	293.6	12385	498	20.0	yes	no	no	no	no
** Li-7 **													
3007.42c	6.9557	endl92	LLNL	<1992	300.0	5834	141	30.0	yes	no	no	no	no
3007.50c	6.9557	endf5p	B-V.0	1972	293.6	4864	343	20.0	yes	no	no	no	no
3007.50d	6.9557	dre5	B-V.0	1972	293.6	4935	263	20.0	yes	no	no	no	no
3007.55c	6.9557	rmccs	B-V.2	1979	293.6	13171	328	20.0	yes	no	no	no	no
3007.55d	6.9557	drmccs	B-V.2	1979	293.6	12647	263	20.0	yes	no	no	no	no
3007.60c	6.9557	endf60	B-VI.0	1988	293.6	14567	387	20.0	yes	no	no	no	no
Z = 4 ***** Beryllium *****													
** Be-7 **													
4007.35c	6.9567	endl85	LLNL	<1985	0.0	1834	180	20.0	no	no	no	no	no
4007.42c	6.9567	endl92	LLNL	<1992	300.0	1544	127	30.0	yes	no	no	no	no
** Be-9 **													
4009.21c	8.9348	100xs[3]	T-2:X-5	1989	300.0	28964	316	100.0	yes	no	no	no	no
4009.50c	8.9348	rmccs	B-V.0	1976	293.6	8886	329	20.0	yes	no	no	no	no
4009.50d	8.9348	drmccs	B-V.0	1976	293.6	8756	263	20.0	yes	no	no	no	no
4009.60c	8.9348	endf60	B-VI.0	1986	293.6	64410	276	20.0	yes	no	no	no	no
Z = 5 ***** Boron *****													

APPENDIX G MCNP NEUTRON CROSS-SECTION LIBRARIES

TABLE G-2 (Cont.)
Continuous-Energy and Discrete Neutron Data Libraries Maintained by X-5

ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	$\bar{U}$	CP	DN	UR
** B-10 **													
5010.42c	9.9269	endl92	LLNL	<1992	300.0	4733	175	30.0	yes	no	no	no	no
5010.50c	9.9269	rmccs	B-V.0	1977	293.6	20200	514	20.0	yes	no	no	no	no
5010.50d	9.9269	drmcscs	B-V.0	1977	293.6	12322	263	20.0	yes	no	no	no	no
5010.53c	9.9269	endl5mt[1]	B-V.0	1977	587.2	23676	700	20.0	yes	no	no	no	no
5010.60c	9.9269	endlf60	B-VI.1	1989	293.6	27957	673	20.0	yes	no	no	no	no
** B-11 **													
5011.35c	10.9147	endl85	LLNL	<1985	0.0	4289	247	20.0	yes	no	no	no	no
5011.42c	10.9147	endl92	LLNL	<1992	300.0	4285	244	30.0	yes	no	no	no	no
5011.50c	10.9150	endlf5p	B-V.0	1974	293.6	4344	487	20.0	no	no	no	no	no
5011.50d	10.9150	drf5	B-V.0	1974	293.6	2812	263	20.0	no	no	no	no	no
5011.55c	10.9150	rmccsa	B-V.0:T-2	1971[4]	293.6	12254	860	20.0	yes	no	no	no	no
5011.55d	10.9150	drmcscs	B-V.0:T-2	1971[4]	293.6	7106	263	20.0	yes	no	no	no	no
5011.56c	10.9147	ncwxs	T-2	1986	293.6	56929	1762	20.0	yes	no	no	no	no
5011.56d	10.9147	newxsd	T-2	1986	293.6	17348	263	20.0	yes	no	no	no	no
5011.60c	10.9147	endlf60	B-VI.0	1989	293.6	108351	2969	20.0	yes	no	no	no	no
Z = 6 ***** Carbon *****													
** C-nat **													
6000.50c	11.8969	rmccs	B-V.0	1977	293.6	23326	875	20.0	yes	no	no	no	no
6000.50d	11.8969	drmcscs	B-V.0	1977	293.6	16844	263	20.0	yes	no	no	no	no
6000.60c	11.8980	endlf60	B-VI.1	1989	293.6	22422	978	32.0	yes	no	no	no	no
** C-12 **													
6012.21c	11.8969	100xs[3]	T-2:X-5	1989	300.0	28809	919	100.0	yes	no	no	no	no
6012.35c	11.8969	endl85	LLNL	<1985	0.0	5154	225	20.0	yes	no	no	no	no
6012.42c	11.8969	endl92	LLNL	<1992	300.0	6229	191	30.0	yes	no	no	no	no
6012.50c	11.8969	rmccs[5]	B-V.0	1977	293.6	23326	875	20.0	yes	no	no	no	no
6012.50d	11.8969	drmcscs[5]	B-V.0	1977	293.6	16844	263	20.0	yes	no	no	no	no
** C-13 **													
6013.35c	12.8916	endl85	LLNL	<1985	0.0	4886	395	20.0	yes	no	no	no	no
6013.42c	12.8916	endl92	LLNL	<1992	300.0	5993	429	30.0	yes	no	no	no	no
Z = 7 ***** Nitrogen *****													
** N-14 **													
7014.42c	13.8828	endl92	LLNL	<1992	300.0	20528	770	30.0	yes	no	no	no	no
7014.50c	13.8830	rmccs	B-V.0	1973	293.6	45457	1196	20.0	yes	no	no	no	no
7014.50d	13.8830	drmcscs	B-V.0	1973	293.6	26793	263	20.0	yes	no	no	no	no
7014.60c	13.8828	endlf60	T-2	1992	293.6	60397	1379	20.0	yes	no	no	no	no
** N-15 **													
7015.42c	14.8713	endl92	LLNL	<1992	300.0	22590	352	30.0	yes	no	no	no	no
7015.55c	14.8710	rmccsa	T-2	1983	293.6	20920	744	20.0	yes	no	no	no	no
7015.55d	14.8710	drmcscs	T-2	1983	293.6	15273	263	20.0	yes	no	no	no	no
7015.60c	14.8710	endlf60	B-VI.0	1993	293.6	24410	653	20.0	yes	no	no	no	no
Z = 8 ***** Oxygen *****													
** O-16 **													
8016.21c	15.8575	100xs[3]	T-2:X-5	1989	300.0	45016	1427	100.0	yes	no	no	no	no
8016.35c	15.8575	endl85	LLNL	<1985	0.0	10357	465	20.0	yes	no	no	no	no
8016.42c	15.8575	endl92	LLNL	<1992	300.0	9551	337	30.0	yes	no	no	no	no
8016.50c	15.8580	rmccs	B-V.0	1972	293.6	37942	1391	20.0	yes	no	no	no	no
8016.50d	15.8580	drmcscs	B-V.0	1972	293.6	20455	263	20.0	yes	no	no	no	no
8016.53c	15.8580	endl5mt[1]	B-V.0	1972	587.2	37989	1398	20.0	yes	no	no	no	no
8016.54c	15.8580	endl5mt[1]	B-V.0	1972	880.8	38017	1402	20.0	yes	no	no	no	no
8016.60c	15.8532	endlf60	B-VI.0	1990	293.6	58253	1609	20.0	yes	no	no	no	no
** O-17 **													
8017.60c	16.8531	endlf60	B-VI.0	1978	293.6	4200	335	20.0	no	no	no	no	no
Z = 9 ***** Fluorine *****													
** F-19 **													
9019.35c	18.8352	endl85	LLNL	<1985	0.0	31547	1452	20.0	yes	no	no	no	no
9019.42c	18.8352	endl92	LLNL	<1992	300.0	37814	1118	30.0	yes	no	no	no	no
9019.50c	18.8350	endlf5p	B-V.0	1976	293.6	44130	1569	20.0	yes	no	no	no	no

APPENDIX G MCNP NEUTRON CROSS-SECTION LIBRARIES

TABLE G-2. (Cont.)
Continuous-Energy and Discrete Neutron Data Libraries Maintained by X-5

ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	$\bar{U}$	CP	DN	UR
9019.50d	18.8350	dre5	B-V.0	1976	293.6	23156	263	20.0	yes	no	no	no	no
9019.51c	18.8350	rmccs	B-V.0	1976	293.6	41442	1541	20.0	yes	no	no	no	no
9019.51d	18.8350	drmcsc	B-V.0	1976	293.6	23156	263	20.0	yes	no	no	no	no
9019.60c	18.8350	endf60	B-VI.0	1990	300.0	93826	1433	20.0	yes	no	no	no	no
Z = 10 ***** Neon *****													
** Ne-20 **													
10020.42c	19.8207	endf92	LLNL	<1992	300.0	14286	1011	30.0	yes	no	no	no	no
Z = 11 ***** Sodium *****													
** Na-23 **													
11023.35c	22.7923	endf85	LLNL	<1985	0.0	22777	1559	20.0	yes	no	no	no	no
11023.42c	22.7923	endf92	LLNL	<1992	300.0	19309	1163	30.0	yes	no	no	no	no
11023.50c	22.7920	endf5p	B-V.0	1977	293.6	52252	2703	20.0	yes	no	no	no	no
11023.50d	22.7920	dre5	B-V.0	1977	293.6	41665	263	20.0	yes	no	no	no	no
11023.51c	22.7920	rmccs	B-V.0	1977	293.6	48863	2228	20.0	yes	no	no	no	no
11023.51d	22.7920	drmcsc	B-V.0	1977	293.6	41665	263	20.0	yes	no	no	no	no
11023.60c	22.7920	endf60	B-VI.1	1977	293.6	50294	2543	20.0	yes	no	no	no	no
Z = 12 ***** Magnesium *****													
** Mg-nat **													
12000.35c	24.0962	endf85	LLNL	<1985	0.0	9686	675	20.0	yes	no	no	no	no
12000.42c	24.0962	endf92	LLNL	<1992	300.0	9288	468	30.0	yes	no	no	no	no
12000.50c	24.0963	endf5u	B-V.0	1978	293.6	56334	2430	20.0	yes	no	no	no	no
12000.50d	24.0963	dre5	B-V.0	1978	293.6	14070	263	20.0	yes	no	no	no	no
12000.51c	24.0963	rmccs	B-V.0	1978	293.6	48917	1928	20.0	yes	no	no	no	no
12000.51d	24.0963	drmcsc	B-V.0	1978	293.6	14070	263	20.0	yes	no	no	no	no
12000.60c	24.0963	endf60	B-VI.0	1978	293.6	55776	2525	20.0	yes	no	no	no	no
Z = 13 ***** Aluminum *****													
** Al-27 **													
13027.21c	26.7498	100xs[3]	T-2:X-5	1989	300.0	35022	1473	100.0	yes	no	no	no	no
13027.35c	26.7498	endf85	LLNL	<1985	0.0	36895	2038	20.0	yes	no	no	no	no
13027.42c	26.7498	endf92	LLNL	<1992	300.0	32388	1645	30.0	yes	no	no	no	no
13027.50c	26.7500	rmccs	B-V.0	1973	293.6	54162	2028	20.0	yes	no	no	no	no
13027.50d	26.7500	drmcsc	B-V.0	1973	293.6	41947	263	20.0	yes	no	no	no	no
13027.60c	26.7500	endf60	B-VI.0	1973	293.6	55427	2241	20.0	yes	no	no	no	no
Z = 14 ***** Silicon *****													
** Si-nat **													
14000.21c	27.8440	100xs[3]	T-2:X-5	1989	300.0	76399	2883	100.0	yes	no	no	no	no
14000.35c	27.8442	endf85	LLNL	<1985	0.0	19016	1012	20.0	yes	no	no	no	no
14000.42c	27.8442	endf92	LLNL	<1992	300.0	16696	855	30.0	yes	no	no	no	no
14000.50c	27.8440	endf5p	B-V.0	1976	293.6	98609	2440	20.0	yes	no	no	no	no
14000.50d	27.8440	dre5	B-V.0	1976	293.6	69498	263	20.0	yes	no	no	no	no
14000.51c	27.8440	rmccs	B-V.0	1976	293.6	88129	1887	20.0	yes	no	no	no	no
14000.51d	27.8440	drmcsc	B-V.0	1976	293.6	69498	263	20.0	yes	no	no	no	no
14000.60c	27.8440	endf60	B-VI.0	1976	293.6	104198	2824	20.0	yes	no	no	no	no
Z = 15 ***** Phosphorus *****													
** P-31 **													
15031.35c	30.7077	endf85	LLNL	<1985	0.0	5875	303	20.0	yes	no	no	no	no
15031.42c	30.7077	endf92	LLNL	<1992	300.0	6805	224	30.0	yes	no	no	no	no
15031.50c	30.7080	endf5u	B-V.0	1977	293.6	5733	326	20.0	yes	no	no	no	no
15031.50d	30.7080	dre5	B-V.0	1977	293.6	5761	263	20.0	yes	no	no	no	no
15031.51c	30.7080	rmccs	B-V.0	1977	293.6	5732	326	20.0	yes	no	no	no	no
15031.51d	30.7080	drmcsc	B-V.0	1977	293.6	5761	263	20.0	yes	no	no	no	no
15031.60c	30.7080	endf60	B-VI.0	1977	293.6	6715	297	20.0	yes	no	no	no	no

APPENDIX G MCNP NEUTRON CROSS-SECTION LIBRARIES

TABLE G-2 (Cont.)
Continuous-Energy and Discrete Neutron Data Libraries Maintained by X-5

ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	$\bar{v}$	CP	DN	UR
Z = 16 ***** Sulfur *****													
** S-nat **													
16000.60c	31.7882	endf60	B-VI.0	1979	293.6	108683	8382	20.0	yes	no	no	no	no
** S-32 **													
16032.35c	31.6974	endf85	LLNL	<1985	0.0	7054	357	20.0	yes	no	no	no	no
16032.42c	31.6974	endf92	LLNL	<1992	300.0	6623	307	30.0	yes	no	no	no	no
16032.50c	31.6970	endf5u	B-V.0	1977	293.6	6789	363	20.0	yes	no	no	no	no
16032.50d	31.6970	dre5	B-V.0	1977	293.6	6302	263	20.0	yes	no	no	no	no
16032.51c	31.6970	rmccs	B-V.0	1977	293.6	6780	362	20.0	yes	no	no	no	no
16032.51d	31.6970	drmcscs	B-V.0	1977	293.6	6302	263	20.0	yes	no	no	no	no
16032.60c	31.6970	endf60	B-VI.0	1977	293.6	7025	377	20.0	yes	no	no	no	no
Z = 17 ***** Chlorine *****													
** Cl-nat **													
17000.35c	35.1484	endf85	LLNL	<1985	0.0	12903	1014	20.0	yes	no	no	no	no
17000.42c	35.1484	endf92	LLNL	<1992	300.0	12012	807	30.0	yes	no	no	no	no
17000.50c	35.1480	endf5p	B-V.0	1967	293.6	23313	1499	20.0	yes	no	no	no	no
17000.50d	35.1480	dre5	B-V.0	1967	293.6	18209	263	20.0	yes	no	no	no	no
17000.51c	35.1480	rmccs	B-V.0	1967	293.6	21084	1375	20.0	yes	no	no	no	no
17000.51d	35.1480	drmcscs	B-V.0	1967	293.6	18209	263	20.0	yes	no	no	no	no
17000.60c	35.1480	endf60	B-VI.0	1967	293.6	24090	1816	20.0	yes	no	no	no	no
Z = 18 ***** Argon *****													
** Ar-nat **													
18000.35c	39.6048	rmccsa	LLNL	<1985	0.0	5585	259	20.0	yes	no	no	no	no
18000.35d	39.6048	drmcscs	LLNL	<1985	0.0	14703	263	20.0	yes	no	no	no	no
18000.42c	39.6048	endf92	LLNL	<1992	300.0	5580	152	30.0	yes	no	no	no	no
18000.59c	39.6048	misc5xs[6,7]	T-2	1982	293.6	3473	252	20.0	yes	no	no	no	no
Z = 19 ***** Potassium *****													
** K-nat **													
19000.35c	38.7624	endf85	LLNL	<1985	0.0	11130	714	20.0	yes	no	no	no	no
19000.42c	38.7624	endf92	LLNL	<1992	300.0	11060	544	30.0	yes	no	no	no	no
19000.50c	38.7660	endf5u	B-V.0	1974	293.6	22051	1243	20.0	yes	no	no	no	no
19000.50d	38.7660	dre5	B-V.0	1974	293.6	23137	263	20.0	yes	no	no	no	no
19000.51c	38.7660	rmccs	B-V.0	1974	293.6	18798	1046	20.0	yes	no	no	no	no
19000.51d	38.7660	drmcscs	B-V.0	1974	293.6	23137	263	20.0	yes	no	no	no	no
19000.60c	38.7660	endf60	B-VI.0	1974	293.6	24482	1767	20.0	yes	no	no	no	no
Z = 20 ***** Calcium *****													
** Ca-nat **													
20000.35c	39.7357	endf85	LLNL	<1985	0.0	12933	974	20.0	yes	no	no	no	no
20000.42c	39.7357	endf92	LLNL	<1992	300.0	13946	1002	30.0	yes	no	no	no	no
20000.50c	39.7360	endf5u	B-V.0	1976	293.6	62624	2394	20.0	yes	no	no	no	no
20000.50d	39.7360	dre5	B-V.0	1976	293.6	29033	263	20.0	yes	no	no	no	no
20000.51c	39.7360	rmccs	B-V.0	1976	293.6	53372	1796	20.0	yes	no	no	no	no
20000.51d	39.7360	drmcscs	B-V.0	1976	293.6	29033	263	20.0	yes	no	no	no	no
20000.60c	39.7360	endf60	B-VI.0	1980	293.6	76468	2704	20.0	yes	no	no	no	no
** Ca-40 **													
20040.21c	39.6193	100xs[3]	T-2:X-5	1989	300.0	53013	2718	100.0	yes	no	no	no	no
Z = 21 ***** Scandium *****													
** Sc-45 **													
21045.60c	44.5679	endf60	B-VI.2	1992	293.6	105627	10639	20.0	yes	no	no	no	no
Z = 22 ***** Titanium *****													
** Ti-nat **													
22000.35c	47.4885	endf85	LLNL	<1985	0.0	13421	1337	20.0	yes	no	no	no	no
22000.42c	47.4885	endf92	LLNL	<1992	300.0	8979	608	30.0	yes	no	no	no	no

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TABLE G-2 (Cont.)
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ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	$\bar{U}$	CP	DN	UR
22000.50c	47.4676	endf5u	B-V.0	1977	293.6	54801	4434	20.0	yes	no	no	no	no
22000.50d	47.4676	dre5	B-V.0	1977	293.6	10453	263	20.0	yes	no	no	no	no
22000.51c	47.4676	rmccs	B-V.0	1977	293.6	31832	1934	20.0	yes	no	no	no	no
22000.51d	47.4676	drmcscs	B-V.0	1977	293.6	10453	263	20.0	yes	no	no	no	no
22000.60c	47.4676	endf60	B-VI.0	1977	293.6	76454	7761	20.0	yes	no	no	no	no
Z = 23 ***** Vanadium *****													
** V-nat **													
23000.50c	50.5040	endf5u	B-V.0	1977	293.6	38312	2265	20.0	yes	no	no	no	no
23000.50d	50.5040	dre5	B-V.0	1977	293.6	8868	263	20.0	yes	no	no	no	no
23000.51c	50.5040	rmccs	B-V.0	1977	293.6	34110	1899	20.0	yes	no	no	no	no
23000.51d	50.5040	drmcscs	B-V.0	1977	293.6	8868	263	20.0	yes	no	no	no	no
23000.60c	50.5040	endf60	B-VI.0	1988	293.6	167334	8957	20.0	yes	no	no	no	no
** V-51 **													
23051.42c	50.5063	endl92	LLNL	<1992	300.0	94082	5988	30.0	yes	no	no	no	no
Z = 24 ***** Chromium *****													
** Cr-nat **													
24000.35c	51.5493	endl85	LLNL	<1985	0.0	9218	358	20.0	yes	no	no	no	no
24000.42c	51.5493	endl92	LLNL	<1992	300.0	12573	377	30.0	yes	no	no	no	no
24000.50c	51.5490	rmccs	B-V.0	1977	293.6	134454	11050	20.0	yes	no	no	no	no
24000.50d	51.5490	drmcscs	B-V.0	1977	293.6	30714	263	20.0	yes	no	no	no	no
** Cr-50 **													
24050.60c	49.5170	endf60	B-VI.1	1989	293.6	119178	11918	20.0	yes	no	no	no	no
** Cr-52 **													
24052.60c	51.4940	endf60	B-VI.1	1989	293.6	117680	10679	20.0	yes	no	no	no	no
** Cr-53 **													
24053.60c	52.4860	endf60	B-VI.1	1989	293.6	114982	10073	20.0	yes	no	no	no	no
** Cr-54 **													
24054.60c	53.4760	endf60	B-VI.1	1989	293.6	98510	9699	20.0	yes	no	no	no	no
Z = 25 ***** Manganese *****													
** Mn-55 **													
25055.35c	54.4661	endl85	LLNL	<1985	0.0	7493	446	20.0	yes	no	no	no	no
25055.42c	54.4661	endl92	LLNL	<1992	300.0	10262	460	30.0	yes	no	no	no	no
25055.50c	54.4661	endf5u	B-V.0	1977	293.6	105093	12525	20.0	yes	no	no	no	no
25055.50d	54.4661	dre5	B-V.0	1977	293.6	9681	263	20.0	yes	no	no	no	no
25055.51c	54.4661	rmccs	B-V.0	1977	293.6	25727	1578	20.0	yes	no	no	no	no
25055.51d	54.4661	drmcscs	B-V.0	1977	293.6	9681	263	20.0	yes	no	no	no	no
25055.60c	54.4661	endf60	B-VI.0	1988	293.6	184269	8207	20.0	yes	no	no	no	no
Z = 26 ***** Iron *****													
** Fe-nat **													
26000.21c	55.3650	100xs[3]	T-2:X-5	1989	300.0	149855	15598	100.0	yes	no	no	no	no
26000.35c	55.3672	endl85	LLNL	<1985	0.0	30983	2772	20.0	yes	no	no	no	no
26000.42c	55.3672	endl92	LLNL	<1992	300.0	38653	3385	30.0	yes	no	no	no	no
26000.50c	55.3650	endf5p	B-V.0	1978	293.6	115447	10957	20.0	yes	no	no	no	no
26000.50d	55.3650	dre5	B-V.0	1978	293.6	33896	263	20.0	yes	no	no	no	no
26000.55c	55.3650	rmccs	T-2	1986	293.6	178392	6899	20.0	yes	no	no	no	no
26000.55d	55.3650	drmcscs	T-2	1986	293.6	72632	263	20.0	yes	no	no	no	no
** Fe-54 **													
26054.60c	53.4760	endf60	B-VI.1	1989	293.6	121631	10701	20.0	yes	no	no	no	no
** Fe-56 **													
26056.60c	55.4540	endf60	B-VI.1	1989	293.6	174517	11618	20.0	yes	no	no	no	no
** Fe-57 **													
26057.60c	56.4460	endf60	B-VI.1	1989	293.6	133995	7606	20.0	yes	no	no	no	no
** Fe-58 **													
26058.60c	57.4360	endf60	B-VI.1	1989	293.6	93450	6788	20.0	yes	no	no	no	no
Z = 27 ***** Cobalt *****													
** Co-59 **													

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TABLE G-2 (Cont.)
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ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	$\bar{U}$	CP	DN	UR
27059.35c	58.4269	endl85	LLNL	<1985	0.0	38958	4177	20.0	yes	no	no	no	no
27059.42c	58.4269	endl92	LLNL	<1992	300.0	119231	13098	30.0	yes	no	no	no	no
27059.50c	58.4269	endf5u	B-V.0	1977	293.6	117075	14502	20.0	yes	no	no	no	no
27059.50d	58.4269	dre5	B-V.0	1977	293.6	11769	263	20.0	yes	no	no	no	no
27059.51c	58.4269	rmccs	B-V.0	1977	293.6	28355	1928	20.0	yes	no	no	no	no
27059.51d	58.4269	drmcsc	B-V.0	1977	293.6	11769	263	20.0	yes	no	no	no	no
27059.60c	58.4269	endf60	B-VI.2	1992	293.6	186618	11838	20.0	yes	no	no	no	no
Z = 28 ***** Nickel *****													
** Ni-nat **													
28000.42c	58.1957	endl92	LLNL	<1992	300.0	44833	3116	30.0	yes	no	no	no	no
28000.50c	58.1826	rmccs	B-V.0	1977	293.6	139913	8927	20.0	yes	no	no	no	no
28000.50d	58.1826	drmcsc	B-V.0	1977	293.6	21998	263	20.0	yes	no	no	no	no
** Ni-58 **													
28058.35c	57.4376	endl85	LLNL	<1985	0.0	42744	4806	20.0	yes	no	no	no	no
28058.42c	57.4376	endl92	LLNL	<1992	300.0	38930	4914	30.0	yes	no	no	no	no
28058.60c	57.4380	endf60	B-VI.1	1989	293.6	172069	16445	20.0	yes	no	no	no	no
** Ni-60 **													
28060.60c	59.4160	endf60	B-VI.1	1991	293.6	110885	10055	20.0	yes	no	no	no	no
** Ni-61 **													
28061.60c	60.4080	endf60	B-VI.1	1989	293.6	93801	5882	20.0	yes	no	no	no	no
** Ni-62 **													
28062.60c	61.3960	endf60	B-VI.1	1989	293.6	82085	7230	20.0	yes	no	no	no	no
** Ni-64 **													
28064.60c	63.3790	endf60	B-VI.1	1989	293.6	66656	6144	20.0	yes	no	no	no	no
Z = 29 ***** Copper *****													
** Cu-nat **													
29000.35c	63.0001	endl85	LLNL	<1985	0.0	7039	293	20.0	yes	no	no	no	no
29000.50c	63.5460	rmccs	B-V.0	1978	293.6	51850	3435	20.0	yes	no	no	no	no
29000.50d	63.5460	drmcsc	B-V.0	1978	293.6	12777	263	20.0	yes	no	no	no	no
** Cu-63 **													
29063.60c	62.3890	endf60	B-VI.2	1989	293.6	119097	11309	20.0	yes	no	no	no	no
** Cu-65 **													
29065.60c	64.3700	endf60	B-VI.2	1989	293.6	118385	11801	20.0	yes	no	no	no	no
Z = 30 ***** Zinc *****													
** Zn-nat **													
30000.40c	64.8183	endl92	LLNL	<1992	300.0	271897	33027	30.0	yes	no	no	no	no
30000.42c	64.8183	endl92	LLNL:X-5	<1992	300.0	271897	33027	30.0	yes	no	no	no	no
Z = 31 ***** Gallium *****													
** Ga-nat **													
31000.35c	69.1211	endl85	LLNL	<1985	0.0	7509	469	20.0	yes	no	no	no	no
31000.42c	69.1211	endl92	LLNL	<1992	300.0	6311	219	30.0	yes	no	no	no	no
31000.50c	69.1211	rmccs	B-V.0	1980	293.6	7928	511	20.0	yes	no	no	no	no
31000.50d	69.1211	drmcsc	B-V.0	1980	293.6	6211	263	20.0	yes	no	no	no	no
31000.60c	69.1211	endf60	B-VI.0	1980	293.6	9228	566	20.0	yes	no	no	no	no
Z = 33 ***** Arsenic *****													
** As-74 **													
33074.35c	73.2889	endl85	LLNL	<1985	0.0	50881	6424	20.0	yes	no	no	no	no
33074.42c	73.2889	endl92	LLNL	<1992	300.0	55752	6851	30.0	yes	no	no	no	no
** As-75 **													
33075.35c	74.2780	rmccsa	B-V.0	1974	0.0	50931	6421	20.0	yes	no	no	no	no
33075.35d	74.2780	drmcsc	B-V.0	1974	0.0	8480	263	20.0	yes	no	no	no	no
33075.42c	74.2780	endl92	LLNL	<1992	300.0	56915	6840	30.0	yes	no	no	no	no
Z = 35 ***** Bromine *****													
** Br-79 **													

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ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	$\bar{U}$	CP	DN	UR
35079.55c	78.2404	misc5xs[6,8]	T-2	1982	293.6	10431	1589	20.0	no	no	no	no	no
** Br-81 **													
35081.55c	80.2212	misc5xs[6,8]	T-2	1982	293.6	5342	831	20.0	no	no	no	no	no
Z = 36 ***** Krypton *****													
** Kr-78 **													
36078.50c	77.2510	rmccsa	B-V.0	1978	293.6	9057	939	20.0	no	no	no	no	no
36078.50d	77.2510	drmcscs	B-V.0	1978	293.6	4358	263	20.0	no	no	no	no	no
** Kr-80 **													
36080.50c	79.2298	rmccsa	B-V.0	1978	293.6	10165	1108	20.0	no	no	no	no	no
36080.50d	79.2298	drmcscs	B-V.0	1978	293.6	4276	263	20.0	no	no	no	no	no
** Kr-82 **													
36082.50c	81.2098	rmccsa	B-V.0	1978	293.6	7220	586	20.0	no	no	no	no	no
36082.50d	81.2098	drmcscs	B-V.0	1978	293.6	4266	263	20.0	no	no	no	no	no
36082.59c	81.2098	misc5xs[6,7]	T-2	1982	293.6	7010	499	20.0	yes	no	no	no	no
** Kr-83 **													
36083.50c	82.2018	rmccsa	B-V.0	1978	293.6	8078	811	20.0	no	no	no	no	no
36083.50d	82.2018	drmcscs	B-V.0	1978	293.6	4359	263	20.0	no	no	no	no	no
36083.59c	82.2018	misc5xs[6,7]	T-2	1982	293.6	8069	704	20.0	yes	no	no	no	no
** Kr-84 **													
36084.50c	83.1906	rmccsa	B-V.0	1978	293.6	9364	944	20.0	no	no	no	no	no
36084.50d	83.1906	drmcscs	B-V.0	1978	293.6	4463	263	20.0	no	no	no	no	no
36084.59c	83.1906	misc5xs[6,7]	T-2	1982	293.6	10370	954	20.0	yes	no	no	no	no
** Kr-86 **													
36086.50c	85.1726	rmccsa	B-V.0	1975	293.6	10416	741	20.0	no	no	no	no	no
36086.50d	85.1726	drmcscs	B-V.0	1975	293.6	4301	263	20.0	no	no	no	no	no
36086.59c	85.1726	misc5xs[6,7]	T-2	1982	293.6	8740	551	20.0	yes	no	no	no	no
Z = 37 ***** Rubidium *****													
** Rb-85 **													
37085.55c	84.1824	misc5xs[6,8]	T-2	1982	293.6	27304	4507	20.0	no	no	no	no	no
** Rb-87 **													
37087.55c	86.1626	misc5xs[6,8]	T-2	1982	293.6	8409	1373	20.0	no	no	no	no	no
Z = 39 ***** Yttrium *****													
** Y-88 **													
39088.35c	87.1543	endl85	LLNL	<1985	0.0	11299	272	20.0	yes	no	no	no	no
39088.42c	87.1543	endl92	LLNL	<1992	300.0	11682	181	30.0	yes	no	no	no	no
** Y-89 **													
39089.35c	88.1421	misc5xs[6]	LLNL	<1985	0.0	49885	6154	20.0	yes	no	no	no	no
39089.42c	88.1421	endl92	LLNL	<1992	300.0	69315	8771	30.0	yes	no	no	no	no
39089.50c	88.1421	endl5u	B-V.0[9]	1985	293.6	18631	3029	20.0	no	no	no	no	no
39089.50d	88.1421	dre5	B-V.0[9]	1985	293.6	2311	263	20.0	no	no	no	no	no
39089.60c	88.1420	endl60	B-V.0	1986	293.6	86556	9567	20.0	yes	no	no	no	no
Z = 40 ***** Zirconium *****													
** Zr-nat **													
40000.35c	90.4364	endl85	LLNL	<1985	0.0	14738	1292	20.0	yes	no	no	no	no
40000.42c	90.4364	endl92	LLNL	<1992	300.0	131855	17909	30.0	yes	no	no	no	no
40000.56c	90.4360	misc5xs[6,10]	B-V:X-5	1976	300.0	52064	7944	20.0	no	no	no	no	no
40000.56d	90.4360	misc5xs[6,10]	B-V:X-5	1976	300.0	5400	263	20.0	no	no	no	no	no
40000.57c	90.4360	misc5xs[6,10]	B-V:X-5	1976	300.0	16816	2116	20.0	no	no	no	no	no
40000.57d	90.4360	misc5xs[6,10]	B-V:X-5	1976	300.0	5400	263	20.0	no	no	no	no	no
40000.58c	90.4360	misc5xs[6,10]	B-V:X-5	1976	587.2	57528	8777	20.0	no	no	no	no	no
40000.60c	90.4360	endl60	B-V.0	1976[10]	293.6	66035	10298	20.0	no	no	no	no	no
** Zr-93 **													
40093.50c	92.1083	kidman	B-V.0	1974	293.6	2579	236	20.0	no	no	no	no	no
Z = 41 ***** Niobium *****													
** Nb-93 **													
41093.35c	92.1083	endl85	LLNL	<1985	0.0	50441	6095	20.0	yes	no	no	no	no

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ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	U	CP	DN	UR
41093.42c	92.1083	endl92	LLNL	<1992	300.0	73324	9277	30.0	yes	no	no	no	no
41093.50c	92.1051	endl5p	B-V.0	1974	293.6	128960	17279	20.0	yes	no	no	no	no
41093.50d	92.1051	dre5	B-V.0	1974	293.6	10332	263	20.0	yes	no	no	no	no
41093.51c	92.1051	rmccs	B-V.0	1974	293.6	14675	963	20.0	yes	no	no	no	no
41093.51d	92.1051	drmcscs	B-V.0	1974	293.6	10332	263	20.0	yes	no	no	no	no
41093.60c	92.1051	endlf60	B-VI.1	1990	293.6	110269	10678	20.0	yes	no	no	no	no
Z = 42 ***** Molybdenum *****													
** Mo-nat **													
42000.35c	95.1158	endl85	LLNL	<1985	0.0	8628	573	20.0	yes	no	no	no	no
42000.42c	95.1158	endl92	LLNL	<1992	300.0	9293	442	30.0	yes	no	no	no	no
42000.50c	95.1160	endl5u	B-V.0	1979	293.6	35634	4260	20.0	yes	no	no	no	no
42000.50d	95.1160	dre5	B-V.0	1979	293.6	7754	263	20.0	yes	no	no	no	no
42000.51c	95.1160	rmccs	B-V.0	1979	293.6	10139	618	20.0	yes	no	no	no	no
42000.51d	95.1160	drmcscs	B-V.0	1979	293.6	7754	263	20.0	yes	no	no	no	no
42000.60c	95.1160	endlf60	B-VI.0	1979	293.6	45573	5466	20.0	yes	no	no	no	no
** Mo-95 **													
42095.50c	94.0906	kidman	B-V.0	1980	293.6	15411	2256	20.0	no	no	no	no	no
Z = 43 ***** Technetium *****													
** Tc-99 **													
43099.50c	98.1500	kidman	B-V.0	1978	293.6	12152	1640	20.0	no	no	no	no	no
43099.60c	98.1500	endlf60	B-VI.0	1978	293.6	54262	8565	20.0	no	no	no	no	no
Z = 44 ***** Ruthenium *****													
** Ru-101 **													
44101.50c	100.0390	kidman	B-V.0	1980	293.6	5299	543	20.0	no	no	no	no	no
** Ru-103 **													
44103.50c	102.0220	kidman	B-V.0	1974	293.6	3052	235	20.0	no	no	no	no	no
Z = 45 ***** Rhodium *****													
** Rh-103 **													
45103.50c	102.0210	rmccsa	B-V.0	1978	293.6	18870	2608	20.0	no	no	no	no	no
45103.50d	102.0210	drmcscs	B-V.0	1974	293.6	4663	263	20.0	no	no	no	no	no
** Rh-105 **													
45105.50c	104.0050	kidman	B-V.0	1974	293.6	1591	213	20.0	no	no	no	no	no
Z = 45 ***** Average fission product from Uranium-235 *****													
** U-235 fp **													
45117.90c	115.5446	rmccs	T-2	1982	293.6	10314	399	20.0	yes	no	no	no	no
45117.90d	115.5446	drmcscs	T-2	1982	293.6	9507	263	20.0	yes	no	no	no	no
Z = 46 ***** Palladium *****													
** Pd-105 **													
46105.50c	104.0040	kidman	B-V.0	1980	293.6	4647	505	20.0	no	no	no	no	no
** Pd-108 **													
46108.50c	106.9770	kidman	B-V.0	1980	293.6	4549	555	20.0	no	no	no	no	no
Z = 46 ***** Average fission product from Plutonium-239 *****													
** Pu-239 fp **													
46119.90c	117.5255	rmccs	T-2	1982	293.6	10444	407	20.0	yes	no	no	no	no
46119.90d	117.5255	drmcscs	T-2	1982	293.6	9542	263	20.0	yes	no	no	no	no
Z = 47 ***** Silver *****													
** Ag-nat **													
47000.55c	106.9420	rmccsa	T-2	1984	293.6	29092	2350	20.0	yes	no	no	no	no
47000.55d	106.9420	drmcscs	T-2	1984	293.6	12409	263	20.0	yes	no	no	no	no

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TABLE G-2 (Cont.)
Continuous-Energy and Discrete Neutron Data Libraries Maintained by X-5

ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	$\bar{v}$	CP	DN	UR
** Ag-107 **													
47107.35c	105.9867	endl85	LLNL	<1985	0.0	13134	994	20.0	yes	no	no	no	no
47107.42c	105.9867	endl92	LLNL	<1992	300.0	27108	2885	30.0	yes	no	no	no	no
47107.50c	105.9870	rmccsa	B-V.0	1978	293.6	12111	1669	20.0	no	no	no	no	no
47107.50d	105.9870	drmcscs	B-V.0	1978	293.6	4083	263	20.0	no	no	no	no	no
47107.60c	105.9870	endlf60	B-VI.0	1983	293.6	64008	10101	20.0	no	no	no	no	no
** Ag-109 **													
47109.35c	107.9692	endl85	LLNL	<1985	0.0	13452	1094	20.0	yes	no	no	no	no
47109.42c	107.9692	endl92	LLNL	<1992	300.0	33603	3796	30.0	yes	no	no	no	no
47109.50c	107.9690	rmccsa	B-V.0	1978	293.6	14585	2120	20.0	no	no	no	no	no
47109.50d	107.9690	drmcscs	B-V.0	1978	293.6	3823	263	20.0	no	no	no	no	no
47109.60c	107.9690	endlf60	B-VI.0	1983	293.6	76181	11903	20.0	no	no	no	no	no
Z = 48 ***** Cadmium *****													
** Cd-nat **													
48000.35c	111.4443	endl85	LLNL	<1985	0.0	12283	1115	20.0	yes	no	no	no	no
48000.42c	111.4443	endl92	LLNL	<1992	300.0	211537	29369	30.0	yes	no	no	no	no
48000.50c	111.4600	endlf5u	B-V.0	1974	293.6	19714	2981	20.0	no	no	no	no	no
48000.50d	111.4600	dre5	B-V.0	1974	293.6	3026	263	20.0	no	no	no	no	no
48000.51c	111.4600	rmccs	B-V.0	1974	293.6	6734	818	20.0	no	no	no	no	no
48000.51d	111.4600	drmcscs	B-V.0	1974	293.6	3026	263	20.0	no	no	no	no	no
Z = 49 ***** Indium *****													
** In-nat **													
49000.42c	113.8336	endl92	LLNL	<1992	300.0	65498	7870	30.0	yes	no	no	no	no
49000.60c	113.8340	endlf60	B-VI.0	1990	293.6	93662	10116	20.0	yes	no	no	no	no
Z = 49-50 ***** Fission products *****													
** Ave fp **													
49120.42c	116.4906	endl92fp[11]	LLNL	<1992	300.0	12755	164	30.0	yes	no	no	no	no
49125.42c	116.4906	endl92fp[11]	LLNL	<1992	300.0	9142	119	30.0	yes	no	no	no	no
50120.35c	116.4906	rmccs	LLNL	<1985	0.0	8366	232	20.0	yes	no	no	no	no
50120.35d	116.4906	drmcscs	LLNL	<1985	0.0	8963	263	20.0	yes	no	no	no	no
Z = 50 ***** Tin *****													
** Sn-nat **													
50000.35c	117.6704	endl85	LLNL	<1985	0.0	5970	205	20.0	yes	no	no	no	no
50000.40c	117.6704	endl92	LLNL	<1992	300.0	248212	34612	30.0	yes	no	no	no	no
50000.42c	117.6704	endl92	LLNL:X-5	<1992	300.0	248212	34612	30.0	yes	no	no	no	no
Z = 51 ***** Antimony *****													
** Sb-nat **													
51000.42c	120.7041	endl92	LLNL	<1992	300.0	95953	10721	30.0	yes	no	no	no	no
Z = 53 ***** Iodine *****													
** I-127 **													
53127.42c	125.8143	endl92	LLNL	<1992	300.0	76321	10	30.0	yes	no	no	no	no
53127.55c	125.8140	misc5xs[6,8]	T-2	1982	293.6	59725	9423	20.0	no	no	no	no	no
53127.60c	125.8143	endlf60[12]	T-2	1991	293.6	399760	7888	30.0	yes	no	no	no	no
** I-129 **													
53129.60c	127.7980	endlf60	B-VI.0	1980	293.6	8792	1237	20.0	no	no	no	no	no
** I-135 **													
53135.50c	133.7510	kidman	B-V.0	1974	293.6	1232	194	20.0	no	no	no	no	no
Z = 54 ***** Xenon *****													
** Xe-nat **													
54000.35c	130.1721	endl85	LLNL	<1985	0.0	41432	5228	20.0	yes	no	no	no	no
54000.42c	130.1721	endl92	LLNL	<1992	300.0	43411	5173	30.0	yes	no	no	no	no

APPENDIX G

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TABLE G-2 (Cont.)
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ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	U	CP	DN	UR
** Xe-131 **													
54131.50c	129.7810	kidman	B-V.0	1978	293.6	22572	3376	20.0	no	no	no	no	no
** Xe-134 **													
54134.35c	132.7551	endl85	LLNL	<1985	0.0	7463	359	20.0	yes	no	no	no	no
54134.42c	132.7551	endl92	LLNL	<1992	300.0	8033	192	30.0	yes	no	no	no	no
** Xe-135 **													
54135.50c	133.7480	endf5mt[1]	B-V	1975	293.6	5529	704	20.0	no	no	no	no	no
54135.53c	133.7480	endf5mt[1]	B-V	1975	587.2	5541	706	20.0	no	no	no	no	no
54135.54c	133.7480	endf5mt[1]	B-V	1975	880.8	5577	712	20.0	no	no	no	no	no
Z = 55 ***** Cesium *****													
** Cs-133 **													
55133.50c	131.7640	kidman	B-V.0	1978	293.6	26713	4142	20.0	no	no	no	no	no
55133.55c	131.7640	misc5xs[6,8]	T-2	1982	293.6	67893	11025	20.0	no	no	no	no	no
55133.60c	131.7640	endf60	B-VI.0	1978	293.6	54723	8788	20.0	no	no	no	no	no
** Cs-134 **													
55134.60c	132.7570	endf60	B-VI.0	1988	293.6	10227	1602	20.0	no	no	no	no	no
** Cs-135 **													
55135.50c	133.7470	kidman	B-V.0	1974	293.6	1903	199	20.0	no	no	no	no	no
55135.60c	133.7470	endf60	B-VI.0	1974	293.6	3120	388	20.0	no	no	no	no	no
** Cs-136 **													
55136.60c	134.7400	endf60	B-VI.0	1974	293.6	10574	1748	20.0	no	no	no	no	no
** Cs-137 **													
55137.60c	135.7310	endf60	B-VI.0	1974	293.6	2925	369	20.0	no	no	no	no	no
Z = 56 ***** Barium *****													
** Ba-138 **													
56138.35c	136.7206	endl85	LLNL	<1985	0.0	5985	262	20.0	yes	no	no	no	no
56138.50c	136.7150	rncs	B-V.0	1978	293.6	6018	292	20.0	yes	no	no	no	no
56138.50d	136.7150	drncs	B-V.0	1978	293.6	6320	263	20.0	yes	no	no	no	no
56138.60c	136.7150	endf60	B-VI.0	1978	293.6	7347	267	20.0	yes	no	no	no	no
Z = 59 ***** Praseodymium *****													
** Pr-141 **													
59141.50c	139.6970	kidman	B-V.0	1980	293.6	15620	1354	20.0	no	no	no	no	no
Z = 60 ***** Neodymium *****													
** Nd-143 **													
60143.50c	141.6820	kidman	B-V.0	1980	293.6	17216	1701	20.0	no	no	no	no	no
** Nd-145 **													
60145.50c	143.6680	kidman	B-V.0	1980	293.6	38473	3985	20.0	no	no	no	no	no
** Nd-147 **													
60147.50c	145.6540	kidman	B-V.0	1979	293.6	1816	251	20.0	no	no	no	no	no
** Nd-148 **													
60148.50c	146.6460	kidman	B-V.0	1980	293.6	10867	1054	20.0	no	no	no	no	no
Z = 61 ***** Promethium *****													
** Pm-147 **													
61147.50c	145.6530	kidman	B-V.0	1980	293.6	9152	825	20.0	no	no	no	no	no
** Pm-148 **													
61148.50c	146.6470	kidman	B-V.0	1979	293.6	1643	257	20.0	no	no	no	no	no
** Pm-149 **													
61149.50c	147.6390	kidman	B-V.0	1979	293.6	2069	238	20.0	no	no	no	no	no
Z = 62 ***** Samarium *****													
** Sm-147 **													
62147.50c	145.6530	kidman	B-V.0	1980	293.6	33773	2885	20.0	no	no	no	no	no
** Sm-149 **													
62149.49c	147.6380	ures	B-VI.0	1978	300.0	57787	7392	20.0	no	no	no	no	yes
62149.50c	147.6380	endf5u	B-V.0	1978	293.6	15662	2008	20.0	no	no	no	no	no

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TABLE G-2 (Cont.)
Continuous-Energy and Discrete Neutron Data Libraries Maintained by X-5

ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	$\bar{U}$	CP	DN	UR
62149.50d ** Sm-150 **	147.6380	dre5	B-V.0	1978	293.6	4429	263	20.0	no	no	no	no	no
62150.49c	148.6290	ures	B-V1.2	1992	300.0	60992	8183	20.0	no	no	no	no	yes
62150.50c ** Sm-151 **	148.6290	kidman	B-V.0	1974	293.6	9345	1329	20.0	no	no	no	no	no
62151.50c ** Sm-152 **	149.6230	kidman	B-V.0	1980	293.6	7303	605	20.0	no	no	no	no	no
62152.49c	150.6150	ures	B-V1.2	1992	300.0	203407	19737	20.0	no	no	no	no	yes
62152.50c	150.6150	kidman	B-V.0	1980	293.6	41252	4298	20.0	no	no	no	no	no
Z = 63 ***** Europium *****													
** Eu-nat **													
63000.35c	150.6546	rmccsa	LLNL	<1985	0.0	6926	364	20.0	yes	no	no	no	no
63000.35d	150.6546	drmcscs	LLNL	<1985	0.0	6654	263	20.0	yes	no	no	no	no
63000.42c ** Eu-151 **	150.6546	endf92	LLNL	<1992	300.0	37421	4498	30.0	yes	no	no	no	no
63151.49c	149.6230	ures	B-V1.0	1986	300.0	147572	10471	20.0	yes	no	no	no	yes
63151.50c	149.6230	rmccs	B-V.0	1977	293.6	68057	5465	20.0	yes	no	no	no	no
63151.50d	149.6230	drmcscs	B-V.0	1977	293.6	10013	263	20.0	yes	no	no	no	no
63151.55c	149.6230	newxs	T-2	1986	293.6	86575	4749	20.0	yes	no	no	no	no
63151.55d	149.6230	newxs	T-2	1986	293.6	35199	263	20.0	yes	no	no	no	no
63151.60c ** Eu-152 **	149.6230	endf60	B-V1.0	1986	293.6	96099	7394	20.0	yes	no	no	no	no
63152.49c	150.6200	ures	B-V1.0	1975	300.0	81509	6540	20.0	no	no	no	no	yes
63152.50c	150.6200	endf5u	B-V.0	1975	293.6	49313	4553	20.0	no	no	no	no	no
63152.50d ** Eu-153 **	150.6200	dre5	B-V.0	1975	293.6	5655	263	20.0	no	no	no	no	no
63153.49c	151.6080	ures	B-V1.0	1986	300.0	129446	8784	20.0	yes	no	no	no	yes
63153.50c	151.6070	rmccs	B-V.0	1978	293.6	55231	4636	20.0	yes	no	no	no	no
63153.50d	151.6070	drmcscs	B-V.0	1978	293.6	11244	263	20.0	yes	no	no	no	no
63153.55c	151.6080	newxs	T-2	1986	293.6	72971	4174	20.0	yes	no	no	no	no
63153.55d	151.6080	newxs	T-2	1986	293.6	36372	263	20.0	yes	no	no	no	no
63153.60c ** Eu-154 **	151.6080	endf60	B-V1.0	1986	293.6	86490	6198	20.0	yes	no	no	no	no
63154.49c	152.6000	ures	B-V1.0	1975	300.0	72804	6627	20.0	no	no	no	no	yes
63154.50c	152.6000	endf5u	B-V.0	1975	293.6	37008	4030	20.0	no	no	no	no	no
63154.50d ** Eu-155 **	152.6000	dre5	B-V.0	1975	293.6	5458	263	20.0	no	no	no	no	no
63155.50c	153.5920	kidman	B-V.0	1974	293.6	4532	273	20.0	no	no	no	no	no
Z = 64 ***** Gadolinium *****													
** Gd-nat **													
64000.35c	155.8991	rmccsa	LLNL	<1985	0.0	7878	454	20.0	yes	no	no	no	no
64000.35d ** Gd-152 **	155.8991	drmcscs	LLNL	<1985	0.0	6833	263	20.0	yes	no	no	no	no
64152.50c	150.6150	endf5u	B-V.0	1977	293.6	26251	3285	20.0	no	no	no	no	no
64152.50d	150.6150	dre5	B-V.0	1977	293.6	5899	263	20.0	no	no	no	no	no
64152.55c	150.6150	misc5xs[6,13]	B-V.0:T-2	1986	293.6	32590	3285	20.0	yes	no	no	no	no
64152.60c ** Gd-154 **	150.6150	endf60	B-V1.0	1977	293.6	32760	4391	20.0	no	no	no	no	no
64154.50c	152.5990	endf5u	B-V.0	1977	293.6	49572	7167	20.0	no	no	no	no	no
64154.50d	152.5990	dre5	B-V.0	1977	293.6	5930	263	20.0	no	no	no	no	no
64154.55c	152.5990	misc5xs[6,13]	B-V.0:T-2	1986	293.6	59814	7167	20.0	yes	no	no	no	no
64154.60c ** Gd-155 **	152.5990	endf60	B-V1.0	1977	293.6	67662	10189	20.0	no	no	no	no	no
64155.50c	153.5920	endf5u	B-V.0	1977	293.6	44965	6314	20.0	no	no	no	no	no
64155.50d	153.5920	dre5	B-V.0	1977	293.6	6528	263	20.0	no	no	no	no	no
64155.55c	153.5920	misc5xs[6,13]	B-V.0:T-2	1986	293.6	54346	6314	20.0	yes	no	no	no	no
64155.60c ** Gd-156 **	153.5920	endf60	B-V1.0	1977	293.6	61398	9052	20.0	no	no	no	no	no
64156.50c	154.5830	endf5u	B-V.0	1977	293.6	37371	3964	20.0	no	no	no	no	no
64156.50d	154.5830	dre5	B-V.0	1977	293.6	6175	263	20.0	no	no	no	no	no
64156.55c	154.5830	misc5xs[6,13]	B-V.0:T-2	1986	293.6	44391	3964	20.0	yes	no	no	no	no
64156.60c	154.5830	endf60	B-V1.0	1977	293.6	42885	5281	20.0	no	no	no	no	no

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TABLE G-2 (Cont.)
Continuous-Energy and Discrete Neutron Data Libraries Maintained by X-5

ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	$\bar{U}$	CP	DN	UR
** Gd-157 **													
64157.50c	155.5760	endf5u	B-V.0	1977	293.6	38975	5370	20.0	no	no	no	no	no
64157.50d	155.5760	dre5	B-V.0	1977	293.6	6346	263	20.0	no	no	no	no	no
64157.55c	155.5760	misc5xs[6,13]	B-V.0:T-2	1986	293.6	47271	5370	20.0	yes	no	no	no	no
64157.60c	155.5760	endf60	B-VI.0	1977	293.6	56957	8368	20.0	no	no	no	no	no
** Gd-158 **													
64158.50c	156.5670	endf5u	B-V.0	1977	293.6	95876	15000	20.0	no	no	no	no	no
64158.50d	156.5670	dre5	B-V.0	1977	293.6	5811	263	20.0	no	no	no	no	no
64158.55c	156.5670	misc5xs[6,13]	B-V.0:T-2	1986	293.6	113916	15000	20.0	yes	no	no	no	no
64158.60c	156.5670	endf60	B-VI.0	1977	293.6	59210	8909	20.0	no	no	no	no	no
** Gd-160 **													
64160.50c	158.5530	endf5u	B-V.0	1977	293.6	53988	8229	20.0	no	no	no	no	no
64160.50d	158.5530	dre5	B-V.0	1977	293.6	5030	263	20.0	no	no	no	no	no
64160.55c	158.5530	misc5xs[6,13]	B-V.0:T-2	1986	293.6	65261	8229	20.0	yes	no	no	no	no
64160.60c	158.5530	endf60	B-VI.0	1977	293.6	54488	8304	20.0	no	no	no	no	no
Z = 67 ***** Holmium *****													
** Ho-165 **													
67165.35c	163.5135	rmccsa	LLNL	<1985	0.0	54279	7075	20.0	yes	no	no	no	no
67165.35d	163.5135	drmcscs	LLNL	<1985	0.0	7019	263	20.0	yes	no	no	no	no
67165.42c	163.5135	endf92	LLNL	<1992	300.0	103467	13884	30.0	yes	no	no	no	no
67165.55c	163.5130	newxs	T-2	1986	293.6	56605	2426	30.0	yes	no	no	no	no
67165.55d	163.5130	newxsd	T-2	1986	293.6	42266	263	20.0	yes	no	no	no	no
67165.60c	163.5130	endf60	B-VI.0	1988	293.6	75307	4688	30.0	yes	no	no	no	no
Z = 69 ***** Thulium *****													
** Tm-169 **													
69169.55c	167.4830	misc5xs[6]	T-2	1986	300.0	47941	4738	20.0	no	no	no	no	no
Z = 72 ***** Hafnium *****													
** Hf-nat **													
72000.35c	176.9567	endf85	LLNL	<1985	0.0	75862	9636	20.0	yes	no	no	no	no
72000.42c	176.9567	endf92	LLNL	<1992	300.0	108989	14113	30.0	yes	no	no	no	no
72000.50c	176.9540	newxs	B-V.0	1976	293.6	52231	8270	20.0	no	no	no	no	no
72000.50d	176.9540	newxsd	B-V.0	1976	293.6	4751	263	20.0	no	no	no	no	no
72000.60c	176.9540	endf60	B-VI.0	1976	293.6	84369	13634	20.0	no	no	no	no	no
Z = 73 ***** Tantalum *****													
** Ta-181 **													
73181.35c	179.3936	endf85	LLNL	<1985	0.0	33547	2812	20.0	yes	no	no	no	no
73181.42c	179.3936	endf92	LLNL	<1992	300.0	47852	4927	30.0	yes	no	no	no	no
73181.50c	179.4000	endf5u	B-V.0	1972	293.6	60740	6341	20.0	yes	no	no	no	no
73181.50d	179.4000	dre5	B-V.0	1972	293.6	16361	263	20.0	yes	no	no	no	no
73181.51c	179.4000	rmccs	B-V.0	1972	293.6	21527	753	20.0	yes	no	no	no	no
73181.51d	179.4000	drmcscs	B-V.0	1972	293.6	16361	263	20.0	yes	no	no	no	no
73181.60c	179.4000	endf60	B-VI.0	1972	293.6	91374	10352	20.0	yes	no	no	no	no
** Ta-182 **													
73182.49c	180.3870	ures	B-VI.0	1971	300.0	20850	2463	20.0	no	no	no	no	yes
73182.60c	180.3870	endf60	B-VI.0	1971	293.6	12085	1698	20.0	no	no	no	no	no
Z = 74 ***** Tungsten *****													
** W-nat **													
74000.21c	182.2706	100xs[3]	T-2:X-5	1989	300.0	194513	21386	100.0	yes	no	no	no	no
74000.55c	182.2770	rmccs	B-V.2	1982	293.6	50639	1816	20.0	yes	no	no	no	no
74000.55d	182.2770	drmcscs	B-V.2	1982	293.6	34272	263	20.0	yes	no	no	no	no
** W-182 **													
74182.49c	180.3900	ures	B-VI.0	1980	300.0	150072	16495	20.0	yes	no	no	no	yes
74182.50c	180.3900	endf5p	B-V.0	1973	293.6	94367	11128	20.0	yes	no	no	no	no
74182.50d	180.3900	dre5	B-V.0	1973	293.6	17729	263	20.0	yes	no	no	no	no
74182.55c	180.3900	rmccsa	B-V.2	1980	293.6	122290	13865	20.0	yes	no	no	no	no
74182.55d	180.3900	drmcscs	B-V.2	1980	293.6	26387	263	20.0	yes	no	no	no	no

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TABLE G-2 (Cont.)
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ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	$\bar{U}$	CP	DN	UR
74182.60c	180.3900	endf60	B-VI.0	1980	293.6	113177	12283	20.0	yes	no	no	no	no
** W-183 **													
74183.49c	181.3800	ures	B-VI.0	1980	300.0	119637	12616	20.0	yes	no	no	no	yes
74183.50c	181.3800	endf5p	B-V.0	1973	293.6	58799	5843	20.0	yes	no	no	no	no
74183.50d	181.3800	dre5	B-V.0	1973	293.6	19443	263	20.0	yes	no	no	no	no
74183.55c	181.3800	rmccsa	B-V.2	1980	293.6	79534	8083	20.0	yes	no	no	no	no
74183.55d	181.3800	drmcscs	B-V.2	1980	293.6	26320	263	20.0	yes	no	no	no	no
74183.60c	181.3800	endf60	B-VI.0	1980	293.6	89350	9131	20.0	yes	no	no	no	no
** W-184 **													
74184.49c	182.3700	ures	B-VI.0	1980	300.0	97118	9794	20.0	yes	no	no	no	yes
74184.50c	182.3700	endf5p	B-V.0	1973	293.6	58870	6173	20.0	yes	no	no	no	no
74184.50d	182.3700	dre5	B-V.0	1973	293.6	17032	263	20.0	yes	no	no	no	no
74184.55c	182.3700	rmccsa	B-V.2	1980	293.6	80006	7835	20.0	yes	no	no	no	no
74184.55d	182.3700	drmcscs	B-V.2	1980	293.6	26110	263	20.0	yes	no	no	no	no
74184.60c	182.3700	endf60	B-VI.0	1980	293.6	78809	7368	20.0	yes	no	no	no	no
** W-186 **													
74186.49c	184.3600	ures	B-VI.0	1980	300.0	102199	10485	20.0	yes	no	no	no	yes
74186.50c	184.3600	endf5p	B-V.0	1973	293.6	63701	6866	20.0	yes	no	no	no	no
74186.50d	184.3600	dre5	B-V.0	1973	293.6	17018	263	20.0	yes	no	no	no	no
74186.55c	184.3600	rmccsa	B-V.2	1980	293.6	83618	8342	20.0	yes	no	no	no	no
74186.55d	184.3600	drmcscs	B-V.2	1980	293.6	26281	263	20.0	yes	no	no	no	no
74186.60c	184.3600	endf60	B-VI.0	1980	293.6	82010	7793	20.0	yes	no	no	no	no
Z = 75 ***** Rhenium *****													
** Re-185 **													
75185.32c	183.3612	misc5xs[6]	LLNL	<1985	0.0	13650	1488	20.0	yes	no	no	no	no
75185.35c	183.3641	endf85	LLNL	<1985	0.0	16038	1487	20.0	yes	no	no	no	no
75185.42c	183.3641	endf92	LLNL	<1992	300.0	23715	2214	30.0	yes	no	no	no	no
75185.50c	183.3640	rmccsa	B-V.0	1968	293.6	9190	1168	20.0	no	no	no	no	no
75185.50d	183.3640	drmcscs	B-V.0	1968	293.6	4252	263	20.0	no	no	no	no	no
75185.60c	183.3640	endf60	B-VI.0	1990	293.6	102775	16719	20.0	no	no	no	no	no
** Re-187 **													
75187.32c	185.3539	misc5xs[6]	LLNL	<1985	0.0	12318	1296	20.0	yes	no	no	no	no
75187.35c	185.3497	endf85	LLNL	<1985	0.0	14769	1295	20.0	yes	no	no	no	no
75187.42c	185.3497	endf92	LLNL	<1992	300.0	20969	1821	30.0	yes	no	no	no	no
75187.50c	185.3500	rmccsa	B-V.0	1968	293.6	8262	959	20.0	no	no	no	no	no
75187.50d	185.3500	drmcscs	B-V.0	1968	293.6	4675	263	20.0	no	no	no	no	no
75187.60c	185.3500	endf60	B-VI.0	1990	293.6	96989	15624	20.0	no	no	no	no	no
Z = 77 ***** Iridium *****													
** Ir-nat **													
77000.55c	190.5630	misc5xs[6]	T-2	1986	300.0	43071	3704	20.0	no	no	no	no	no
** Ir-191 **													
77191.49c	189.3200	ures	B-VI.4	1995	300.0	83955	8976	20.0	yes	no	no	no	yes
** Ir-193 **													
77193.49c	191.3050	ures	B-VI.4	1995	300.0	82966	8943	20.0	yes	no	no	no	yes
Z = 78 ***** Platinum *****													
** pt-nat **													
78000.35c	193.4141	rmccsa	LLNL	<1985	0.0	15371	1497	20.0	yes	no	no	no	no
78000.35d	193.4141	drmcscs	LLNL	<1985	0.0	6933	263	20.0	yes	no	no	no	no
78000.40c	193.4141	endf92	LLNL	<1992	300.0	43559	5400	30.0	yes	no	no	no	no
78000.42c	193.4141	endf92	LLNL:X-5	<1992	300.0	43559	5400	30.0	yes	no	no	no	no
Z = 79 ***** Gold *****													
** Au-197 **													
79197.35c	195.2745	endf85	LLNL	<1985	0.0	31871	3781	20.0	yes	no	no	no	no
79197.50c	195.2740	endf5p	B-V.0	1977	293.6	139425	22632	20.0	no	no	no	no	no
79197.50d	195.2740	dre5	B-V.0	1977	293.6	4882	263	20.0	no	no	no	no	no
79197.55c	195.2740	rmccsa	T-2	1983[4]	293.6	134325	17909	20.0	yes	no	no	no	no
79197.55d	195.2740	drmcscs	T-2	1983[4]	293.6	7883	263	20.0	yes	no	no	no	no
79197.56c	195.2740	newxs	T-2	1984	293.6	122482	14823	30.0	yes	no	no	no	no

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TABLE G-2 (Cont.)
Continuous-Energy and Discrete Neutron Data Libraries Maintained by X-5

ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	$\bar{v}$	CP	DN	UR
79197.56d	195.2740	newxsd	T-2	1984	293.6	38801	263	20.0	yes	no	no	no	no
79197.60c	195.2740	endf60	B-VI.1	1984	293.6	161039	17724	30.0	yes	no	no	no	no
Z = 80 ***** Mercury *****													
** Hg-nat **													
80000.40c	198.8668	endf92	LLNL	<1992	300.0	29731	2507	30.0	yes	no	no	no	no
80000.42c	198.8668	endf92	LLNL:X-5	<1992	300.0	29731	2507	30.0	yes	no	no	no	no
Z = 82 ***** Lead *****													
** Pb-nat **													
82000.35c	205.4200	endf85	LLNL	<1985	0.0	6639	349	20.0	yes	no	no	no	no
82000.42c	205.4200	endf92	LLNL	<1992	300.0	270244	18969	30.0	yes	no	no	no	no
82000.50c	205.4300	rmccs	B-V.0	1976	293.6	37633	1346	20.0	yes	no	no	no	no
82000.50d	205.4300	drmcscs	B-V.0	1976	293.6	20649	263	20.0	yes	no	no	no	no
** Pb-206 **													
82206.60c	204.2000	endf60	B-VI.0	1989	293.6	148815	12872	20.0	yes	no	no	no	no
** Pb-207 **													
82207.60c	205.2000	endf60	B-VI.1	1991	293.6	111750	7524	20.0	yes	no	no	no	no
** Pb-208 **													
82208.60c	206.1900	endf60	B-VI.0	1989	293.6	70740	5105	20.0	yes	no	no	no	no
Z = 83 ***** Bismuth *****													
** Bi-209 **													
83209.35c	207.1851	endf85	LLNL	<1985	0.0	18316	1303	20.0	yes	no	no	no	no
83209.42c	207.1851	endf92	LLNL	<1992	300.0	20921	1200	30.0	yes	no	no	no	no
83209.50c	207.1850	endf5u	B-V.0	1980	293.6	14939	1300	20.0	yes	no	no	no	no
83209.50d	207.1850	dre5	B-V.0	1980	293.6	7516	263	20.0	yes	no	no	no	no
83209.51c	207.1850	rmccs	B-V.0	1980	293.6	13721	1186	20.0	yes	no	no	no	no
83209.51d	207.1850	drmcscs	B-V.0	1980	293.6	7516	263	20.0	yes	no	no	no	no
83209.60c	207.1850	endf60	B-VI.0	1989	293.6	100138	8427	20.0	yes	no	no	no	no
Z = 90 ***** Thorium *****													
** Th-230 **													
90230.60c	228.0600	endf60	B-VI.0	1977	293.6	35155	5533	20.0	no	tot	no	no	no
** Th-231 **													
90231.35c	229.0516	endf85	LLNL	<1985	0.0	9157	308	20.0	yes	pr	no	no	no
90231.42c	229.0516	endf92	LLNL	<1992	300.0	15712	187	30.0	yes	both	no	no	no
** Th-232 **													
90232.35c	230.0447	endf85	LLNL	<1985	0.0	56091	6169	20.0	yes	pr	no	no	no
90232.42c	230.0447	endf92	LLNL	<1992	300.0	109829	13719	30.0	yes	both	no	no	no
90232.49c	230.0400	ures	B-VI.0	1977	300.0	305942	41414	20.0	yes	both	no	no	yes
90232.50c	230.0400	endf5u	B-V.0	1977	293.6	152782	17901	20.0	yes	both	no	no	no
90232.50d	230.0400	dre5	B-V.0	1977	293.6	11937	263	20.0	yes	both	no	no	no
90232.51c	230.0400	rmccs	B-V.0	1977	293.6	17925	1062	20.0	yes	both	no	no	no
90232.51d	230.0400	drmcscs	B-V.0	1977	293.6	11937	263	20.0	yes	both	no	no	no
90232.60c	230.0400	endf60	B-VI.0	1977	293.6	127606	16381	20.0	yes	both	no	no	no
90232.61c	230.0400	endf6dn	B-VI.0	1977	293.6	132594	16381	20.0	yes	both	no	yes	no
** Th-233 **													
90233.35c	231.0396	endf85	LLNL	<1985	0.0	9352	348	20.0	yes	pr	no	no	no
90233.42c	231.0396	endf92	LLNL	<1992	300.0	16015	206	30.0	yes	both	no	no	no
Z = 91 ***** Protactinium *****													
** Pa-231 **													
91231.60c	229.0500	endf60	B-VI.0	1977	293.6	19835	2610	20.0	no	both	no	no	no
91231.61c	229.0500	endf6dn	B-VI.0	1977	293.6	24733	2610	20.0	no	both	no	yes	no
** Pa-233 **													
91233.35c	231.0383	endf85	LLNL	<1985	0.0	19170	1910	20.0	yes	pr	no	no	no
91233.42c	231.0383	endf92	LLNL	<1992	300.0	27720	1982	30.0	yes	both	no	no	no
91233.50c	231.0380	endf5u	B-V.0	1974	293.6	19519	2915	20.0	no	tot	no	no	no
91233.50d	231.0380	dre5	B-V.0	1974	293.6	3700	263	20.0	no	tot	no	no	no
91233.51c	231.0380	rmccs	B-V.0	1974	293.6	5641	637	20.0	no	tot	no	no	no

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ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	U	CP	DN	UR
91233.51d	231.0380	drmcscs	B-V.0	1974	293.6	3700	263	20.0	no	tot	no	no	no
Z = 92 ***** Uranium *****													
** U-232 **													
92232.49c	230.0400	ures	B-VI.0	1977	300.0	21813	2820	20.0	no	both	no	no	yes
92232.60c	230.0400	endf60	B-VI.0	1977	293.6	13839	1759	20.0	no	both	no	no	yes
92232.61c	230.0400	endf6dn	B-VI.0	1977	293.6	18734	1759	20.0	no	both	no	yes	no
** U-233 **													
92233.35c	231.0377	endl85	LLNL	<1985	0.0	29674	2924	20.0	yes	pr	no	no	no
92233.42c	231.0377	endl92	LLNL	<1992	300.0	29521	2163	30.0	yes	both	no	no	no
92233.49c	231.0430	ures	B-VI.0	1978	300.0	47100	4601	20.0	yes	both	no	no	yes
92233.50c	231.0430	rmccs	B-V.0	1978	293.6	18815	2293	20.0	no	both	no	no	no
92233.50d	231.0430	drmcscs	B-V.0	1978	293.6	4172	263	20.0	no	both	no	no	no
92233.60c	231.0430	endf60[14]	B-VI.0	1978	293.6	32226	3223	20.0	yes	both	no	no	no
92233.61c	231.0430	endf6dn	B-VI.0	1978	293.6	37218	3223	20.0	yes	both	no	yes	no
** U-234 **													
92234.35c	232.0304	endl85	LLNL	<1985	0.0	8557	237	20.0	yes	pr	no	no	no
92234.42c	232.0304	endl92	LLNL	<1992	300.0	13677	149	30.0	yes	both	no	no	no
92234.49c	232.0300	ures	B-VI.0	1978	300.0	161296	22539	20.0	no	both	no	no	yes
92234.50c	232.0300	endf5p	B-V.0	1978	293.6	89433	12430	20.0	no	tot	no	no	no
92234.50d	232.0300	dre5	B-V.0	1978	293.6	4833	263	20.0	no	tot	no	no	no
92234.51c	232.0300	rmccs	B-V.0	1978	293.6	6426	672	20.0	no	tot	no	no	no
92234.51d	232.0300	drmcscs	B-V.0	1978	293.6	4833	263	20.0	no	tot	no	no	no
92234.60c	232.0300	endf60	B-VI.0	1978	293.6	77059	10660	17.5	no	both	no	no	no
92234.61c	232.0300	endf6dn	B-VI.0	1978	293.6	82047	10660	17.5	no	both	no	yes	no
** U-235 **													
92235.01c	233.0250	endfht	B-VI.2	1989	1.2e4	234381	18913	20.0	yes	both	no	no	no
92235.02c	233.0250	endfht	B-VI.2	1989	1.2e5	138369	8245	20.0	yes	both	no	no	no
92235.03c	233.0250	endfht	B-VI.2	1989	1.2e6	102567	4267	20.0	yes	both	no	no	no
92235.04c	233.0250	endfht	B-VI.2	1989	1.2e7	85917	2417	20.0	yes	both	no	no	no
92235.05c	233.0250	endfht	B-VI.2	1989	1.2e8	79635	1719	20.0	yes	both	no	no	no
92235.06c	233.0250	endfht	B-V.0	1977	1.2e4	47562	3712	20.0	yes	both	no	no	no
92235.07c	233.0250	endfht	B-V.0	1977	1.2e5	32721	2063	20.0	yes	both	no	no	no
92235.08c	233.0250	endfht	B-V.0	1977	1.2e6	28905	1639	20.0	yes	both	no	no	no
92235.09c	233.0250	endfht	B-V.0	1977	1.2e7	27627	1497	20.0	yes	both	no	no	no
92235.10c	233.0250	endfht	B-V.0	1977	1.2e8	27312	1462	20.0	yes	both	no	no	no
92235.11c	233.0250	endf62mt[15]	B-VI.2	1989	77.0	696398	78912	20.0	yes	both	no	no	no
92235.12c	233.0250	endf62mt[15]	B-VI.2	1989	400.0	411854	43344	20.0	yes	both	no	no	no
92235.13c	233.0250	endf62mt[15]	B-VI.2	1989	500.0	379726	39328	20.0	yes	both	no	no	no
92235.14c	233.0250	endf62mt[15]	B-VI.2	1989	600.0	353678	36072	20.0	yes	both	no	no	no
92235.15c	233.0250	endf62mt[15]	B-VI.2	1989	800.0	316622	31440	20.0	yes	both	no	no	no
92235.16c	233.0250	endf62mt[15]	B-VI.2	1989	900.0	300278	29397	20.0	yes	both	no	no	no
92235.17c	233.0250	endf62mt[15]	B-VI.2	1989	1200	269062	25495	20	yes	both	no	no	no
92235.42c	233.0248	endl92	LLNL	<1992	300.0	72790	5734	30.0	yes	both	no	no	no
92235.49c	233.0250	ures	B-VI.4	1996	300.0	647347	72649	20.0	yes	both	no	no	yes
92235.50c	233.0250	rmccs	B-V.0	1977	293.6	60489	5725	20.0	yes	both	no	no	no
92235.50d	233.0250	drmcscs	B-V.0	1977	293.6	11788	263	20.0	yes	both	no	no	no
92235.52c	233.0250	endf5mt[1]	B-V.0	1977	587.2	65286	6320	20.0	yes	both	no	no	no
92235.53c	233.0250	endf5mt[1]	B-V.0	1977	587.2	36120	2685	20.0	yes	both	no	no	no
92235.54c	233.0250	endf5mt[1]	B-V.0	1977	880.8	36008	2671	20.0	yes	both	no	no	no
92235.56c	233.0250	endf5ht	B-V.0	1977	1.2e4	28494	1729	20.0	yes	both	no	no	no
92235.57c	233.0250	endf5ht	B-V.0	1977	1.2e5	25214	1319	20.0	yes	both	no	no	no
92235.58c	233.0250	endf5ht	B-V.0	1977	1.2e6	22966	1038	20.0	yes	both	no	no	no
92235.59c	233.0250	endf5ht	B-V.0	1977	1.2e7	22406	968	20.0	yes	both	no	no	no
92235.60c	233.0250	endf60	B-VI.2	1989	293.6	289975	28110	20.0	yes	both	no	no	no
92235.61c	233.0250	endf6dn	B-VI.2	1989	293.6	294963	28110	20.0	yes	both	no	yes	no
** U-236 **													
92236.35c	234.0178	endl85	LLNL	<1985	0.0	8699	224	20.0	yes	pr	no	no	no
92236.42c	234.0178	endl92	LLNL	<1992	300.0	14595	311	30.0	yes	both	no	no	no
92236.49c	234.0180	ures	B-VI.0	1989	300.0	159074	20865	20.0	no	both	no	no	yes
92236.50c	234.0180	endf5p	B-V.0	1978	293.6	138715	19473	20.0	no	tot	no	no	no
92236.50d	234.0180	dre5	B-V.0	1978	293.6	4838	263	20.0	no	tot	no	no	no
92236.51c	234.0180	rmccs	B-V.0	1978	293.6	7302	800	20.0	no	tot	no	no	no
92236.51d	234.0180	drmcscs	B-V.0	1978	293.6	4838	263	20.0	no	tot	no	no	no
92236.60c	234.0180	endf60	B-VI.0	1989	293.6	82819	10454	20.0	no	both	no	no	no

APPENDIX G

MCNP NEUTRON CROSS-SECTION LIBRARIES

TABLE G-2 (Cont.)
Continuous-Energy and Discrete Neutron Data Libraries Maintained by X-5

ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	$\bar{U}$	CP	DN	UR
92236.61c ** U-237 **	234.0180	endf6dn	B-VI.0	1989	293.6	87807	10454	20.0	no	both	no	yes	no
92237.35c	235.0123	endl85	LLNL	<1985	0.0	9364	353	20.0	yes	pr	no	no	no
92237.42c	235.0123	endl92	LLNL	<1992	300.0	13465	210	30.0	yes	both	no	no	no
92237.50c	235.0120	endf5p	B-V.0	1976	293.6	32445	3293	20.0	yes	tot	no	no	no
92237.50d	235.0120	dre5	B-V.0	1976	293.6	8851	263	20.0	yes	tot	no	no	no
92237.51c	235.0120	rmccs	B-V.0	1976	293.6	10317	527	20.0	yes	tot	no	no	no
92237.51d ** U-238 **	235.0120	drmcscs	B-V.0	1976	293.6	8851	263	20.0	yes	tot	no	no	no
92238.01c	236.0060	endfht	B-VI.2	1993	1.2e4	296788	30203	20.0	yes	both	no	no	no
92238.02c	236.0060	endfht	B-VI.2	1993	1.2e5	138937	12664	20.0	yes	both	no	no	no
92238.03c	236.0060	endfht	B-VI.2	1993	1.2e6	77638	5853	20.0	yes	both	no	no	no
92238.04c	236.0060	endfht	B-VI.2	1993	1.2e7	54625	3296	20.0	yes	both	no	no	no
92238.05c	236.0060	endfht	B-VI.2	1993	1.2e8	44356	2155	20.0	yes	both	no	no	no
92238.06c	236.0060	endfht	B-V.0	1979	1.2e4	185164	18732	20.0	yes	both	no	no	no
92238.07c	236.0060	endfht	B-V.0	1979	1.2e5	85705	7681	20.0	yes	both	no	no	no
92238.08c	236.0060	endfht	B-V.0	1979	1.2e6	46123	3283	20.0	yes	both	no	no	no
92238.09c	236.0060	endfht	B-V.0	1979	1.2e7	34774	2022	20.0	yes	both	no	no	no
92238.10c	236.0060	endfht	B-V.0	1979	1.2e8	30193	1513	20.0	yes	both	no	no	no
92238.11c	236.0060	endf62mt[15]	B-VI.2	1993	77.0	621385	74481	20.0	yes	both	no	no	no
92238.12c	236.0060	endf62mt[15]	B-VI.2	1993	400.0	456593	53882	20.0	yes	both	no	no	no
92238.13c	236.0060	endf62mt[15]	B-VI.2	1993	500.0	433681	51018	20.0	yes	both	no	no	no
92238.14c	236.0060	endf62mt[15]	B-VI.2	1993	600.0	414185	48581	20.0	yes	both	no	no	no
92238.15c	236.0060	endf62mt[15]	B-VI.2	1993	800.0	386305	45096	20.0	yes	both	no	no	no
92238.16c	236.0060	endf62mt[15]	B-VI.2	1993	900.0	372625	43386	20.0	yes	both	no	no	no
92238.17c	236.0060	endf62mt[15]	B-VI.2	1993	1200.0	348137	40325	20.0	yes	both	no	no	no
92238.21c	236.0060	100xs[3]	T-2:X-5	1989	300.0	279245	30911	100.0	yes	both	no	no	no
92238.35c	236.0058	endl85	LLNL	<1985	0.0	27168	1845	20.0	yes	pr	no	no	no
92238.42c	236.0058	endl92	LLNL	<1992	300.0	107739	7477	30.0	yes	both	no	no	no
92238.49c	236.0060	ures	B-VI.2	1993	300.0	705623	85021	20.0	yes	both	no	no	yes
92238.50c	236.0060	rmccs	B-V.0	1979	293.6	88998	9285	20.0	yes	both	no	no	no
92238.50d	236.0060	drmcscs	B-V.0	1979	293.6	16815	263	20.0	yes	both	no	no	no
92238.52c	236.0060	endf5mt[1]	B-V.0	1979	587.2	123199	8454	20.0	yes	both	no	no	no
92238.53c	236.0060	endf5mt[1]	B-V.0	1979	587.2	160107	17876	20.0	yes	both	no	no	no
92238.54c	236.0060	endf5mt[1]	B-V.0	1979	880.8	160971	17984	20.0	yes	both	no	no	no
92238.56c	233.0250	endf5ht	B-V.0	1979	1.2e4	82470	8176	20.0	yes	both	no	no	no
92238.57c	233.0250	endf5ht	B-V.0	1979	1.2e5	47206	3768	20.0	yes	both	no	no	no
92238.58c	233.0250	endf5ht	B-V.0	1979	1.2e6	27814	1344	20.0	yes	both	no	no	no
92238.59c	233.0250	endf5ht	B-V.0	1979	1.2e7	22078	627	20.0	yes	both	no	no	no
92238.60c	236.0060	endf60	B-VI.2	1993	293.6	206322	22600	20.0	yes	both	no	no	no
92238.61c ** U-239 **	236.0060	endf6dn	B-VI.2	1993	293.6	211310	22600	20.0	yes	both	no	yes	no
92239.35c	237.0007	rmccsa	LLNL	<1985	0.0	9809	394	20.0	yes	pr	no	no	no
92239.35d	237.0007	drmcscs	LLNL	<1985	0.0	9286	263	20.0	yes	pr	no	no	no
92239.42c	237.0007	endl92	LLNL	<1992	300.0	14336	205	30.0	yes	both	no	no	no
** U-240 **													
92240.35c	237.9944	endl85	LLNL	<1985	0.0	8495	218	20.0	yes	pr	no	no	no
92240.42c	237.9944	endl92	LLNL	<1992	300.0	14000	128	30.0	yes	both	no	no	no
Z = 93 ***** Neptunium *****													
** Np-235 **													
93235.35c	233.0249	endl85	LLNL	<1985	0.0	9490	364	20.0	yes	pr	no	no	no
93235.42c	233.0249	endl92	LLNL	<1992	300.0	17717	660	30.0	yes	both	no	no	no
** Np-236 **													
93236.35c	234.0188	endl85	LLNL	<1985	0.0	8821	284	20.0	yes	pr	no	no	no
93236.42c	234.0188	endl92	LLNL	<1992	300.0	13464	179	30.0	yes	both	no	no	no
** Np-237 **													
93237.35c	235.0118	endl85	LLNL	<1985	0.0	20225	1678	20.0	yes	pr	no	no	no
93237.42c	235.0118	endl92	LLNL	<1992	300.0	31966	2477	30.0	yes	both	no	no	no
93237.50c	235.0120	endf5p	B-V.0	1978	293.6	63223	8519	20.0	no	tot	no	no	no
93237.50d	235.0120	dre5	B-V.0	1978	293.6	5267	263	20.0	no	tot	no	no	no
93237.55c	235.0120	rmccsa	T-2	1984	293.6	32558	1682	20.0	no	both	no	no	no
93237.55d	235.0120	drmcscs	T-2	1984	293.6	20484	263	20.0	no	both	no	no	no
93237.60c	235.0118	endf60	B-VI.1	1990	293.6	105150	7218	20.0	yes	both	no	no	no
93237.61c	235.0118	endf6dn	B-VI.1	1990	293.6	110048	7218	20.0	yes	both	no	yes	no

APPENDIX G MCNP NEUTRON CROSS-SECTION LIBRARIES

TABLE G-2 (Cont.)
Continuous-Energy and Discrete Neutron Data Libraries Maintained by X-5

ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	U	CP	DN	UR
** Np-238 **													
93238.35c	236.0060	endl85	LLNL	<1985	0.0	8878	282	20.0	yes	pr	no	no	no
93238.42c	236.0060	endl92	LLNL	<1992	300.0	13445	165	30.0	yes	both	no	no	no
** Np-239 **													
93239.60c	236.9990	endf60	B-VI.0	1988	293.6	7406	562	20.0	no	tot	no	no	no
Z = 94 ***** Plutonium *****													
** Pu-236 **													
94236.60c	234.0180	endf60	B-VI.0	1978	293.6	33448	4610	20.0	no	tot	no	no	no
** Pu-237 **													
94237.35c	235.0120	endl85	LLNL	<1985	0.0	11300	202	20.0	yes	pr	no	no	no
94237.42c	235.0120	endl92	LLNL	<1992	300.0	17284	279	30.0	yes	both	no	no	no
94237.60c	235.0120	endf60	B-VI.0	1978	293.6	3524	257	20.0	no	tot	no	no	no
** Pu-238 **													
94238.35c	236.0046	endl85	LLNL	<1985	0.0	15619	958	20.0	yes	pr	no	no	no
94238.42c	236.0046	endl92	LLNL	<1992	300.0	30572	2177	30.0	yes	both	no	no	no
94238.49c	236.0045	ures	B-VI.0	1978	300.0	41814	5337	20.0	no	both	no	no	yes
94238.50c	236.1670	endf5p	B-V.0	1978	293.6	18763	2301	20.0	no	tot	no	no	no
94238.50d	236.1670	dre5	B-V.0	1978	293.6	5404	263	20.0	no	tot	no	no	no
94238.51c	236.1670	rmccs	B-V.0	1978	293.6	6067	537	20.0	no	tot	no	no	no
94238.51d	236.1670	drnccs	B-V.0	1978	293.6	5404	263	20.0	no	tot	no	no	no
94238.60c	236.0045	endf60	B-VI.0	1978	293.6	29054	3753	20.0	no	both	no	no	no
94238.61c	236.0045	endf6dn	B-VI.0	1978	293.6	33952	3753	20.0	no	both	no	yes	no
** Pu-239 **													
94239.01c	236.9986	endfht	B-VI.2	1993	1.2e4	229878	18004	20.0	yes	both	no	no	no
94239.02c	236.9986	endfht	B-VI.2	1993	1.2e5	126018	6464	20.0	yes	both	no	no	no
94239.03c	236.9986	endfht	B-VI.2	1993	1.2e6	97362	3280	20.0	yes	both	no	no	no
94239.04c	236.9986	endfht	B-VI.2	1993	1.2e7	85788	1994	20.0	yes	both	no	no	no
94239.05c	236.9986	endfht	B-VI.2	1993	1.2e8	81423	1509	20.0	yes	both	no	no	no
94239.06c	236.9990	endfht	B-V.2	1983	1.2e4	76790	6005	20.0	yes	both	no	no	no
94239.07c	236.9990	endfht	B-V.2	1983	1.2e5	45461	2524	20.0	yes	both	no	no	no
94239.08c	236.9990	endfht	B-V.2	1983	1.2e6	36236	1499	20.0	yes	both	no	no	no
94239.09c	236.9990	endfht	B-V.2	1983	1.2e7	33797	1228	20.0	yes	both	no	no	no
94239.10c	236.9990	endfht	B-V.2	1983	1.2e8	33230	1165	20.0	yes	both	no	no	no
94239.11c	236.9986	endf62mt[15]	B-VI.2	1993	77.0	568756	62522	20.0	yes	both	no	no	no
94239.12c	236.9986	endf62mt[15]	B-VI.2	1993	400.0	418556	43747	20.0	yes	both	no	no	no
94239.13c	236.9986	endf62mt[15]	B-VI.2	1993	500.0	395964	40923	20.0	yes	both	no	no	no
94239.14c	236.9986	endf62mt[15]	B-VI.2	1993	600.0	377116	38567	20.0	yes	both	no	no	no
94239.15c	236.9986	endf62mt[15]	B-VI.2	1993	800.0	350292	35214	20.0	yes	both	no	no	no
94239.16c	236.9986	endf62mt[15]	B-VI.2	1993	900.0	338236	33707	20.0	yes	both	no	no	no
94239.17c	236.9986	endf62mt[15]	B-VI.2	1993	1200.0	312572	30499	20	yes	both	no	no	no
94239.42c	236.9986	endl92	LLNL	<1992	300.0	93878	6827	30.0	yes	both	no	no	no
94239.49c	236.9986	ures	B-VI.2	1993	300.0	595005	64841	20.0	yes	both	no	no	yes
94239.50c	236.9990	endf5p	B-V.0	1976	293.6	74049	7809	20.0	yes	both	no	no	no
94239.50d	236.9990	dre5	B-V.0	1976	293.6	12631	263	20.0	yes	both	no	no	no
94239.55c	236.9990	rmccs	B-V.2	1983	293.6	102099	10318	20.0	yes	both	no	no	no
94239.55d	236.9990	drnccs	B-V.2	1983	293.6	20727	263	20.0	yes	both	no	no	no
94239.56c	236.9990	endf5ht	B-V.2	1983	1.2e4	45529	2547	20.0	yes	both	no	no	no
94239.57c	236.9990	endf5ht	B-V.2	1983	1.2e5	36201	1381	20.0	yes	both	no	no	no
94239.58c	236.9990	endf5ht	B-V.2	1983	1.2e6	31049	737	20.0	yes	both	no	no	no
94239.59c	236.9990	endf5ht	B-V.2	1983	1.2e7	29761	576	20.0	yes	both	no	no	no
94239.60c	236.9986	endf60	B-VI.2	1993	293.6	283354	26847	20.0	yes	both	no	no	no
94239.61c	236.9986	endf6dn	B-VI.2	1993	293.6	288252	26847	20.0	yes	both	no	yes	no
** Pu-240 **													
94240.42c	237.9916	endl92	LLNL	<1992	300.0	198041	16626	30.0	yes	both	no	no	no
94240.49c	237.9920	ures	B-VI.2	1986	300.0	341542	41596	20.0	yes	both	no	no	yes
94240.50c	237.9920	rmccs	B-V.0	1977	293.6	58917	6549	20.0	yes	both	no	no	no
94240.50d	237.9920	drnccs	B-V.0	1977	293.6	9569	263	20.0	yes	both	no	no	no
94240.60c	237.9920	endf60	B-VI.2	1986	293.6	133071	15676	20.0	yes	both	no	no	no
94240.61c	237.9920	endf6dn	B-VI.2	1986	293.6	137969	15676	20.0	yes	both	no	yes	no
** Pu-241 **													
94241.35c	238.9860	endl85	LLNL	<1985	0.0	8844	257	20.0	yes	pr	no	no	no
94241.42c	238.9860	endl92	LLNL	<1992	300.0	14108	203	30.0	yes	both	no	no	no
94241.49c	238.9780	ures	B-VI.3	1994	300.0	155886	17753	20.0	yes	both	no	no	yes
94241.50c	238.9780	endf5p	B-V.0	1977	293.6	38601	3744	20.0	yes	both	no	no	no

APPENDIX G

MCNP NEUTRON CROSS-SECTION LIBRARIES

TABLE G-2 (Cont.)
Continuous-Energy and Discrete Neutron Data Libraries Maintained by X-5

ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	U	CP	DN	UR
94241.50d	238.9780	dre5	B-V.0	1977	293.6	11575	263	20.0	yes	both	no	no	no
94241.51c	238.9780	rmccs	B-V.0	1977	293.6	13403	623	20.0	yes	both	no	no	no
94241.51d	238.9780	drmcscs	B-V.0	1977	293.6	11575	263	20.0	yes	both	no	no	no
94241.60c	238.9780	endf60	B-VI.1	1988	293.6	76453	8112	20.0	yes	both	no	no	no
94241.61c	238.9780	endf6dn	B-VI.1	1988	293.6	81351	8112	20.0	yes	both	no	yes	no
** Pu-242 **													
94242.35c	239.9793	endl85	LLNL	<1985	0.0	21159	1724	20.0	yes	pr	no	no	no
94242.42c	239.9793	endl92	LLNL	<1992	300.0	48688	4287	30.0	yes	both	no	no	no
94242.49c	239.9790	ures	B-VI.0	1978	300.0	130202	14922	20.0	yes	both	no	no	no
94242.50c	239.9790	endf5p	B-V.0	1978	293.6	71429	7636	20.0	yes	both	no	no	yes
94242.50d	239.9790	dre5	B-V.0	1978	293.6	12463	263	20.0	yes	both	no	no	no
94242.51c	239.9790	rmccs	B-V.0	1978	293.6	15702	728	20.0	yes	both	no	no	no
94242.51d	239.9790	drmcscs	B-V.0	1978	293.6	12463	263	20.0	yes	both	no	no	no
94242.60c	239.9790	endf60	B-VI.0	1978	293.6	73725	7896	20.0	yes	both	no	no	no
94242.61c	239.9790	endf6dn	B-VI.0	1978	293.6	78623	7896	20.0	yes	both	no	yes	no
** Pu-243 **													
94243.35c	240.9740	endl85	LLNL	<1985	0.0	10763	485	20.0	yes	pr	no	no	no
94243.42c	240.9740	endl92	LLNL	<1992	300.0	20253	745	30.0	yes	both	no	no	no
94243.60c	240.9740	endf60	B-VI.2	1976	293.6	45142	4452	20.0	yes	tot	no	no	no
** Pu-244 **													
94244.60c	241.9680	endf60	B-VI.0	1978	293.6	23654	3695	20.0	no	tot	no	no	no
Z = 95 ***** Americium *****													
** Am-241 **													
95241.35c	238.9860	endl85	LLNL	<1985	0.0	25290	1982	20.0	yes	pr	no	no	no
95241.42c	238.9860	endl92	LLNL	<1992	300.0	32579	2011	30.0	yes	both	no	no	no
95241.50c	238.9860	endf5u	B-V.0	1978	293.6	42084	4420	20.0	yes	tot	no	no	no
95241.50d	238.9860	dre5	B-V.0	1978	293.6	9971	263	20.0	yes	tot	no	no	no
95241.51c	238.9860	rmccs	B-V.0	1978	293.6	12374	713	20.0	yes	tot	no	no	no
95241.51d	238.9860	drmcscs	B-V.0	1978	293.6	9971	263	20.0	yes	tot	no	no	no
95241.60c	238.9860	endf60	T-2	1994	300.0	168924	13556	30.0	yes	both	no	no	no
95241.61c	238.9860	endf6dn	T-2	1994	300.0	173822	13556	30.0	yes	both	no	yes	no
** Am-242 ms **													
95242.35c	239.9801	endl85	LLNL	<1985	0.0	20908	1817	20.0	yes	pr	no	no	no
95242.42c	239.9801	endl92	LLNL	<1992	300.0	21828	1368	20.0	yes	both	no	no	no
95242.50c	239.9800	endf5u	B-V.0	1978	293.6	8593	323	20.0	yes	tot	no	no	no
95242.50d	239.9800	dre5	B-V.0	1978	293.6	9048	263	20.0	yes	tot	no	no	no
95242.51c	239.9800	rmccs	B-V.0	1978	293.6	8502	317	20.0	yes	tot	no	no	no
95242.51d	239.9800	drmcscs	B-V.0	1978	293.6	9048	263	20.0	yes	tot	no	no	no
** Am-243 **													
95243.35c	240.9733	endl85	LLNL	<1985	0.0	39400	4093	20.0	yes	pr	no	no	no
95243.42c	240.9733	endl92	LLNL	<1992	300.0	52074	4867	30.0	yes	both	no	no	no
95243.50c	240.9730	endf5u	B-V.0	1978	293.6	92015	11921	20.0	yes	tot	no	no	no
95243.50d	240.9730	dre5	B-V.0	1978	293.6	11742	263	20.0	yes	tot	no	no	no
95243.51c	240.9730	rmccs	B-V.0	1978	293.6	13684	757	20.0	yes	tot	no	no	no
95243.51d	240.9730	drmcscs	B-V.0	1978	293.6	11742	263	20.0	yes	tot	no	no	no
95243.60c	240.9730	endf60	B-VI.0	1988	293.6	104257	11984	20.0	yes	both	no	no	no
95243.61c	240.9730	endf6dn	B-VI.0	1988	293.6	109155	11984	20.0	yes	both	no	yes	no
Z = 96 ***** Curium *****													
** Cm-241 **													
96241.60c	238.9870	endf60	B-VI.0	1978	293.6	3132	278	20.0	no	tot	no	no	no
** Cm-242 **													
96242.35c	239.9794	endl85	LLNL	<1985	0.0	21653	1891	20.0	yes	pr	no	no	no
96242.42c	239.9794	endl92	LLNL	<1992	300.0	37766	3141	30.0	yes	both	no	no	no
96242.50c	239.9790	endf5u	B-V.0	1978	293.6	30897	3113	20.0	yes	tot	no	no	no
96242.50d	239.9790	dre5	B-V.0	1978	293.6	8903	263	20.0	yes	tot	no	no	no
96242.51c	239.9790	rmccs	B-V.0	1978	293.6	9767	472	20.0	yes	tot	no	no	no
96242.51d	239.9790	drmcscs	B-V.0	1978	293.6	8903	263	20.0	yes	tot	no	no	no
96242.60c	239.9790	endf60	B-VI.0	1978	293.6	34374	3544	20.0	yes	both	no	no	no
96242.61c	239.9790	endf6dn	B-VI.0	1978	293.6	39269	3544	20.0	yes	both	no	yes	no
** Cm-243 **													
96243.35c	240.9733	endl85	LLNL	<1985	0.0	21577	1880	20.0	yes	pr	no	no	no
96243.42c	240.9733	endl92	LLNL	<1992	300.0	21543	1099	30.0	yes	both	no	no	no

APPENDIX G MCNP NEUTRON CROSS-SECTION LIBRARIES

TABLE G-2 (Cont.)
Continuous-Energy and Discrete Neutron Data Libraries Maintained by X-5

ZAID	AWR	Library Name	Source	Eval Date	Temp (°K)	Length words	NE	E _{max} MeV	GPD	$\bar{U}$	CP	DN	UR
96243.60c	240.9730	endf60	B-VI.0	1978	293.6	18860	1445	20.0	yes	tot	no	no	no
** Cm-244 **													
96244.35c	241.9661	endl85	LLNL	<1985	0.0	21196	1815	20.0	yes	pr	no	no	no
96244.42c	241.9661	endl92	LLNL	<1992	300.0	46590	4198	30.0	yes	both	no	no	no
96244.49c	241.9660	ures	B-VI.0	1978	300.0	97975	11389	20.0	yes	pr	no	no	yes
96244.50c	241.9660	endf5u	B-V.0	1978	293.6	45991	4919	20.0	yes	tot	no	no	no
96244.50d	241.9660	dre5	B-V.0	1978	293.6	9509	263	20.0	yes	tot	no	no	no
96244.51c	241.9660	rmccs	B-V.0	1978	293.6	10847	566	20.0	yes	tot	no	no	no
96244.51d	241.9660	drmcscs	B-V.0	1978	293.6	9509	263	20.0	yes	tot	no	no	no
96244.60c	241.9660	endf60	B-VI.0	1978	293.6	73001	8294	20.0	yes	tot	no	no	no
** Cm-245 **													
96245.35c	242.9602	endl85	LLNL	<1985	0.0	24128	2230	20.0	yes	pr	no	no	no
96245.42c	242.9602	endl92	LLNL	<1992	300.0	25678	1564	30.0	yes	both	no	no	no
96245.60c	242.9600	endf60	B-VI.2	1979	293.6	29535	2636	20.0	yes	both	no	no	no
96245.61c	242.9600	endf6dn	B-VI.2	1979	293.6	34433	2636	20.0	yes	both	no	yes	no
** Cm-246 **													
96246.35c	243.9534	endl85	LLNL	<1985	0.0	12489	711	20.0	yes	pr	no	no	no
96246.42c	243.9534	endl92	LLNL	<1992	300.0	24550	1376	30.0	yes	both	no	no	no
96246.60c	243.9530	endf60	B-VI.2	1976	293.6	37948	3311	20.0	yes	tot	no	no	no
** Cm-247 **													
96247.35c	244.9479	endl85	LLNL	<1985	0.0	20265	1654	20.0	yes	pr	no	no	no
96247.42c	244.9479	endl92	LLNL	<1992	300.0	39971	3256	30.0	yes	both	no	no	no
96247.60c	244.9500	endf60	B-VI.2	1976	293.6	38800	3679	20.0	yes	tot	no	no	no
** Cm-248 **													
96248.35c	245.9413	endl85	LLNL	<1985	0.0	18178	1425	20.0	yes	pr	no	no	no
96248.42c	245.9413	endl92	LLNL	<1992	300.0	40345	3355	30.0	yes	both	no	no	no
96248.60c	245.9410	endf60	B-VI.0	1978	293.6	83452	9706	20.0	yes	tot	no	no	no
Z = 97 ***** Berkelium *****													
** Bk-249 **													
97249.35c	246.9353	endl85	LLNL	<1985	0.0	11783	633	20.0	yes	pr	no	no	no
97249.42c	246.9353	endl92	LLNL	<1992	300.0	19573	809	30.0	yes	both	no	no	no
97249.60c	246.9400	endf60	B-VI.X-5	1986	293.6	50503	5268	20.0	no	both	no	no	no
Z = 98 ***** Californium *****													
** Cf-249 **													
98249.35c	246.9352	endl85	LLNL	<1985	0.0	28055	2659	20.0	yes	pr	no	no	no
98249.42c	246.9352	endl92	LLNL	<1992	300.0	49615	4554	30.0	yes	both	no	no	no
98249.60c	246.9400	endf60	B-VI.X-5	1989	293.6	41271	4329	20	no	both	no	no	no
98249.61c	246.9400	endf6dn	B-VI.X-5	1989	293.6	46154	4329	20.0	no	both	no	yes	no
** Cf-250 **													
98250.35c	247.9281	endl85	LLNL	<1985	0.0	10487	457	20.0	yes	pr	no	no	no
98250.42c	247.9281	endl92	LLNL	<1992	300.0	17659	574	30.0	yes	both	no	no	no
98250.60c	247.9280	endf60	B-VI.2	1976	293.6	47758	5554	20.0	yes	tot	no	no	no
** Cf-251 **													
98251.35c	248.9227	endl85	LLNL	<1985	0.0	10969	516	20.0	yes	pr	no	no	no
98251.42c	248.9227	endl92	LLNL	<1992	300.0	17673	545	30.0	yes	both	no	no	no
98251.60c	248.9230	endf60	B-VI.2	1976	293.6	42817	4226	20.0	yes	both	no	no	no
98251.61c	248.9230	endf6dn	B-VI.2	1976	293.6	47715	4226	20.0	yes	both	no	yes	no
** Cf-252 **													
98252.35c	249.9161	endl85	LLNL	<1985	0.0	17908	1535	20.0	yes	pr	no	no	no
98252.42c	249.9161	endl92	LLNL	<1992	300.0	21027	1210	30.0	yes	both	no	no	no
98252.60c	249.9160	endf60	B-VI.2	1976	293.6	49204	5250	20.0	yes	both	no	no	no

Not all libraries listed in this table are publically available.

APPENDIX G

MCNP NEUTRON CROSS-SECTION LIBRARIES

SPECIAL NOTES

- note 1. The data libraries previously known as EPRIXS and U600K are now a part of the data library ENDF5MT.
- note 2. Data translated to ENDF/B-VI format with some modifications by LANL.
- note 3. The 100XS data library contains data for 9 nuclides up to 100 MeV. Heating numbers on this data library are known to be incorrect, overestimating the energy deposition.⁴
- note 4. Photon production data were added to the existing ENDF evaluation in 1984. A complete new evaluation was performed in 1986.
- note 5. The natural carbon data 6000.50c are repeated here with the ZAID of 6012.50c for the user's convenience. Both are based on the natural carbon ENDF/B-V.0 evaluation.
- note 6. The data libraries previously known as ARKRC, GDT2GP, IRNAT, MISCXS, TM169, and T2DDC are now a part of the data library MISC5XS.
- note 7. Photon production added to ENDF/B-V.0 neutron files by T-2, with the intent to estimate photon heating roughly.⁵
- note 8. These data were taken from incomplete fission-product evaluations.⁶
- note 9. This is ENDF/B-V.0 after modification by evaluator to get better agreement with ENDL85.^{7,8}
- note 10. The following files for Zr have been replaced by the indicated ZAID, eliminating the rare problem of having a secondary neutron energy greater than the incident neutron energy caused by an ENDF/B-V.0 evaluation problem.⁹ Note that this correction has been made for the ENDF/B-VI evaluation.

40000.50c	rmccs	->	40000.56c	misc5xs
40000.50d	drmcscs	->	40000.56d	misc5xs
40000.51c	endf5p	->	40000.57c	misc5xs
40000.51d	dre5	->	40000.57d	misc5xs
40000.53c	eprixs	->	40000.58c	misc5xs

- note 11. The ZAIDs for ENDL-based average fission product data files have been changed for the latest library, ENDL92, to 49120.42c and 49125.42c. Z is now set to 49 to ensure that the appropriate atomic fraction and photon transport library is used. You may need to update the atomic weight ratio table in your XSDIR file to include these entries.^{10,11} The ENDL92FP library is not publically available.
- note 12. The LANL/T-2 evaluation for I-127 was accepted for ENDF/B-VI.2 with modifications. These data are processed from the original LANL/T-2 evaluation.
- note 13. Photon production data for Gd were added to the ENDF/B-V.0 neutron cross sections by T-2. These data are valid only to 1 MeV.¹²
- note 14. Photon production data added to original evaluation in 1981 by LANL.
- note 15. The multitemperature data library ENDF62MT is still under development and is not publically available.¹³

IV. MULTIGROUP DATA FOR MCNP

Currently, only one coupled neutron-photon multigroup library is supported by X-5, MGXSNP.¹⁴ MGXSNP is comprised of 30-group neutron and 12-group photon data primarily based on ENDF/B-V for 95 nuclides. The MCNP-compatible multigroup data library was produced from the original Sn multigroup libraries MENDF5 and MENDF5G using the code CRSRD in April 1987.^{15,16} The original neutron data library MENDF5 was produced using the "TD-Division Weight Function," also called "CLAW" by the processing code NJOY.^{17,18,19} This weight function is a combination of a Maxwellian thermal + 1/E + fission + fusion peak at 14.0 MeV. The data library contains no upscatter groups or self-shielding, and is most applicable for fast systems. All cross-sections are for room temperature, 300°K. P0 through P4 scattering matrices from the original library were processed by CRSRD into angular distributions for MCNP using the Carter-Forest equiprobable bin treatment. When available, both total and prompt nubar data are provided. The edit reactions available for each ZAIID are fully described in reference 14.

TABLE G-3 describes the MGXSNP data library. The ZAIIDs used for this library correspond to the source evaluation in the same manner as the ZAIID for the continuous-energy and discrete data; as an example the same source evaluation for natural iron was used to produce 26000.55c, 26000.55d and 26000.55m. For coupled neutron-photon problems, specifying a particular isotope on a material card will invoke the neutron set for that isotope and the corresponding photon set for that element. For example, an entry of "1003" on a material card will cause MCNP to use ZAIID=1003.50m for neutron data and 1000.01g for photon data.

TABLE G-3
MGXSNP: A Coupled Neutron-Photon Multigroup Data Library

<u>ZAIID</u>	<u>Neutron</u> <u>AWR</u>	<u>Length</u>	<u>ZAIID</u>	<u>Photon</u> <u>AWR</u>	<u>Length</u>
1001.50m	0.999172	3249	1000.01g	0.999317	583
1002.55m	1.996810	3542			
1003.50m	2.990154	1927			
2003.50m	2.990134	1843	2000.01g	3.968217	583
2004.50m	3.968238	1629			
3006.50m	5.963479	3566	3000.01g	6.881312	583
3007.55m	6.955768	3555			
4007.35m	6.949815	1598	4000.01g	8.934763	557
4009.50m	8.934807	3014			
5010.50m	9.926970	3557	5000.01g	10.717168	583
5011.56m	10.914679	2795			
6000.50m [1]	11.896972	2933	6000.01g	11.907955	583
6012.50m [1]	11.896972	2933			
7014.50m	13.882849	3501	7000.01g	13.886438	583

APPENDIX G
MULTIGROUP DATA FOR MCNP

TABLE G-3 (Cont.)
MGXSNP: A Coupled Neutron-Photon Multigroup Data Library

<u>ZAID</u>	<u>Neutron AWR</u>	<u>Length</u>	<u>ZAID</u>	<u>Photon AWR</u>	<u>Length</u>
7015.55m	14.871314	2743			
8016.50m	15.857588	3346	8000.01g	15.861942	583
9019.50m	18.835289	3261	9000.01g	18.835197	583
11023.50m	22.792388	2982	11000.01g	22.792275	583
12000.50m	24.096375	3802	12000.01g	24.096261	583
13027.50m	26.749887	3853	13000.01g	26.749756	583
14000.50m	27.844378	3266	14000.01g	27.844241	583
15031.50m	30.707833	2123	15000.01g	30.707682	583
16032.50m	31.697571	2185	16000.01g	31.788823	583
17000.50m	35.148355	2737	17000.01g	35.148180	583
18000.35m	39.605021	2022	18000.01g	39.604489	557
19000.50m	38.762616	2833	19000.01g	38.762423	583
20000.50m	39.734053	3450	20000.01g	39.733857	583
22000.50m	47.455981	3015	22000.01g	47.455747	583
23000.50m	50.504104	2775	23000.01g	50.503856	583
24000.50m	51.549511	3924	24000.01g	51.549253	583
25055.50m	54.466367	2890	25000.01g	54.466099	583
26000.55m	55.366734	4304	26000.01g	55.366466	583
27059.50m	58.427218	2889	27000.01g	58.426930	583
28000.50m	58.182926	3373	28000.01g	58.182641	583
29000.50m	62.999465	2803	29000.01g	62.999157	583
31000.50m	69.124611	2084	31000.01g	69.124270	583
33075.35m	74.278340	2022	33000.01g	74.277979	557
36078.50m	77.251400	2108	36000.01g	83.080137	583
36080.50m	79.230241	2257			
36082.50m	81.210203	2312			
36083.50m	82.202262	2141			
36084.50m	83.191072	2460			
36086.50m	85.173016	2413			
40000.50m	90.440039	2466	40000.01g	90.439594	583
41093.50m	92.108717	2746	41000.01g	92.108263	583
42000.50m	95.107162	1991	42000.01g	95.106691	583
45103.50m	102.021993	2147	45000.01g	102.021490	583
45117.90m	115.544386	2709			
46119.90m	117.525231	2629	46000.01g	105.513949	557
47000.55m	106.941883	2693	47000.01g	106.941685	583
47107.50m	105.987245	2107			
47109.50m	107.969736	1924			
48000.50m	111.442911	1841	48000.01g	111.442363	583
50120.35m	115.995479	1929	50000.01g	117.667336	557
50998.99m	228.025301	1382			
50999.99m	228.025301	1413			
54000.35m	130.171713	1929	54000.01g	130.165202	557
56138.50m	136.721230	2115	56000.01g	136.146809	583

APPENDIX G
MULTIGROUP DATA FOR MCNP

TABLE G-3 (Cont.)
MGXSNP: A Coupled Neutron-Photon Multigroup Data Library

<u>ZAID</u>	<u>Neutron AWR</u>	<u>Length</u>	<u>ZAID</u>	<u>Photon AWR</u>	<u>Length</u>
63000.35m	150.654333	1933	63000.01g	150.657141	557
63151.55m	149.623005	2976			
63153.55m	151.608005	2691			
64000.35m	155.898915	1929	64000.01g	155.900158	557
67165.55m	163.512997	2526	67000.01g	163.513493	583
73181.50m	179.394458	2787	73000.01g	179.393456	583
74000.55m	182.270446	4360	74000.01g	182.269548	583
74182.55m	180.386082	3687			
74183.55m	181.379499	3628			
74184.55m	182.371615	3664			
74186.55m	184.357838	3672			
75185.50m	183.365036	1968	75000.01g	184.607108	583
75187.50m	185.350629	2061			
78000.35m	193.415026	1929	78000.01g	193.404225	557
79197.56m	195.274027	3490	79000.01g	195.274513	583
82000.50m	205.437162	3384	82000.01g	205.436151	583
83209.50m	207.186158	2524	83000.01g	207.185136	583
90232.50m	230.045857	2896	90000.01g	230.044724	583
91233.50m	231.039442	1970	91000.01g	229.051160	479
92233.50m	231.038833	1988	92000.01g	235.984125	583
92234.50m	232.031554	2150			
92235.50m	233.025921	3164			
92236.50m	234.018959	2166			
92237.50m	235.013509	2174			
92238.50m	236.006966	3553			
92239.35m	236.997601	2147			
93237.55m	235.012957	2812	93000.01g	235.011799	479
94238.50m	236.005745	2442	94000.01g [2]	241.967559	583
94239.55m	236.999740	3038			
94240.50m	237.992791	3044			
94241.50m	238.987218	2856			
94242.50m	239.980508	2956			
95241.50m	238.987196	2535			
95242.50m	239.981303	2284			
95243.50m	240.974535	2480			
96242.50m	239.980599	1970			
96244.50m	241.967311	1950			

note 1. The neutron transport data for ZAID's 6012.50m and 6000.50m are the same.

note 2. Photon transport data are not provided for Z>94.

APPENDIX G

DOSIMETRY DATA FOR MCNP

V. DOSIMETRY DATA FOR MCNP

The tally multiplier (FM) feature in MCNP allows users to calculate quantities of the form: $C \int \phi(E) R(E) dE$, where C is a constant, $\phi(E)$ is the fluence (n/cm^2), and $R(E)$ is a response function. If $R(E)$ is a cross section, and with the appropriate choice of units for C [atom/b-cm], the quantity calculated becomes the total number of some type of reaction per unit volume. If the tally is made over a finite time interval, it becomes a reaction rate per unit volume. In addition to using the standard reaction cross-section information available in our neutron transport libraries, dosimetry or activation reaction data may also be used as a response function. Often only dosimetry data is available for rare nuclides.

A full description of the use of dosimetry data can be found in reference 20. This memorandum also gives a listing of all reaction data that is available for each ZAID. There have been no major revisions of the LLNL/ACTL data since LLLDOS was produced. Users need to remember that dosimetry data libraries are appropriate only when used as a source of $R(E)$ for FM tally multipliers. Dosimetry data libraries can not be used as a source of data for materials through which actual transport is required. TABLE G-4 lists the available dosimetry data libraries for use with MCNP, the evaluation source and date, and the length of the data in words.

TABLE G-4
Dosimetry Data Libraries for MCNP Tallies

ZAID	AWR	Library	Source	Date	Length
Z = 1 ***** Hydrogen *****					
1001.30y	1.00782	llldos	LLNL/ACTL	<1983	209
1002.30y	2.01410	llldos	LLNL/ACTL	<1983	149
1003.30y	3.01605	llldos	LLNL/ACTL	<1983	27
Z = 2 ***** Helium *****					
2003.30y	3.01603	llldos	LLNL/ACTL	<1983	267
Z = 3 ***** Lithium *****					
3006.24y	5.96340	531dos	ENDF/B-V	1978	735
3006.26y	5.96340	532dos	ENDF/B-V	1977	713
3006.30y	6.01512	llldos	LLNL/ACTL	<1983	931
3007.26y	6.95570	532dos	ENDF/B-V	1972	733
3007.30y	7.01601	llldos	LLNL/ACTL	<1983	201
Z = 4 ***** Beryllium *****					
4007.30y	7.01693	llldos	LLNL/ACTL	<1983	253
4009.30y	9.01218	llldos	LLNL/ACTL	<1983	335
Z = 5 ***** Boron *****					

APPENDIX G
DOSIMETRY DATA FOR MCNP

TABLE G-4 (Cont.)
Dosimetry Data Libraries for MCNP Tallies

ZAID	AWR	Library	Source	Date	Length
5010.24y	9.92690	531dos	ENDF/B-V	1979	769
5010.26y	9.92690	532dos	ENDF/B-V	1976	589
5010.30y	10.01290	llldos	LLNL/ACTL	<1983	381
5011.30y	11.00930	llldos	LLNL/ACTL	<1983	119
Z = 6 ***** Carbon *****					
6012.30y	12.00000	llldos	LLNL/ACTL	<1983	97
6013.30y	13.00340	llldos	LLNL/ACTL	<1983	479
6014.30y	14.00320	llldos	LLNL/ACTL	<1983	63
Z = 7 ***** Nitrogen *****					
7014.26y	13.88300	532dos	ENDF/B-V	1973	1013
7014.30y	14.00310	llldos	LLNL/ACTL	<1983	915
Z = 8 ***** Oxygen *****					
8016.26y	15.85800	532dos	ENDF/B-V	1973	95
8016.30y	15.99490	llldos	LLNL/ACTL	<1983	215
8017.30y	16.99910	llldos	LLNL/ACTL	<1983	239
Z = 9 ***** Fluorine *****					
9019.26y	18.83500	532dos	ENDF/B-V	1979	31
9019.30y	18.99840	llldos	LLNL/ACTL	<1983	517
Z = 11 ***** Sodium *****					
11023.30y	22.98980	llldos	LLNL/ACTL	<1983	621
Z = 12 ***** Magnesium *****					
12023.30y	22.99410	llldos	LLNL/ACTL	<1983	333
12024.26y	23.98500	532dos	ENDF/B-V	1979	165
12024.30y	23.98500	llldos	LLNL/ACTL	<1983	309
12025.30y	24.98580	llldos	LLNL/ACTL	<1983	309
12026.30y	25.98260	llldos	LLNL/ACTL	<1983	321
12027.30y	26.98430	llldos	LLNL/ACTL	<1983	309
Z = 13 ***** Aluminum *****					
13026.30y	25.98690	llldos	LLNL/ACTL	<1983	447
13027.24y	26.75000	531dos	ENDF/B-V	1973	1165
13027.26y	26.75000	532dos	ENDF/B-V	1973	1753
13027.30y	26.98150	llldos	LLNL/ACTL	<1983	491

APPENDIX G
DOSIMETRY DATA FOR MCNP

TABLE G-4 (Cont.)
Dosimetry Data Libraries for MCNP Tallies

ZAID	AWR	Library	Source	Date	Length
Z = 14 ***** Silicon *****					
14027.30y	26.98670	llldos	LLNL/ACTL	<1983	401
14028.30y	27.97690	llldos	LLNL/ACTL	<1983	377
14029.30y	28.97650	llldos	LLNL/ACTL	<1983	389
14030.30y	29.97380	llldos	LLNL/ACTL	<1983	395
14031.30y	30.97540	llldos	LLNL/ACTL	<1983	337
Z = 15 ***** Phosphorus *****					
15031.26y	30.70800	532dos	ENDF/B-V	1977	65
15031.30y	30.97380	llldos	LLNL/ACTL	<1983	263
Z = 16 ***** Sulfur *****					
16031.30y	30.97960	llldos	LLNL/ACTL	<1983	393
16032.24y	31.69740	531dos	ENDF/B-V	1979	145
16032.26y	31.69700	532dos	ENDF/B-V	1977	35
16032.30y	31.97210	llldos	LLNL/ACTL	<1983	417
16033.30y	32.97150	llldos	LLNL/ACTL	<1983	435
16034.30y	33.96790	llldos	LLNL/ACTL	<1983	437
16035.30y	34.96900	llldos	LLNL/ACTL	<1983	339
16036.30y	35.96710	llldos	LLNL/ACTL	<1983	293
16037.30y	36.97110	llldos	LLNL/ACTL	<1983	279
Z = 17 ***** Chlorine *****					
17034.30y	33.97380	llldos	LLNL/ACTL	<1983	401
17035.30y	34.96890	llldos	LLNL/ACTL	<1983	459
17036.30y	35.96830	llldos	LLNL/ACTL	<1983	563
17037.30y	36.96590	llldos	LLNL/ACTL	<1983	407
7038.30y	37.96800	llldos	LLNL/ACTL	<1983	33
Z = 18 ***** Argon *****					
18036.30y	35.96750	llldos	LLNL/ACTL	<1983	309
18037.30y	36.96680	llldos	LLNL/ACTL	<1983	311
18038.30y	37.96270	llldos	LLNL/ACTL	<1983	311
18039.30y	38.96430	llldos	LLNL/ACTL	<1983	337
18040.26y	39.61910	532dos	ENDF/B-V	1979	3861
18040.30y	39.96240	llldos	LLNL/ACTL	<1983	347
18041.30y	40.96450	llldos	LLNL/ACTL	<1983	317
18042.30y	41.96300	llldos	LLNL/ACTL	<1983	291
18043.30y	42.96570	llldos	LLNL/ACTL	<1983	295
Z = 19 ***** Potassium *****					
19038.30y	37.96910	llldos	LLNL/ACTL	<1983	603
19039.30y	38.96370	llldos	LLNL/ACTL	<1983	405

APPENDIX G
DOSIMETRY DATA FOR MCNP

TABLE G-4 (Cont.)
Dosimetry Data Libraries for MCNP Tallies

ZAID	AWR	Library	Source	Date	Length
19040.30y	39.96400	llldos	LLNL/ACTL	<1983	675
19041.26y	40.60990	532dos	ENDF/B-V	1979	33
19041.30y	40.96180	llldos	LLNL/ACTL	<1983	369
19042.30y	41.96240	llldos	LLNL/ACTL	<1983	343
19043.30y	42.96070	llldos	LLNL/ACTL	<1983	277
19044.30y	43.96160	llldos	LLNL/ACTL	<1983	275
19045.30y	44.96070	llldos	LLNL/ACTL	<1983	283
19046.30y	45.96200	llldos	LLNL/ACTL	<1983	283
Z = 20 ***** Calcium *****					
20039.30y	38.97070	llldos	LLNL/ACTL	<1983	601
20040.30y	39.96260	llldos	LLNL/ACTL	<1983	309
20041.30y	40.96230	llldos	LLNL/ACTL	<1983	313
20042.30y	41.95860	llldos	LLNL/ACTL	<1983	285
20043.30y	42.95880	llldos	LLNL/ACTL	<1983	295
20044.30y	43.95550	llldos	LLNL/ACTL	<1983	269
20045.30y	44.95620	llldos	LLNL/ACTL	<1983	271
20046.30y	45.95370	llldos	LLNL/ACTL	<1983	255
20047.30y	46.95450	llldos	LLNL/ACTL	<1983	243
20048.30y	47.95250	llldos	LLNL/ACTL	<1983	239
20049.30y	48.95570	llldos	LLNL/ACTL	<1983	229
Z = 21 ***** Scandium *****					
21044.30y	43.95940	llldos	LLNL/ACTL	<1983	313
21044.31y	43.95940	llldos	LLNL/ACTL	<1983	311
21045.24y	44.56790	531dos	ENDF/B-V	1979	20179
21045.26y	44.56790	532dos	ENDF/B-V	1979	20211
21045.30y	44.95590	llldos	LLNL/ACTL	<1983	547
21046.30y	45.95520	llldos	LLNL/ACTL	<1983	323
21046.31y	45.95520	llldos	LLNL/ACTL	<1983	323
21047.30y	46.95240	llldos	LLNL/ACTL	<1983	331
21048.30y	47.95220	llldos	LLNL/ACTL	<1983	325
Z = 22 ***** Titanium *****					
22045.30y	44.95810	llldos	LLNL/ACTL	<1983	449
22046.24y	45.55780	531dos	ENDF/B-V	1977	53
22046.26y	45.55780	532dos	ENDF/B-V	1977	53
22046.30y	45.95260	llldos	LLNL/ACTL	<1983	391
22047.24y	46.54800	531dos	ENDF/B-V	1977	209
22047.26y	46.54800	532dos	ENDF/B-V	1977	209
22047.30y	46.95180	llldos	LLNL/ACTL	<1983	419
22048.24y	47.53600	531dos	ENDF/B-V	1977	145
22048.26y	47.53600	532dos	ENDF/B-V	1977	177
22048.30y	47.94790	llldos	LLNL/ACTL	<1983	415
22049.30y	48.94790	llldos	LLNL/ACTL	<1983	409
22050.26y	49.57000	532dos	ENDF/B-V	1979	33

APPENDIX G
DOSIMETRY DATA FOR MCNP

TABLE G-4 (Cont.)
Dosimetry Data Libraries for MCNP Tallies

ZAID	AWR	Library	Source	Date	Length
22050.30y	49.94480	llldos	LLNL/ACTL	<1983	345
22051.30y	50.94660	llldos	LLNL/ACTL	<1983	389
Z = 23 ***** Vanadium *****					
23047.30y	46.95490	llldos	LLNL/ACTL	<1983	209
23048.30y	47.95230	llldos	LLNL/ACTL	<1983	399
23049.30y	48.94850	llldos	LLNL/ACTL	<1983	423
23050.30y	49.94720	llldos	LLNL/ACTL	<1983	407
23051.30y	50.94400	llldos	LLNL/ACTL	<1983	357
23052.30y	51.94480	llldos	LLNL/ACTL	<1983	401
Z = 24 ***** Chromium *****					
24049.30y	48.95130	llldos	LLNL/ACTL	<1983	377
24050.26y	49.51650	532dos	ENDF/B-V	1979	7405
24050.30y	49.94600	llldos	LLNL/ACTL	<1983	435
24051.30y	50.94480	llldos	LLNL/ACTL	<1983	377
24052.26y	51.49380	532dos	ENDF/B-V	1979	27
24052.30y	51.94050	llldos	LLNL/ACTL	<1983	417
24053.30y	52.94060	llldos	LLNL/ACTL	<1983	425
24054.30y	53.93890	llldos	LLNL/ACTL	<1983	461
24055.30y	54.94080	llldos	LLNL/ACTL	<1983	419
24056.30y	55.94070	llldos	LLNL/ACTL	<1983	297
Z = 25 ***** Manganese *****					
25051.30y	50.94820	llldos	LLNL/ACTL	<1983	417
25052.30y	51.94560	llldos	LLNL/ACTL	<1983	379
25053.30y	52.94130	llldos	LLNL/ACTL	<1983	425
25054.30y	53.94040	llldos	LLNL/ACTL	<1983	391
25055.24y	54.46610	531dos	ENDF/B-V	1977	119
25055.30y	54.93800	llldos	LLNL/ACTL	<1983	435
25056.30y	55.93890	llldos	LLNL/ACTL	<1983	423
25057.30y	56.93830	llldos	LLNL/ACTL	<1983	419
25058.30y	57.93970	llldos	LLNL/ACTL	<1983	285
Z = 26 ***** Iron *****					
26053.30y	52.94530	llldos	LLNL/ACTL	<1983	387
26054.24y	53.47620	531dos	ENDF/B-V	1979	517
26054.26y	53.47600	532dos	ENDF/B-V	1978	21563
26054.30y	53.93960	llldos	LLNL/ACTL	<1983	457
26055.30y	54.93830	llldos	LLNL/ACTL	<1983	373
26056.24y	55.45400	531dos	ENDF/B-V	1978	449
26056.26y	55.45400	532dos	ENDF/B-V	1978	581
26056.30y	55.93490	llldos	LLNL/ACTL	<1983	415
26057.30y	56.93540	llldos	LLNL/ACTL	<1983	447
26058.24y	57.43560	531dos	ENDF/B-V	1979	7077

APPENDIX G
DOSIMETRY DATA FOR MCNP

TABLE G-4 (Cont.)
Dosimetry Data Libraries for MCNP Tallies

ZAID	AWR	Library	Source	Date	Length
26058.26y	57.43560	532dos	ENDF/B-V	1979	7097
26058.30y	57.93330	llldos	LLNL/ACTL	<1983	431
26059.30y	58.93490	llldos	LLNL/ACTL	<1983	397
26060.30y	59.93400	llldos	LLNL/ACTL	<1983	285
Z = 27 ***** Cobalt *****					
27057.30y	56.93630	llldos	LLNL/ACTL	<1983	629
27058.30y	57.93580	llldos	LLNL/ACTL	<1983	531
27058.31y	57.93580	llldos	LLNL/ACTL	<1983	569
27059.30y	58.93320	llldos	LLNL/ACTL	<1983	657
27060.30y	59.93380	llldos	LLNL/ACTL	<1983	435
27060.31y	59.93380	llldos	LLNL/ACTL	<1983	499
27061.30y	60.93250	llldos	LLNL/ACTL	<1983	613
27062.30y	61.93400	llldos	LLNL/ACTL	<1983	463
27062.31y	61.93400	llldos	LLNL/ACTL	<1983	519
27063.30y	62.93360	llldos	LLNL/ACTL	<1983	339
27064.30y	63.93580	llldos	LLNL/ACTL	<1983	323
Z = 28 ***** Nickel *****					
28057.30y	56.93980	llldos	LLNL/ACTL	<1983	441
28058.24y	57.43760	531dos	ENDF/B-V	1977	411
28058.26y	57.43760	532dos	ENDF/B-V	1978	4079
28058.30y	57.93530	llldos	LLNL/ACTL	<1983	509
28059.30y	58.93430	llldos	LLNL/ACTL	<1983	513
28060.24y	59.41590	531dos	ENDF/B-V	1977	435
28060.26y	59.41590	532dos	ENDF/B-V	1978	479
28060.30y	59.93080	llldos	LLNL/ACTL	<1983	503
28061.30y	60.93110	llldos	LLNL/ACTL	<1983	489
28062.26y	61.39630	532dos	ENDF/B-V	1978	3847
28062.30y	61.92830	llldos	LLNL/ACTL	<1983	459
28063.30y	62.92970	llldos	LLNL/ACTL	<1983	375
28064.30y	63.92800	llldos	LLNL/ACTL	<1983	397
28065.30y	64.93010	llldos	LLNL/ACTL	<1983	345
Z = 29 ***** Copper *****					
29062.30y	61.93260	llldos	LLNL/ACTL	<1983	507
29063.24y	62.93000	531dos	ENDF/B-V	1978	3375
29063.26y	62.93000	532dos	ENDF/B-V	1978	3615
29063.30y	62.92960	llldos	LLNL/ACTL	<1983	513
29064.30y	63.92980	llldos	LLNL/ACTL	<1983	437
29065.24y	64.92800	531dos	ENDF/B-V	1978	49
29065.26y	64.92800	532dos	ENDF/B-V	1978	49
29065.30y	64.92780	llldos	LLNL/ACTL	<1983	563
29066.30y	65.92890	llldos	LLNL/ACTL	<1983	397
Z = 30 ***** Zinc *****					

APPENDIX G
DOSIMETRY DATA FOR MCNP

TABLE G-4 (Cont.)
Dosimetry Data Libraries for MCNP Tallies

ZAID	AWR	Library	Source	Date	Length
30064.30y	63.92910	llldos	LLNL/ACTL	<1983	555
30066.30y	65.92600	llldos	LLNL/ACTL	<1983	561
30067.30y	66.92710	llldos	LLNL/ACTL	<1983	411
30068.30y	67.92480	llldos	LLNL/ACTL	<1983	643
30070.30y	69.92530	llldos	LLNL/ACTL	<1983	619
Z = 31 ***** Gallium *****					
31069.30y	68.92560	llldos	LLNL/ACTL	<1983	197
31071.30y	70.92470	llldos	LLNL/ACTL	<1983	419
Z = 32 ***** Germanium *****					
32070.30y	69.92420	llldos	LLNL/ACTL	<1983	405
32072.30y	71.92210	llldos	LLNL/ACTL	<1983	423
32073.30y	72.92350	llldos	LLNL/ACTL	<1983	431
32074.30y	73.92120	llldos	LLNL/ACTL	<1983	629
32076.30y	75.92140	llldos	LLNL/ACTL	<1983	623
Z = 33 ***** Arsenic *****					
33075.30y	74.92160	llldos	LLNL/ACTL	<1983	987
Z = 34 ***** Selenium *****					
34074.30y	73.92250	llldos	LLNL/ACTL	<1983	159
34076.30y	75.91920	llldos	LLNL/ACTL	<1983	177
34080.30y	79.91650	llldos	LLNL/ACTL	<1983	205
34082.30y	81.91670	llldos	LLNL/ACTL	<1983	223
Z = 35 ***** Bromine *****					
35079.30y	78.91830	llldos	LLNL/ACTL	<1983	263
35081.30y	80.91630	llldos	LLNL/ACTL	<1983	695
Z = 37 ***** Rubidium *****					
37085.30y	84.91180	llldos	LLNL/ACTL	<1983	193
37087.30y	86.90920	llldos	LLNL/ACTL	<1983	199
Z = 38 ***** Strontium *****					
38084.30y	83.91340	llldos	LLNL/ACTL	<1983	163
38086.30y	85.90930	llldos	LLNL/ACTL	<1983	33
Z = 39 ***** Yttrium *****					
39089.30y	88.90590	llldos	LLNL/ACTL	<1983	419

APPENDIX G
DOSIMETRY DATA FOR MCNP

TABLE G-4 (Cont.)
Dosimetry Data Libraries for MCNP Tallies

ZAID	AWR	Library	Source	Date	Length
Z = 40 ***** Zirconium *****					
40089.30y	88.90890	llldos	LLNL/ACTL	<1983	321
40090.26y	89.13200	532dos	ENDF/B-V	1976	37
40090.30y	89.90470	llldos	LLNL/ACTL	<1983	385
40091.30y	90.90560	llldos	LLNL/ACTL	<1983	407
40092.26y	91.11200	532dos	ENDF/B-V	1976	3821
40092.30y	91.90500	llldos	LLNL/ACTL	<1983	431
40093.30y	92.90650	llldos	LLNL/ACTL	<1983	371
40094.26y	93.09600	532dos	ENDF/B-V	1976	5255
40094.30y	93.90630	llldos	LLNL/ACTL	<1983	417
40095.30y	94.90800	llldos	LLNL/ACTL	<1983	375
40096.30y	95.90830	llldos	LLNL/ACTL	<1983	57
40097.30y	96.91090	llldos	LLNL/ACTL	<1983	339
Z = 41 ***** Niobium *****					
41091.30y	90.90700	llldos	LLNL/ACTL	<1983	491
41091.31y	90.90700	llldos	LLNL/ACTL	<1983	491
41092.30y	91.90720	llldos	LLNL/ACTL	<1983	285
41092.31y	91.90720	llldos	LLNL/ACTL	<1983	285
41093.30y	92.90640	llldos	LLNL/ACTL	<1983	493
41094.30y	93.90730	llldos	LLNL/ACTL	<1983	331
41095.30y	94.90680	llldos	LLNL/ACTL	<1983	333
41096.30y	95.90810	llldos	LLNL/ACTL	<1983	335
41097.30y	96.90810	llldos	LLNL/ACTL	<1983	339
41098.30y	97.91030	llldos	LLNL/ACTL	<1983	341
41100.30y	99.91420	llldos	LLNL/ACTL	<1983	349
Z = 42 ***** Molybdenum *****					
42090.30y	89.91390	llldos	LLNL/ACTL	<1983	261
42091.30y	90.91180	llldos	LLNL/ACTL	<1983	281
42092.26y	91.21000	532dos	ENDF/B-V	1980	7815
42092.30y	91.90680	llldos	LLNL/ACTL	<1983	537
42093.30y	92.90680	llldos	LLNL/ACTL	<1983	429
42093.31y	92.90680	llldos	LLNL/ACTL	<1983	461
42094.30y	93.90510	llldos	LLNL/ACTL	<1983	443
42095.30y	94.90580	llldos	LLNL/ACTL	<1983	523
42096.30y	95.90470	llldos	LLNL/ACTL	<1983	501
42097.30y	96.90600	llldos	LLNL/ACTL	<1983	427
42098.26y	97.06440	532dos	ENDF/B-V	1980	6489
42098.30y	97.90540	llldos	LLNL/ACTL	<1983	421
42099.30y	98.90770	llldos	LLNL/ACTL	<1983	445
42100.26y	99.04920	532dos	ENDF/B-V	1980	4971
42100.30y	99.90750	llldos	LLNL/ACTL	<1983	427
42101.30y	100.91000	llldos	LLNL/ACTL	<1983	447

APPENDIX G
DOSIMETRY DATA FOR MCNP

TABLE G-4 (Cont.)
Dosimetry Data Libraries for MCNP Tallies

ZAID	AWR	Library	Source	Date	Length
Z = 43 ***** Technetium *****					
43099.30y	98.90620	lildos	LLNL/ACTL	<1983	469
43099.31y	98.90620	lildos	LLNL/ACTL	<1983	469
Z = 45 ***** Rhodium *****					
45103.30y	102.90600	lildos	LLNL/ACTL	<1983	275
Z = 46 ***** Palladium *****					
46110.30y	109.90500	lildos	LLNL/ACTL	<1983	417
Z = 47 ***** Silver *****					
47106.30y	105.90700	lildos	LLNL/ACTL	<1983	263
47106.31y	105.90700	lildos	LLNL/ACTL	<1983	265
47107.30y	106.90500	lildos	LLNL/ACTL	<1983	517
47108.30y	107.90600	lildos	LLNL/ACTL	<1983	275
47108.31y	107.90600	lildos	LLNL/ACTL	<1983	275
47109.30y	108.90500	lildos	LLNL/ACTL	<1983	583
47110.30y	109.90600	lildos	LLNL/ACTL	<1983	277
47110.31y	109.90600	lildos	LLNL/ACTL	<1983	281
Z = 48 ***** Cadmium *****					
48106.30y	105.90600	lildos	LLNL/ACTL	<1983	177
48111.30y	110.90400	lildos	LLNL/ACTL	<1983	317
48112.30y	111.90300	lildos	LLNL/ACTL	<1983	221
48116.30y	115.90500	lildos	LLNL/ACTL	<1983	231
Z = 49 ***** Indium *****					
49113.30y	112.90400	lildos	LLNL/ACTL	<1983	861
49115.24y	113.92000	531dos	ENDF/B-V	1978	26009
49115.26y	113.92000	532dos	ENDF/B-V	1978	26009
49115.30y	114.90400	lildos	LLNL/ACTL	<1983	1265
Z = 50 ***** Tin *****					
50112.30y	111.90500	lildos	LLNL/ACTL	<1983	789
50114.30y	113.90300	lildos	LLNL/ACTL	<1983	435
50115.30y	114.90300	lildos	LLNL/ACTL	<1983	389
50116.30y	115.90200	lildos	LLNL/ACTL	<1983	603
50117.30y	116.90300	lildos	LLNL/ACTL	<1983	313
50118.30y	117.90200	lildos	LLNL/ACTL	<1983	745
50119.30y	118.90300	lildos	LLNL/ACTL	<1983	311
50120.26y	118.87200	532dos	ENDF/B-V	1974	12881
50120.30y	119.90200	lildos	LLNL/ACTL	<1983	309

APPENDIX G
DOSIMETRY DATA FOR MCNP

TABLE G-4 (Cont.)
Dosimetry Data Libraries for MCNP Tallies

ZAID	AWR	Library	Source	Date	Length
50122.26y	120.85600	532dos	ENDF/B-V	1974	1891
50122.30y	121.90300	lildos	LLNL/ACTL	<1983	275
50124.26y	122.84100	532dos	ENDF/B-V	1974	1693
50124.30y	123.90500	lildos	LLNL/ACTL	<1983	485
Z = 51 ***** Antimony *****					
51121.30y	120.90400	lildos	LLNL/ACTL	<1983	811
51123.30y	122.90400	lildos	LLNL/ACTL	<1983	1013
Z = 53 ***** Iodine *****					
53127.24y	125.81400	531dos	ENDF/B-V	1972	115
53127.26y	125.81400	532dos	ENDF/B-V	1980	14145
53127.30y	126.90400	lildos	LLNL/ACTL	<1983	221
Z = 55 ***** Cesium *****					
55133.30y	132.90500	lildos	LLNL/ACTL	<1983	215
Z = 57 ***** Lanthanum *****					
57139.26y	137.71300	532dos	ENDF/B-V	1980	15475
Z = 58 ***** Cerium *****					
58140.30y	139.90500	lildos	LLNL/ACTL	<1983	427
58142.30y	141.90900	lildos	LLNL/ACTL	<1983	265
Z = 59 ***** Praseodymium *****					
59141.30y	140.90800	lildos	LLNL/ACTL	<1983	215
Z = 60 ***** Neodymium *****					
60142.30y	141.90800	lildos	LLNL/ACTL	<1983	207
60148.30y	147.91700	lildos	LLNL/ACTL	<1983	255
60150.30y	149.92100	lildos	LLNL/ACTL	<1983	259
Z = 62 ***** Samarium *****					
62144.30y	143.91200	lildos	LLNL/ACTL	<1983	189
62148.30y	147.91500	lildos	LLNL/ACTL	<1983	245
62152.30y	151.92000	lildos	LLNL/ACTL	<1983	237
62154.30y	153.92200	lildos	LLNL/ACTL	<1983	247
Z = 63 ***** Europium *****					
63151.30y	150.92000	lildos	LLNL/ACTL	<1983	731

APPENDIX G
DOSIMETRY DATA FOR MCNP

TABLE G-4 (Cont.)
Dosimetry Data Libraries for MCNP Tallies

ZAID	AWR	Library	Source	Date	Length
63153.30y	152.92100	lildos	LLNL/ACTL	<1983	565
Z = 64 ***** Gadolinium *****					
64150.30y	149.91900	lildos	LLNL/ACTL	<1983	237
64151.30y	150.92000	lildos	LLNL/ACTL	<1983	241
Z = 66 ***** Dysprosium *****					
66164.26y	162.52000	532dos	ENDF/B-V	1967	581
Z = 67 ***** Holmium *****					
67163.30y	162.92900	lildos	LLNL/ACTL	<1983	533
67164.30y	163.93000	lildos	LLNL/ACTL	<1983	327
67164.31y	163.93000	lildos	LLNL/ACTL	<1983	327
67165.30y	164.93000	lildos	LLNL/ACTL	<1983	589
67166.30y	165.93200	lildos	LLNL/ACTL	<1983	333
67166.31y	165.93200	lildos	LLNL/ACTL	<1983	333
Z = 69 ***** Thulium *****					
69169.30y	168.93400	lildos	LLNL/ACTL	<1983	453
Z = 71 ***** Lutetium *****					
71173.30y	172.93900	lildos	LLNL/ACTL	<1983	587
71174.30y	173.94000	lildos	LLNL/ACTL	<1983	417
71174.31y	173.94000	lildos	LLNL/ACTL	<1983	465
71175.30y	174.94100	lildos	LLNL/ACTL	<1983	559
71176.30y	175.94300	lildos	LLNL/ACTL	<1983	621
71176.31y	175.94300	lildos	LLNL/ACTL	<1983	637
71177.30y	176.94400	lildos	LLNL/ACTL	<1983	573
71177.31y	176.94400	lildos	LLNL/ACTL	<1983	573
Z = 72 ***** Hafnium *****					
72174.30y	173.94000	lildos	LLNL/ACTL	<1983	147
72175.30y	174.94100	lildos	LLNL/ACTL	<1983	121
72176.30y	175.94100	lildos	LLNL/ACTL	<1983	153
72177.30y	176.94300	lildos	LLNL/ACTL	<1983	157
72178.30y	177.94400	lildos	LLNL/ACTL	<1983	153
72179.30y	178.94600	lildos	LLNL/ACTL	<1983	433
72180.30y	179.94700	lildos	LLNL/ACTL	<1983	409
72181.30y	180.94900	lildos	LLNL/ACTL	<1983	365
72183.30y	182.95400	lildos	LLNL/ACTL	<1983	373
Z = 73 ***** Tantalum *****					

APPENDIX G
DOSIMETRY DATA FOR MCNP

TABLE G-4 (Cont.)
Dosimetry Data Libraries for MCNP Tallies

ZAID	AWR	Library	Source	Date	Length
73179.30y	178.94600	llldos	LLNL/ACTL	<1983	629
73180.30y	179.94700	llldos	LLNL/ACTL	<1983	523
73180.31y	179.94700	llldos	LLNL/ACTL	<1983	435
73181.30y	180.94800	llldos	LLNL/ACTL	<1983	715
73182.30y	181.95000	llldos	LLNL/ACTL	<1983	435
73182.31y	181.95000	llldos	LLNL/ACTL	<1983	447
73183.30y	182.95100	llldos	LLNL/ACTL	<1983	425
73184.30y	183.95400	llldos	LLNL/ACTL	<1983	371
73186.30y	185.95900	llldos	LLNL/ACTL	<1983	377
Z = 74 ***** Tungsten *****					
74179.30y	178.94700	llldos	LLNL/ACTL	<1983	263
74180.30y	179.94700	llldos	LLNL/ACTL	<1983	397
74181.30y	180.94800	llldos	LLNL/ACTL	<1983	263
74182.30y	181.94800	llldos	LLNL/ACTL	<1983	415
74183.30y	182.95000	llldos	LLNL/ACTL	<1983	499
74184.30y	183.95100	llldos	LLNL/ACTL	<1983	443
74185.30y	184.95300	llldos	LLNL/ACTL	<1983	267
74186.30y	185.95400	llldos	LLNL/ACTL	<1983	413
74187.30y	186.95700	llldos	LLNL/ACTL	<1983	279
74188.30y	187.95800	llldos	LLNL/ACTL	<1983	271
Z = 75 ***** Rhenium *****					
75184.30y	183.95300	llldos	LLNL/ACTL	<1983	331
75184.31y	183.95300	llldos	LLNL/ACTL	<1983	335
75185.30y	184.95300	llldos	LLNL/ACTL	<1983	373
75186.30y	185.95500	llldos	LLNL/ACTL	<1983	381
75187.30y	186.95600	llldos	LLNL/ACTL	<1983	547
75188.30y	187.95800	llldos	LLNL/ACTL	<1983	339
75188.31y	187.95800	llldos	LLNL/ACTL	<1983	341
Z = 77 ***** Iridium *****					
77191.30y	190.96100	llldos	LLNL/ACTL	<1983	237
77193.30y	192.96300	llldos	LLNL/ACTL	<1983	243
77194.30y	193.96500	llldos	LLNL/ACTL	<1983	421
Z = 78 ***** Platinum *****					
78190.30y	189.96000	llldos	LLNL/ACTL	<1983	151
78192.30y	191.96100	llldos	LLNL/ACTL	<1983	153
78193.30y	192.96300	llldos	LLNL/ACTL	<1983	123
78193.31y	192.96300	llldos	LLNL/ACTL	<1983	123
78194.30y	193.96300	llldos	LLNL/ACTL	<1983	211
78195.30y	194.96500	llldos	LLNL/ACTL	<1983	157
78196.30y	195.96500	llldos	LLNL/ACTL	<1983	157
78197.30y	196.96700	llldos	LLNL/ACTL	<1983	427

APPENDIX G
DOSIMETRY DATA FOR MCNP

TABLE G-4 (Cont.)
Dosimetry Data Libraries for MCNP Tallies

ZAID	AWR	Library	Source	Date	Length
78197.31y	196.96700	llldos	LLNL/ACTL	<1983	129
78198.30y	197.96800	llldos	LLNL/ACTL	<1983	183
78199.30y	198.97100	llldos	LLNL/ACTL	<1983	99
78199.31y	198.97100	llldos	LLNL/ACTL	<1983	99
Z = 79 ***** Gold *****					
79193.30y	192.96400	llldos	LLNL/ACTL	<1983	209
79194.30y	193.96500	llldos	LLNL/ACTL	<1983	261
79195.30y	194.96500	llldos	LLNL/ACTL	<1983	261
79196.30y	195.96700	llldos	LLNL/ACTL	<1983	265
79196.31y	195.96700	llldos	LLNL/ACTL	<1983	265
79197.30y	196.96700	llldos	LLNL/ACTL	<1983	307
79198.30y	197.96800	llldos	LLNL/ACTL	<1983	265
79199.30y	198.96900	llldos	LLNL/ACTL	<1983	269
79200.30y	199.97100	llldos	LLNL/ACTL	<1983	39
Z = 80 ***** Mercury *****					
80202.30y	201.97100	llldos	LLNL/ACTL	<1983	381
80203.30y	202.97300	llldos	LLNL/ACTL	<1983	379
80204.30y	203.97300	llldos	LLNL/ACTL	<1983	365
Z = 81 ***** Thallium *****					
81202.30y	201.97200	llldos	LLNL/ACTL	<1983	377
81203.30y	202.97200	llldos	LLNL/ACTL	<1983	375
81204.30y	203.97400	llldos	LLNL/ACTL	<1983	373
81205.30y	204.97400	llldos	LLNL/ACTL	<1983	369
Z = 82 ***** Lead *****					
82203.30y	202.97300	llldos	LLNL/ACTL	<1983	257
82204.30y	203.97300	llldos	LLNL/ACTL	<1983	405
82205.30y	204.97400	llldos	LLNL/ACTL	<1983	257
82206.30y	205.97400	llldos	LLNL/ACTL	<1983	347
82207.30y	206.97600	llldos	LLNL/ACTL	<1983	333
82208.30y	207.97700	llldos	LLNL/ACTL	<1983	263
82209.30y	208.98100	llldos	LLNL/ACTL	<1983	279
82210.30y	209.98400	llldos	LLNL/ACTL	<1983	351
Z = 83 ***** Bismuth *****					
83208.30y	207.98000	llldos	LLNL/ACTL	<1983	409
83209.30y	208.98000	llldos	LLNL/ACTL	<1983	551
83210.30y	209.98400	llldos	LLNL/ACTL	<1983	421
83210.31y	209.98400	llldos	LLNL/ACTL	<1983	421

APPENDIX G
DOSIMETRY DATA FOR MCNP

TABLE G-4 (Cont.)
Dosimetry Data Libraries for MCNP Tallies

ZAID	AWR	Library	Source	Date	Length
Z = 84 ***** Polonium *****					
84210.30y	209.98300	lldos	LLNL/ACTL	<1983	441
Z = 90 ***** Thorium *****					
90230.30y	230.03300	lldos	LLNL/ACTL	<1983	209
90231.30y	231.03600	lldos	LLNL/ACTL	<1983	599
90232.30y	232.03800	lldos	LLNL/ACTL	<1983	347
90233.30y	233.04200	lldos	LLNL/ACTL	<1983	561
90234.30y	234.04400	lldos	LLNL/ACTL	<1983	37
Z = 91 ***** Protactinium *****					
91231.26y	229.05000	532dos	ENDF/B-V	1978	2861
91233.26y	231.03800	532dos	ENDF/B-V	1978	73
91233.30y	233.04000	lldos	LLNL/ACTL	<1983	361
Z = 92 ***** Uranium *****					
92233.26y	231.04300	532dos	ENDF/B-V	1978	75
92233.30y	233.04000	lldos	LLNL/ACTL	<1983	461
92234.30y	234.04100	lldos	LLNL/ACTL	<1983	393
92235.30y	235.04400	lldos	LLNL/ACTL	<1983	4629
92236.30y	236.04600	lldos	LLNL/ACTL	<1983	395
92237.30y	237.04900	lldos	LLNL/ACTL	<1983	609
92238.30y	238.05100	lldos	LLNL/ACTL	<1983	3103
92239.30y	239.05400	lldos	LLNL/ACTL	<1983	825
92240.30y	240.05700	lldos	LLNL/ACTL	<1983	389
Z = 93 ***** Neptunium *****					
93237.30y	237.04800	lldos	LLNL/ACTL	<1983	629
Z = 94 ***** Plutonium *****					
94237.30y	237.04800	lldos	LLNL/ACTL	<1983	487
94238.30y	238.05000	lldos	LLNL/ACTL	<1983	459
94239.30y	239.05200	lldos	LLNL/ACTL	<1983	497
94240.30y	240.05400	lldos	LLNL/ACTL	<1983	479
94241.30y	241.05700	lldos	LLNL/ACTL	<1983	559
94242.30y	242.05900	lldos	LLNL/ACTL	<1983	505
94243.30y	243.06200	lldos	LLNL/ACTL	<1983	511
Z = 95 ***** Americium *****					
95241.30y	241.05700	lldos	LLNL/ACTL	<1983	673
95242.30y	242.06000	lldos	LLNL/ACTL	<1983	473
95243.30y	243.06100	lldos	LLNL/ACTL	<1983	431

APPENDIX G REFERENCES

TABLE G-4 (Cont.)
Dosimetry Data Libraries for MCNP Tallies

ZAID	AWR	Library	Source	Date	Length
Z = 96 ***** Curium *****					
96242.30y	242.05900	lildos	LLNL/ACTL	<1983	467
96243.30y	243.06100	lildos	LLNL/ACTL	<1983	465
96244.30y	244.06300	lildos	LLNL/ACTL	<1983	483
96245.30y	245.06500	lildos	LLNL/ACTL	<1983	465
96246.30y	246.06700	lildos	LLNL/ACTL	<1983	491
96247.30y	247.07000	lildos	LLNL/ACTL	<1983	491
96248.30y	248.07200	lildos	LLNL/ACTL	<1983	495
Z = 97 ***** Berkelium *****					
97249.30y	249.07500	lildos	LLNL/ACTL	<1983	545
Z = 98 ***** Californium *****					
98249.30y	249.07500	lildos	LLNL/ACTL	<1983	491
98250.30y	250.07600	lildos	LLNL/ACTL	<1983	335
98251.30y	251.08000	lildos	LLNL/ACTL	<1983	485
98252.30y	252.08200	lildos	LLNL/ACTL	<1983	467

VI. REFERENCES

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S E C T I O N 2

Readme_LA150N

This 150-MeV neutron library designated LA150N was released July 2000 and updated slightly in May 2002.

There are data tables for 42 isotopes on the library. Forty-one are based on recent T-16 work and extend to 150 MeV. These are all ENDF/B-VI Release 6 evaluations, with the exception of Si-28, Nb-93, Bi-209, Pb-208, and the Hg isotopes. These sets were either not submitted for ENDF/B-VI Release 6, or have undergone minor modifications since then. The 42nd isotope is Be-9, which extends to 100 MeV and is based on work performed in T-2 over 10 years ago.

The modification in May 2002 was made to Pb-208 to correct a small error in the original release. The ZAID for Pb-208 has been changed from 82208.24c to 82208.25c to differentiate from the earlier version.

The isotopes and ZAIIDs are as follows:
(all ZAIIDs end in .24c except for Pb-208)

H-1	1001.24c
H-2	1002.24c
Be-9	4009.24c
C-nat	6000.24c
N-14	7014.24c
O-16	8016.24c
Al-27	13027.24c
Si-28	14028.24c
Si-29	14029.24c
Si-30	14030.24c
P-31	15031.24c
Ca-nat	20000.24c
Cr-50	24050.24c
Cr-52	24052.24c
Cr-53	24053.24c
Cr-54	24054.24c
Fe-54	26054.24c
Fe-56	26056.24c
Fe-57	26057.24c
Ni-58	28058.24c
Ni-60	28060.24c
Ni-61	28061.24c
Ni-62	28062.24c
Ni-64	28064.24c
Cu-63	29063.24c
Cu-65	29065.24c
Nb-93	41093.24c
W-182	74182.24c
W-183	74183.24c
W-184	74184.24c
W-186	74186.24c
Hg-196	80196.24c
Hg-198	80198.24c
Hg-199	80199.24c
Hg-200	80200.24c
Hg-201	80201.24c
Hg-202	80202.24c
Hg-204	80204.24c
Pb-206	82206.24c

Readme_la150n.txt

Pb-207 82207.24c

Pb-208 82208.25c

Bi-209 83209.24c

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Robert C. Little

Los Alamos National Laboratory

Readme_LA150H

The LA150H library contains data for incident protons. Data for 41 target isotopes are included. All ZAIDs end in .24h. The LA150H library is compatible with version 2.3.0 of MCNPX and subsequent releases (also with earlier versions released to the Beta Testing community).

Data are provided up to an incident proton energy of 150 MeV. The minimum energy varies from 1 KeV to 3 MeV. The data include production cross sections and energy-angle correlated spectra of secondary light particles (neutrons, protons, deuterons, tritons, and alphas) and production cross sections and energy distributions of gamma rays. Recoil data from elastic scattering are not included on the library; the intention is to update MCNPX to handle such effects within the code for light targets.

The data contained in LA150H are based on evaluations performed in Group T-16 at Los Alamos. The evaluations are discussed in further detail in Nucl. Sci. Eng., 131, 293 (1999).

The list of target materials and ZAIDs found on LA150H follows:

H-1	1001.24h
H-2	1002.24h
C-12	6012.24h
N-14	7014.24h
O-16	8016.24h
Al-27	13027.24h
Si-28	14028.24h
Si-29	14029.24h
Si-30	14030.24h
P-31	15031.24h
Ca-40	20040.24h
Cr-50	24050.24h
Cr-52	24052.24h
Cr-53	24053.24h
Cr-54	24054.24h
Fe-54	26054.24h
Fe-56	26056.24h
Fe-57	26057.24h
Ni-58	28058.24h
Ni-60	28060.24h
Ni-61	28061.24h
Ni-62	28062.24h
Ni-64	28064.24h
Cu-63	29063.24h
Cu-65	29065.24h
Nb-93	41093.24h
W-182	74182.24h
W-183	74183.24h
W-184	74184.24h
W-186	74186.24h
Hg-196	80196.24h
Hg-198	80198.24h
Hg-199	80199.24h
Hg-200	80200.24h
Hg-201	80201.24h
Hg-202	80202.24h

Hg-204	80204.24h
Pb-206	82206.24h
Pb-207	82207.24h
Pb-208	82208.24h
Bi-209	83209.24h

Robert C. Little
Los Alamos National Laboratory

Readme_la150u.txt

The LA150U photonuclear file was added to the MCNPXDATA package in May 2002. These data are on the distribution CD in file d205tar2.gz.

Seven files related to the new photonuclear data are included in this package.

Readme_pn.txt - This readme file
mcw-00-86.pdf - ACE Class 'u' Format Documentation
mcw-00-87.pdf - Release documentation for LA150u
mcw-00-88.pdf - User Interface Guide for PN Physics
mcw-00-89.pdf - Example calculation using PN Physics
la150u - The photonuclear data library with 12 tables
xsdir - includes XSDIR entries for the 12 photonuclear tables

LANL Memorandum X-5:MCW-00-86 (public release under LA-UR-01-1597) This memo documents the ACE format for photonuclear data. It is useful to those persons wishing to create their own data sets for use in MCNP(X). It is similar to information found in Appendix F of the MCNP manual.

LANL Memorandum X-5:MCW-00-87 (public release under LA-UR-01-1596)

This memo documents the creation and release of the LA150u library. It includes details of how the library was created as well as summary information appropriate to Appendix G of the MCNP manual.

LANL Memorandum X-5:MCW-00-88 (public release under LA-UR-01-1598)

This memo documents the changes to the MCNP(X) user interface including updates to the material (M) and photon physics (phys:p) cards and new information on the photonuclear isotope override (MPN) card. The default is for photonuclear physics to be off due to the sparse availability of tables. This memo provides necessary information for setting up materials to load appropriate photonuclear tables and for turning on and/or biasing photonuclear interactions. It also includes samples of the new information provided in the summary tables. The user interface information is appropriate to Chapter 3 of the MCNP manual.

LANL Memorandum X-5:MCW-00-89 (public release under LA-UR-01-1599)

This memo provides a primer for doing simple calculations using the photonuclear physics capability. It is a good starting place for anyone interested in this new capability and contains several lessons learned during development and validation. It is appropriate to Chapter 5 of the MCNP manual.

Information in these memorandum will be updated within the MCNP and MCNPX manuals.

la150u

la150u is the data library containing photonuclear tables for the following isotopes: 12-C-6, 16-O-8, 27-Al-13, 28-Si-14, 40-Ca-20, 56-Fe-26, 63-Cu-29, 181-Ta-73, 184-W-74, 206-Pb-82, 207-Pb-82, and 208-Pb-82.

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3**

This new package of data libraries contains all publically available data libraries supported by X-5 for use with MCNP. The package contains the following 28 neutron data libraries:

Continuous-energy

ENDF60 RMCCS RMCCSA ENDF5U ENDF5P
NEWXS ENDF5MT* MISC5XS** ENDL85 KIDMAN
100XS URES ENDF62MT ENDF6DN ENDL92

Discrete

NEWXSD DRMCCS DRE5

Multigroup

MGXSNP

S(alpha,beta)

TMCCS THERXS

Dosimetry

531DOS 532DOS LLLDOS

the photon libraries: MCPLIB1 MCPLIB02

and the electron libraries: EL1 EL03

* The data library ENDF5MT contains data previously available in the library EPRIXS, along with the U600K data library.

** The data library MISC5XS contains corrected data for ENDF/B-V based Zr as described below, and the libraries IRNAT, MISCXS, ARKRC, TM169, GDT2GP, and T2DDC. None of these data libraries have been previously distributed through RSICC. The ENDF/B-V Zr data has been corrected for five ZAID's from the libraries RMCCS, DRMCCS, ENDF5P, DRE5, and EPRIXS. [Ref. 1-2] Below is a summary of the changes that have been made:

(Previous)	(Corrected)
RMCCS 40000.51c -->	40000.57c MISC5XS 300K
DRMCCS 40000.51d -->	40000.57d MISC5XS 300K
ENDF5P 40000.50c -->	40000.56c MISC5XS 300K
DRE5 40000.50d -->	40000.56d MISC5XS 300K
EPRIXS 40000.53c -->	40000.58c MISC5XS 600K

Additionally, the corresponding XSDIR file for all of these libraries is also distributed with this package.

A SPECS file which preserves the default libraries is also provided for the MCNP user for producing Type 2 (Binary) versions of the data libraries to conserve disk space and speed up calculations.

ADDITIONS AND CHANGES IN DLC-200 RELATIVE TO DLC-189

Five new libraries have been made available in DLC-200 that were not generally distributed earlier. Also, the Atomic Weight Ratios have been updated in the XSDIR file.

Documentation is available at:

<http://www-xdiv.lanl.gov/XCI/PROJECTS/DATA/nuclear/avdoc.htm>

and an updated version of Appendix G is available at:

http://www-xdiv.lanl.gov/XCI/PROJECTS/DATA/nuclear/doc/Appendix_G.html

We now summarize briefly the additions and changes.

Atomic Weight Ratios

New atomic weight ratios (AWRs) have been obtained from the latest set of atomic mass data by G. Audi and A. H. Wapstra (1995) and included in the first section of the XSDIR file. These AWRs are used to convert weight fraction and densities into atomic fractions and densities. They are also used to compute an average Z for electron stopping powers. Hence slight differences in calculations can be expected if the user specified the materials with weight fractions and densities. The new ratios are documented in XCI:TKG-99-53(U).

ENDF6DN

ENDF6DN is a room-temperature neutron data library for 20 actinides that contains detailed delayed-neutron data. The data tables on ENDF6DN are referenced by the ZAID ending of '.61c'. This library supplemented data from the ENDF60 neutron data library (in other words, data from .61c are identical to .60c except for the additional detailed delayed-neutron data provided), and is based on Release 2 of ENDF/B-VI. More information for this library can be obtained from XCI:CJW-99-25(U).

Note: Utilization of detailed delayed-neutron data is a new MCNP4C feature. These data tables may be used by MCNP4B and earlier versions; however the detailed delayed-neutron data will be ignored.

ENDL92

ENDL92 is a room-temperature neutron data library from LLNL based upon Livermore neutron evaluations circa 1992. Data for 100 isotopes are included, with ZAID endings of '.42c'. Included among these isotopes are Ne-20, Zn, Sb, and Hg, which were previously unavailable in MCNP libraries. Two sets of data exist for Zn, Sn, Pt, and Hg. The original LLNL data for these isotopes have ZAID endings of '.40c'; sets modified at LANL to correct non-physical total and elastic cross sections have ZAID endings of '.42c'. This library is considered a replacement for ENDL85 and ENDL85 will be retired during the next update. Users wishing to contact LLNL with questions about ENDL92 may send requests to nuclear-data@lists.llnl.gov.

EL03

EL03 is the newest electron data library, and becomes the default library for electron transport. The data in EL03 are referenced by the ZAID ending of '.03e' and cannot be used in combination with data from the EL library (.01e). EL03 is roughly equivalent to the data used in ITS Version 3.0 and is documented in X-5:KJA-00-34(U).

Note: EL03 requires new functionality in MCNP4C. EL03 cannot be used with MCNP4B and earlier versions.

ENDF62MT

ENDF62MT is a multi-temperature neutron library, with data for U-235, U-238, and Pu-239 at 77, 400, 500, 600, 800, 900, and 1200 degrees K. The same evaluations sources were used for this library as for the room-temperature sets found on ENDF60; however the NJOY processing tolerance was significantly tighter for ENDF62MT than for ENDF60.

The library is documented in XTM:96-153.

URES

URES is the only library distributed with DLC-200 that contains unresolved resonance probability-table data. Data are provided for 27 isotopes based on ENDF/B-VI Release 4 (note - other ENDF/B-VI libraries in this package are based on Release 2 of ENDF/B-VI). ZAIID endings on URES are '.49c'. The library is documented in LA-UR-98-5718.

Note: Utilization of probability-table data is a new MCNP4C feature. These data tables may be used by MCNP4B and earlier versions; however the detailed probability-table data will be ignored.

The user must be aware of the limitations for some of these continuous-energy neutron data libraries.

ENDF60: This library is described in the accompanying file README_ENDF60, and other documentation in pdf format.

100XS: The 100XS library contains 9 data files nuclides each having an evaluation extending to 100 MeV. The 9 materials are Be-9, C-12, O-16, Al-27, Si, Ca- 40, Fe, W, and U-238. Modifications were made to the original evaluations and resulting data files to make them more compatible with MCNP, and version 4B of MCNP is needed to use these libraries. These files have been given the ZAIID ending of ".21c". Heating numbers on this library are known to overestimate the energy deposition. The report "Summary Documentation for the 100XS Neutron Cross Section Library (Release 1)" LA-UR-96-24 [Ref. 3] describes the general characteristics of the library.

KIDMAN: This library contains data for a number of fission-product nuclides at 300K. There are no photon production available for nuclides on this library, and therefore the neutron heating numbers correspond to total heating (neutron + photon). Total charged particle production MT's (MT=203-207) are available where appropriate. [Ref. 4]

MISC5XS: The data previously released as GDT2GP provide ENDF/B-V based Gd cross-section with photon production data from T-2 (LANL). The photon-production data are only valid up to 1 MeV. Since the photon production data were added to an existing set (ENDF5U), the neutron heating information is incorrect for this library and should not be used. [Ref. 5]

The data previously released as ARKRC are from both ENDF and ENDL sources. The neutron cross-section information for the Kr isotopes are from ENDF, while the cross-sections are from ENDL for Ar. Photon production data were added by T-2 (LANL). Photon production data were included to provide the user with rough estimates for quantities such as photon heating, not for detailed calculations. [Ref. 6]

The data previously released as T2DDC were provided by T-2 (LANL) for Br-79, Br-81, Rb-85, Rb-87, I-127 and Cs-133. All evaluations, with the exception of Cs-133, are incomplete having no photon production, heating numbers, or angular distribution information. [Ref. 7]

The thermal neutron libraries, TMCCS and THERXS, contain S(alpha,beta) data tables for MCNP. TMCCS is a library based on ENDF/B-V (which was unchanged from ENDF/B-III) and provides data for several materials at a number of temperatures. [Ref. 8] THERXS provides data for a few additional moderators based on later T-2 (LANL) evaluations.

The three dosimetry libraries, 531DOS, 532DOS, and LLLDOS, provide (energy, cross-section) pairs for a large number of reactions that are used in various dosimetry or activation applications. The ZAIID's on these libraries can only be used with the FM card option in MCNP, not for actual transport. [Ref. 9]

The photon data library MCPLIB02 is described in the pdf formatted document mcplib02.pdf.

REFERENCES

1. "Repairing ENDF/B-V Zirconium Cross Sections," R.E. Seamon, Los Alamos National Laboratory internal memorandum, X-6:RES-91-358 (1991).
2. "Revised ENDF/B-V Zirconium Cross Sections," R.E. Seamon, Los Alamos National Laboratory internal memorandum, X-6:RES-92-324 (1992).
3. "Summary Documentation for the 100XS Neutron Cross Section Library (Release 1)," Los Alamos National Laboratory internal memorandum, XTM:RCL-95-259 and LA-UR-96-24 (1995).
5. "ENDF/B-V Gd Cross Sections with Photon Production," R. C. Little and R. E. Seamon, Los Alamos National Laboratory internal memorandum, X-6:RCL:RES-86-30 (1986).
6. "Argon and Krypton Cross-section Files," R. C. Little, Los Alamos National Laboratory internal memorandum, June 30, 1982.
7. "Cross Sections in ACE Format for Various IP Target Materials," R. C. Little, Los Alamos National Laboratory internal memorandum, August 19, 1982.
8. "New Thermal Library for MCNP (Update)," R. C. Little, Los Alamos National Laboratory internal memorandum, X-6:RCL-86-400 (1986).
9. "Dosimetry/Activation Cross Sections for MCNP," R. C. Little, Los Alamos National Laboratory internal memorandum, March 13, 1984.

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The following README file describes the ENDF60 library which was released as DLC-181 in October, 1994. To make it easier to obtain all of the X-5 currently supported data libraries for MCNP, we have combined all of these data libraries into one package, DLC-200, for release through RSICC. This file includes all of the information from the original README and README2 files for ENDF60, along with updated information as it pertains to the new release DLC-200.

The ENDF60 library includes data for 124 nuclides. The ENDF60 library is composed of type 1 files and requires approximately 200 megabytes of disk space for the complete library.

The rest of this ENDF60 README file first describes the data library characteristics, and then briefly describes code modifications for NJOY and MCNP for the new scattering laws of ENDF/B-VI and the testing that the library has undergone. The testing of the data library included infinite medium simulations, the LLNL pulsed-sphere benchmarks, criticality simulations, photon production assessment at thermal and 14-MeV incident neutron energies, and a number of iron benchmarks. A more detailed description of the ENDF60 library has been published as a Los Alamos National Laboratory report (LA-12891). You can also access additional (up-to-date) documentation from our WWW site at:

<http://www-xdiv.lanl.gov/XCI/PROJECTS/DATA/nuclear/avdoc.htm>

Users who have further questions should contact
nucldata@lanl.gov

DATA LIBRARY CHARACTERISTICS:

The ENDF60 library was processed with NJOY91 at room temperature (300 K), using flat weighting, and thinned such that most nuclides had no more than 400,000 words (or used approximately 3.2 megabytes). The largest data file is i53127 which requires 8.1 megabytes of disk space. Each of the library files has the unique ZAID ending of .60c.

For the Z>94 nuclides with photon production, MCNP will not run in mode [np] as photon transport libraries are not available. MCNP can use the data libraries in mode [n] for neutron transport. It is anticipated that a new photon library for Z's up to 100 will be released during 1995. Additionally, charged-particle production cross-sections in the MT=200 series are not provided.

There are a few common questions that have been asked. First, the upper energy limit for the data files of the ENDF60 library are 20 MeV with the following exceptions:

am95241	30.0
au79197	30.0
c6000	32.0
ho67165	30.0
h1001	100.0
i53127	30.0
np93238	17.5
u92234	17.5

Secondly, the ENDF60 library does not contain the 200 series of MT, or R, values for tally on the total charged-particle production.

Table 1 describes the individual data files of the ENDF60 library. The ZAID, evaluation group, evaluation type, release number, and photon production are given for each nuclide. The evaluation group is identified by the laboratory participants who performed the evaluation, and the evaluation type indicates whether the ENDF/B-VI evaluation is a new evaluation (N) or is a translation of the ENDF/B-V evaluation (T).

New evaluations comprise 48% of the total in the ENDF60 library. If the data file is a new evaluation, the corresponding revision number for the ENDF/B-VI release is specified, where release 6.0 may correspond to a new or translated evaluation. If the evaluation has been modified at Los Alamos, to add in photon production for instance, the revision number has been specified as LANL. Photon production availability is indicated in the last column. **The specification of a new or translated library is given with respect to the ENDF/B-V Release 0 evaluations on which most of the '.50c' data files are based. The changes in Release # indicate those LANL modifications which were accepted as part of ENDF/B-VI Release 2. Errata Sheets for Table 1 of LA-12887 and LA-12891 are available. **

Table 1: ENDF60 Library for Use with MCNP

ZAID	Source	Evaluation Type	Release #	Photon
1001.60c	LANL	N	6.1	yes
1002.60c	LANL, AWRE	N	6	yes
1003.60c	LANL	T	6	no
2003.60c	LANL	N	6.1	no
2004.60c	LANL	T	6	no
3006.60c	LANL	N	6.1	yes
3007.60c	LANL	N	6	yes
4009.60c	LLNL	N	6	yes
5010.60c	LANL	N	6.1	yes
5011.60c	LANL	N	6	yes
6000.60c	ORNL	N	6.1	yes
7014.60c	LANL	N	LANL	yes
7015.60c	LANL	N	6	yes
8016.60c	LANL	N	6	yes
8017.60c	BNL	T	6	no
9019.60c	ORNL	N	6	yes
11023.60c	ORNL	T	6.1	yes
12000.60c	ORNL	T	6	yes
13027.60c	LANL	T	6	yes
14000.60c	ORNL	T	6	yes
15031.60c	LLNL	T	6	yes
16000.60c	BNL	T	6	yes
16032.60c	LLNL	T	6	yes
17000.60c	GGA	T	6	yes
19000.60c	GGA	T	6	yes
20000.60c	ORNL	N*	6	yes
21045.60c	BNL	N*	6.2	yes
22000.60c	BRC, ANL	T	6	yes
23000.60c	ANL, LLNL, +	N	6	yes
24050.60c	ORNL	N	6.1	yes
24052.60c	ORNL	N	6.1	yes
24053.60c	ORNL	N	6.1	yes
24054.60c	ORNL	N	6.1	yes
25055.60c	ORNL	N	6	yes
26054.60c	ORNL	N	6.1	yes
26056.60c	ORNL	N	6.1	yes
26057.60c	ORNL	N	6.1	yes
26058.60c	ORNL	N	6.1	yes
27059.60c	ANL	N	6.2	yes
28058.60c	ORNL	N	6.1	yes
28060.60c	ORNL	N	6.1	yes
28061.60c	ORNL	N	6.1	yes
28062.60c	ORNL	N	6.1	yes
28064.60c	ORNL	N	6.1	yes
29063.60c	ORNL	N	6.2	yes

			Readme_endf60.txt	
29065.60c	ORNL	N	6.2	yes
31000.60c	LLNL, LANL	T	6	yes
39089.60c	ANL, LLNL	N*	6	yes
40000.60c	SAI, BNL	T	6.1	no
41093.60c	ANL, LLL	N	6.1	yes
42000.60c	LLNL, HEDL	T	6	yes
43099.60c	HEDL, BAW	T	6	no
47107.60c	BNL, HEDL	N*	6	no
47109.60c	BNL, HEDL	N*	6	no
49000.60c	ANL	N	6	yes
53127.60c	HEDL, RCN	N*	LANL	yes
53129.60c	HEDL, RCN	T	6	no
55133.60c	HEDL, BNL, +	T	6	no
55134.60c	ORNL, HEDL	N	6	no
55135.60c	HEDL	T	6	no
55136.60c	HEDL	T	6	no
55137.60c	HEDL	T	6	no
56138.60c	ORNL, HEDL	T	6	yes
63151.60c	LANL	N	6	yes
63153.60c	LANL	N	6	yes
64152.60c	BNL	T	6	no
64154.60c	BNL	T	6	no
64155.60c	BNL	T	6	no
64156.60c	BNL	T	6	no
64157.60c	BNL	T	6	no
64158.60c	BNL	T	6	no
64160.60c	BNL	T	6	no
67165.60c	LANL	N	6	yes
72000.60c	SAI	T	6	no
73181.60c	LLNL	T	6	yes
73182.60c	AI	T	6	no
74182.60c	LANL, ANL, +	N*	6	yes
74183.60c	LANL, ANL, +	N*	6	yes
74184.60c	LANL, ANL, +	N*	6	yes
74186.60c	LANL, ANL, +	N*	6	yes
75185.60c	ORNL, LANL	N	6	no
75187.60c	ORNL, LANL	N	6	no
79197.60c	LANL	N	6.1	yes
82206.60c	ORNL	N	6	yes
82207.60c	ORNL	N	6.1	yes
82208.60c	ORNL	N	6	yes
83209.60c	ANL	N	6	yes
90230.60c	HEDL	T	6	no
90232.60c	BNL, ANL, +	T	6	yes
91231.60c	HEDL	T	6	no
92232.60c	HEDL	T	6	no
92233.60c	LANL, ORNL	T	6	yes
92234.60c	BNL, GGA	T	6	no
92235.60c	ORNL, LANL	N	6.2*	yes
92236.60c	HEDL	N	6	no
92238.60c	ORNL, LANL, +	N	6.2	yes
93237.60c	LANL	N	6.1	yes
93239.60c	ORNL	N	6	no
94236.60c	HEDL, SRL	N	6	no
94237.60c	HEDL	N	6	no
94238.60c	HEDL, AI, +	T	6	no
94239.60c	LANL	N	6.2	yes
94240.60c	ORNL	N	6.2*	yes
94241.60c	ORNL	N	6.1	yes
94242.60c	HEDL, SRL, +	T	6	yes
94243.60c	BNL, SRL, +	T	6.2*	yes
94244.60c	HEDL, SRL	T	6	no
95241.60c	CNDC	N	LANL	yes
95243.60c	ORNL, HEDL, +	N	6	yes
96241.60c	HEDL	T	6	no

			Readme_endf60.txt	
96242.60c	HEDL, SRL, +	T	6	yes
96243.60c	HEDL, SRL, +	T	6	yes
96244.60c	HEDL, SRL, +	T	6	yes
96245.60c	SRL, LLNL	T	6.2	yes
96246.60c	BNL, SRL, +	T	6.2	yes
96247.60c	BNL, SRL, +	T	6.2	yes
96248.60c	HEDL, SRL, +	T	6	yes
97249.60c	CNDC	N	6	no
98249.60c	CNDC	N	LANL	no
98250.60c	BNL, SRL, +	T	6.2	yes
98251.60c	BNL, SRL, +	T	6.2	yes
98252.60c	BNL, SRL, +	T	6.2*	yes

* The data files np93238 and am95242 have been removed from distribution. While the data files accurately reflected the evaluations, we believe that the evaluations are too incomplete to be used for most transport calculations.

NEW SCATTERING LAWS IN ENDF/B-VI:

Three new scattering laws of the ENDF/B-VI evaluations have been implemented in NJOY and MCNP. This required new data formats and processing, transport physics, and next-event estimator sampling schemes. The three laws are the Kalbach-87 formalism (file 6, law=1, lang=2 : MCNP law 44), correlated angle-energy scattering (file 6, law=7 : MCNP law 67), and correlated angle-energy scattering phase-space law (file 6 law=6 : MCNP law 66). In addition to the algorithms to sample these laws for neutron transport, new algorithms were developed for next-event estimators so that collisions using the new laws now contribute to point and ring detectors, as well as to the DXTRAN variance reduction method.

The Kalbach-87 formalism requires sampling the cosine between the incident and outgoing particle directions from a density function. This formalism was also used to approximate the new ORNL evaluations (F, Cr, Mn, Fe, Ni, Cu and Pb) which used file 6, law=1, lang=1. An approximation was required since MCNP cannot as yet sample from a Legendre polynomial expansion. The major error in this approximation for the ORNL evaluations is that the tabulated outgoing laboratory energy is used in the formulation as if it were in the center-of-mass. This error is not substantial for the heavier elements, but was thought to be significant for the F evaluation. Therefore, F was approximated using file 6, law=7 instead.

The correlated angle-energy scattering law first samples the angle μ from 32-equiprobable cosine bins between the incident and outgoing particle directions with interpolation for a given incident energy E . With μ and E fixed, the outgoing energy, E' , is sampled from tables interpolated on the μ and E tables. This law is used for the 9-Be and F evaluations only.

The correlated angle-energy scattering phase-space law first samples the outgoing particle direction cosine isotropically in the center-of-mass system. Then given the number of bodies in phase space and the total mass ratio for all the bodies, the outgoing energy is sampled from a density function. This law is used for the 2-H evaluation only.

DATA LIBRARY TESTING AND ERRORS:

In the process of data testing, several problems were encountered and corrected. The 16-0 data file was modified such that exponents less than -37 were increased to -30 to prevent MCNP from crashing on HP workstations. The natural Cd file was not released due to a processing

error by NJOY when photon production was added. As the Cd evaluation is simply a translation from ENDF/B-V, this was not considered critical to the library's release. 249-Cf was also modified to correct an evaluation problem concerning center-of-mass to lab energy conversion. Prior to this correction, neutrons could up-scatter above 20 MeV and cause MCNP to crash.

All of the new ORNL evaluations (F, Cr, Mn, Fe, Ni, Cu, and Pb) use the Legendre expansion description for energy-angle distributions. Because MCNP does not yet handle this scattering law, these were approximated with the Kalbach-87 formalism. The interpolation scheme for INT=12 in the Kalbach-87 formalism has been replaced with INT=2 for neutrons in the data library. For photons, INT=12 remains specified in the data library but is treated as INT=2 by MCNP.

There are two files which have problems with TYR values (see Appendix F of the MCNP4A manual for discussion of these values). The first file is c6000, natural carbon. The (n,2n)p reaction, MT=41, should have a TYR value of 2 instead of the 1 contained in the data file. To correct, go to line number 1256 of the data file 'c6000' and replace the 1 in column 40 with a 2. The threshold for this reaction is $E_n=29.7$ MeV and hence not many applications will be affected by this error.

The second file which has errors in the TYR values is sc21045. In this particular case, the errors occur in the evaluation of Sc-45 and are carried through the processing. The Cross Section Evaluation Working Group (CSEWG) has been notified of these problems and the next release of Sc-45 will hopefully have this corrected. The TYR values for MT=16, 22, 28, and 91 should be positive values indicating the distributions are for the laboratory system, not the center-of-mass system. To correct these values, one should do the following to the sc21045 file:

```
go to line number: 13323   change the '-' to ' ' in column: 79
                   13324                                     19 and 39
                   13328                                     19
```

The space, ' ', is important as the TYR values must be in the proper column.

Further information on issues relating to the QA of ENDF60 can be found in XTM:SCF-96-363.

BENCHMARKING:

There were five sets of MCNP simulations performed for the ENDF60 library: (1) infinite medium simulations, (2) LLNL pulsed sphere benchmarks, (3) criticality simulations, (4) iron benchmarks, and (5) photon production assessment. For all tests, both the ENDF/B-VI and B-V libraries were used and compared with one another. Additionally, the recommended data library was used when it differed from the ENDF/B-V library. The results of each of these is briefly described below.

(1) Infinite Medium Simulations:

The infinite medium simulations used a 20-MeV neutron source in an infinite medium of the nuclide. This source was modified so that the reactions for the heaviest elements spanned the entire neutron energy range from 0-20 MeV. The neutron flux and heating, as well as the photon flux and heating, were tallied and coplots of the results were made for the different libraries. These simulations showed that the flux tallies were basically equivalent for the two libraries, with the obvious exception of when ENDF/B-VI has photon production while ENDF/B-V did not. NJOY has made improvements in calculating the neutron heating since the previous libraries were processed. NJOY now takes into account the recoil of the resulting nucleus as well as the outgoing particle's energy. Corrections to the Q-values of the B-V translated evaluations

also improved the neutron heating results. This work will be documented in an LA-report.

(2) LLNL Pulsed Sphere Benchmarks:

Simulations were performed for 28 neutron pulsed-sphere experiments originally designed to benchmark several neutron transport codes at LLNL in the 1960's. The experiments used a pulsed-deuterium beam on a tritium target centered within a spherical shell of material. The pulsed beam produced 14-MeV neutrons and time-of-flight was used to measure the resulting neutron flux spectrum. The spheres of material were 0.5 to 4.8 mean free paths thick. The results were compared with the experimental value. Both the B-VI and B-V libraries gave similar results for O, Mg, Al, Ti, H₂O, D₂O, and concrete. The B-V library more closely matched the experimental value for 6-Li and C, while the B-VI library more closely matched the experimental value for 7-Li, Be, N, F, Pb, CF₂, and CH₂. The special T-2 evaluation for Fe did marginally better than the B-VI library, both of which were better than the previous B-V library. This work is documented in LA-12885. * These benchmarks have been revised and expanded. To be documented in a forthcoming LA report.

(3) Criticality Simulations:

The MCNP4 validation benchmarks, LA-12212 and LA-12415, which used the ENDF/B-V and recommended data libraries, were repeated using MCNP4A with all data libraries. The first set of benchmarks consisted of 9 critical assemblies used to benchmark the COG code, while the second set of benchmarks consisted of 25 simulations problems used to benchmark the KENO code. The results indicate very little difference between the libraries for these simulations. * These benchmarks have been revised and expanded, XTM-RN(U)96-030. To be documented in a forthcoming LA report.

(4) Iron Benchmarks:

Four sets of Iron benchmarks were performed; (1) the LLNL pulsed spheres, (2) ORNL Fusion Reactor Shielding benchmark, (3) a 76-cm diameter iron sphere benchmark from University of Illinois, and (4) the ORNL benchmark for neutron transport through iron using a fission reactor source. The special T-2 library did marginally better than the ENDF/B-VI data library for the experiments. In all cases, the ENDF/B-VI library did better than the B-V data library, both of which did better than the B-IV data library. This work is documented in LA-12884.

(5) Photon Production Assessment:

These simulations compared photon production for the data libraries using a thermal and 14-MeV isotropic neutron source in a thin shell of material. This comparison was made for natural elements, or isotopes for non-naturally occurring nuclides. The neutron and photon fluxes were tallied in 20-keV and 5-keV wide energy bins from 0 to 14 MeV. In general, photon production is represented by discrete energy gamma-rays or by wide histogram energy bins. For translated evaluations, photon production was equivalent. For the new evaluations, photon production generally changed substantially. There is often a combination of both discrete-energy and wide energy-bin photon production, particularly at higher energies. Some decrease in energy-bin width was also made for a number of nuclides for the ENDF60 library. All changes in photon production were verified by hand checking against the evaluation. Which data library to use will be dependent upon the specific application. This work will be documented in an LA-report. * This report has been published as LA-13092-MS.

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ERRATA

LA-12887: MCNPTM ENDF/B-VI Validation:
Infinite Media Comparisons of ENDF/B-VI and ENDF/B-V
and
LA-12891: ENDF/B-VI Data for MCNPTM

{CORRECTIONS ARE INDICATED BY A *}

TABLE I. The MCNP ENDF60 Library.

Material	ZAID	Filename	Evaluation	Release	Type	Photon
¹ H	1001.60c	h1001	LANL	6.1 ^a	New ^b	Yes ^c
² H	1002.60c	d1002	LANL, AWRE	-	New	-
³ H	1003.60c	t1003	LANL	-	-	No
³ He	2003.60c	he2003	LANL	6.1	New	No
⁴ He	2004.60c	he2004	LANL	-	-	No
⁶ Li	3006.60c	li3006	LANL	6.1	New	-
⁷ Li	3007.60c	li3007	LANL	-	New	-
⁹ Be	4009.60c	be4009	LLNL	-	New	-
¹⁰ B	5010.60c	b5010	LANL	6.1	New	-
¹¹ B	5011.60c	b5011	LANL	-	New	-
C	6000.60c	c6000	ORNL	6.1	New	-
¹⁴ N	7014.60c	n7014	LANL	LANL	New	-
¹⁵ N	7015.60c	n7015	LANL	-	New	-
¹⁶ O	8016.60c	o8016	LANL	-	New	-
¹⁷ O	8017.60c	o8017	BNL	-	-	No
¹⁹ F	9019.60c	f9019	ORNL	-	New	-
²³ Na	11023.60c	na11023	ORNL	6.1	-	-
Mg	12000.60c	mg12000	ORNL	-	-	-
²⁷ Al	13027.60c	al13027	LANL	-	-	-
Si	14000.60c	si14000	ORNL	-	-	-
³¹ P	15031.60c	p15031	LLNL	-	-	-
S	16000.60c	s16000	BNL	-	-	-
³² S	16032.60c	s16032	LLNL	-	-	-
Cl	17000.60c	cl17000	GGA	-	-	-
K	19000.60c	k19000	GGA	-	-	-
Ca	20000.60c	ca20000	ORNL	-	New*	-
⁴⁵ Sc	21045.60c	sc21045	BNL	6.2	New*	-
Ti	22000.60c	ti22000	BRC, ANL	-	-	-
V	23000.60c	v23000	ANL, LLNL, +	-	New	-
⁵⁰ Cr	24050.60c	cr24050	ORNL	6.1	New	-

^aAll revisions are revision 6.0 of ENDF/B-VI unless otherwise noted. LANL indicates modifications were performed.

^bAll types are translations from ENDF/B-V, unless otherwise noted.

^cAll nuclides have photon production, unless otherwise noted.

TABLE I (cont.) The MCNP ENDF60 Library.

Material	ZAID	Filename	Evaluation	Release	Type	Photon
⁵² Cr	24052.60c	cr24052	ORNL	6.1	New	-
⁵³ Cr	24053.60c	cr24053	ORNL	6.1	New	-
⁵⁴ Cr	24054.60c	cr24054	ORNL	6.1	New	-
⁵⁵ Mn	25055.60c	mn25055	ORNL	-	New	-
⁵⁴ Fe	26054.60c	fe26054	ORNL	6.1	New	-
⁵⁶ Fe	26056.60c	fe26056	ORNL	6.1	New	-
⁵⁷ Fe	26057.60c	fe26057	ORNL	6.1	New	-
⁵⁸ Fe	26058.60c	fe26058	ORNL	6.1	New	-
⁵⁹ Co	27059.60c	co27059	ANL	6.2	New	-
⁵⁸ Ni	28058.60c	ni28058	ORNL	6.1	New	-
⁶⁰ Ni	28060.60c	ni28060	ORNL	6.1	New	-
⁶¹ Ni	28061.60c	ni28061	ORNL	6.1	New	-
⁶² Ni	28062.60c	ni28062	ORNL	6.1	New	-
⁶⁴ Ni	28064.60c	ni28062	ORNL	6.1	New	-
⁶³ Cu	29063.60c	cu29063	ORNL	6.2	New	-
⁶⁵ Cu	29065.60c	cu29065	ORNL	6.2	New	-
Ga	31000.60c	ga31000	LLNL, LANL	-	-	-
⁸⁹ Y	39089.60c	y39089	ANL, LLNL	-	New*	-
Zr	40000.60c	zr40000	SAI, BNL	6.1	-	No
⁹³ Nb	41093.60c	nb41093	ANL, LLNL	6.1	New	-
Mo	42000.60c	mo42000	LLNL, HEDL	-	-	-
⁹⁹ Tc	43099.60c	tc43099	HEDL, BAW	-	-	No
¹⁰⁷ Ag	47107.60c	ag47107	BNL, HEDL	-	New*	No
¹⁰⁹ Ag	47109.60c	ag47109	BNL, HEDL	-	New*	No
In	49000.60c	in49000	ANL	-	New	-
¹²⁷ I	53127.60c	i53127	HEDL, RCN	LANL	New*	-
¹²⁹ I	53129.60c	i53129	HEDL, RCN	-	-	No
¹³³ Cs	55133.60c	cs55133	HEDL, BNL, +	-	-	No
¹³⁴ Cs	55134.60c	cs55134	ORNL, HEDL	-	New	No
¹³⁵ Cs	55135.60c	cs55135	HEDL	-	-	No
¹³⁶ Cs	55136.60c	cs55136	HEDL	-	-	No
¹³⁷ Cs	55137.60c	cs55137	HEDL	-	-	No

All revisions are revision 6.0 of ENDF/B-VI unless otherwise noted. LANL indicates modifications were performed.

All types are translations from ENDF/B-V, unless otherwise noted.

All nuclides have photon production, unless otherwise noted.

TABLE I (cont.) The MCNP ENDF60 Library.

Material	ZAID	Filename	Evaluation	Release	Type	Photon
¹³⁸ Ba	56138.60c	ba56138	ORNL, HEDL	-	-	-
¹⁵¹ Eu	63151.60c	eu63151	LANL	-	New	-
¹⁵³ Eu	63153.60c	eu63153	LANL	-	New	-
¹⁵² Gd	64152.60c	gd64152	BNL	-	-	No
¹⁵⁴ Gd	64154.60c	gd64154	BNL	-	-	No
¹⁵⁵ Gd	64155.60c	gd64155	BNL	-	-	No
¹⁵⁶ Gd	64156.60c	gd64156	BNL	-	-	No
¹⁵⁷ Gd	64157.60c	gd64157	BNL	-	-	No
¹⁵⁸ Gd	64158.60c	gd64158	BNL	-	-	No
¹⁶⁰ Gd	64160.60c	gd64160	BNL	-	-	No
¹⁶⁵ Ho	67165.60c	ho67165	LANL	-	New	-
Hf	72000.60c	hf72000	SAI	-	-	No
¹⁸¹ Ta	73181.60c	ta73181	LLNL	-	-	-
¹⁸² Ta	73182.60c	ta73182	AI	-	-	No
¹⁸² W	74182.60c	w74182	LANL, ANL, +	-	New*	-
¹⁸³ W	74183.60c	w74183	LANL, ANL, +	-	New*	-
¹⁸⁴ W	74184.60c	w74184	LANL, ANL, +	-	New*	-
¹⁸⁶ W	74186.60c	w74186	LANL, ANL, +	-	New*	-
¹⁸⁵ Re	75185.60c	re75185	ORNL, LANL	-	New	No
¹⁸⁷ Re	75187.60c	re75187	ORNL, LANL	-	New	No
¹⁹⁷ Au	79197.60c	au79197	LANL	6.1	New	-
²⁰⁶ Pb	82206.60c	pb82206	ORNL	-	New	-
²⁰⁷ Pb	82207.60c	pb82207	ORNL	6.1	New	-
²⁰⁸ Pb	82208.60c	pb82208	ORNL	-	New	-
²⁰⁹ Bi	83209.60c	bi83209	ANL	-	New	-
²³⁰ Th	90230.60c	th90230	HEDL	-	-	No
²³² Th	90232.60c	th90232	BNL, ANL, +	-	-	-
²³¹ Pa	91231.60c	pa91231	HEDL	-	-	No
²³² U	92232.60c	u92232	HEDL	-	-	No
²³³ U	92233.60c	u92233	LANL, ORNL	-	-	-
²³⁴ U	92234.60c	u92234	BNL, GGA	-	-	No

All revisions are revision 6.0 of ENDF/B-VI unless otherwise noted. LANL indicates modifications were performed.

All types are translations from ENDF/B-V, unless otherwise noted.

All nuclides have photon production, unless otherwise noted.

TABLE I (cont.) The MCNP ENDF60 Library.

Material	ZAID	Filename	Evaluation	Release	Type	Photon
²³⁵ U	92235.60c	u92235	ORNL, LANL	6.2*	New	-
²³⁶ U	92236.60c	u92236	HEDL	-	New	No
²³⁸ U	92238.60c	u92238	ORNL, LANL, +	6.2	New	-
²³⁷ Np	93237.60c	np93237	LANL	6.1	New	-
²³⁸ Np ^d	93238.60c	np93238	SRL	6.2*	New	No
²³⁹ Np	93239.60c	np93239	ORNL	-	New	No
²³⁶ Pu	94236.60c	pu94236	HEDL, SRL	-	.*	No
²³⁷ Pu	94237.60c	pu94237	HEDL	-	.*	No
²³⁸ Pu	94238.60c	pu94238	HEDL, AI, +	-	-	No
²³⁹ Pu	94239.60c	pu94239	LANL	6.2	New	-
²⁴⁰ Pu	94240.60c	pu94240	ORNL	6.2*	New	-
²⁴¹ Pu	94241.60c	pu94241	ORNL	6.1	New	-
²⁴² Pu	94242.60c	pu94242	HEDL, SRL, +	-	-	-
²⁴³ Pu	94243.60c	pu94243	BNL, SRL, +	6.2*	-	-
²⁴⁴ Pu	94244.60c	pu94244	HEDL, SRL	-	-	No
²⁴¹ Am	95241.60c	am95241	CNDC	LANL	New	-
²⁴² Am ^d	95242.60c	am95242	SRL	6.1	-	No
²⁴³ Am	95243.60c	am95243	ORNL, HEDL, +	-	New	-
²⁴¹ Cm	96241.60c	cm96241	HEDL	-	-	No
²⁴² Cm	96242.60c	cm96242	HEDL, SRL, +	-	-	-
²⁴³ Cm	96243.60c	cm96243	HEDL, SRL, +	-	-	-
²⁴⁴ Cm	96244.60c	cm96244	HEDL, SRL, +	-	-	-
²⁴⁵ Cm	96245.60c	cm96245	SRL, LLNL	6.2	-	-
²⁴⁶ Cm	96246.60c	cm96246	BNL, SRL, +	6.2	-	-
²⁴⁷ Cm	96247.60c	cm96247	BNL, SRL, +	6.2	-	-
²⁴⁸ Cm	96248.60c	cm96248	HEDL, SRL, +	-	-	-
²⁴⁹ Bk	97249.60c	bk97249	CNDC	-	New	No
²⁴⁹ Cf	98249.60c	cf98249	CNDC	LANL	New	No
²⁵⁰ Cf	98250.60c	cf98250	BNL, SRL, +	6.2	-	-
²⁵¹ Cf	98251.60c	cf98251	BNL, SRL, +	6.2	-	-
²⁵² Cf	98252.60c	cf98252	BNL, SRL, +	6.2*	-	-

All revisions are revision 6.0 of ENDF/B-VI unless otherwise noted. LANL indicates modifications were performed.

All types are translations from ENDF/B-V, unless otherwise noted.

All nuclides have photon production, unless otherwise noted.

^dThe data files are not recommended for use due to the evaluations being incomplete.

These are currently being removed from distribution. Additionally, 95242.60c represents the ground state of ²⁴²Am, not the metastable state.

ENDF/B-VI Data for MCNP™

*John S. Hendricks
Stephanie C. Frankle
John D. Court*

ENDF/B-VI Data for MCNP™

by

John S. Hendricks, Stephanie C. Frankle and John D. Court

ABSTRACT

Nuclear and atomic data are the foundation upon which the radiation transport codes are built. For neutron transport the international standard is the Evaluated Nuclear Data File from Brookhaven National Laboratory. The latest version, ENDF/B-VI release 2, has recently become available for use in the Monte Carlo N-Particle (MCNP) radiation transport code. These neutron cross-section data are designated by ZAIID identifiers ending in .60c and are referred to as the ENDF60 library. The ENDF60 data library was processed from the ENDF/B-VI evaluations using the NJOY code. Fifty-two percent of the data evaluations are translations from ENDF/B-V. The remaining 48% are new evaluations which have sometimes changed significantly. The RSIC release package contains the ENDF60 neutron library, a new photon library MCPLIB02, the electron library EL1, and an updated XSDIR file. We report here the work done by the LANL Radiation Transport Group (X-6) in testing and validating the ENDF60 data library and in developing the necessary new sampling and detector schemes. When the ENDF60 library should be used in preference to the previous libraries, is also considered. The development of the new photon library MCPLIB02 is also discussed.

Table 1 (cont.) The MCNP ENDF60 Library.

Material	ZAID	Filename	Evaluation	Revision	Type	Photon
⁵² Cr	24052.60c	cr24052	ORNL	6.1	New	-
⁵³ Cr	24053.60c	cr24053	ORNL	6.1	New	-
⁵⁴ Cr	24054.60c	cr24054	ORNL	6.1	New	-
⁵⁵ Mn	25055.60c	mn25055	ORNL	-	New	-
⁵⁴ Fe	26054.60c	fe26054	ORNL	6.1	New	-
⁵⁶ Fe	26056.60c	fe26056	ORNL	6.1	New	-
⁵⁷ Fe	26057.60c	fe26057	ORNL	6.1	New	-
⁵⁸ Fe	26058.60c	fe26058	ORNL	6.1	New	-
⁵⁹ Co	27059.60c	co27059	ANL	6.2	New	-
⁵⁸ Ni	28058.60c	ni28058	ORNL	6.1	New	-
⁶⁰ Ni	28060.60c	ni28060	ORNL	6.1	New	-
⁶¹ Ni	28061.60c	ni28061	ORNL	6.1	New	-
⁶² Ni	28062.60c	ni28062	ORNL	6.1	New	-
⁶⁴ Ni	28064.60c	ni28062	ORNL	6.1	New	-
⁶³ Cu	29063.60c	cu29063	ORNL	6.2	New	-
⁶⁵ Cu	29065.60c	cu29065	ORNL	6.2	New	-
Ga	31000.60c	ga31000	LLNL, LANL	-	-	-
⁸⁹ Y	39089.60c	y39089	ANL, LLNL	-	-	-
Zr	40000.60c	zr40000	SAI, BNL	6.1	-	No
⁹³ Nb	41093.60c	nb41093	ANL, LLNL	6.1	New	-
Mo	42000.60c	mo42000	LLNL, HEDL	-	-	-
⁹⁹ Tc	43099.60c	tc43099	HEDL, BAW	-	-	No
¹⁰⁷ Ag	47107.60c	ag47107	BNL, HEDL	-	-	No
¹⁰⁹ Ag	47109.60c	ag47109	BNL, HEDL	-	-	No
In	49000.60c	in49000	ANL	-	New	-
¹²⁷ I	53127.60c	i53127	HEDL, RCN	LANL	-	-
¹²⁹ I	53129.60c	i53129	HEDL, RCN	-	-	No
¹³³ Cs	55133.60c	cs55133	HEDL, BNL, +	-	-	No
¹³⁴ Cs	55134.60c	cs55134	ORNL, HEDL	-	New	No
¹³⁵ Cs	55135.60c	cs55135	HEDL	-	-	No
¹³⁶ Cs	55136.60c	cs55136	HEDL	-	-	No
¹³⁷ Cs	55137.60c	cs55137	HEDL	-	-	No

All revisions are revision 6.0 of ENDF/B-VI unless otherwise noted. LANL indicates modifications were performed.

All types are translations from ENDF/B-V, unless otherwise noted.

All nuclides have photon production, unless otherwise noted.

Table 1 (cont.) The MCNP ENDF60 Library.

Material	ZAID	Filename	Evaluation	Revision	Type	Photon
¹³⁸ Ba	56138.60c	ba56138	ORNL, HEDL	-	-	-
¹⁵¹ Eu	63151.60c	eu63151	LANL	-	New	-
¹⁵³ Eu	63153.60c	eu63153	LANL	-	New	-
¹⁵² Gd	64152.60c	gd64152	BNL	-	-	No
¹⁵⁴ Gd	64154.60c	gd64154	BNL	-	-	No
¹⁵⁵ Gd	64155.60c	gd64155	BNL	-	-	No
¹⁵⁶ Gd	64156.60c	gd64156	BNL	-	-	No
¹⁵⁷ Gd	64157.60c	gd64157	BNL	-	-	No
¹⁵⁸ Gd	64158.60c	gd64158	BNL	-	-	No
¹⁶⁰ Gd	64160.60c	gd64160	BNL	-	-	No
¹⁶⁵ Ho	67165.60c	ho67165	LANL	-	New	-
Hf	72000.60c	hf72000	SAI	-	-	No
¹⁸¹ Ta	73181.60c	ta73181	LLNL	-	-	-
¹⁸² Ta	73182.60c	ta73182	AI	-	-	No
¹⁸² W	74182.60c	w74182	LANL, ANL, +	-	-	-
¹⁸³ W	74183.60c	w74183	LANL, ANL, +	-	-	-
¹⁸⁴ W	74184.60c	w74184	LANL, ANL, +	-	-	-
¹⁸⁶ W	74186.60c	w74186	LANL, ANL, +	-	-	-
¹⁸⁵ Re	75185.60c	re75185	ORNL, LANL	-	New	No
¹⁸⁷ Re	75187.60c	re75187	ORNL, LANL	-	New	No
¹⁹⁷ Au	79197.60c	au79197	LANL	6.1	New	-
²⁰⁶ Pb	82206.60c	pb82206	ORNL	-	New	-
²⁰⁷ Pb	82207.60c	pb82207	ORNL	6.1	New	-
²⁰⁸ Pb	82208.60c	pb82208	ORNL	-	New	-
²⁰⁹ Bi	83209.60c	bi83209	ANL	-	New	-
²³⁰ Th	90230.60c	th90230	HEDL	-	-	No
²³² Th	90232.60c	th90232	BNL, ANL, +	-	-	-
²³¹ Pa	91231.60c	pa91231	HEDL	-	-	No
²³² U	92232.60c	u92232	HEDL	-	-	No
²³³ U	92233.60c	u92233	LANL, ORNL	-	-	-
²³⁴ U	92234.60c	u92234	BNL, GGA	-	-	No

All revisions are revision 6.0 of ENDF/B-VI unless otherwise noted. LANL indicates modifications were performed.

All types are translations from ENDF/B-V, unless otherwise noted.

All nuclides have photon production, unless otherwise noted.

Table 1 (cont.) The MCNP ENDF60 Library.

Material	ZAID	Filename	Evaluation	Revision	Type	Photon
²³⁵ U	92235.60c	u92235	ORNL, LANL	LANL	New	-
²³⁶ U	92236.60c	u92236	HEDL	-	New	No
²³⁸ U	92238.60c	u92238	ORNL, LANL, +	6.2	New	-
²³⁷ Np	93237.60c	np93237	LANL	6.1	New	-
²³⁸ Np	93238.60c	np93238	SRL	LANL	New	No
²³⁹ Np	93239.60c	np93239	ORNL	-	New	No
²³⁶ Pu	94236.60c	pu94236	HEDL, SRL	-	New	No
²³⁷ Pu	94237.60c	pu94237	HEDL	-	New	No
²³⁸ Pu	94238.60c	pu94238	HEDL, AI, +	-	-	No
²³⁹ Pu	94239.60c	pu94239	LANL	6.2	New	-
²⁴⁰ Pu	94240.60c	pu94240	ORNL	LANL	New	-
²⁴¹ Pu	94241.60c	pu94241	ORNL	6.1	New	-
²⁴² Pu	94242.60c	pu94242	HEDL, SRL, +	-	-	-
²⁴³ Pu	94243.60c	pu94243	BNL, SRL, +	LANL	-	-
²⁴⁴ Pu	94244.60c	pu94244	HEDL, SRL	-	-	No
²⁴¹ Am	95241.60c	am95241	CNDC	LANL	New	-
²⁴² Am	95242.60c	am95242	SRL	6.1	-	No
²⁴³ Am	95243.60c	am95243	ORNL, HEDL, +	-	New	-
²⁴¹ Cm	96241.60c	cm96241	HEDL	-	-	No
²⁴² Cm	96242.60c	cm96242	HEDL, SRL, +	-	-	-
²⁴³ Cm	96243.60c	cm96243	HEDL, SRL, +	-	-	-
²⁴⁴ Cm	96244.60c	cm96244	HEDL, SRL, +	-	-	-
²⁴⁵ Cm	96245.60c	cm96245	SRL, LLNL	6.2	-	-
²⁴⁶ Cm	96246.60c	cm96246	BNL, SRL, +	6.2	-	-
²⁴⁷ Cm	96247.60c	cm96247	BNL, SRL, +	6.2	-	-
²⁴⁸ Cm	96248.60c	cm96248	HEDL, SRL, +	-	-	-
²⁴⁹ Bk	97249.60c	bk97249	CNDC	-	New	No
²⁴⁹ Cf	98249.60c	cf98249	CNDC	LANL	New	No
²⁵⁰ Cf	98250.60c	cf98250	BNL, SRL, +	6.2	-	-
²⁵¹ Cf	98251.60c	cf98251	BNL, SRL, +	6.2	-	-
²⁵² Cf	98252.60c	cf98252	BNL, SRL, +	LANL	-	-

All revisions are revision 6.0 of ENDF/B-VI unless otherwise noted. LANL indicates modifications were performed.

All types are translations from ENDF/B-V, unless otherwise noted.

All nuclides have photon production, unless otherwise noted.

Information Center (RSIC) in October, 1994. To obtain the library package, contact RSIC, P. O. Box 2008, Oak Ridge, Tennessee, 37831-6362, USA. Phone (615) 574-6176, Fax (615) 574-6182, E-mail: PDC@ORNL.GOV.

II. NEW SCATTERING LAWS

MCNP4A added three new ENDF/B-VI scattering laws complete with new data formats and processing, transport physics, and next-event estimator (point detector) sampling schemes. These three laws are the Kalbach-87 formalism (ENDF/B-VI file 6 LAW=1, LANG=2), correlated angle-energy scattering (ENDF/B-VI file 6 LAW=7), and phase-space law (ENDF/B-VI file 6 LAW=6.)

The Kalbach-87 formalism, ENDF/B-VI file 6 LAW=1, LANG=2 or MCNP Law 44, requires sampling the cosine between the incident and outgoing particle directions, μ , from the density function:

$$p(\mu, E, E') = \frac{1}{2} \frac{A}{\sinh(A)} [\cosh(A\mu) + R \sinh(A\mu)], \quad (1)$$

where E is the incident energy, E' is the exiting energy, and μ is the cosine between the incident and outgoing particle directions. The R and A values are found from tabulated values with an R and A value for each E' , and a table of E' values for each E . The Kalbach-87 formalism is used for many of the new ENDF/B-VI evaluations, and it is also used as an approximation for the new ORNL evaluations with the exception of F.

The correlated angle-energy scattering law, ENDF/B-VI file 6 LAW=7 or MCNP Law 67, first samples μ from 32-equiprobable cosine bins between incident and outgoing particle direction with interpolation for a given incident energy, E . Then, with μ and E known, the outgoing energy, E' , is sampled from tables interpolated on the μ and E tables. That is, μ and E' are sampled from a three-dimensional table in E , μ , and E' , with full interpolation. This tabular correlated angle-energy scattering law is used for ^9Be only in ENDF/B-VI. This law is also used as an approximation for the new ENDF/B-VI Legendre expansion law in the case of F only.

The correlated angle-energy scattering phase-space law, ENDF/B-VI file 6 LAW= 6 or MCNP Law 66, first samples the outgoing particle direction cosine, μ , isotropically in the center-of-mass system. Then, given the number of bodies in phase space n , and the total mass ratio A_p for n particles,

$$E' = TE_i^{\max}, \quad (2)$$

where T is sampled from the density function:

$$p(\mu, E, T) = C_n \sqrt{T} (E_i^{\max} - T)^{3n/2-4}, \quad (3)$$

and where E_i^{max} is the maximum possible energy for particle i , μ , and E' :

$$E_i^{max} = \frac{A_p - 1}{A_p} \left(\frac{A}{A + 1} E + Q \right). \quad (4)$$

Here C_n is a normalization constant, A is the atomic weight ratio, and Q is the Q -value of the reaction. This law is used for ENDF/B-VI deuterium only with $n = 3$.

In addition to the algorithms to sample these laws for neutron transport, new algorithms were developed for next-event estimators so that collisions using these new laws now contribute to point and ring detectors as well as the DXTRAN variance reduction method. The NJOY code was also modified by T-2 to provide data in these new formats.

There is an additional new law in the ENDF/B-VI evaluations. The ORNL evaluators produced evaluations of F, Cr, Mn, Fe, Ni, Cu, and Pb isotopes using Legendre expansions to represent energy-angle distribution in the laboratory system. Whereas MCNP4A does not yet have a sampling scheme for this representation, NJOY reformulates these scattering distributions according to the Kalbach-87 distribution for use in MCNP for all except F which uses the correlated angle-energy law. A further discussion of these approximations to the Legendre expansion law is given in Section XII.

III. OTHER MCNP ENDF/B-VI LIBRARIES

Since the release of NJOY91.91, dozens of ENDF/B-VI MCNP compatible data libraries have been generated around the world. These libraries have been generated with different assumptions suitable for the sites where they were produced. In particular, the continuous-energy neutron data of MCNP requires a pointwise-linear unionized grid; that is, all reaction cross sections are on the same energy grid with linear-linear interpolation. Because an infinite number of points is required to give an exact continuous-energy representation, which is impossible, the energy grid must be thinned. This thinning is accomplished by preserving the cross-section integral weighting, to some specified tolerance. For example, fusion applications would use constant weighting, whereas reactor applications would use $1/E$ weighting. Different thinning tolerances would be used depending upon the computer storage capacities at different sites. Further, different ways of arranging the data into libraries were used. Consequently there are many MCNP compatible libraries of ENDF/B-VI data around the world and it is very confusing as to which data set is which. The MCNP developers felt there was a need for a standard, or "official", set of ENDF/B-VI data, just as there are standard sets of ENDF/B-V and other data.

To provide a standard MCNP ENDF/B-VI library, the Radiation Transport group engaged the Nuclear Theory and Applications Group to construct a complete library based on ENDF/B-VI through revision 2. This ENDF60 library contains 70 revision 0, 26 revision 1, and only 11 revision 2 data. Special modifications to 9 nuclides were made by T-2 and are indicated as

LANL in Table 1. This library was thinned such that most nuclides were no more than 400,000 words in storage length (3.2 megabytes), used flat weighting, and were at room temperature. The unique identifiers assigned to this library were .60C such that any reference to the ZAID=ZZZAAA.60C in MCNP would denote this library and nothing else. This new library is henceforth referred to as the ENDF60 library. The ENDF60 library is stored by nuclide rather than grouped into a larger library as was done previously for RMCCS. For example, the data file h1001 is hydrogen, pu94239 is ^{239}Pu , etc. This allows the user to keep only the necessary data files, thereby reducing the required storage space.

The ENDF60 library is made up of 124 nuclides in MCNP Type 1 (ascii) format, which can be used on any computer platform. It requires approximately 200 megabytes of disk space for the complete library, with an average data file size of 1.6 megabytes. The largest nuclide is ^{127}I which takes 8.1 megabytes, considerably more than the 400,000 word guideline. The RSIC release package includes the ENDF60 neutron library, the new photon library MCPLIB02, the electron library EL1, and a new XSDIR file. The new photon data library MCPLIB02 extends the photon interaction data up to 100 GeV and is described in detail in the following section. As in the original MCPLIB library, MCPLIB02 only includes data for $Z \leq 94$. Therefore, coupled neutron-photon problems cannot be performed using ENDF60 for $Z > 94$. It is anticipated that a new photon library which will include $Z \geq 95$ will be released within a year. In addition to the XSDIR entries for the new data libraries, this updated XSDIR file has been further modified such that the default data libraries are now the MCNP recommended libraries as indicated in Appendix G of the MCNP4A manual, as well as defining MCPLIB02 to be the default photon library.

IV. THE MCPLIB02 PHOTON LIBRARY

MCNP is unable to run coupled neutron-photon problems without photon data, and electron data for the thick target bremsstrahlung treatment in the default case. Consequently, the ENDF60 library release to RSIC includes a new photon library, MCPLIB02, and the electron library, EL1. The MCPLIB02 library consists of photon cross sections with ZAID identifiers ending in .02P. The capability to utilize more than one evaluation of photons or electrons was added to MCNP4A using the PLIB or ELIB parameters on the materials (MAT) cards. The MCPLIB02 library is an extension of the original MCPLIB library which has been in use for over 15 years, extending the photon interaction data from 100 MeV to 100 GeV based upon the Lawrence Livermore National Laboratory (LLNL) Evaluated Photon Data Library (EPDL).

EPDL contains considerably more detail than MCNP4A is currently prepared to use. For example, the EPDL distinguishes between pair- and triplet-production cross sections, tabulates real and imaginary anomalous scattering factors, and includes photoelectric subshell cross sections for shells as high-lying as Q1. As a long-term project, it would be desirable to incorporate all of this detail into MCNP. For the first implementation of EPDL data, the energy

range for the photon interaction data was extended leaving the sampling algorithms unchanged. Specifically, EPDL data is incorporated above the maximum energy available in MCPLIB, and a smooth transition is made from MCPLIB to EPDL between these limiting energies. The maximum energy in MCPLIB is usually 100 MeV, but for seven elements ($Z = 84, 85, 87, 88, 89, 91, \text{ and } 93$) it is 15 MeV. For each element, five blocks of MCPLIB data have been extended. These are the incoherent, coherent, photoelectric, and pair production cross sections, and the heating numbers. The remaining blocks of data (incoherent, coherent, and integrated coherent form factors, and fluorescence data) remain unchanged. Since MCPLIB data are used at low energies, the new MCPLIB02 data are backward-compatible with most existing MCNP photon calculations; but high-energy applications up to the limit of the electron data (1 GeV) are now possible.

Unlike the MCPLIB data, the EPDL uses a different energy grid for tabulating each separate photon cross section. For MCPLIB02 a common grid was chosen on which to tabulate all new data above 10 MeV. This grid is based on the EPDL grid for pair production cross sections, both because the pair-production energy grid has the greatest resolution, and because pair production is the most important process at high energies. Both the EPDL and the MCPLIB cross-section data (and the MCPLIB heating numbers) rely on logarithmic interpolation to evaluate quantities at energies between tabulated points:

$$\sigma = F(E_1, E_2, \sigma_1, \sigma_2, E) = \exp \left(\frac{ \left\{ \ln \left[\frac{E}{E_1} \right] \ln[\sigma_2] + \ln \left[\frac{E_2}{E} \right] \ln[\sigma_1] \right\} }{ \ln \left[\frac{E_2}{E_1} \right] } \right). \quad (5)$$

The EPDL also calls for the use of linear interpolation for certain quantities. The important quantities relying on linear interpolation are the average energies ΔE of secondary particles.

The procedure for merging MCPLIB and EPDL data for each element into the new library was as follows. For energies between 1 keV and 10 MeV, the MCPLIB energy grid and data were included unchanged. The maximum energy ϵ available in MCPLIB was identified. (ϵ is either 15 MeV or 100 MeV). For energies E in the transition range $10 \text{ MeV} < E < \epsilon$, the MCPLIB cross section σ_E was determined for each process. An interpolated cross section

$$\sigma = F(10.0, \epsilon, \sigma_M, \sigma_E, E) \quad (6)$$

was used for that process in MCPLIB02. Above ϵ , the EPDL cross section was determined, interpolating as needed, and used unchanged. Whenever a pair cross section had to be extracted from the EPDL, the sum of the pair and triplet cross sections was used, since MCPLIB does not separate the two processes.

There is no quantity comparable to the MCPLIB heating number in the EPDL, so heating numbers were constructed according to the formula

$$H(E) = \frac{\sum_i \Delta E_i(E) \sigma_i(E)}{\sigma_{total}(E)}, \quad (7)$$

where $H(E)$ is an MCPLIB-style heating number as a function of energy E ; the summation index i labels the incoherent, photoelectric, pair, and triplet processes; and $\Delta E_i(E)$ is the average secondary electron energy from process i , except in the case of photoelectric absorption, when $\Delta E_i(E)$ also includes the energy of those fluorescent X-rays not explicitly accounted for in the EPDL. Since the EPDL accounts for X-rays from many more shells than does MCPLIB, Eq. 7 is not strictly consistent with the MCPLIB heating numbers. Fortunately, at the high energies above MCPLIB this inconsistency is completely negligible.

For 87 of the 94 elements in MCPLIB02, an artificial, small pair-production cross section (10^{-6} barn) is included at the pair production threshold (1.022 MeV). This prevents the MCNP interpolation scheme from returning zero for the pair-production cross section all the way up to the first real data point (1.5 MeV). The seven elements whose maximum MCPLIB energies were only 15 MeV also lacked this special starting point for the pair cross section. An energy point at 1.022 MeV for these seven elements was inserted, so that the treatment of pair production is now consistent across the entire database. Other cross sections and heating numbers at this point were derived by logarithmic interpolation, so that the sampled values for these quantities will not be affected.

V. TESTING AND RELEASE OF THE ENDF60 LIBRARY

In order to get the ENDF60 library out in a timely fashion, the testing and validation phase was radically changed from in the past. In particular, the former manual checking used in previous libraries was not implemented. Therefore, a new set of quality assurance tests was established for NJOY and carried out by T-2. In addition to these tests, X-6 subjected the data to a number of customized checks, ENDF/B-V comparisons, and experimental benchmark comparisons. Extensive tests were carried out to ensure that the new scattering laws were properly implemented in MCNP and that both transport and next-event estimator solutions agreed.

The ENDF60 library was compared with a number of other data libraries. In particular, infinite medium simulations were performed for all nuclides in the ENDF60 data library, comparing them to the recommended data libraries from Appendix G of the MCNP4A manual, and to the ENDL-90 data library when appropriate. In general, the MCNP recommended data libraries are based on ENDF/B-V evaluations. When this is not the case, the recommended data library has often been produced by T-2 at LANL and is indicated as the T-2 recommended data library in the discussion. For nuclides where ENDF/B-V, ENDF/B-VI, and T-2 recommended

libraries are available, all three were compared. THE ENDF60 library was also benchmarked against experimental data for the 47 nuclides listed in Table 2. These experimental benchmarks consisted of the LLNL pulsed sphere experiments, a set of iron benchmark experiments, and a set of criticality experiments.

VI. INFINITE MEDIUM TESTS

All ENDF60 nuclides were run with MCNP and compared to ENDF/B-V in infinite medium simulations. Comparisons were also made with the MCNP recommended data libraries when these were not ENDF/B-V. The infinite medium simulations consisted of a 20-MeV neutron source in an 100-m radius sphere of the nuclide. For light materials this was sufficient to downscatter throughout the entire energy range. As materials got heavier, the source was modified so that for the heaviest elements the source spanned the entire energy range so as to sample collisions for the entire energy range. The neutron flux and heating were then tallied along with the induced photon flux and heating, and coplots were made for the ENDF/B-VI, ENDF/B-V, and the T-2 recommended libraries. All these coplots are published in a LANL Report LA-12287.²

In general, these infinite medium simulations demonstrated that for most materials ENDF/B-V and ENDF/B-VI give similar results. The neutron and photon fluxes are close, indicating that there are no gross errors in the ENDF/B-VI collision processes or reaction cross sections. The ENDF/B-VI neutron heating numbers are generally better, often being extended in range. The results for an example nuclide are provided in Figures 1-4. The ENDF/B-V and ENDF/B-VI results for ^1H are coplotted for the neutron flux, neutron heating, neutron-induced photon flux, and photon heating respectively. ENDF/B-V and ENDF/B-VI agree for all but the neutron heating. As shown in Figure 2, the neutron heating is higher at low energies for ENDF/B-VI because of improvements in modeling the nucleus recoil effects that have been made in the NJOY code. These improvements also appear for some translated evaluations as illustrated in Figure 5 for Mg, and they can be quite dramatic in the case of Cl and K. Figure 6 illustrates the changes in neutron heating due to modifications in the NJOY code for Cl.

In cases where there is no photon production in ENDF/B-V but it is now present in ENDF/B-VI, the heating numbers are also again lower for ENDF/B-VI as shown in Figure 7 for ^{197}Au . When photon production is not present, the photon energy is deposited as heating of the material and therefore added to the neutron heating tally. Note how the T-2 MCNP recommended library agrees with the ENDF/B-IV data because it also had photon production available. In general, the infinite media simulations for the new ENDF60 library indicate that ENDF/B-VI is an improvement over ENDF/B-V.

Table 2 MCNP ENDF60 Library Experimental Benchmark Coverage.

ZAID	Livermore Ref.[5]	Iron Ref.[7]	Critical
¹ H	✓	✓	✓
² H	✓		
⁶ Li	✓	✓	
⁷ Li	✓	✓	
⁹ Be	✓		
¹⁰ B		✓	✓
¹¹ B		✓	✓
C	✓	✓	✓
¹⁴ N	✓	✓	✓
¹⁶ O	✓	✓	✓
¹⁹ F	✓		✓
²³ Na	✓	✓	✓
Mg	✓	✓	✓
²⁷ Al	✓	✓	✓
Si	✓	✓	✓
³¹ P	✓	✓	✓
³² S	✓	✓	✓
K		✓	
Ca	✓	✓	
Ti	✓		✓
⁵⁰ Cr	✓	✓	✓
⁵² Cr	✓	✓	✓
⁵³ Cr	✓	✓	✓
⁵⁴ Cr	✓	✓	✓

Table 2 (cont) MCNP ENDF60 Library Experimental Benchmark Coverage.

ZAID	Livermore Ref.[5]	Iron Ref.[7]	Critical
⁵⁵ Mn	✓	✓	✓
⁵⁴ Fe	✓	✓	✓
⁵⁶ Fe	✓	✓	✓
⁵⁷ Fe	✓	✓	✓
⁵⁸ Fe	✓	✓	✓
⁵⁸ Ni	✓	✓	
⁶⁰ Ni	✓	✓	
⁶¹ Ni	✓	✓	
⁶² Ni	✓	✓	
⁶⁴ Ni	✓	✓	
⁶³ Cu			✓
⁶⁵ Cu			✓
²⁰⁶ Pb	✓		
²⁰⁷ Pb	✓		
²⁰⁸ Pb	✓		
²³⁴ U			✓
²³⁵ U			✓
²³⁶ U			✓
²³⁸ U			✓
²³⁹ Pu			✓
²⁴⁰ Pu			✓
²⁴¹ Pu			✓
²⁴² Pu			✓

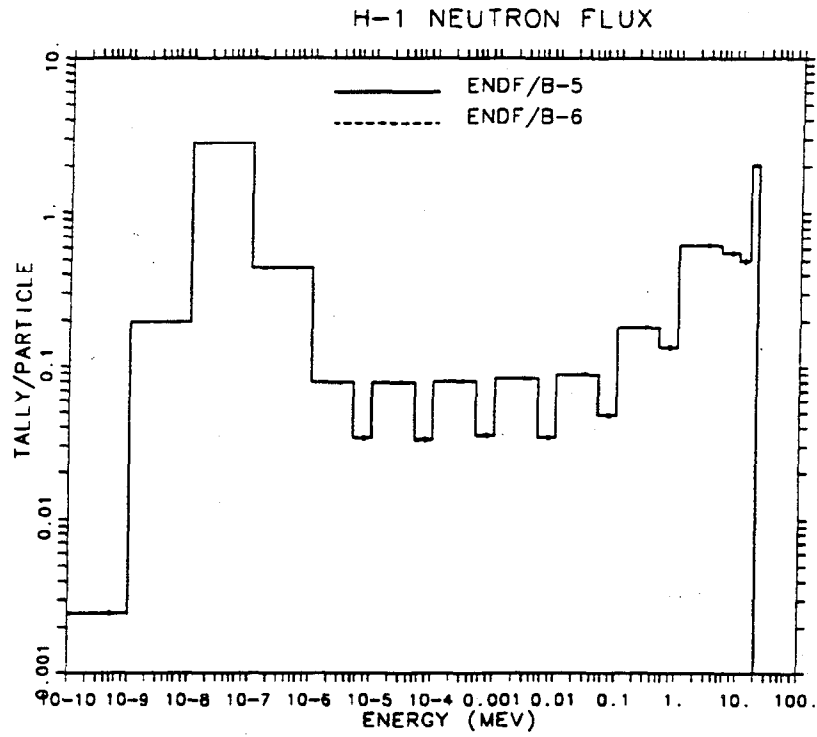


Figure 1: Plot of ENDF/B-V and ENDF/B-VI calculated neutron flux for a 10000 cm radius ^1H sphere.

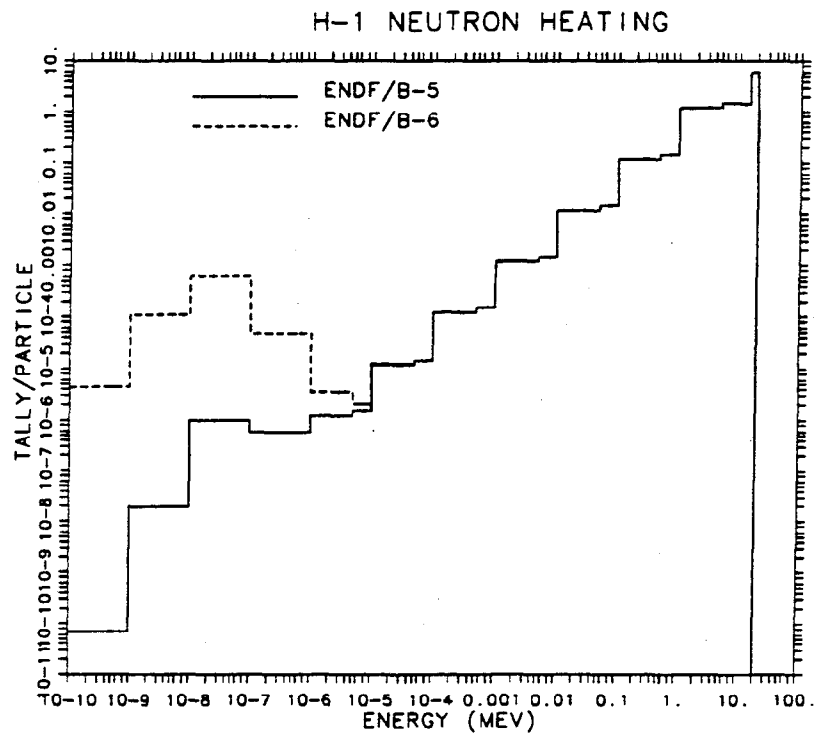


Figure 2: Plot of ENDF/B-V and ENDF/B-VI calculated neutron heating for the ^1H sphere.

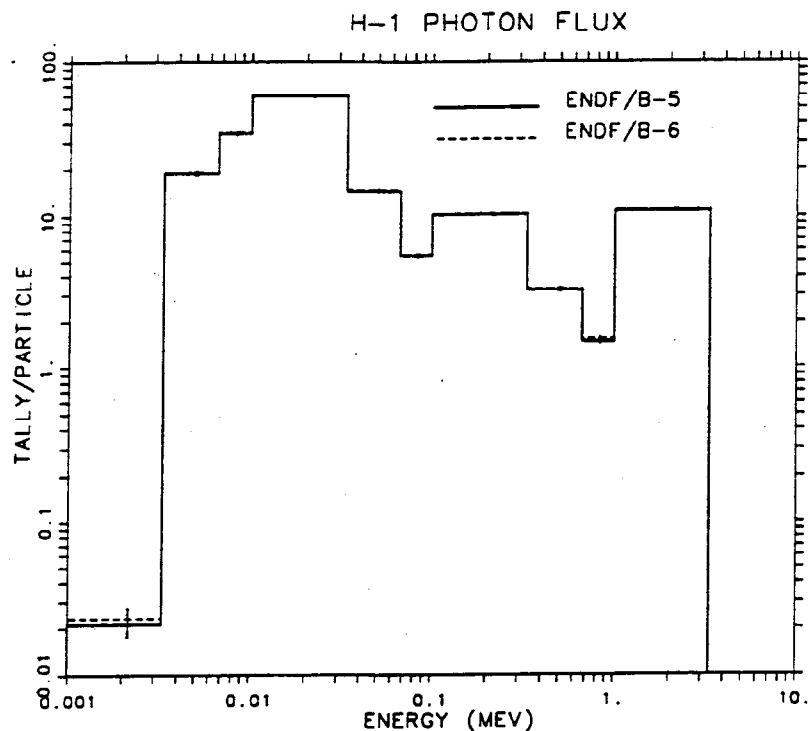


Figure 3: Plot of ENDF/B-V and ENDF/B-VI calculated photon flux for the ^1H sphere.

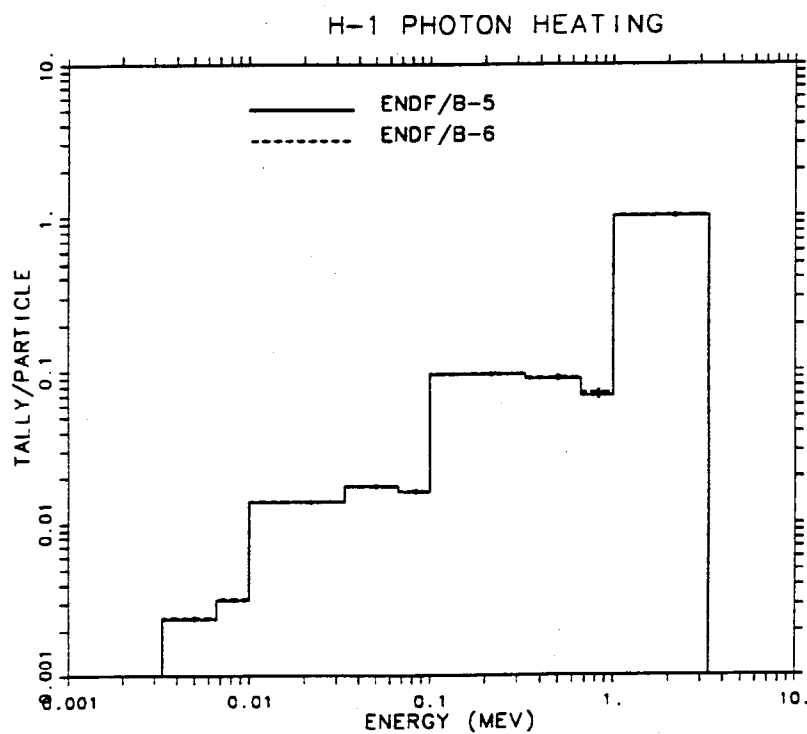


Figure 4: Plot of ENDF/B-V and ENDF/B-VI calculated photon heating for the ^1H sphere.

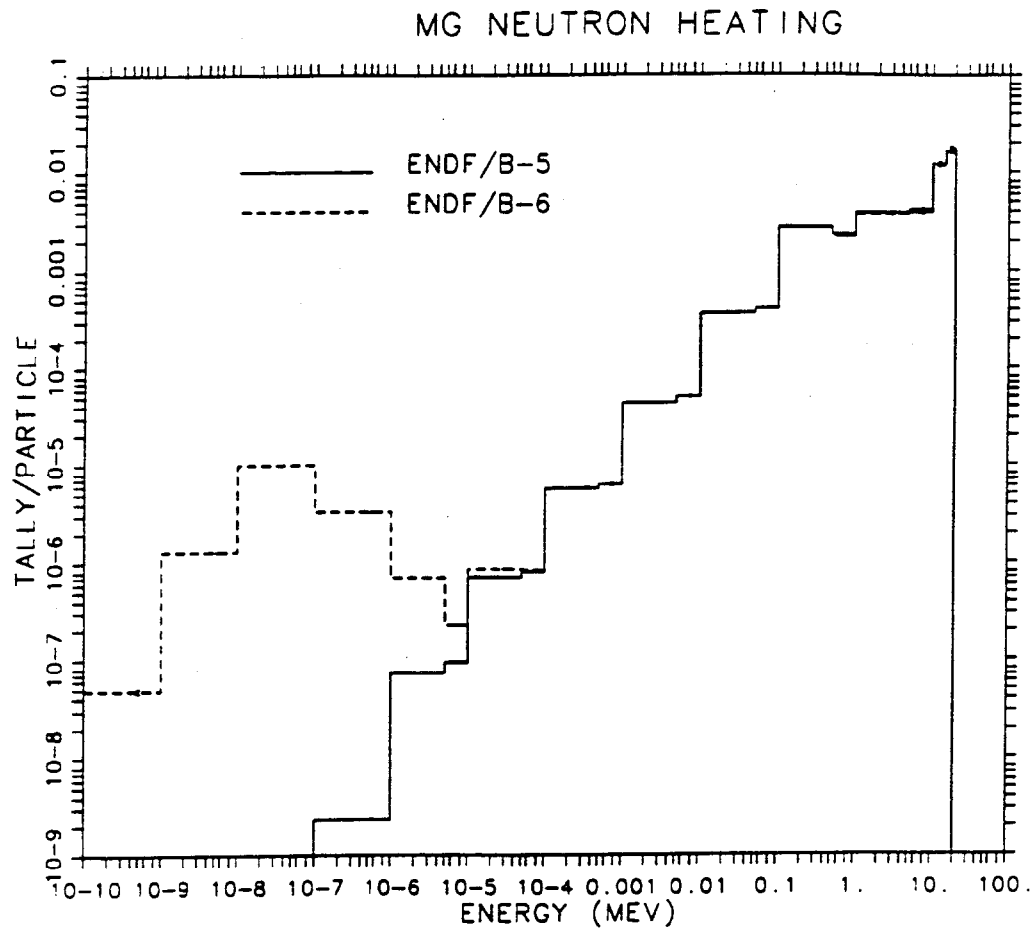


Figure 5: Plot of ENDF/B-V and ENDF/B-VI calculated neutron heating for the magnesium sphere.

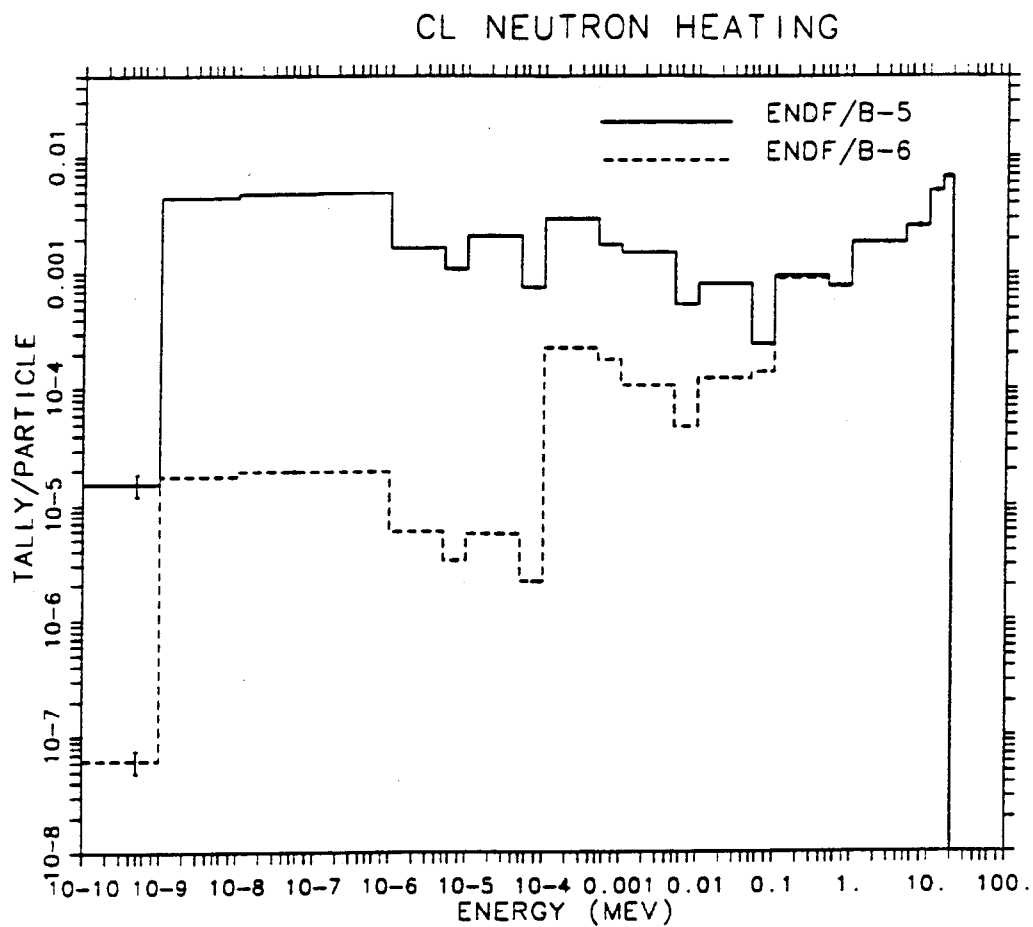


Figure 6: Plot of ENDF/B-V and ENDF/B-VI calculated neutron heating for the chlorine sphere.

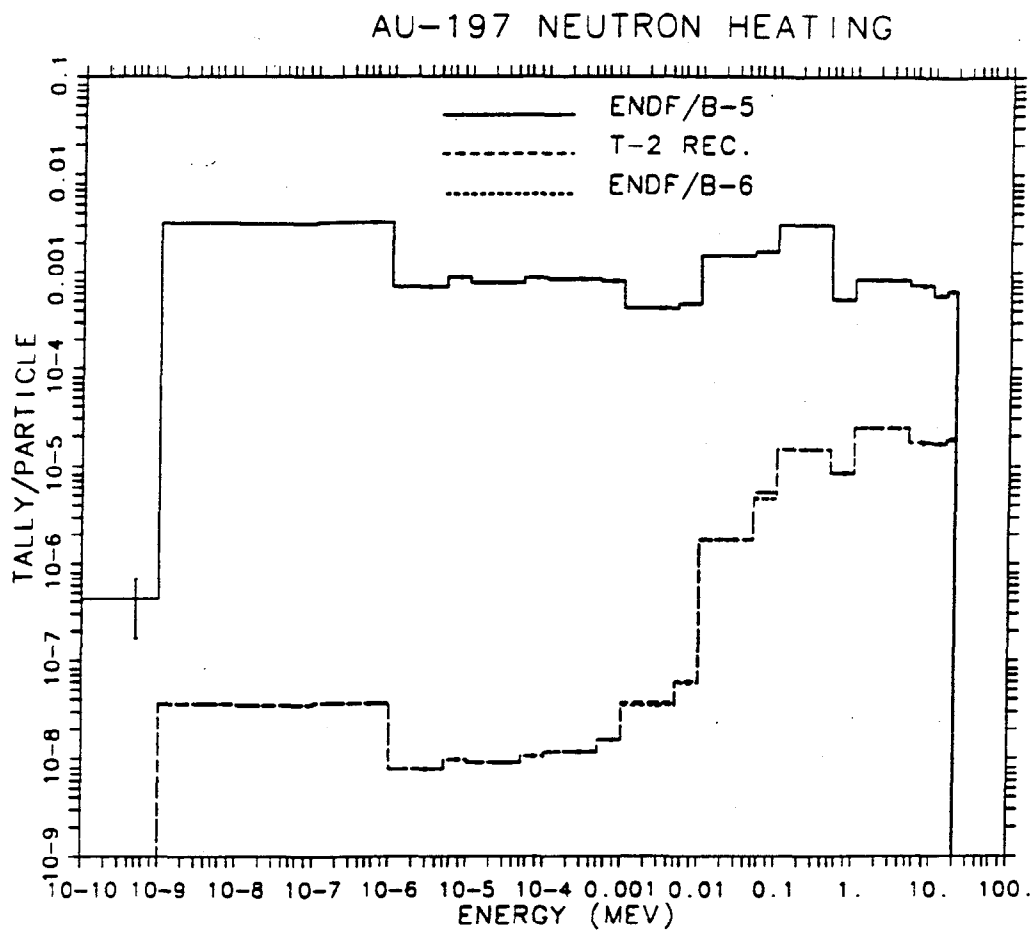


Figure 7: Plot of ENDF/B-V, MCNP recommended, and ENDF/B-VI calculated neutron heating for the ^{197}Au sphere.

VII. PHOTON PRODUCTION ASSESSMENT

Another part of the validation program was photon production at incident thermal and 14 MeV neutron energies. Again, these are not comparisons to measured benchmarks, but rather an ENDF/B-V to ENDF/B-VI comparison. These simulations used an isotropic point source inside a thin spherical shell of material. The photon and neutron fluxes were tallied from $E_\gamma = 0.014$ MeV in 5 keV bins and 20 keV bins respectively.

These results, also to be published,³ are illustrated in Figures 8-13. For evaluations which were translated from ENDF/B-V, the photon production assessment showed no differences between the two data libraries as shown for Si in Figures 8 and 9. For new evaluations, there were often substantial changes for photon production at both thermal and 14-MeV incident neutron energies. In general, photon production is represented by discrete energy gamma-ray lines, wide histogram energy bins, or a combination of the two. The new ORNL evaluations generally use histogram energy bins for photon production from thermal neutrons, while usually using a combination of the two for photon production from 14-MeV neutrons.

The comparisons for photon production from Cr, using the new ORNL evaluations, are illustrated in Figures 10 and 11. The energy resolution for photon production at thermal energies has been improved from 250 keV to 100 keV. The energy resolution for photon production at 14 MeV shows substantial improvement; however, the combination of discrete photon energies with histogram energy bins may make spectral analysis difficult for some applications. The new evaluations were verified by hand checking to be properly represented. The comparisons for photon production from F are illustrated in Figures 12 and 13. The energy resolution for photon production at thermal energies has been substantially decreased, thereby losing the majority of the discrete photons that are present in the ENDF/B-V libraries. The reverse is true for photon production at 14 MeV from F.

Substantial improvements in energy resolution have been made in the ENDF/B-VI evaluations, though the combination of histogram and discrete energy photon representations may make spectral analysis difficult as for Cr. The recommended ENDF library to use for applications requiring detailed photon production treatment will therefore be dependent upon the neutron energy range of interest and the specific requirements of the application.

VIII. LIVERMORE PULSED SPHERE BENCHMARKS

Thirty-eight neutron pulsed sphere experiments were performed at Lawrence Livermore National Laboratory in the late 1960's to evaluate the performance of several neutron transport codes.⁴ The experiments involved placement of a tritium target in the center of a sphere of a given material. The target was bombarded with deuterium ions to induce the reaction $T(d,n)^4\text{He}$, which produces a nearly isotropic source of 14 MeV neutrons. Then the time-of-flight energy spectrum of the neutrons was measured at particular angles. The spheres were

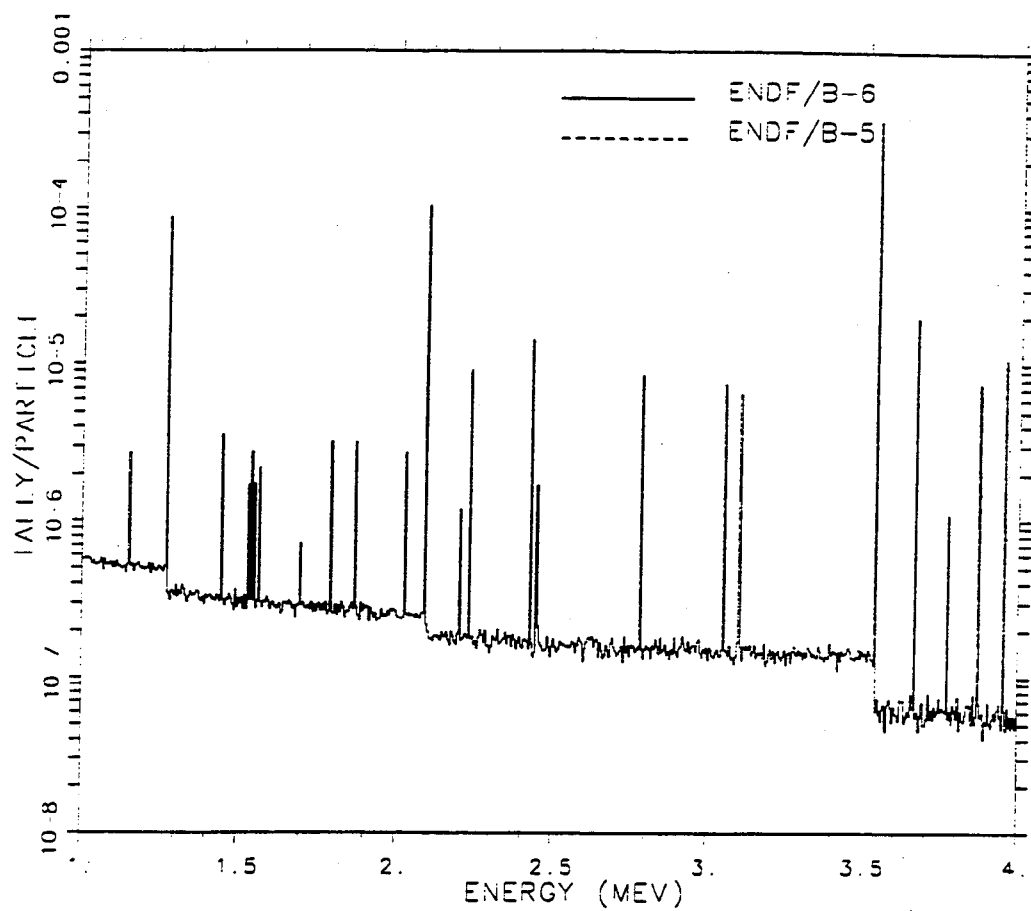


Figure 8: Plot of ENDF/B-V and ENDF/B-VI simulated photon production spectrum from Si at thermal neutron energies.

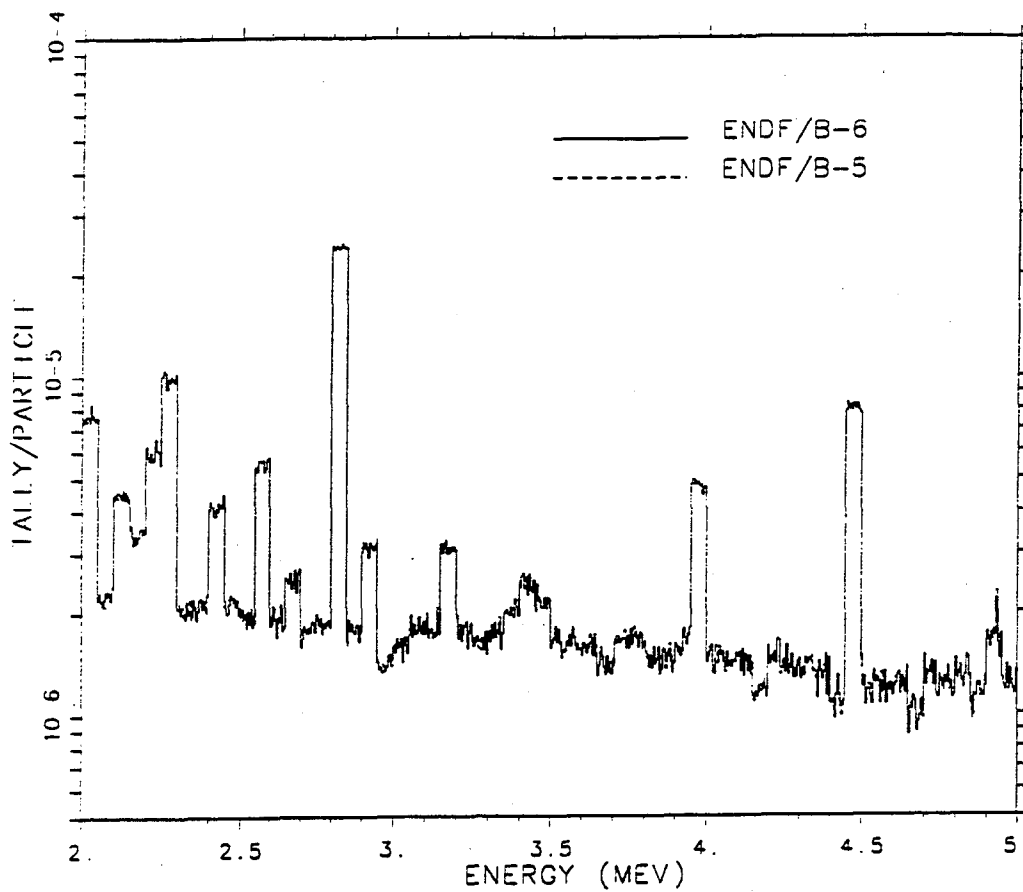


Figure 9: Plot of ENDF/B-V and ENDF/B-VI simulated photon production spectrum from Si for 14 MeV neutrons.

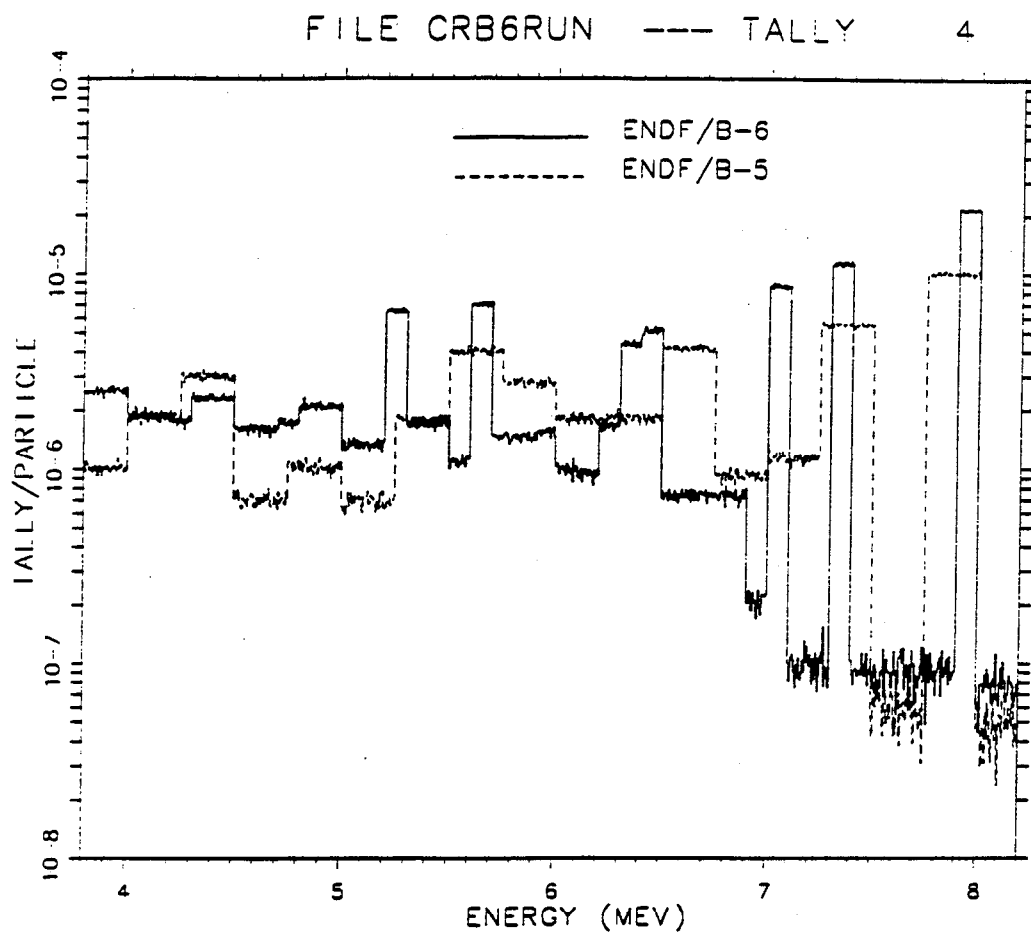


Figure 10: Plot of ENDF/B-V and ENDF/B-VI simulated photon production spectrum from Cr at thermal neutron energies.

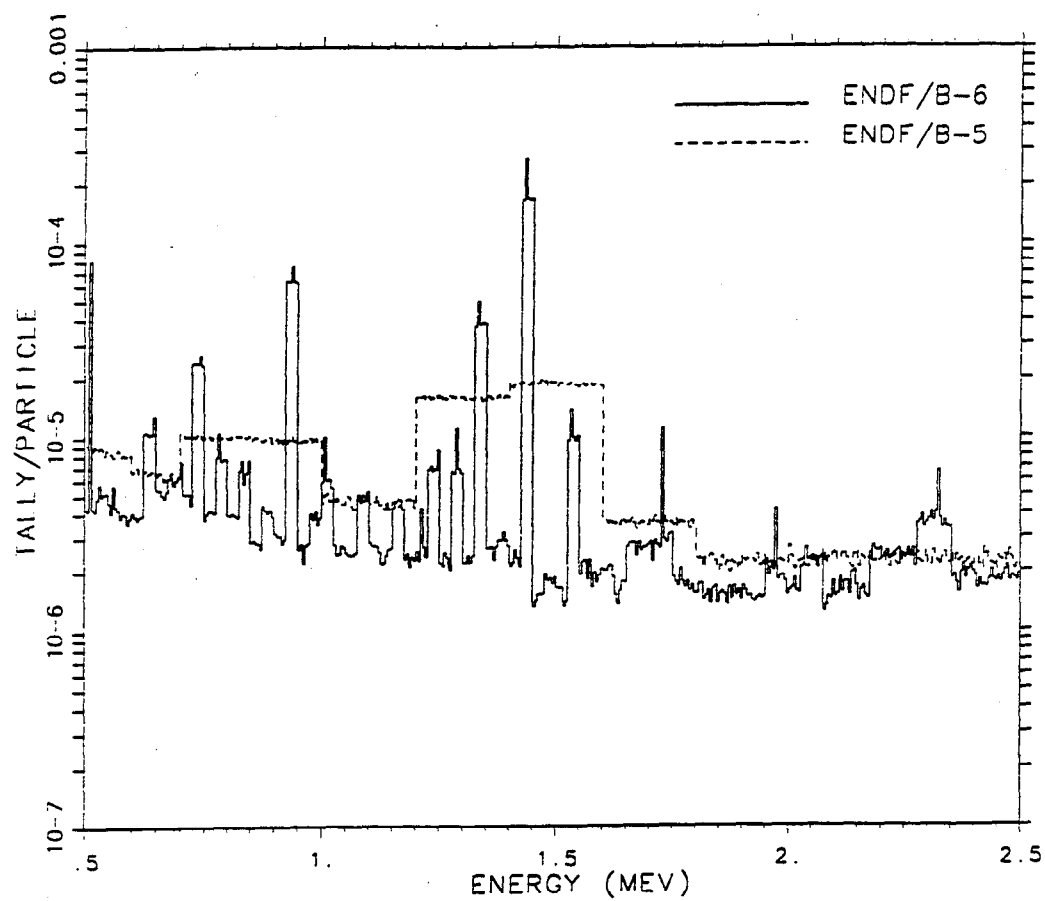


Figure 11: Plot of ENDF/B-V and ENDF/B-VI simulated photon production spectrum from Cr for 14 MeV neutrons.

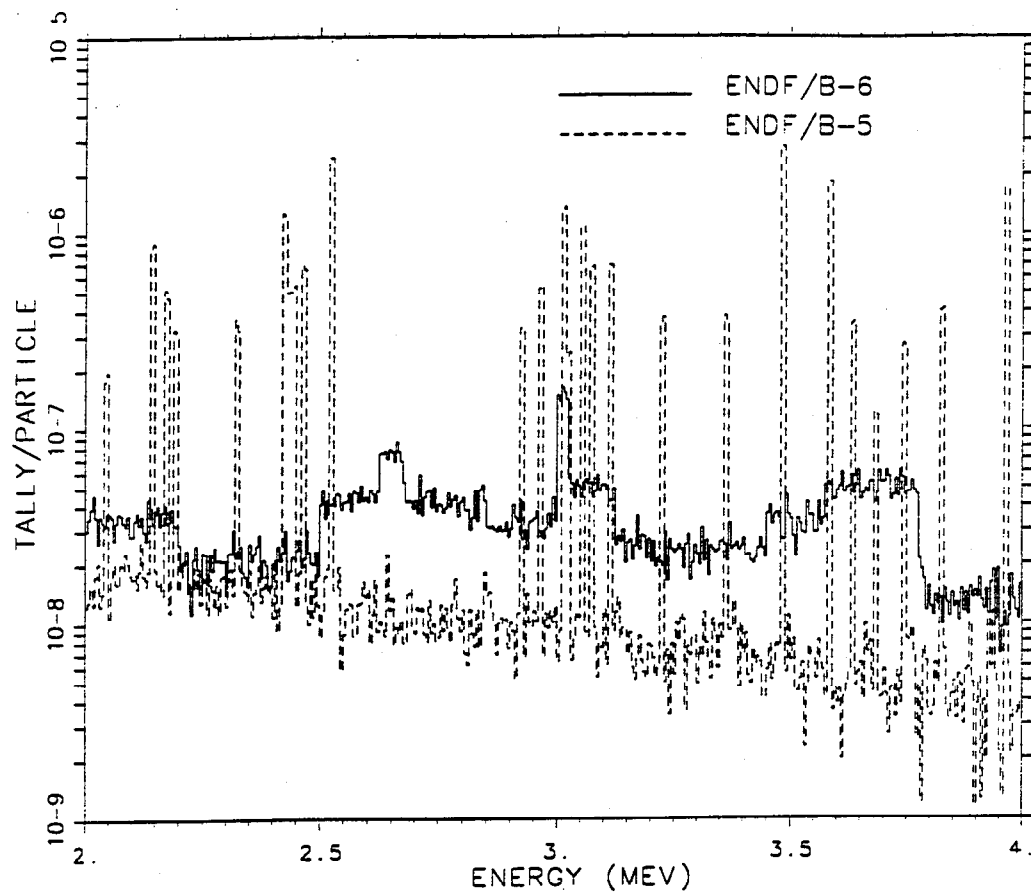


Figure 12: Plot of ENDF/B-V and ENDF/B-VI simulated photon production spectrum from F at thermal neutron energies.

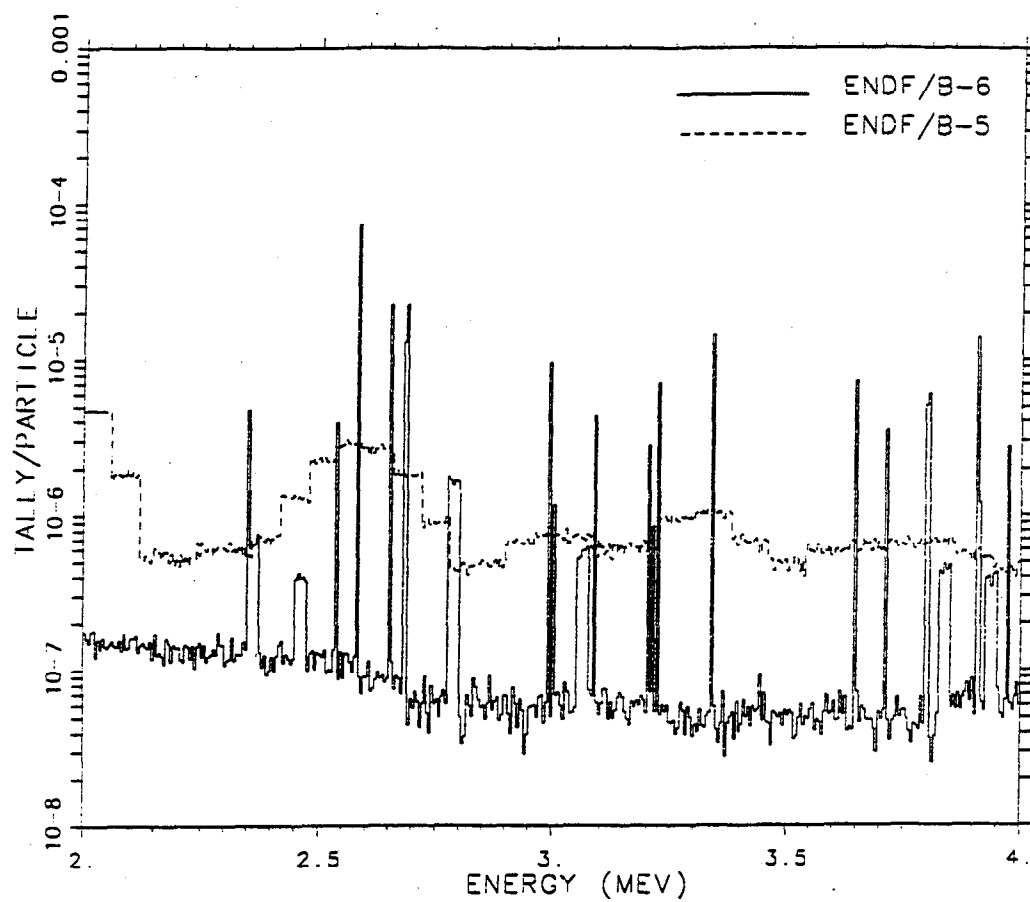


Figure 13: Plot of ENDF/B-V and ENDF/B-VI simulated photon production spectrum from F for 14 MeV neutrons.

from 0.5 to 4.8 mean free paths thick. Twenty-eight of these experimental measurements were simulated with MCNP4A. Table 3 lists the 38 neutron pulsed sphere experiments and indicates which were simulated. In general, when three different thicknesses were measured, only the smallest and largest were simulated. The ratios of calculated-to-experimental results are given in Table 4 for two energy ranges, $E_n=12.0-16.0$ MeV and $E_n=2.0-16.0$ MeV, for both ENDF/B-V and ENDF/B-VI.

The results from these calculations⁵ differ slightly from the independent preliminary results using MCNP4A with ENDF/B-VI published earlier⁶ because we are using the final ENDF60 library rather than a preliminary library and using different assumptions in the problem input, particularly the time-of-flight times to neutron energy correspondence. The data sets were different because in some cases newer revisions of ENDF/B-VI are used in ENDF60, as well as changes between data sets due to modifications in the processing code NJOY. Also, the earlier comparison used some T-2 recommended data, particularly for iron, instead of ENDF/B-V. But in both the preliminary and final studies, the results are consistent.

The ENDF/B-V results more closely represent the experimental results for C and ^6Li as seen in Figures 14-17. However, ENDF/B-V and ENDF/B-VI look about the same in the integral results of Table 4, and we have no evidence from other experiments that C and ^6Li have problems. In particular, ENDF/B-VI results for the CF_2 and CH_2 pulsed spheres were better than ENDF/B-V. The poor results of C and ^6Li in these pulsed sphere experiments may be due to poor experimental measurements or to other materials such as N which was in the air and Fe in the structural material of the experiment.

The ENDF/B-V and ENDF/B-VI results are about the same for O, Mg, Al, Ti, H_2O , D_2O , and concrete. The ENDF/B-VI results appear to match the measurements better for ^7Li , Be, N, Fe, Pb, CF_2 and CH_2 . The results for N are plotted in Figures 18-19. The T-2 recommended library for Fe appears better than either ENDF/B-V or ENDF/B-VI.

IX. IRON BENCHMARKS

The ENDF/B-VI isotopic iron cross section data were subjected to four benchmark studies as part of the Hiroshima/Nagasaki dose re-evaluation for the National Academy of Sciences and the Defense Nuclear Agency.⁷ The four benchmark studies were (1) the iron sphere benchmarks from the Lawrence Livermore Pulsed Spheres which are repeated from the above study; (2) the ORNL Fusion Reactor Shielding Benchmark; (3) a 76-cm diameter iron sphere benchmark done at the University of Illinois; and (4) the ORNL Benchmark for Neutron Transport through Iron which used a fission reactor source. In all cases, the ENDF60 library was compared to the ENDF/B-V and T-2 recommended data libraries. Although experimental data were available for all four benchmarks, the description of the experimental setup and interpretation of the experimental results were inadequate in two cases to obtain reasonable agreement between measurements and either ENDF/B-V or ENDF/B-VI. But for the Hiroshima/Nagasaki dose re-

Table 3: Livermore Pulsed Spheres Target Material Outline.

Material	Radius (mfp)	Used for ENDF/B-VI
⁶ Li	0.5	✓
	1.1	
	1.6	✓
⁷ Li	0.5	✓
	1.0	
	1.6	✓
Beryllium	0.8	✓
Carbon	0.5	✓
	1.3	
	2.9	✓
Nitrogen	1.1	✓
	3.1	✓
Oxygen	0.7	✓
Magnesium	0.7	✓
	1.2	
	1.9	✓
Aluminum	0.9	✓
	1.6	
	2.6	✓
Titanium	1.2	✓
	2.2	
	3.5	✓
Iron	0.9	✓
	2.9	
	4.8	✓
Lead	1.4	✓
H ₂ O	1.1	✓
	1.9	✓

Table 3 (cont.) Livermore Pulsed Spheres Target Material Outline.

Material	Radius (mfp)	Used for ENDF/B-VI
D ₂ O	1.2	✓
	2.1	✓
CH ₂	0.7	✓
	1.6	
	3.0	✓
CF ₂	0.9	✓
	1.8	
	2.9	✓
Concrete	2.0	✓
	3.8	

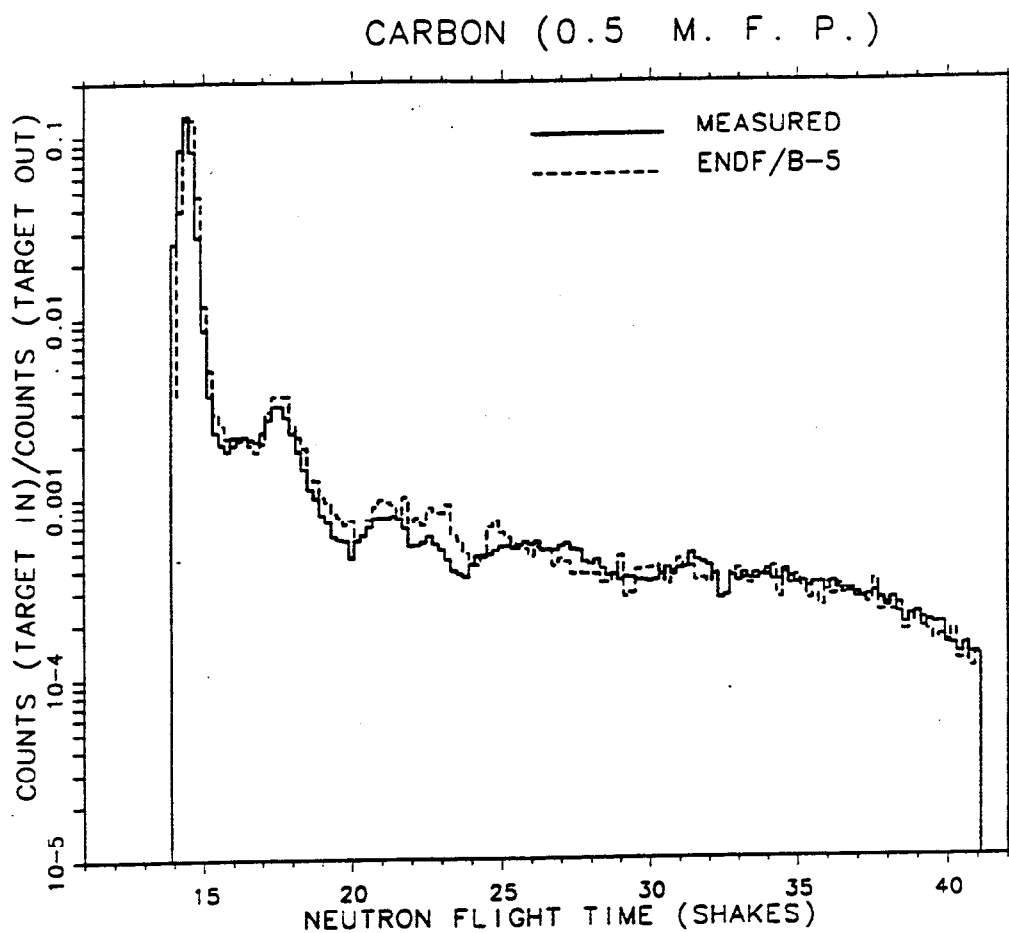


Figure 14: Plot of experimental and ENDF/B-V calculated count rates for the carbon sphere of 0.5 mean free path radius.

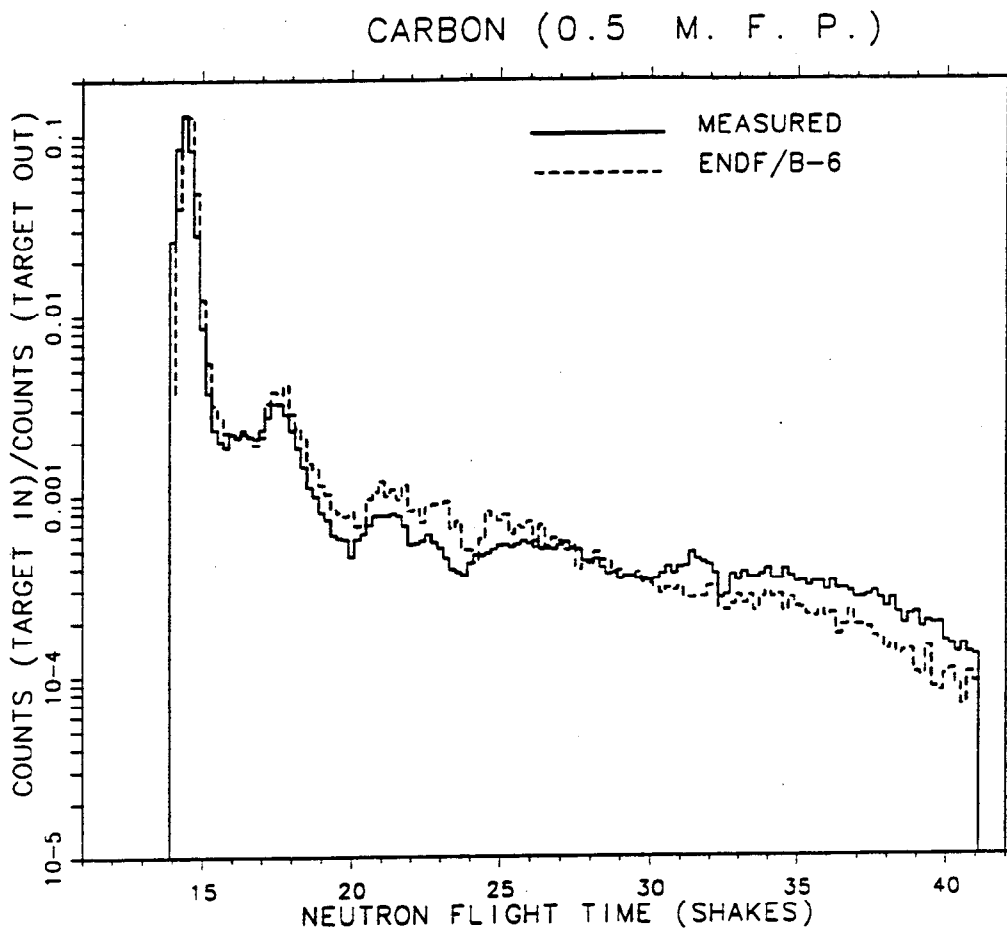


Figure 15: Plot of experimental and ENDF/B-VI calculated count rates for the carbon sphere of 0.5 mean free path radius.

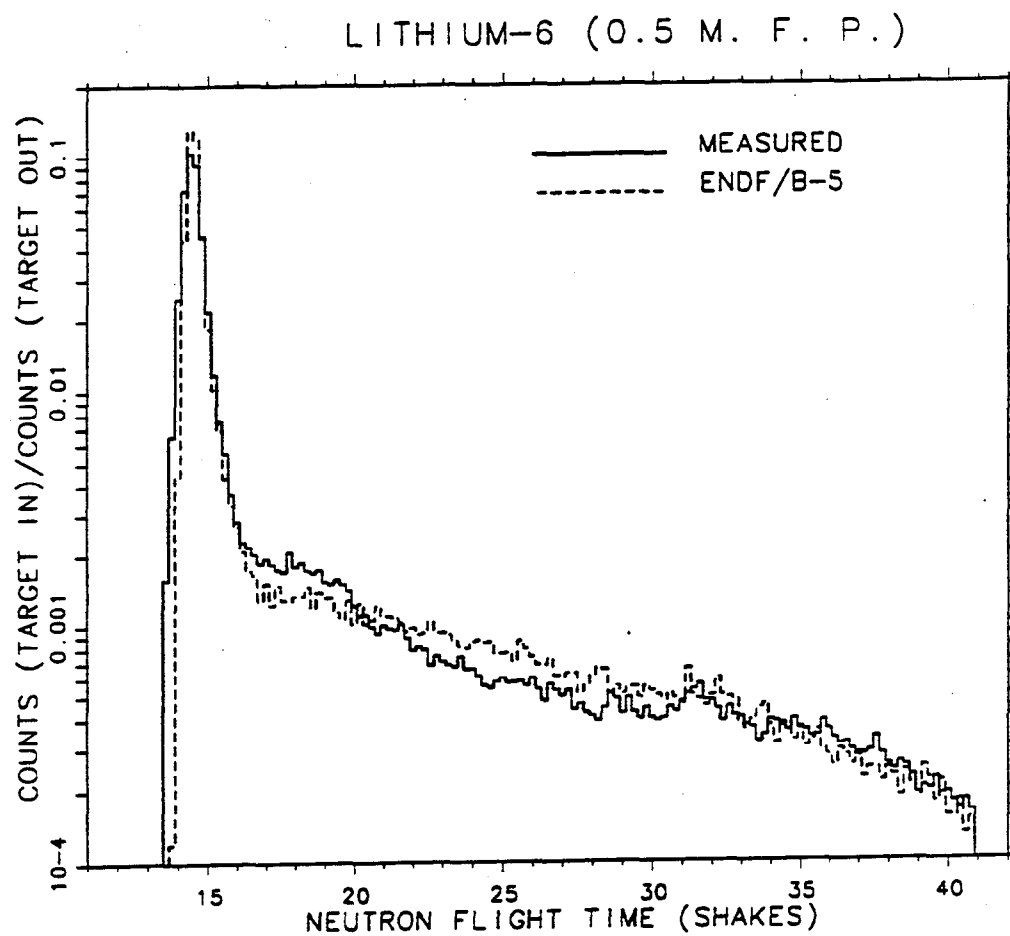


Figure 16: Plot of experimental and ENDF/B-V calculated count rates for the ^6Li sphere of 0.5 mean free path radius.

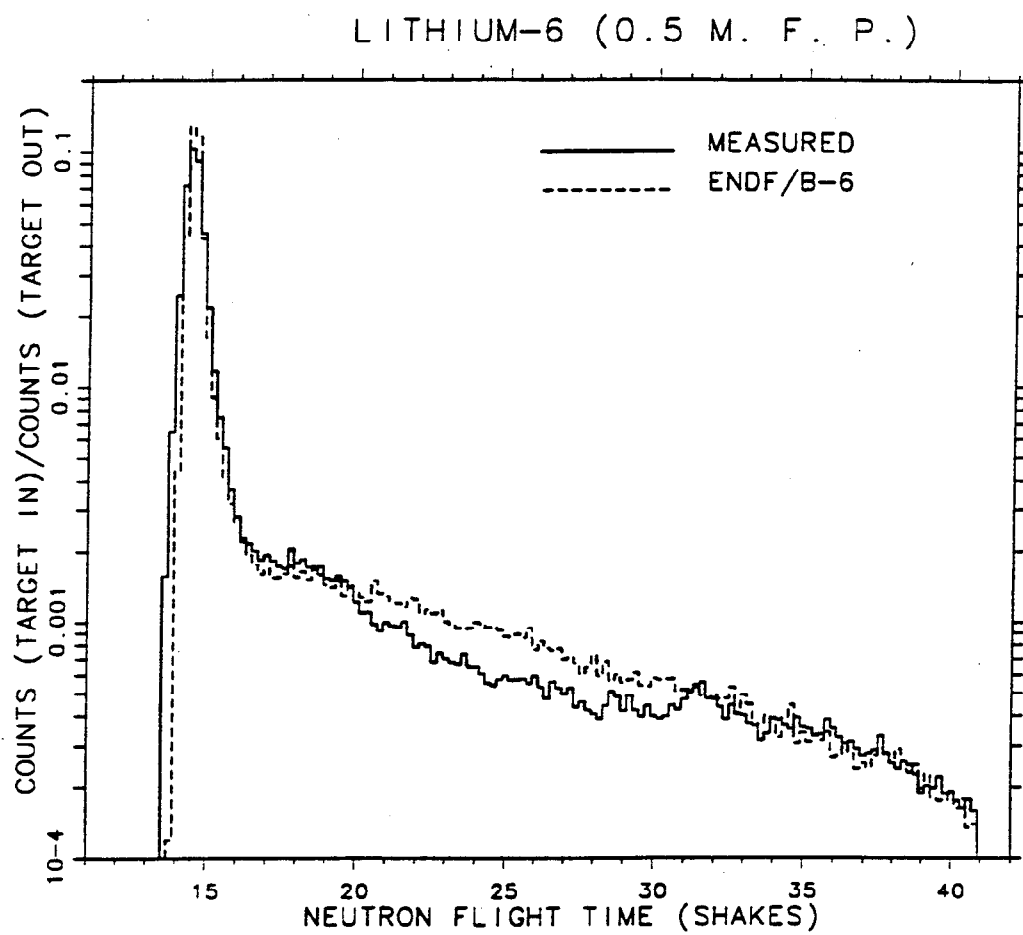


Figure 17: Plot of experimental and ENDF/B-VI calculated count rates for the ${}^6\text{Li}$ sphere of 0.5 mean free path radius.

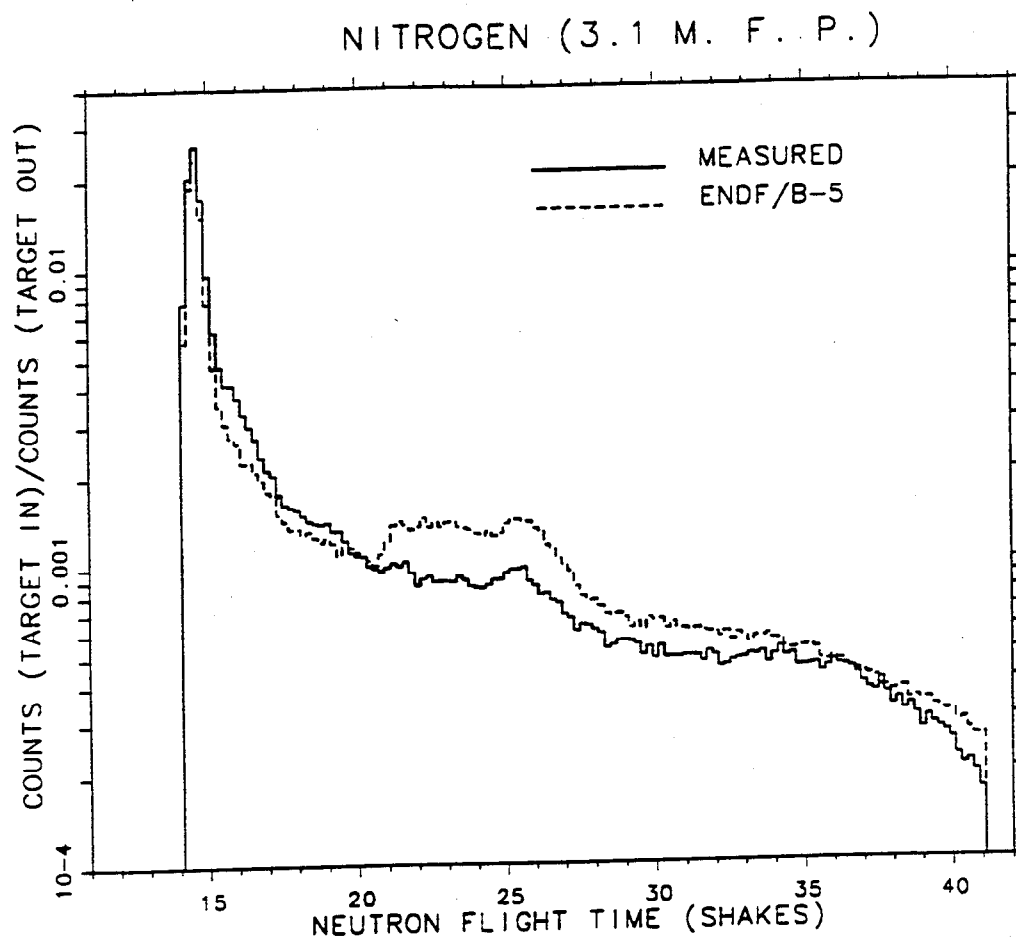


Figure 18: Plot of experimental and ENDF/B-V calculated count rates for the nitrogen sphere of 3.1 mean free path radius.

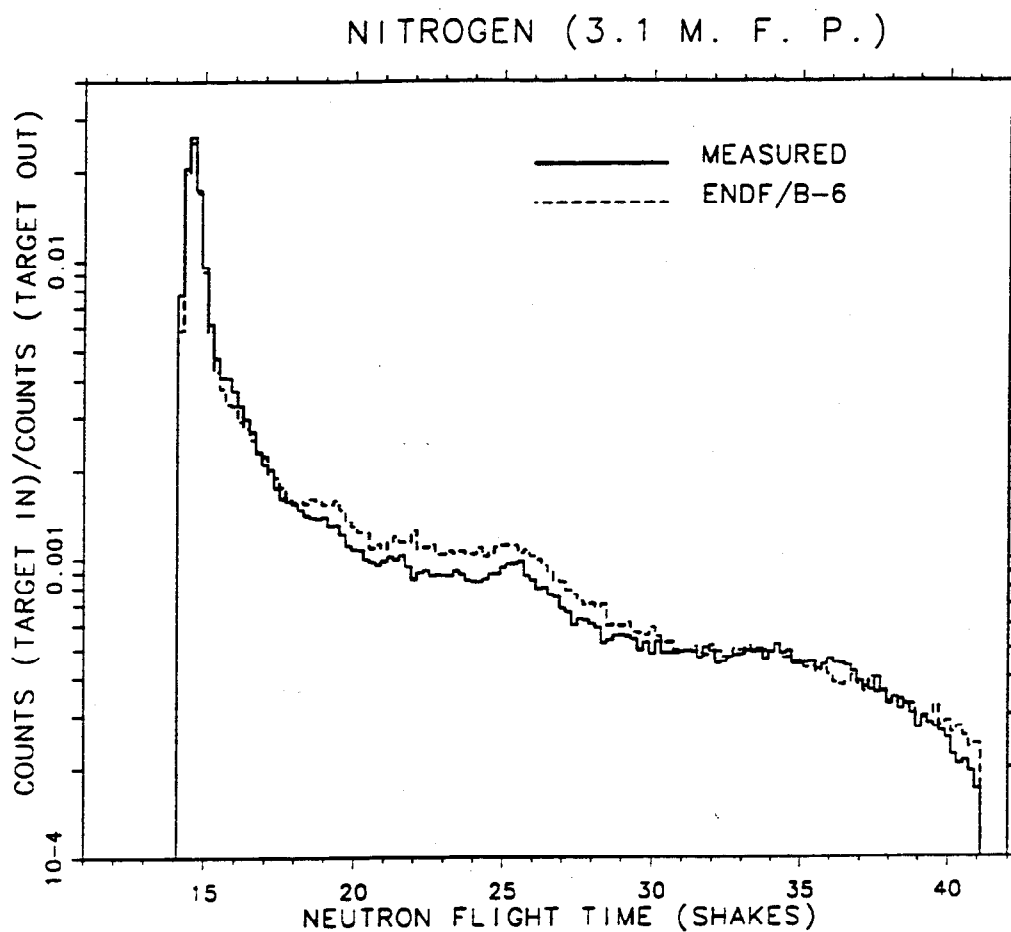


Figure 19: Plot of experimental and ENDF/B-VI calculated count rates for the nitrogen sphere of 3.1 mean free path radius.

evaluation project, the primary purpose was to make a comparison between ENDF/B-VI and ENDF/B-V and the T-2 recommended data libraries.

The Livermore Pulsed Sphere experiments have already been described.

Oak Ridge National Laboratory Fusion Reactor Shielding Benchmark was undertaken to simulate the radiation exposure that would be present at the first wall of a fusion reactor. Again, the $T(d,n)^4\text{He}$ reaction was utilized by bombarding a tritiated target with a deuteron beam, resulting in a nearly-isotropic source of 14 MeV neutrons. The tritiated titanium target was imbedded inside an iron cavity liner which served as a spectrum modifier and made the neutron spectrum incident on the experimental shield configuration similar to the spectrum incident on the first wall of a fusion reactor. The 14 MeV neutrons passed through several different configurations of laminated slabs composed of SS-304 and borated polyethylene. Four configurations were modeled: configurations 1, 2, 3, and 7. Table 5 shows the shielding configuration compositions. Table 6 shows the ratio of calculated to experimental flux transmissions for the two detector positions, on and off axis, and for the 5 neutron energy regions, $E_n=0-2, 2-5, 5-10, 10-13$, and >13 MeV. Generally ENDF/B-VI and the T-2 MCNP recommended iron cross sections agree and give larger flux transmissions than the ENDF/B-V data. A comparison of the calculated to experimental results shows that agreement is best for the T-2 recommended library 18 times, the ENDF/B-VI library 12 times, and the ENDF/B-V library only 7 times. The experimental data are given in the form of experimental upper and lower bounds. In Figures 20 and 21, the results for the thickest shielding are shown. ENDF/B-VI agrees much more closely to the experiment than ENDF/B-V for configuration 7.

The University of Illinois conducted neutron leakage experiments on a 76-cm-diameter iron sphere in the early 1970's. The sphere contained a 15-cm void in the center, where two types of sources were used in the measurements: 14 MeV neutrons from the $T(d,n)^4\text{He}$ reaction and a ^{252}Cf source. Comparisons were made between ENDF/B-VI and ENDF/B-IV, ENDF/B-V, and the T-2 recommended library. Comparisons to experimental results are not presented because the description of the experimental setup and results was insufficient to obtain reasonable agreement. The results are detailed in Table 7 and show that the ENDF/B-VI and T-2 recommended libraries are in good agreement, while the ENDF/B-IV and ENDF/B-V libraries tend to show reduced flux transmission,

The ORNL Benchmark for Neutron Transport through Iron used a highly collimated beam of fission-spectrum neutrons that was nearly point mono-directional on 4 thicknesses of iron. The flux at various angles and distances from the back face of the iron slab was measured with three types of detectors. The results of the simulations are presented in Tables 8-11. The percentage deviations from experiment indicate that our model is not an accurate representation of the experimental setup. Nevertheless, the results are useful in that they again demonstrate that the T-2 recommended library is generally close to the ENDF/B-VI iron in results, and that both

Table 5: Composition and Thickness of SS-304 and Borated Polyethylene Slabs

Configuration	Composition						Total Slab Thickness (cm)
	SS-304 ^a	SS-304	BP ^b	SS-304	BP	SS-304	
	Slab Thickness (cm)						
1	0						0
2	15.24						15.24
3	30.48						30.48
4	30.48	5.08	5.08				40.64
5	30.48	5.08	5.08	5.08			45.72
6	30.48	5.08	5.08	5.08	5.08		50.80
7	30.49	5.08	5.08	5.08	5.08	5.08	55.88
^a Stainless Steel Type 304				^b Borated Polyethylene			

Table 6: Fusion Reactor Shielding Integrated Energy Group Comparisons

Configuration	Detector Position	Energy Group (MeV)	ENDF/B-V	T-2 Rec.	ENDF/B-VI
1	On	0-2	1.249	1.157	1.164
		2-5	1.140	1.143	1.140
		5-10	1.000	1.000	1.000
		10-13	0.934	0.955	1.109
		> 13	1.116	1.100	0.949
2	On	0-2	1.003	0.997	0.982
		2-5	0.981	1.000	1.012
		5-10	0.865	1.000	1.000
		10-13	1.000	1.056	1.054
		> 13	1.035	1.085	1.043
3	On	0-2	0.979	1.000	0.970
		2-5	0.875	0.999	1.000
		5-10	0.770	0.970	1.000
		10-13	0.949	1.000	1.000
		> 13	0.953	1.039	0.972
	Off	0-2	1.258	1.301	1.255
		2-5	1.000	1.145	1.127
		5-10	0.869	1.000	1.018
		10-13	0.910	1.000	1.000
		> 13	1.170	1.356	1.217
7	On	0-2	0.798	0.901	0.842
		2-5	0.823	0.997	0.961
		5-10	0.848	1.000	1.000
		10-13	0.910	1.000	1.000
		> 13	1.000	1.097	1.038
	Off	0-2	0.736	0.822	0.793
		2-5	0.757	0.948	0.921
		5-10	0.765	0.951	1.000
		10-13	0.870	1.000	1.000
		> 13	0.907	1.016	0.977

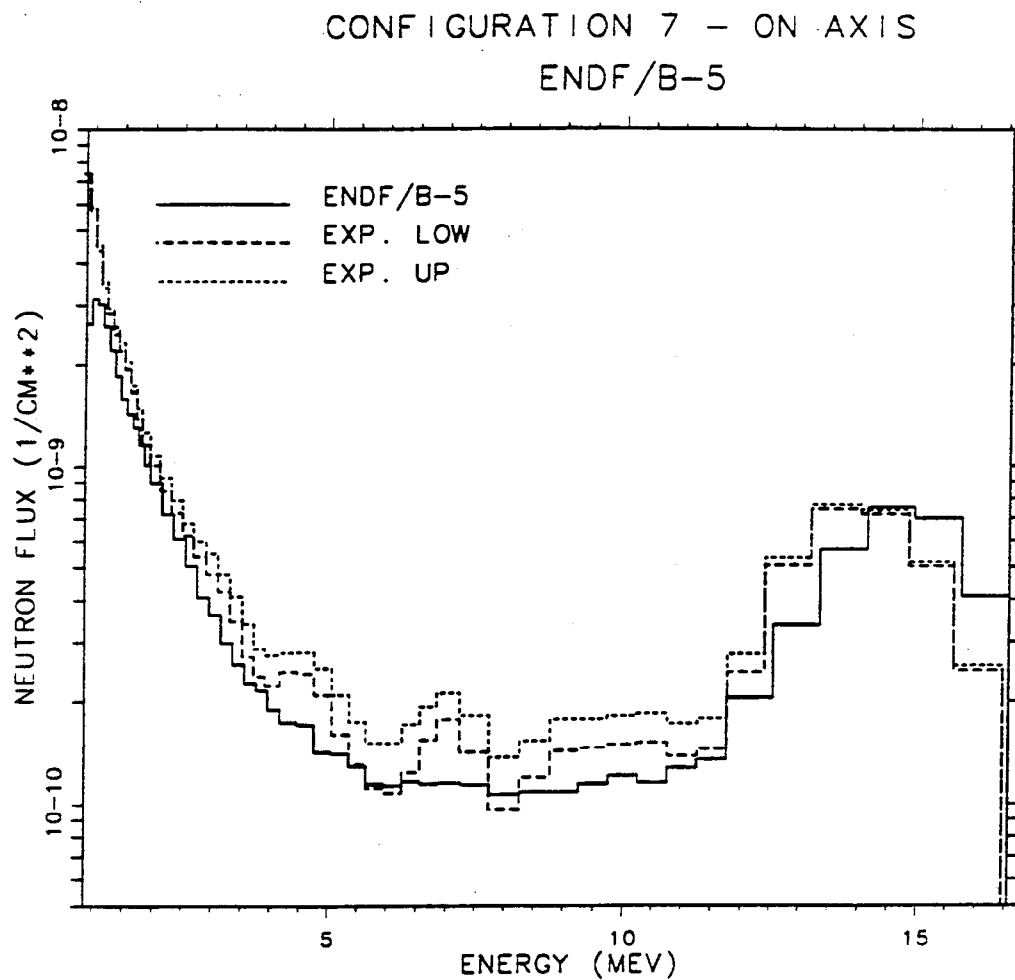


Figure 20: Plot of ENDF/B-V calculated and experimental upper and lower confidence levels for configuration 7 with the detector on axis.

CONFIGURATION 7 - ON AXIS
ENDF/B-6

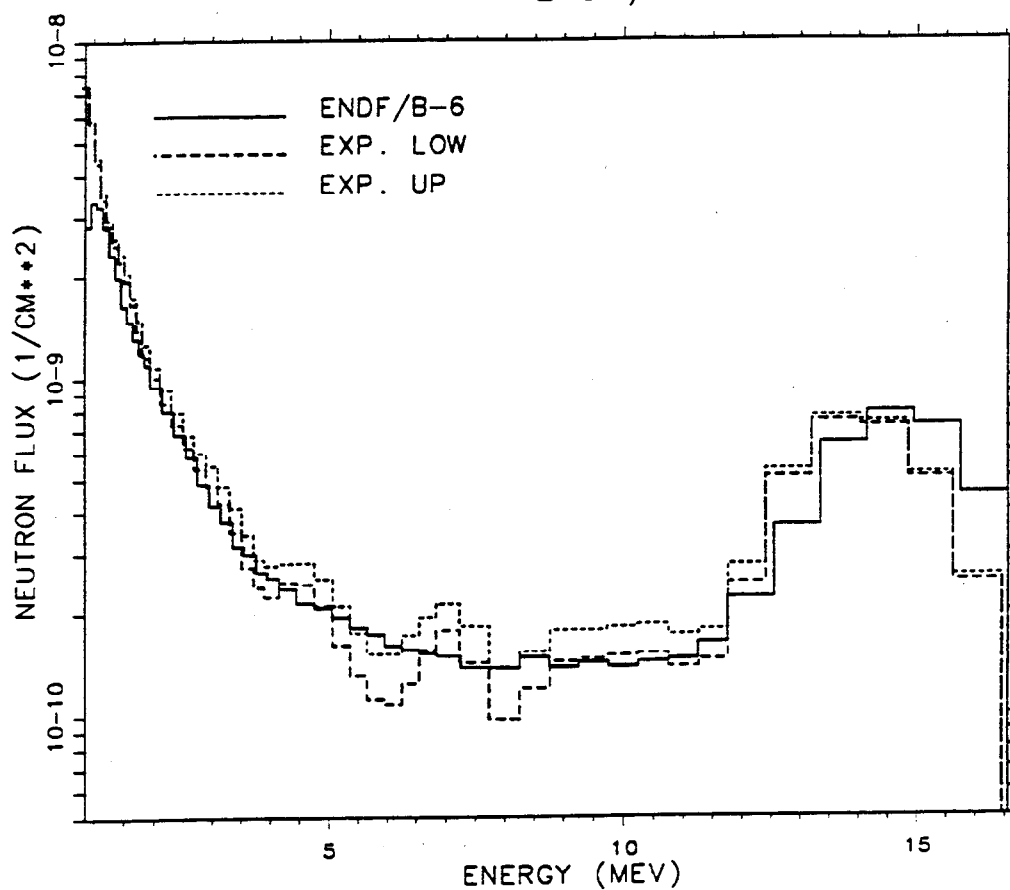


Figure 21: Plot of ENDF/B-VI calculated and experimental upper and lower confidence levels for configuration 7 with the detector on axis.

Table 7: 76 cm. Iron Sphere Results: ENDF/B-VI Data Set Differences

Composition	Source	Energy Group (MeV)	% Difference ($\frac{ENDF/B-VI-ENDF/B-z}{ENDF/B-z}$) $\times$ 100		
			ENDF/B-IV	ENDF/B-V	T-2 Rec.
Pure Iron	^{252}Cf	0.5-2	+4.54	+0.342	-2.02
		2-5	+17.48	+14.11	+1.739
		5-10	+21.0	+22.8	+10.48
		10-13	+82.2	+128.2	+16.5
		> 13	+41.6	-56.8	-21.6
	$\text{T(d,n)}^4\text{He}$	0.5-2	+0.819	+1.017	-2.76
		2-5	+20.8	+19.22	-1.580
		5-10	+36.9	+39.2	+14.44
		10-13	+12.62	+11.93	-0.769
		> 13	+2.33	+3.32	-10.81
High Carbon Steel	^{252}Cf	0.5-2	*	+3.21	-0.723
		2-5	*	+11.11	+2.91
		5-10	*	+18.22	+19.48
		10-13	*	-17.72	-31.9
		> 13	*	-16.95	-1.211
	$\text{T(d,n)}^4\text{He}$	0.5-2	*	+2.53	-0.802
		2-5	*	+21.8	+1.913
		5-10	*	+43.3	+15.05
		10-13	*	+19.07	+2.86
		> 13	*	+5.22	-8.22
*Contains isotopes not available in ENDF/B-IV					

Table 8: Comparison to Experiment for the 3.94 cm Iron Slab.

Data Set	Observation Angle (deg)	% Deviation from Experiment		
		3.09 inch	5.88 inch	9.86 inch
ENDF/B-IV	0	2.0	-3.3	0.1
	15	10.3	0.9	-0.3
	45	10.9	8.0	7.5
ENDF/B-V	0	1.7	-4.4	-0.7
	15	10.1	1.1	0.3
	45	9.7	7.6	8.0
T-2 Rec.	0	2.4	-3.2	0.1
	15	9.5	0.8	1.1
	45	8.9	6.6	8.0
ENDF/B-VI	0	2.4	-2.5	0.6
	15	10.7	0.8	1.6
	45	10.4	6.3	7.9

Table 9: Comparison to Experiment for the 30.81 cm Iron Slab.

Data Set	Observation Angle (deg)	% Deviation from Experiment		
		3.09 inch	5.88 inch	9.86 inch
ENDF/B-IV	0	-4.7	-16.7	-18.5
	15	11.3	6.4	4.3
	45	12.1	10.4	7.5
ENDF/B-V	0	-6.1	-23.5	-28.9
	15	11.0	4.6	0.1
	45	11.4	8.0	2.3
T-2 Rec.	0	-4.6	-17.2	-19.3
	15	9.8	5.6	3.6
	45	10.5	9.3	6.4
ENDF/B-VI	0	1.5	-3.1	-3.1
	15	10.2	5.7	5.3
	45	10.6	9.9	9.2

Table 10: Comparison to Experiment for the 62.00 cm Iron Slab.

Data Set	Observation Angle (deg)	% Deviation from Experiment		
		3.09 inch	5.88 inch	9.86 inch
ENDF/B-IV	0	6.7	-9.3	-18.3
	15	17.8	9.9	-0.1
	45	40.0	27.6	10.6
ENDF/B-V	0	13.6	-6.2	-21.7
	15	20.3	9.4	-5.4
	45	42.9	27.3	5.7
T-2 Rec.	0	10.0	-6.1	-16.4
	15	18.0	9.2	-2.3
	45	38.1	26.1	7.9
ENDF/B-VI	0	16.6	15.4	15.8
	15	12.1	9.6	4.3
	45	32.0	26.9	16.1

Table 11: Comparison to Experiment for the 92.86 cm Iron Slab.

Data Set	Observation Angle (deg)	% Deviation from Experiment		
		3.09 inch	5.88 inch	9.86 inch
ENDF/B-IV	0	18.5	5.2	-7.7
	15	34.2	26.4	8.4
	45	77.4	70.6	47.3
ENDF/B-V	0	31.0	17.9	0.1
	15	40.3	30.7	8.4
	45	81.7	73.5	45.7
T-2 Rec.	0	23.8	11.8	-2.4
	15	35.4	27.3	7.5
	45	77.2	69.4	43.8
ENDF/B-VI	0	28.0	36.9	40.6
	15	29.3	31.0	20.9
	45	81.6	86.1	69.8

allow greater transmission than ENDF/B-V or ENDF/B-IV for iron.

In summary, these 4 sets of iron benchmark calculations showed that when experimental data were available, the T-2 recommended library gave the best agreement, followed by ENDF/B-VI, ENDF/B-V, and ENDF/B-IV. Generally, the T-2 recommended library and ENDF60 tended to agree, whereas the ENDF/B-V and ENDF/B-IV libraries gave lower transmissions. Thus these studies tended to confirm that the T-2 evaluation for iron is still best, followed by ENDF/B-VI, ENDF/B-V, and ENDF/B-IV respectively.

X. CRITICALITY BENCHMARKS

Two sets of criticality calculations were also performed: a set of nine experimental benchmarks for critical assemblies⁸ and a set of 25 benchmark problems for KENO code.⁹ The calculations were performed using both the ENDF60 and ENDF/B-V libraries. The results for the set of nine experimental benchmarks are shown in Table 12. The second set of tests, the KENO benchmarks, consisted of 25 problems, some of which were repeats of each other and some of which were fictitious, such as an infinite cylinder, and are described in Table 13. The results for the KENO benchmarks are shown in Table 14. Again, the ENDF60 library appears to give results as good as ENDF/B-V or better, thus validating the library. Note that these criticality benchmark results with MCNP4A and the ENDF60 library are not published elsewhere in contrast to the other test problems which are described in greater detail in other reports.

XI. DISCUSSION OF THE ENDF60 LIBRARY

In the process of running the above sets of tests, several problems were encountered, and not all could be corrected. Also, ENDF/B-VI is not always an improvement over ENDF/B-V. Here are the changes to the ENDF60 library required before publicly releasing it and some quirks that still remain in the ENDF60 library.

The ¹⁶O evaluation as processed by NJOY had some cross sections with values less than 10⁻³⁷ barns which caused HP workstations to crash. The exponent values were manually increased to 10⁻³⁰ in the ENDF60 library.

All new ORNL isotopic evaluations for F, Cr, Mn, Fe, Ni, Cu and Pb used the Legendre polynomial expansion description for energy-angle distributions. Although MCNP does not yet handle this scattering law, these were approximated by NJOY using the Kalbach-87 formalism, with the exception of F. The Kalbach-87 formalism requires scattering in the center-of-mass system but the Legendre expansion representation is in the laboratory system. The NJOY approach to approximating the Legendre expansion with the Kalbach-87 formalism thus set laboratory and center-of-mass energies for the incident neutron to be equal. Therefore energy is

Table 12: Critical Assembly Benchmarks

	ENDF/B-V	T-2	ENDF/B-VI
Godiva 93.71% Enriched Bare Sphere	1.00008 ± 0.00104	*	0.99525 ± 0.00108
Jezebel 95.5% Enriched ^{239}Pu	1.01510 ± 0.00223	0.99857 ± 0.00212	1.00228 ± 0.00217
Jezebel 80% Enriched ^{239}Pu	1.01595 ± 0.00120	1.00799 ± 0.00121	1.00970 ± 0.00116
Uranium Cylinder 10.9% Enriched ^{235}U	1.00095 ± 0.00054	*	0.99980 ± 0.00049
Uranium Cylinder 14.11% Enriched ^{235}U	1.00087 ± 0.00056	*	0.99715 ± 0.00047
Graphite-Tamped Uranium Sphere	0.98690 ± 0.00102	0.98850 ± 0.00110	0.98998 ± 0.00105
Water-Reflected Uranium Sphere	0.99667 ± 0.00190	*	0.99606 ± 0.00191
Three Cylinders of Uranium Solution	1.00156 ± 0.00126	*	0.99611 ± 0.00136
3×3 Array of Plutonium Fuel Rods	1.00757 ± 0.00169	1.00082 ± 0.00174	0.99916 ± 0.00154

Table 13: The KENO Standard Test Set

KENO 1	Simple unreflected $2 \times 2 \times 2$ array of 93.2% enriched uranium metal cylinders.
KENO 2	Identical to KENO 1, with explicit geometry definition.
KENO 3	$2 \times 2 \times 2$ array of 93.2% enriched uranium metal cylinders reflected by 15.24 cm of paraffin on all six sides.
KENO 4	Identical to KENO 3 with different paraffin specifications.
KENO 5	Identical to KENO 3 and 4 except with 30.48 cm of paraffin.
KENO 6	Single unreflected uranium cylinder from KENO 1.
KENO 7	Identical to KENO 1 and 2 but using specular reflection.
KENO 8	Infinitely long uranium cylinder using the materials and radius of KENO 1.
KENO 9	Infinite array of KENO 1 units through the use of specular reflection.
KENO 10	Identical to KENO 1 except set up to write restart information on every fifth cycle.
KENO 11	Restart of KENO 10 from the 50th cycle. (KENO 10 and 11 utilize a KENO feature not needed by MCNP.)
KENO 12	Composite array of 4 93.2% enriched uranium cylinders and 4 Plexiglas containers filled with 92.6% enriched uranyl-nitrite solution.
KENO 13	Two 93.2% enriched uranium cuboids in a uranium metal cylindrical annulus.
KENO 14	One 93.2% enriched uranium cylinder in a uranium metal cylindrical annulus.
KENO 15	Small 97.6% enriched uranium metal sphere supported by a Plexiglass doughnut in a tank of water.
KENO 16	Infinite number of slabs of uranyl-fluoride solution contained in Pyrex glass and separated by borated uranyl-fluoride solution.
KENO 17	Single 93% enriched uranyl-fluoride sphere.
KENO 18	Reflected cubic array of 27 cylinders of aqueous uranyl-nitrate in Plexiglas bottles.
KENO 19	Identical to KENO 12 but using repeated structures.
KENO 20	Critical experiment consisting of seven cylinders in a triangular pitched array.
KENO 21	Critical experiment of an aluminum spherical container, 98% filled with 4.89% enriched uranyl-fluoride.
KENO 22	Identical to KENO 1 using nested holes in a void spacing cuboid.
KENO 23	Identical to KENO 1 using hemi-cylinders.
KENO 24	Identical to KENO 23 but with the hemi-cylinders aligned with the x-axis.
KENO 25	Identical to KENO 23 but with the hemi-cylinders aligned with the y-axis.

Table 14: Keno Benchmarks

	ENDF/B-V	ENDF/B-VI	Change
KENO 1	0.99987 ± 0.00093	0.99365 ± 0.00087	-0.006
KENO 2	0.99987 ± 0.00093	0.99365 ± 0.00087	-0.006
KENO 3	0.99933 ± 0.00112	1.00015 ± 0.00109	+0.001
KENO 4	1.00084 ± 0.00282	0.99983 ± 0.00258	-0.001
KENO 5	1.00042 ± 0.00281	1.00441 ± 0.00293	+0.004
KENO 6	0.74606 ± 0.00074	0.74257 ± 0.00071	-0.003
KENO 7	1.00022 ± 0.00083	0.99536 ± 0.00079	-0.005
KENO 8	0.94036 ± 0.00085	0.93807 ± 0.00070	-0.002
KENO 9	2.29097 ± 0.00100	2.25973 ± 0.00093	-0.031
KENO 10	0.99987 ± 0.00093	0.99365 ± 0.00087	-0.006
KENO 11	0.99987 ± 0.00093	0.99365 ± 0.00087	-0.006
KENO 12	0.99869 ± 0.00121	0.99940 ± 0.00127	+0.001
KENO 13	0.99489 ± 0.00084	0.99141 ± 0.00081	-0.003
KENO 14	0.99849 ± 0.00084	0.99686 ± 0.00082	-0.002
KENO 15	1.00155 ± 0.00097	1.00027 ± 0.00109	-0.001
KENO 16	0.99066 ± 0.00093	0.99235 ± 0.00089	+0.002
KENO 17	1.00290 ± 0.00143	0.99862 ± 0.00152	-0.004
KENO 18	1.02802 ± 0.00126	1.03085 ± 0.00128	+0.003
KENO 19	0.99869 ± 0.00121	0.99940 ± 0.00127	+0.001
KENO 20	0.99707 ± 0.00133	0.99809 ± 0.00147	+0.001
KENO 21	0.99510 ± 0.00082	0.99292 ± 0.00089	-0.002
KENO 22	0.99775 ± 0.00082	0.99551 ± 0.00082	-0.002
KENO 23	0.99987 ± 0.00093	0.99365 ± 0.00087	-0.006
KENO 24	0.99819 ± 0.00081	0.99440 ± 0.00085	-0.004
KENO 25	1.00115 ± 0.00090	0.99516 ± 0.00088	-0.006

not conserved properly. This is a small effect for the heavier elements, but it could have been a significant effect for F.

The original F data as processed by NJOY did not properly sample both spectra given in the evaluation for the (n,2n) reaction. Thus the original F data was replaced with one in which the (n,2n) reaction was broken into two reactions, MT=6 and MT= 46, each with half the true (n,2n) cross section, but one with the first neutron out spectrum and the other with the second neutron-out spectrum. This representation is similar to the treatment in ENDF/B-V of ^9Be . The correlated energy-angle scattering law (MCNP law 67) is used for approximating the Legendre polynomial expansion description for energy-angle distribution by NJOY, thereby correcting the center-of-mass error found in the other new ORNL evaluations discussed above. A more thorough discussion of these approximations for the Legendre polynomial expansion can be found in the following section.

Another modification to the ENDF/B-VI evaluations using the Kalbach-87 formalism replaces the interpolation scheme for INT=12 with INT=2 for neutrons, but it has been left as INT=12 for photons. However, MCNP treats both neutrons and photons as INT=2 (linear-linear interpolation). This detail is another quirk of the ENDF60 library that should some day be corrected but was left in for the present.

No cadmium file is provided in the ENDF60 library due to an NJOY processing problem. Prior to processing by NJOY, an attempt had been made to add photon production to the ENDF/B-VI Cd (nat) evaluation. This modification of the input data set caused errors with the neutron heating numbers while not fully succeeding at implementing photon production. Since ENDF/B-VI cadmium is simply a translation from ENDF/B-V with no photon production, it was felt that correcting the ENDF/B-VI modified cadmium would cause a significant delay in the release of the ENDF60 library.

The data file cf98249 was also modified to correct an evaluation problem concerning center-of-mass to lab energy conversion. Prior to this correction, neutrons could upscatter above 20 MeV and cause MCNP to crash.

XII. APPROXIMATION TO FILE 6, LAW=1, LANG=1

Several new descriptions for energy and angle distributions of secondary particles from nuclear reactions were implemented in ENDF/B-VI. These were briefly described in Section II. Some of the formats of these new descriptions for file 6 in the ENDF/B-IV evaluations are as follows:

- LCT=1, LAW=1, LANG=1: give angular distributions for each $E \rightarrow E'$ using Legendre coefficients in the laboratory frame and are used in the ORNL evaluations.
- LCT=2, LAW=1, LANG=2: give angular distributions for each $E \rightarrow E'$ using Kalbach systematics through the R parameter in the center-of-mass frame and are used in the LANL and Russian evaluations.

- LCT=1, LAW=7: give secondary energy distributions for a set of emission cosines in the laboratory frame and are used in the LLNL and European Fusion File (EFF) evaluations.
- LCT=2, LAW=6: give distributions in either the laboratory or center-of-mass frame using an analytic law based on phase space and are used in the LANL and Russian evaluations.

There are also two other variations that have not been used which provide for tabulated angular distributions in either the laboratory or center-of-mass frame.

Since MCNP does not yet handle data using LCT=1, LAW=1, LANG=1, it was necessary to approximate the energy-angle distributions using other formalisms. At the time when most of the data for ENDF60 library was processed, NJOY approximated these distributions using the Kalbach-87 formalism, which predicts the following differential cross section in the center-of-mass system,

$$P_{cm}(\mu_{cm}, E, E'_{cm}) = P_{cm}(E, E'_{cm}) \frac{1}{2} \frac{A}{\sinh(A)} [\cosh(A\mu) + R \sinh(A\mu)]. \quad (8)$$

The goal was to select $P_{cm}(E, E'_{cm})$, R , and A to obtain reasonable fits to the tabulated laboratory differential cross section given in the ENDF/B-VI evaluations. Unfortunately, full use of this procedure often produces physically unreasonable center-of-mass distributions when applied to the existing data. Therefore, NJOY assumes that the angle-averaged P_{cm} was identical to P_L , the P_0 component of the laboratory differential cross section. The forward and backward ratios in the laboratory system, $P_L(+1, E, E')$ to $P_{cm}(E, E'_{cm})$, were converted approximately into the corresponding center-of-mass ratios, and the R and A parameters were calculated from these ratios where $R \geq 0$. The center-of-mass ratios were typically more isotropic than the laboratory ratios. This procedure is easy to implement, but it is not formally correct. It works well for isotropic distributions (low E' values) and for forward-peaked distributions (high E' values). It provides a fairly good fit to the forward-scattering energy spectrum and to the backward-scattering spectrum, but the P_0 average could be distorted if the R value was not close to either 0 or 1. The approximation works better for heavier targets and was used for the Cr, Mn, Fe, Ni, Cu and Pb nuclides of the ORNL evaluations. MCNP samples from the data by using the energy distribution to choose an outgoing energy E' , and then using the corresponding R and A values produced by NJOY to determine an outgoing cosine from the Kalbach formula. A consequence of this approach sets the laboratory and center-of-mass energies for the incident neutron to be equal, thereby not conserving energy.

As was discussed in Sections II and XI, the original F file was replaced due to an error in representing the (n,2n) reaction. When this ORNL evaluation was reprocessed correcting the (n,2n) error, a new approach for approximating LCT=1, LAW=1, LANG=1 had been

implemented in NJOY. Currently, NJOY uses the energy-angle law, with LCT=1 and LAW=7, for approximating the Legendre polynomial expansion version of the energy-angle distributions and was therefore used for F. As this law is already in the laboratory frame, the difficult conversion to the center-of-mass frame is not required, and the energy error found in the other new ORNL evaluations is not present. The first step in this approximation is to choose a set of angle cosines. Inside a loop on incident energy E , NJOY constructs an outgoing energy spectrum for each of these angles. For use in MCNP, it is also necessary for NJOY to integrate each of the distributions over secondary energy and produce a single-differential distribution $f(E,\mu)$. This distribution is then converted into 32 equally probable cosine bins. MCNP samples from this law by using the 32-bin data to select a cosine value.

XIII. CONCLUSION

A standard, "official" MCNP ENDF/B-VI library, the ENDF60 library, is available and has the unique .60C ZAID designator. The ENDF60 library required the development of new scattering laws and sampling techniques in MCNP. It has been tested with infinite media, LLNL Pulsed Sphere benchmarks, photon production tests, critical assembly benchmarks, and iron benchmarks. It is available to MCNP users worldwide so they do not have to generate their own ENDF/B-VI library. Therefore, when they run MCNP with ENDF60, the same library is available for comparison purposes. A new photon library, MCPLIB02, was released with the RSIC package. This new library extends the photon interaction range from 100 MeV to 100 GeV using the LLNL Evaluated Photon Data Library (EPDL). Additionally, the EL1 electron library and an updated XSDIR file were also released. Unlike previous versions, the new XSDIR file defined the default neutron libraries to be the recommended data library as indicated in Appendix G of the MCNP4A manual, and MCPLIB02 as the default photon library.

XIV. ACKNOWLEDGMENTS

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Los Alamos
NATIONAL LABORATORY
memorandum

*Applied Theoretical & Computational
Physics Division
Radiation Transport Group, X-6
Los Alamos, New Mexico 87545*

To/MS: Distribution
From/MS: H. Grady Hughes, XTM / MS B226
Phone/FAX:
Symbol: X-6:HGH-93-77
Date: January 28, 1993

SUBJECT: Information on the MCPLIB02 Photon Library

A new photon interaction library, MCPLIB02, covering the energy range 1 keV -- 100 GeV for elements $Z = 1$ -- 94 is now available. This library is the first product arising from my investigations of the Livermore Evaluated Photon Data Library (EPDL).[1] With this memorandum, I present the relevant files, give a brief description of the data, and show the results of some comparisons and tests.

1. Contents of the New Library

The EPDL contains considerably more detail than MCNP is currently prepared to use. For example, the EPDL distinguishes between pair and triplet production cross sections, tabulates real and imaginary anomalous scattering factors, and includes photoelectric subshell cross sections for shells as high-lying as Q1. As a long-term project, it would be desirable to incorporate all of this detail into MCNP. For the first implementation of EPDL data, I have chosen to take a more gradual approach, and have used the EPDL only to extend our existing photon library, leaving the sampling algorithms unchanged. Specifically, I have retained MCPLIB data in the energy range 0.001--10.0 MeV, incorporated the new EPDL data above the maximum energy available in MCPLIB, and made a smooth transition from MCPLIB to EPDL between these limiting energies. The maximum energy in MCPLIB is usually 100 MeV, but for seven elements ($Z = 84, 85, 87, 88, 89, 91, \text{ and } 93$) it is 15 MeV.[2] For each element, five blocks of MCPLIB data have been extended. These are the incoherent, coherent, photoelectric, and pair production cross sections, and the heating numbers. The remaining blocks of data (incoherent, coherent, and integrated coherent form factors, and fluorescence data) remain unchanged. Since MCPLIB data are used at low energies, the new database will be backward-compatible with most existing MCNP photon calculations; but high-energy applications up to the limit of the electron data (1 GeV) will now be possible.

Unlike the MCPLIB data, the EPDL uses a different energy grid for tabulating each separate photon cross section. By comparing these various energy grids, I selected an appropriate common grid on which to tabulate all new data above 10 MeV. This grid is based on the EPDL grid for pair production cross sections, both because the pair production energy grid has the greatest resolution, and because pair production is the most important process at high energies. Both the EPDL and the MCPLIB cross-section data (and the MCPLIB heating numbers) rely on logarithmic interpolation to evaluate quantities at energies between tabulated points:

$$\sigma = F(E_1, E_2, \sigma_1, \sigma_2, E) = \exp \left(\frac{\left\{ \ln \left[\frac{E}{E_1} \right] \ln[\sigma_2] + \ln \left[\frac{E_2}{E} \right] \ln[\sigma_1] \right\}}{\ln \left[\frac{E_2}{E_1} \right]} \right). \quad (1)$$

The EPDL also calls for the use of linear interpolation for certain quantities. For present purposes, the important quantities relying on linear interpolation are the average energies ΔE of secondary particles.

My procedure for merging MCPLIB and EPDL data for each element into the new library is as follows. For energies between 1 keV and 10 MeV, the MCPLIB energy grid and data are included unchanged. The maximum energy $E_{\max}$ available in MCPLIB is identified. ($E_{\max}$ is either 15 MeV or 100 MeV.) The maximum energy ϵ available in MCPLIB was identified. (ϵ is either 15 MeV or 100 MeV). For energies E in the transition range $10 \text{ MeV} < E < \epsilon$, the MCPLIB cross section σ_E was determined for each process. An interpolated cross section

$$\sigma = F(10.0, \epsilon, \sigma_M, \sigma_E, E) \quad (2)$$

is used for that process in the new library. Above $E_{\max}$, the EPDL cross section is determined, interpolating as needed, and used unchanged. Whenever a pair cross section must be extracted from the EPDL, the sum of the pair and triplet cross sections is used, since MCPLIB does not separate the two processes. There is no quantity comparable to the MCPLIB heating numbers in the EPDL, so heating numbers are constructed according to the formula

$$H(E) = \frac{\sum_i \Delta E_i(E) \sigma_i(E)}{\sigma_{\text{total}}(E)}, \quad (3)$$

where $H(E)$ is an MCPLIB-style heating number as a function of energy E ; the summation index i labels the incoherent, photoelectric, pair, and triplet processes; and $\Delta E_i(E)$ is the average secondary electron energy from process i , except in the case of photoelectric absorption, when $\Delta E_i(E)$ also includes the energy of those fluorescent X-rays not explicitly accounted for in the EPDL. Since the EPDL accounts for X-rays from many more shells than does MCPLIB, Eq.3 is not strictly consistent with the MCPLIB heating numbers. Fortunately, at the high energies of present interest, this inconsistency is completely negligible.

I have made one additional enhancement to the new library. For 87 of the 94 elements in MCPLIB, an artificial, small pair production cross section (1 microbarn) is included at the pair production threshold (1.022 MeV). This prevents MCNP's interpolation scheme from returning zero for the pair cross section all the way up to the first real data point (1.5 MeV). The seven elements whose maximum MCPLIB energies were only 15 MeV also lacked this special starting point for the pair cross section. I have inserted an energy point at 1.022 MeV for these seven elements, so that the treatment of pair production is now consistent across the entire database. I derived the other cross

sections and heating number at this point by logarithmic interpolation, so that the sampled values for these quantities will not be affected.

II. Comparisons Between MCPLIB and EPDL-based MCPLIB02

In the energy range of interest for the transition from MCPLIB to EPDL, the two photon libraries are in reasonable agreement for the important processes. To make this conclusion more quantitative, I have made a global survey of the two data sets. For each Z 1....94, I identify $E_{\max}$, the maximum energy in the MCPLIB table. Then I distribute 1000 energies, equally spaced in the logarithm, between 10 MeV and $E_{\max}$. At each energy, for each process (coherent, incoherent, photoelectric, and pair production cross section, and heating number), I find the MCPLIB and EPDL values, σ_M and σ_E , respectively. I calculate the magnitude of the relative difference

$$r = |(\sigma_M - \sigma_E) / \sigma_E|$$

and retain the maximum value of r for each process over the entire energy range. This is a severe test of the similarity of the two data sets, since only the overall largest relative difference is reported. The results are given in Table I, which shows the maximum relative difference in the range (10 MeV -- $E_{\max}$) for each process as a function of Z . At high energies, the coherent and photoelectric cross sections are negligible compared with the incoherent scattering and pair (plus triplet) production cross sections. In addition, heating numbers are also needed at high energies. For these three significant categories, the two databases agree quite well, with the largest relative differences being 1.2% for the incoherent cross section, 5.3% for the pair (plus triplet) production cross section, and 0.53% for the heating number. The agreement for the coherent cross section, for the 87 elements with ENDF data, is also satisfactory; the maximum relative difference among these elements is only 0.82%. For the seven elements that rely on older data, the agreement is not quite as good; the maximum relative difference is 16%. The photoelectric cross section fares less well, with relative differences as large as 56%. However, the insignificance of the coherent and photoelectric processes at high energies makes these disagreements unimportant.

To reinforce this last statement, I have created Table II, which again shows maximum differences in the range (10 MeV -- $E_{\max}$) as a function of Z . Here, however, the difference between MCPLIB and EPDL is compared not to the EPDL value for the individual process, but to the total cross section. The tabulated quantity is

$$R = |(\sigma_M - \sigma_E) / \sigma_T|$$

where σ_T is the total cross section from the EPDL library. The relative unimportance of the differences in the coherent and photoelectric cross sections is easily seen in this table. Here the maximum fractional differences are 0.01% for the coherent cross section, 0.786% for the incoherent cross section, 0.0721% for the photoelectric cross section, and 4.37% for the pair plus triplet cross section.

III. Testing the New Library

Since the new photon database is identical to MCPLIB up to 10 MeV (except for the addition of a threshold point for pair production for seven elements), MCNP calculations using the new library should track most old calculations. In fact, using MCPLIB02 with the standard set of 25 test problems results in 24 cases of exact tracking. The sole exception is problem 8, which includes photons with energies higher than 10 MeV. It is initially difficult to judge the consistency of the old and new calculations, because of the poor statistics typical of the short-running test problems. A much longer execution of problems 7 and 8, however, reveals a satisfying similarity between the old and new results. I have shown selected tallies (those dealing with photons and electrons) from the two versions of problem 8 in Table III.

Problem 8 from the test set is not a very severe test of the new library, since very few photons are found in the high-energy regime. It will be important to test the new data as extensively as possible using applications in which the significant aspects of the transport occur at high energies. I am currently undertaking some investigations both to validate the new photon library and to satisfy certain other programmatic tasks, and I hope to report further on these matters in the future.

IV. References

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Table I: Maximal Relative Differences for All Processes.

Z	Coherent	Incoherent	Photoel.	Pair	Heating
1	1.30E-03	1.20E-02	4.80E-01	2.80E-02	3.70E-03
2	9.20E-04	1.10E-02	5.60E-01	2.40E-02	3.70E-03
3	9.20E-04	1.10E-02	4.10E-01	2.20E-02	3.70E-03
4	1.50E-03	1.10E-02	3.40E-01	2.10E-02	3.70E-03
5	4.60E-04	1.20E-02	2.50E-01	2.10E-02	3.70E-03
6	6.90E-04	1.10E-02	2.20E-01	2.00E-02	3.70E-03
7	8.20E-03	1.20E-02	2.00E-01	2.00E-02	3.60E-03
8	7.60E-03	1.10E-02	1.70E-01	2.00E-02	3.60E-03
9	2.10E-03	1.20E-02	1.60E-01	2.00E-02	3.60E-03
10	1.70E-03	1.20E-02	1.50E-01	2.00E-02	3.70E-03
11	1.70E-03	1.20E-02	1.20E-01	2.00E-02	3.90E-03
12	1.40E-03	1.10E-02	1.10E-01	2.10E-02	4.00E-03
13	8.60E-04	1.10E-02	1.40E-01	2.10E-02	4.00E-03
14	6.20E-04	1.20E-02	1.10E-01	2.10E-02	4.10E-03
15	2.70E-04	1.20E-02	1.00E-01	2.10E-02	4.20E-03
16	2.10E-04	1.20E-02	1.00E-01	2.10E-02	4.30E-03
17	4.10E-04	1.20E-02	9.70E-02	2.10E-02	4.30E-03
18	4.60E-04	1.10E-02	8.80E-02	2.10E-02	4.30E-03
19	1.90E-04	1.10E-02	6.10E-02	2.10E-02	4.40E-03
20	5.60E-04	1.10E-02	1.10E-01	2.10E-02	4.40E-03
21	9.30E-04	1.20E-02	5.30E-02	2.10E-02	4.50E-03
22	1.30E-03	1.20E-02	7.70E-02	2.10E-02	4.50E-03
23	1.70E-03	1.10E-02	1.10E-01	2.10E-02	4.50E-03
24	1.80E-03	1.10E-02	1.20E-01	2.00E-02	4.40E-03
25	2.00E-03	1.10E-02	1.00E-01	2.00E-02	4.50E-03
26	2.10E-03	1.10E-02	1.10E-01	2.00E-02	4.50E-03
27	2.20E-03	1.20E-02	7.00E-02	2.00E-02	4.50E-03
28	2.30E-03	1.20E-02	8.30E-02	2.00E-02	4.60E-03
29	2.30E-03	1.20E-02	1.10E-01	2.10E-02	4.60E-03
30	2.10E-03	1.20E-02	6.80E-02	2.10E-02	4.60E-03
31	1.90E-03	1.10E-02	2.20E-02	2.10E-02	4.70E-03
32	1.90E-03	1.10E-02	2.30E-02	2.20E-02	4.70E-03
33	1.50E-03	1.10E-02	2.30E-02	2.30E-02	4.70E-03
34	1.20E-03	1.20E-02	3.00E-02	2.30E-02	4.70E-03
35	1.10E-03	1.20E-02	2.90E-02	2.40E-02	4.80E-03
36	9.60E-04	1.20E-02	2.00E-02	2.40E-02	4.80E-03
37	5.50E-04	1.20E-02	3.00E-02	2.40E-02	4.80E-03
38	4.80E-04	1.20E-02	3.30E-02	2.50E-02	4.80E-03
39	6.50E-04	1.20E-02	2.30E-02	2.50E-02	4.80E-03
40	6.40E-04	1.20E-02	3.40E-02	2.50E-02	4.80E-03
41	5.90E-04	1.10E-02	6.10E-02	2.50E-02	4.80E-03
42	6.60E-04	1.10E-02	3.30E-02	2.60E-02	4.80E-03
43	7.50E-04	1.10E-02	3.50E-02	2.60E-02	4.80E-03
44	7.90E-04	1.20E-02	4.00E-02	2.70E-02	4.80E-03
45	7.20E-04	1.20E-02	4.40E-02	2.80E-02	4.90E-03
46	6.20E-04	1.20E-02	2.80E-02	2.80E-02	4.90E-03
47	4.30E-04	1.20E-02	5.80E-02	2.90E-02	4.90E-03
48	2.90E-04	1.20E-02	3.80E-02	2.90E-02	4.90E-03
49	3.60E-04	1.10E-02	2.80E-02	3.00E-02	4.90E-03
50	4.20E-04	1.10E-02	5.40E-02	3.00E-02	4.90E-03
51	6.10E-04	1.10E-02	7.60E-02	3.10E-02	4.90E-03
52	9.00E-04	1.10E-02	5.60E-02	3.10E-02	4.90E-03
53	1.20E-03	1.10E-02	4.10E-02	3.20E-02	4.90E-03
54	1.20E-03	1.10E-02	5.90E-02	3.20E-02	4.90E-03
55	1.30E-03	1.10E-02	5.20E-02	3.30E-02	4.90E-03

56	1.50E-03	1.10E-02	6.20E-02	3.30E-02	4.90E-03
57	1.50E-03	1.10E-02	4.50E-02	3.40E-02	4.90E-03
58	1.50E-03	1.10E-02	6.50E-02	3.40E-02	4.90E-03
59	1.70E-03	1.10E-02	4.80E-02	3.50E-02	4.90E-03
60	1.60E-03	1.20E-02	4.00E-02	3.50E-02	5.00E-03
61	1.30E-03	1.10E-02	4.60E-02	3.50E-02	4.90E-03
62	1.30E-03	1.10E-02	4.80E-02	3.60E-02	4.90E-03
63	1.30E-03	1.10E-02	4.50E-02	3.60E-02	4.90E-03
64	1.20E-03	1.10E-02	4.90E-02	3.60E-02	4.90E-03
65	8.80E-04	1.20E-02	4.70E-02	3.70E-02	5.00E-03
66	8.10E-04	1.10E-02	4.50E-02	3.70E-02	4.90E-03
67	6.20E-04	1.20E-02	5.90E-02	3.70E-02	4.90E-03
68	5.20E-04	1.10E-02	4.00E-02	3.80E-02	4.90E-03
69	7.80E-04	1.10E-02	3.80E-02	3.80E-02	4.90E-03
70	9.70E-04	1.10E-02	3.00E-02	3.80E-02	4.90E-03
71	1.00E-03	1.10E-02	3.40E-02	3.80E-02	4.90E-03
72	1.20E-03	1.10E-02	6.10E-02	3.80E-02	4.90E-03
73	1.50E-03	1.10E-02	5.60E-02	3.80E-02	4.90E-03
74	1.60E-03	1.10E-02	9.10E-02	3.80E-02	4.90E-03
75	1.80E-03	1.10E-02	6.80E-02	3.80E-02	4.80E-03
76	2.00E-03	1.10E-02	3.20E-02	3.80E-02	4.90E-03
77	1.90E-03	1.20E-02	3.00E-02	3.80E-02	4.80E-03
78	2.10E-03	1.20E-02	3.60E-02	3.80E-02	4.80E-03
79	2.20E-03	1.20E-02	3.40E-02	3.80E-02	4.80E-03
80	2.10E-03	1.20E-02	3.80E-02	3.80E-02	4.80E-03
81	2.10E-03	1.20E-02	5.50E-02	3.80E-02	4.80E-03
82	1.80E-03	1.20E-02	9.00E-02	3.80E-02	4.80E-03
83	1.80E-03	1.10E-02	4.50E-02	3.80E-02	4.70E-03
84	1.30E-01	6.70E-03	1.90E-02	5.20E-02	5.30E-03
85	1.40E-01	7.40E-03	2.20E-02	4.90E-02	5.10E-03
86	1.60E-03	1.20E-02	3.20E-02	3.90E-02	4.70E-03
87	1.40E-01	6.00E-03	2.60E-02	4.80E-02	5.10E-03
88	1.40E-01	6.00E-03	2.10E-02	4.90E-02	5.00E-03
89	1.50E-01	6.70E-03	2.40E-02	4.90E-02	5.00E-03
90	8.30E-04	1.20E-02	4.60E-02	3.90E-02	4.70E-03
91	1.50E-01	8.80E-03	2.90E-02	5.30E-02	5.10E-03
92	5.20E-04	1.10E-02	6.30E-02	3.90E-02	4.60E-03
93	1.60E-01	6.20E-03	2.90E-02	4.70E-02	4.80E-03
94	1.10E-03	1.10E-02	4.30E-02	3.80E-02	4.50E-03

Table II: Maximal Differences as Fractions of the Total Cross Section.

Z	Coherent	Incoherent	Photoel.	Pair
1	1.99E-10	7.86E-03	7.37E-10	8.06E-03
2	2.28E-09	6.52E-03	1.76E-08	7.77E-03
3	3.59E-09	5.86E-03	6.72E-08	5.78E-03
4	1.23E-08	5.36E-03	1.76E-07	5.34E-03
5	2.80E-09	5.14E-03	3.07E-07	5.93E-03
6	1.02E-08	4.51E-03	5.38E-07	6.92E-03
7	1.91E-07	4.28E-03	8.55E-07	7.12E-03
8	2.15E-07	3.82E-03	1.20E-06	7.31E-03
9	7.17E-08	3.86E-03	1.69E-06	7.88E-03
10	6.92E-08	3.57E-03	2.34E-06	8.46E-03
11	7.50E-08	3.69E-03	2.56E-06	9.06E-03
12	6.38E-08	3.15E-03	3.33E-06	9.58E-03
13	3.77E-08	3.11E-03	5.49E-06	1.02E-02
14	2.42E-08	3.18E-03	5.37E-06	1.05E-02
15	2.59E-09	3.04E-03	6.30E-06	1.08E-02
16	1.67E-08	2.94E-03	7.90E-06	1.11E-02
17	3.61E-08	2.84E-03	9.32E-06	1.14E-02
18	4.31E-08	2.61E-03	1.02E-05	1.17E-02
19	6.21E-09	2.51E-03	8.14E-06	1.19E-02
20	3.61E-08	2.46E-03	1.85E-05	1.22E-02
21	7.66E-08	2.61E-03	9.86E-06	1.22E-02
22	1.24E-07	2.50E-03	1.75E-05	1.22E-02
23	1.79E-07	2.24E-03	2.95E-05	1.26E-02
24	2.04E-07	2.18E-03	3.68E-05	1.26E-02
25	2.47E-07	2.16E-03	3.66E-05	1.27E-02
26	2.81E-07	2.14E-03	4.44E-05	1.24E-02
27	3.01E-07	2.14E-03	3.15E-05	1.30E-02
28	3.33E-07	2.16E-03	4.1E-05	1.33E-02
29	3.43E-07	2.05E-03	6.1E-05	1.39E-02
30	3.28E-07	2.15E-03	4.18E-05	1.42E-02
31	3.01E-07	1.94E-03	1.51E-05	1.43E-02
32	2.99E-07	1.83E-03	1.73E-05	1.52E-02
33	2.33E-07	1.81E-03	1.19E-05	1.55E-02
34	1.78E-07	1.91E-03	2.67E-05	1.59E-02
35	1.74E-07	1.85E-03	2.78E-05	1.66E-02
36	1.40E-07	1.81E-03	1.94E-05	1.67E-02
37	5.04E-08	1.77E-03	1.29E-05	1.71E-02
38	4.03E-08	1.79E-03	4.11E-05	1.77E-02
39	3.42E-08	1.76E-03	2.23E-05	1.78E-02
40	3.65E-08	1.64E-03	4.93E-05	1.83E-02
41	3.14E-08	1.54E-03	9.32E-05	1.84E-02
42	5.51E-08	1.52E-03	5.47E-05	1.88E-02
43	8.79E-08	1.55E-03	2.37E-05	1.94E-02
44	1.10E-07	1.54E-03	2.97E-05	1.99E-02
45	1.02E-07	1.57E-03	3.60E-05	2.06E-02
46	8.20E-08	1.58E-03	2.85E-05	2.11E-02
47	3.46E-08	1.57E-03	1.31E-04	2.15E-02
48	1.79E-08	1.56E-03	3.59E-05	2.20E-02
49	5.25E-08	1.35E-03	3.74E-05	2.24E-02
50	5.90E-08	1.33E-03	1.44E-04	2.29E-02
51	1.01E-07	1.32E-03	1.05E-04	2.35E-02
52	1.84E-07	1.32E-03	7.17E-05	2.39E-02
53	2.85E-07	1.28E-03	4.88E-05	2.44E-02

54	2.80E-07	1.26E-03	8.30E-05	2.50E-02
55	3.10E-07	1.26E-03	7.21E-05	2.55E-02
56	3.78E-07	1.24E-03	9.56E-05	2.60E-02
57	3.94E-07	1.21E-03	6.49E-05	2.64E-02
58	3.87E-07	1.18E-03	1.10E-04	2.67E-02
59	4.65E-07	1.18E-03	7.80E-05	2.71E-02
60	4.32E-07	1.27E-03	6.14E-05	2.74E-02
61	3.57E-07	1.15E-03	7.90E-05	2.79E-02
62	3.41E-07	1.14E-03	8.45E-05	2.82E-02
63	3.71E-07	1.13E-03	8.02E-05	2.86E-02
64	3.14E-07	1.15E-03	9.29E-05	2.89E-02
65	2.05E-07	1.19E-03	9.23E-05	2.92E-02
66	1.78E-07	1.11E-03	8.98E-05	2.95E-02
67	1.20E-07	1.18E-03	1.33E-04	2.96E-02
68	9.61E-08	1.08E-03	9.68E-05	3.01E-02
69	4.80E-08	1.08E-03	1.13E-04	3.04E-02
70	1.27E-07	1.07E-03	1.74E-04	3.05E-02
71	1.40E-07	1.06E-03	1.64E-04	3.04E-02
72	2.28E-07	1.07E-03	1.61E-04	3.09E-02
73	3.76E-07	1.04E-03	4.25E-04	3.11E-02
74	4.39E-07	1.05E-03	7.21E-04	3.11E-02
75	5.20E-07	1.03E-03	2.17E-04	3.09E-02
76	6.67E-07	1.03E-03	2.68E-04	3.12E-02
77	6.14E-07	1.08E-03	2.67E-04	3.08E-02
78	7.21E-07	1.06E-03	2.80E-04	3.13E-02
79	7.80E-07	1.06E-03	2.08E-04	3.11E-02
80	7.60E-07	1.09E-03	1.94E-04	3.09E-02
81	7.82E-07	1.07E-03	3.20E-04	3.10E-02
82	6.82E-07	1.10E-03	6.38E-04	3.10E-02
83	6.61E-07	9.74E-04	2.35E-04	3.13E-02
84	7.50E-05	1.13E-03	1.19E-04	4.24E-02
85	7.77E-05	1.15E-03	1.41E-04	3.93E-02
86	6.13E-07	1.05E-03	3.35E-04	3.20E-02
87	8.22E-05	1.01E-03	1.78E-04	3.89E-02
88	8.47E-05	9.09E-04	1.47E-04	3.96E-02
89	8.81E-05	1.01E-03	1.71E-04	4.06E-02
90	2.72E-07	1.02E-03	2.35E-04	3.25E-02
91	9.35E-05	1.31E-03	2.22E-04	4.37E-02
92	1.35E-07	9.41E-04	2.85E-04	3.25E-02
93	1.00E-04	9.84E-04	2.35E-04	3.84E-02
94	4.95E-07	9.07E-04	2.26E-04	3.13E-02

Table III: Selected Tallies from Test Problem 8

Old Photon Library			New Photon Library		
tally 14 cell 7 energy			tally 14 cell 7 energy		
1.0000E-01	1.84487E-10	0.1847	1.0000E-01	1.84506E-10	0.1847
1.0000E+00	5.93596E-09	0.1017	1.0000E+00	6.13097E-09	0.1055
1.0000E+01	1.16635E-08	0.0702	1.0000E+01	1.16579E-08	0.0702
1.0000E+02	2.44636E-11	0.0600	1.0000E+02	2.44809E-11	0.0600
total	1.78084E-08	0.0643	total	1.79978E-08	0.0650
tally 15 total			tally 15 total		
2.30869E-08	0.1569		2.31866E-08	0.1564	
uncollided			uncollided		
1.63878E-08	0.2089		1.65034E-08	0.2077	
tally 26 cell: 6 energy			tally 26 cell: 6 energy		
1.0000E-02	3.97447E-13	0.1417	1.0000E-02	3.97447E-13	0.1417
1.0000E-01	9.27573E-12	0.1049	1.0000E-01	9.27544E-12	0.1049
1.0000E+00	7.97194E-11	0.1094	1.0000E+00	7.97709E-11	0.1094
5.0000E+00	7.09343E-10	0.1067	5.0000E+00	7.35742E-10	0.1093
1.0000E+01	4.28407E-10	0.1145	1.0000E+01	4.28435E-10	0.1145
2.0000E+01	1.59890E-11	0.0395	2.0000E+01	1.59911E-11	0.0395
total	1.24313E-09	0.0808	total	1.26961E-09	0.0822
cell: 7 energy			cell: 7 energy		
1.0000E-02	1.16693E-13	0.3619	1.0000E-02	1.16693E-13	0.3619
1.0000E-01	3.81103E-12	0.2436	1.0000E-01	3.81088E-12	0.2436
1.0000E+00	7.81185E-11	0.0975	1.0000E+00	8.00518E-11	0.0998
5.0000E+00	5.73599E-10	0.1109	5.0000E+00	5.73405E-10	0.1109
1.0000E+01	1.52313E-10	0.1483	1.0000E+01	1.52312E-10	0.1483

2.0000E+01	5.40356E-12	0.0596	2.0000E+01	5.40890E-12	0.0595
total	8.13361E-10	0.0873	total	8.15106E-10	0.0872

tally 24
cell 7

energy		
1.0000E-02	3.85743E-12	0.2196
1.0000E-01	3.05849E-11	0.2015
1.0000E+00	2.37711E-10	0.2580
1.0000E+01	2.71241E-10	0.3894
1.0000E+02	8.94309E-15	0.2392
1.0000E+03	0.00000E+00	0.0000
total	5.43403E-10	0.3132

tally 24
cell 7

energy		
1.0000E-02	3.84604E-12	0.2202
1.0000E-01	3.06428E-11	0.2012
1.0000E+00	2.36864E-10	0.2589
1.0000E+01	2.71270E-10	0.3894
1.0000E+02	8.94309E-15	0.2392
1.0000E+03	0.00000E+00	0.0000
total	5.42631E-10	0.3137

S E C T I O N 7

Summary Documentation
for the
100XS Neutron Cross Section Library
(Release 1.0)

Version I
October 1995
XTM:95-259
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R.C.Little

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Abstract

The 100XS library (Release 1.0) contains continuous-energy neutron cross sections for nine materials for MCNPTM.¹ The nine materials are ⁹Be, ¹²C, ¹⁶O, ²⁷Al, Si, ⁴⁰Ca, Fe, W, and ²³⁸U. The incident neutron energy range for these cross-section tables is extended beyond the traditional upper limit of 20 MeV to a limit of 100 MeV. ENDF6 format² evaluations have been processed with the NJOY nuclear data processing system.³ Some further modifications have been made to increase the robustness of the resulting MCNP data sets. The library has undergone fairly significant QA and limited data testing. ZAIDs assigned to the final data sets all end in "21c." The library is compatible with MCNP Version 4A with the exception of the ²³⁸U data set, which requires an enhancement to 4A. Heating numbers on the library are known to be incorrect, overestimating the energy deposition. The library is currently available only to XTM sponsors and LARAMIE (Los Alamos Radiation Modeling Interactive Environment) customers.

Nuclear Data Evaluations

The nuclear-data evaluations upon which the 100XS library is based were the result of pioneering work performed in the Nuclear Theory and Applications group at Los Alamos. Extensive model and model code developments, as well as improved phenomenological data representations, were required in order to prepare the 100-MeV evaluations. All model improvements were incorporated into the GNASH statistical/preequilibrium/fission theory computer code system.⁴ Full details of the theoretical work may be found in Ref. 5. The resulting evaluations were among the first to fully exercise the new features and capabilities of ENDF6.

The starting points for the present work were the Los Alamos CFS files listed in Table I.

Table 1: Initial ENDF6 Evaluations

Material	ENDF/B MAT	CFS Node	Date Last Written
Be-9	409	/T2/PGYC/EVAL/LA100N/BE100N2	01/27/89
C-12	612	/T2/PGYC/EVAL/LA100N/C100N6	12/21/89
O-16	816	/T2/PGYC/EVAL/LA100N/O100N6	12/21/89
Al-27	1327	/T2/PGYC/EVAL/LA100N/AL100N6	12/21/89
Si	1428	/T2/PGYC/EVAL/LA100N/SI100N6	12/21/89
Ca-40	2040	/ENDF/100U/NEUTRON/CA	07/01/93
Fe	260	/T2/PGYC/EVAL/LA100N/FE100N6	12/21/89
W	7400	/T2/PGYC/EVAL/LA100N/W100N4	12/21/89

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Table 1: Initial ENDF6 Evaluations

U-238	9238	/T2/PGYC/EVAL/LA100N/U100N2	01/11/89
-------	------	-----------------------------	----------

These files represent the most recent ENDF6 format evaluations from P.G. Young. (In the instance of ^{40}Ca , Young's file [/T2/PGYC/EVAL/LA100N/CA100N4, last written 02/17/88] was modified by R.E. MacFarlane for consistency with format requirements of ENDF MF=1.)

Processing Code

The version of NJOY employed in this work was vers.94.10. The UNICOS executable used is archived on CFS as NJOY94/XNJOY in the tar file /NJOY/94/TAR10R (last written 7/22/95).

Several modules of NJOY were executed: MODER, RECONR, BROADR, HEATR, and ACER. The processing strategy followed closely that devised by R.E. MacFarlane for preliminary processing of these evaluations. A representative INPUT file for NJOY (for ^{16}O) follows:

```

0
6
moder
20 -21
reconr
-21 -22
*pendf tape for 100 mev o-16 from o100n6*/
816 2/
.002/
*8-o-16 from lanl 100 mev library (o100n6)*/
*processed with the njoy system (njoy/94/xnjoynr)*/
0/
broadr
-22 -23
816 1/
.002/
300/
0/
heatr
-21 -23 -24 /
816 /
acer
-21 -24 0 31 32
1 1 1/
*8-o-16 from pgc 100 mev library (o100n6)*/
816 300./
.01/
/
stop

```

The original ENDF/B evaluation is TAPE20. TAPE31 is the processed Type-1 MCNP ACE file.

Some of the key parameters are: the fractional reconstruction tolerance in RECONR is 0.002 (0.2%); the fractional tolerance for thinning in BROADR is 0.002; there is no cross-section energy grid thinning in ACER; and the tolerance for thinning secondary distributions in ACER is 0.01.

ZAIDs for MCNP

The MCNP ZAID identifiers for the materials on 100XS have all been selected to end with "21C." All tables were processed at 300°K. The nine tables have been combined onto one Type 1 library called 100XS1. A nine-line directory file, 100DIR1, contains the information required by MCNP to access 100XS1. The nine lines must be added to the appropriate directory section of an existing XSDIR. The access route for 100XS1 is currently entered as a '0' in 100DIR1. The access route may be changed as appropriate once the library is installed on specific systems (of course, one may first use MAKXSF to convert 100XS1 to more compact Type-2 format). Table 2 provides information for each data table of the sort given in Appendix G of Ref. 1.

Table 2: Information About 100XS Data Tables

Material	ZAID	Atomic Weight Ratio	Temperature (°K)	Number of Energy Points	Total File Size	Photon Production Data	Fission Nubar
Be-9	4009.21c	8.934761	300	316	28,964	Yes	No
C-12	6012.21c	11.89691	300	919	28,809	Yes	No
O-16	8016.21c	15.85751	300	1427	45,016	Yes	No
Al-27	13027.21c	26.74975	300	1473	35,022	Yes	No
Si	14000.21c	27.844	300	2883	76,399	Yes	No
Ca-40	20040.21c	39.61929	300	2718	53,013	Yes	No
Fe	26000.21c	55.365	300	15,598	149,855	Yes	No
W	74000.21c	182.2706	300	21,386	194,513	Yes	No
U-238	92238.21c	236.006	300	30,911	279,245	Yes	Both

Caveats

The major known flaw with the processed library involves neutron heating numbers. The heating numbers are known to be too large in certain energy ranges. Why is this so? The difficulty traces back to sophisticated use of a new ENDF6 feature in the evaluations. MT=5 is used (particularly

at higher energies) as a sort of “catch-all” reaction; multiple reaction channels are all lumped into MT=5 in an attempt to economize on the length and complexity of the evaluation. However, it is clearly impossible to define a single Q-value to represent all of the channels over the entire energy range. Instead, the Q-value for MT=5 has been given in the evaluations as either 0 or the Q-value of the lowest threshold reaction channel to be included in MT=5; in any event the absolute magnitude of Q is lower than it should be. NJOY attempts to calculate neutron heating via the “energy-balance” method (see the HEATR chapter of Ref. 3 for a detailed discussion). Namely, the heating (for a particular reaction) is obtained (as a function of neutron energy) by adding the Q-value to the incident energy and subtracting the average total secondary neutron and photon energy. By using the given Q-value for MT=5 the calculated heating is too high. [Note: For ^{238}U , with fission being one of the channels included in MT=5, the problem may be the reverse. Namely, the Q-value may be too small and therefore the heating numbers too low.]

There are (at least) three ways to remedy the problem. All are well outside the scope of this work. The first is to theoretically determine the average Q-value as a function of neutron energy and then to enter this information as allowed by the HEATR module of NJOY. The second is to include data in MF=6, MT=5 so that multiplicities and average energies of all residual reaction products, specifically including the target residuals, could be obtained. Then, NJOY could calculate heating as the summation of the average energies of all charged particles and recoil nuclei (referred to as the “direct method” in Ref. 3). Finally, MT=5 could be abandoned in favor of the brute-force method of including data for each reaction channel individually in the evaluations.

How high are the heating numbers found on 100XS? Preliminary assessments indicate that they may range up to factors of 2-4 too high as the neutron energy approaches 100 MeV. *Users should clearly be very careful if using heating numbers from this library.*

At the other extreme, there are modest problems with negative heating numbers for a few materials: Si (~3-6 MeV), Fe (~3-4 MeV), and W (~0.7-2 MeV).

Overview of Data Content

For the sake of brevity, we begin this section by simply listing the reactions included for each data table. We will then include a very small sample of representative cross-section plots.

A. Neutron Reactions

A quick reference guide to interpreting the MT values is given in Table 3. All of the MTs listed below may be used with the FM feature of MCNP to obtain reaction rates.

Table 3: Quick Reference to MT Values

MT(s)	Description
2	elastic scattering
3	nonelastic [redundant]

Table 3: Quick Reference to MT Values

4	total inelastic [redundant]
5	(n,anything) [“catch-all” reaction]
6-9, 46-49	(n,2n) for ^9Be
16, 17	(n,2n) and (n,3n)
19, 20, 21, 38	first, second, third, and fourth-chance fission
51-90	inelastic scattering to discrete levels
91	continuum inelastic scattering
102	radiative capture (n, γ)
103	(n,p)
104	(n,d)
105	(n,t)
107	(n, α)

^9Be : MT = 2, 5, 6-9, 46-49, 51-56, 102-105, and 107.

^{12}C : MT = 2, 5, 51, and 102.

^{16}O : MT = 2, 5, 51-68, and 102.

^{27}Al : MT = 2, 5, 51-56, and 102.

Si: MT = 2, 5, 51-72, and 102.

^{40}Ca : MT = 2, 5, 51-82, and 102.

Fe: MT = 2, 5, and 102.

W: MT = 2, 5, 51-90, and 102.

^{238}U : MT = 2, 5, 16, 17, 19-21, 38, 51-77, 91, and 102.

B. Photon-Production Reactions

The MCNP scheme for identifying photon-production MTs is as follows: the first photon produced by neutron reaction MT=n has photon-production MT=1000*n+1, the second has photon-production MT=1000*n+2, etc. Thus, for example, the evaluators have included 89 different photons from ^{27}Al radiative capture, each with its own energy-dependent cross section. All of the MTs listed below may also be used with the FM feature of MCNP to obtain information about the production of specific photons.

^9Be : MT = 5001, 102001, and 105001.

^{12}C : MT = 5001, 51001, and 102001-102003.

^{16}O : MT = 4001-4013, 5001, 22001, 102001-102004, 103001-103003, and 107001-107006.

²⁷Al: MT = 4001-4009, 5001, and 102001-102089.

Si: MT = 4001-4007, 5001, and 102001-102042.

⁴⁰Ca: MT = 5001 and 102001.

Fe: MT = 5001 and 102001.

W: MT = 4001, 5001, and 102001.

²³⁸U: MT = 3001-3004, 5001, 18001, and 102001.

C. Cross-Section Plots

Four representative cross-section plots of data from 100XS are included at the end of this report. Fig. 1 and Fig. 2 are plots of the neutron reaction cross sections for ¹²C and Fe respectively from 1 eV to 100 MeV. Fig. 3 compares the neutron total cross section for six of the materials on 100XS over the energy range from 1 to 100 MeV. Fig. 4 shows the total photon-production cross section for the same materials over the same energy range. (Note: In both Figs. 3 and 4, the higher solid curve is W, the lower solid curve is ⁹Be.)

MCNP Compatibility

The 100XS library is designed to take advantage of new ENDF6 capabilities that were included in MCNP Version 4A. As such, with the exception of ²³⁸U, the library is completely compatible with MCNP4A.

In the case of ²³⁸U, neutron upscattering ($E_{\text{scattered}} > E_{\text{incident}}$) is allowed in MT=5. This upscattering can be correct physically, because the evaluators have included fission (a positive Q-value reaction) as one of the reactions lumped into MT=5. When such upscattering is sampled in MCNP4A, a fatal error will result and the calculation will be terminated. J.S. Hendricks has written a patch to MCNP4A (included first in intermediate version MCNP4XL⁶) that allows neutron upscatter in certain cases such as when fission is included in MT=5. Those users having access to intermediate version 4XL or later thus have a version of MCNP that is completely compatible with 100XS. The Hendricks patch is proposed for MCNP4B. Final code for MCNP4B may be different. Availability of the patch is described below under 'Library Availability.'

Library Availability

The 100XS library is stored in Type 1 format on CFS as /X6DATA/CE/SPECIAL/HIENERGY/100XS1 (both open and secure). The corresponding directory file 100DIR1 and the patch to MCNP4A needed to use ²³⁸U are stored under the same directory as 100DIR1 and UPPATCH respectively. All files were last written on 10/18/95 and are currently available only to certain XTM customers, sponsors, and collaborators. We would anticipate general release to RSIC at some time in the future.

Related Libraries

Users of the 100XS library may be interested in other MCNP data sets that extend to neutron energies greater than 20 MeV. The ENDF60 library⁷ includes several such sets. Information is

given in Table 4.

Table 4: Other MCNP Data Tables with $E_n > 20$ MeV

Material	ZAID	Upper Energy (MeV)
H-1	1001.60C	100
C	6000.60C	32
I-127	53127.60C	30
Ho-165	67165.60C	30
Au-197	79197.60C	30
Am-241	95241.60C	30

The 100-MeV evaluation upon which ZAID=1001.60C on ENDF60 is based was actually performed by the Nuclear Theory and Applications group at Los Alamos at the same time as the other 100-MeV evaluations upon which the 100XS library is based.

We should also point out that the evaluations upon which the 100XS library is based are most often built upon ENDF/B-V Revision 2 evaluations up to an energy that is material-dependent. Thus, there is significant overlap with “.50c” data tables at low energies. See Ref. 5 for specific energy ranges that are identical to ENDF/B-V.

Quality Assurance

Several tools were used to QA the 100XS library. All of the modifications discussed below resulted from issues discovered in the QA process. For brevity, we will only list here the types of tools used in our QA.

- NJOY performs several consistency checks in ACER. The code checks several attributes of reaction thresholds, energy grids, and secondary energy and angular distributions. It flags anything known to be wrong or considered to be unreasonable.
- The XDATAP code⁸ was used to coplot all cross sections from the processed files against cross sections from the original evaluations.
- Average cross sections in many energy ranges were calculated for each reaction.⁹ Comparisons were made between the average cross sections from the processed files and those from the original evaluations. Comparisons were also made between the newly-processed results and a previous version of the processed library.
- MF=1 information for each evaluation was studied. Spot checks of the remainder of the evaluations were made as necessary to compare with the detailed outputs from NJOY.
- A host of small, special-purpose checking codes were written, primarily to identify issues on the processed files relating to new ENDF6 features.
- Generic sample MCNP problems were run for each material. The goals were to exercise as much of the data sets as possible and to check for proper flow and reasonableness of results. Also,

a million history, 20-million collision problem was run for 6 hours on a Sun IPX workstation to ensure compatibility with 32-bit machines.

Modifications

As a result of the QA process described above, several modifications were made to either the original evaluations or the processed ACE files. These modifications were largely necessitated by stress imposed on the evaluations, NJOY, and MCNP by the significant use of advanced ENDF6 features. The modifications can be summarized in the following seven categories.

1. Both ^{12}C and ^{16}O evaluations include anisotropic angular distributions for some photons produced in neutron reactions. In particular, the ^{12}C evaluation specifies that photons from the first discrete inelastic level (MT=51, 4.439 MeV) are anisotropic, and the ^{16}O evaluation specifies that photons from two inelastic levels (at 6.131 MeV and 6.917 MeV) are anisotropic. The distributions specified can be quite anisotropic, especially for ^{12}C at neutron energies near the reaction threshold.

NJOY ver.94.10 does not carry forward information about anisotropic neutron-induced photons to ACE files. Therefore, it was necessary to add this information to the NJOY-processed ACE files for ^{12}C and ^{16}O . The angular distributions given in the new evaluations were unchanged (except for the incident neutron energy grid being extended to 100 MeV) from those given in ENDF/B-V. Therefore, we obtained the appropriate data from the corresponding ENDF/B-V ACE files (ZAID=6012.50C and ZAID=8016.50C) and added it to 100XS.

2. Four of the evaluations (^{12}C , ^{16}O , Si, and ^{40}Ca) included negative probabilities of scattering for certain combinations of incident and exiting neutron energies. In all cases, the reaction in question was MT=5 and the scattering data were provided in terms of Law=1, LANG=2 (Kalbach-87) in MF=6.

The absolute values of all offending probabilities were relatively small. Therefore, in consultation with P.G. Young, it was determined that we would simply set the probabilities to zero. This was accomplished by making the appropriate corrections to the ACE files themselves, rather than to the evaluations. The modifications, of course, necessitated that the entire probability density function and cumulative density function be renormalized at incident energies having this problem.

3. All nine evaluations included energy distributions that allow neutrons to be upscattered. This was found most frequently for MT=5, with scattering data provided in terms of Law=1, LANG=2 in MF=6. The circumstance was also found for W for MT=91 where the distributions were specified in the more traditional form of an arbitrary tabulated distribution (LF=1) in MF=5. The frequency with which neutrons would upscatter varied considerably among the materials; for W (MT=5) it was greater than one in a thousand at some incident neutron energies.

For all materials except ^{238}U , it was decided to truncate the energy distributions such that non-

physical upscatter would not occur. Again, this was accomplished by making the appropriate corrections to the ACE files themselves, rather than to the evaluations. The modifications, of course, necessitated that the entire probability density function and cumulative density function be renormalized at incident energies having this problem. When energy distributions were given in the center of mass (COM) the maximum allowable COM energy was determined as that energy for which the laboratory scattering energy equaled the incident energy using $\mu_{\text{COM}}=1$.

^{238}U was not modified. See the discussion under 'MCNP Compatibility' for an explanation.

4. For ^9Be , the evaluators employed a radical shift in specified reactions at 20 MeV. MTs 6-9, 46-49, 102-105, and 107 were given below 20 MeV; MTs 5 and 51-56 were given above 20 MeV. A transition was accomplished in the evaluation by having the cross section for every reaction (except elastic scattering) be either zero at 20.0 MeV and non-zero at 20.00001 MeV or non-zero at 20.0 MeV and zero at 20.00001 MeV. Such a transition should be fine. Unfortunately, however, the processed energy grid did not include both 20.0 and 20.00001 MeV (the 20 MeV point was discarded). This leads to a rather severe discontinuity in the total cross section at 20 MeV.

The solution implemented was to change all occurrences of 20.0 MeV in the original evaluation (in MF=3,6,12) to 19.99999 MeV. The modified evaluation was then completely reprocessed with NJOY. Both 19.99999 and 20.00001 MeV were retained in the processed energy grid, thus successfully implementing the evaluators' transition. The modified evaluation is stored on the open CFS as /090895/HIENERGY/BE100N2_NEW (last written 09/25/95).

5. In the Si evaluation, at neutron energies greater than 3 MeV, the dominant neutron reaction for producing photons is MT=5. The original evaluation led to a processed photon-production cross section from MT=5 several thousand times greater than intended over a limited neutron energy range near the reaction threshold. The origin of the anomaly was traced to an artifact of interpolation. The photon-production cross section from MT=5 is equal to the product of the MT=5 neutron interaction cross section times the MT=5 photon multiplicity. Both quantities are dependent on the neutron energy. It turns out that the evaluated quantities vary extremely rapidly (particularly from 3.75 to 4.0 MeV); so rapidly that the product of the quantities (at interpolated energies) was very ill-behaved.

The solution, arrived at in consultation with P.G. Young, was to add many more points to the photon multiplicity energy grid between threshold and 6 MeV. Multiplicities at the energy points added were chosen so as to preserve the interpolated product of cross section and multiplicity. To further improve the results, the interpolation scheme specified for a portion of the energy range in the multiplicity table was changed from linear-linear to log-log. After these changes were made, the modified evaluation was then completely reprocessed with NJOY. The resulting behavior of the photon-production cross section from MT=5 was judged to be acceptable. The modified evaluation is stored on the open CFS as /090895/HIENERGY/SI100N6_NEW (last written 09/28/95).

6. The ENDF6 format allows secondary energy distributions to be a combination of discrete ener-

gies and a continuous distribution. This capability has been heavily used by the evaluators in this work to describe secondary photons. A complication arises downstream in MCNP in certain circumstances. MCNP would like to use this data in as sophisticated a manner as possible. One aspect of accomplishing this is to sample a secondary photon energy by interpolating between adjacent (in neutron energy) distributions. This becomes a nightmare to code unless one assumes that the number of discrete photon energies is constant (and furthermore, that there is a direct one-to-one correspondence) at adjacent incident neutron energies. Such was not the case for evaluations of ^{16}O , ^{27}Al , ^{40}Ca , Fe, and W.

Rather than complicate MCNP unnecessarily, the approach taken was to modify the processed files (not the evaluations) to ensure that the number of discrete photons was constant for a particular reaction as a function of neutron energy (and that the discrete energies were also identical). All probability densities for energies added to the secondary grids as a result of this modification were zero. The net result is data files that are modestly longer and somewhat less efficient; the plus side is simpler code in MCNP and more robust data libraries.

7. An examination of differential cross-section plots for W led to the obvious conclusion that something had gone wrong in the neutron energy range from 0.02 to 0.1 MeV. After reviewing the details of the evaluation in this energy range, the author incorrectly concluded that the processing code was having difficulty in the unresolved energy region. As later pointed out by R.E. MacFarlane, however, the actual root cause of the problem was that the evaluators had failed to include double-valued energy points in MF=3 at boundaries of various resonance regions. This ultimately caused the processing code to interpolate over a portion of the unresolved resonance region differently than the evaluators intended. MacFarlane's solution, therefore, was to modify the evaluation to include the necessary double-valued energy points in MF=3. Rigorously, this is the correct solution. We had already embarked on a different path, however, and chose not to change directions at the last minute.

We instead relied on two facts. The first was that the new W evaluation is documented to be the same as ENDF/B-V Rev. 2 up to 6 MeV. The second was that we already had well-behaved processed cross sections from ENDF/B-V Rev. 2 W in the energy range of question. We therefore decided to extract the necessary data from ZAID=74000.55C and use this to overwrite the newly-processed data in the energy range from 0.02 to 0.1 MeV. The results are more than adequate to correctly represent the intent of the evaluators.

Data Testing

Only limited data testing was performed to compare calculated results with measurements.

Calculations of high-energy neutron transmission through various thicknesses of concrete were reported in Ref. 10 using a very early version of these cross sections. All calculations associated with that work were repeated with 100XS. Major differences (for example, in unpublished results for induced photons) were at least qualitatively explained by improvements made over the years.

Also, G.P. Estes ran an MCNP calculation of a water-phantom experiment performed by a collaborator of his at the University of Washington. The experiment was designed to be relevant to fast

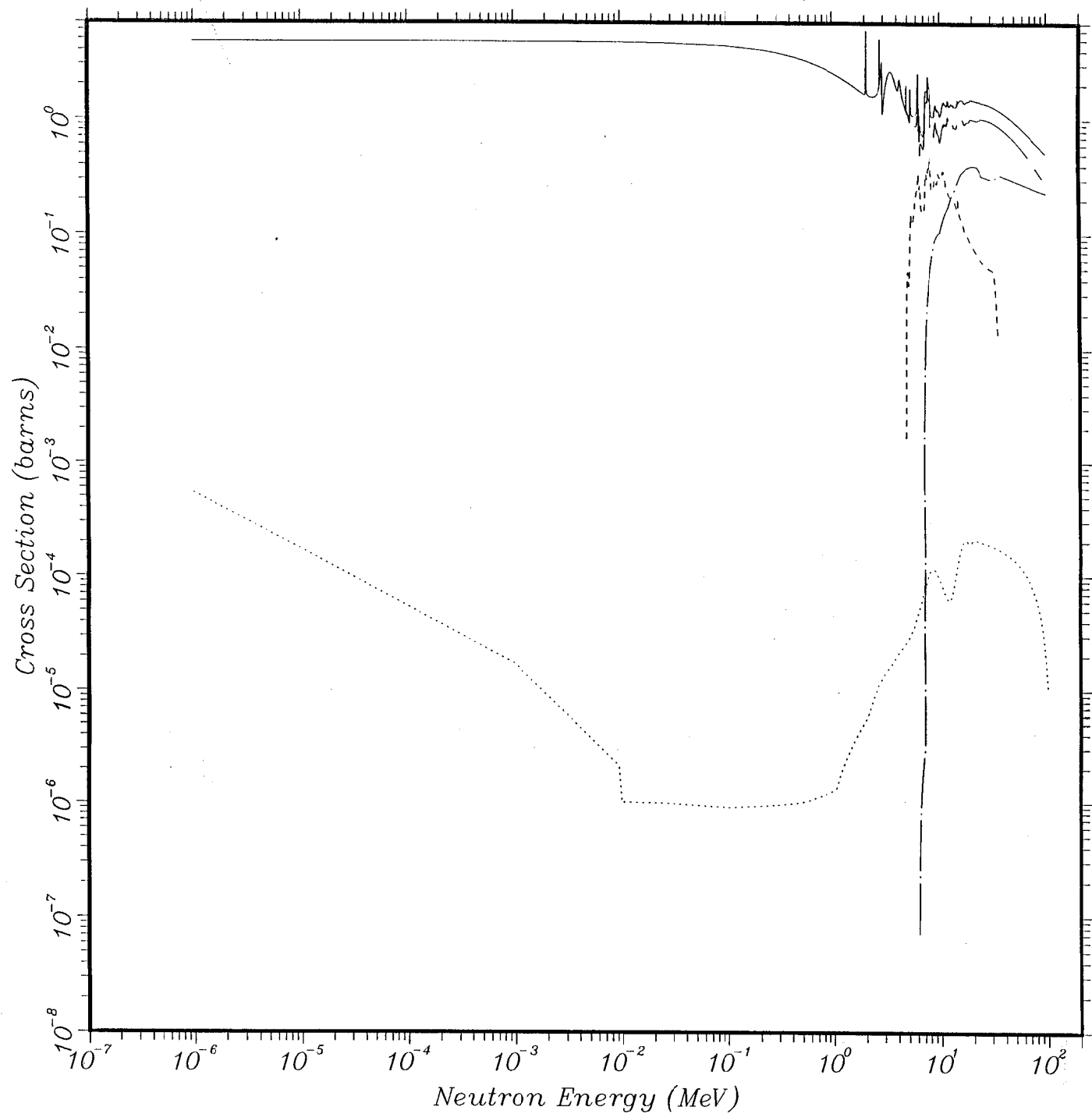
neutron therapy. Once again, differences between new calculations and Estes' previous results were qualitatively understood.

Acknowledgments

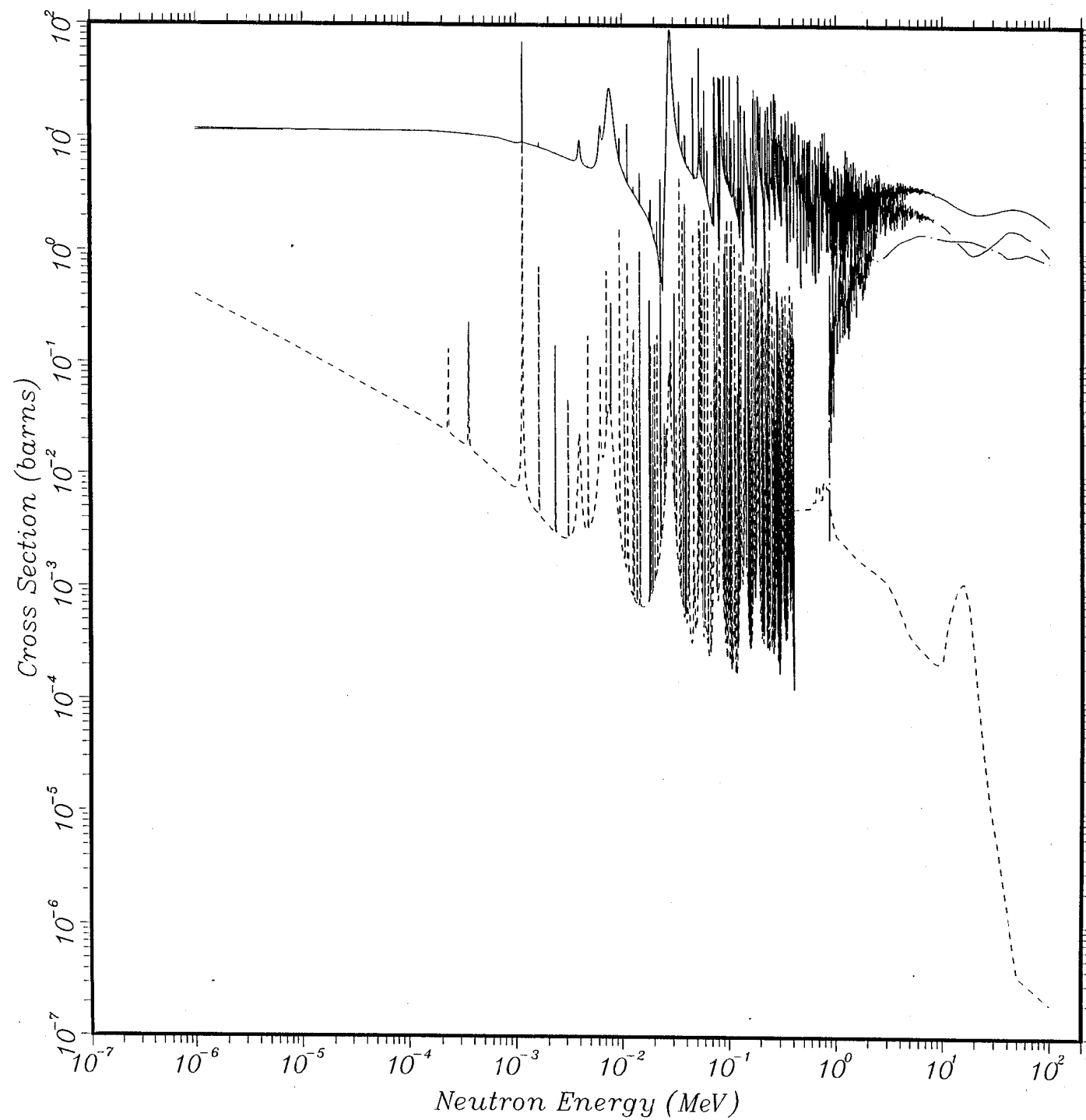
It is a pleasure to acknowledge the invaluable assistance of several colleagues. Phil Young provided much information about the evaluations as well as the strategy to fix the Si photon-production problem. Bob MacFarlane was the source of NJOY facts and guidance, and provided valuable discussions about processed results and subsequent use in MCNP. Stephanie Frankle assisted in many ways and actually performed some of the initial work on this project. John Hendricks provided patches to MCNP and consulting on code strategy. Guy Estes helped by using the new library in an MCNP calculation and providing the results for comparison with those obtained using previous cross-section sets.

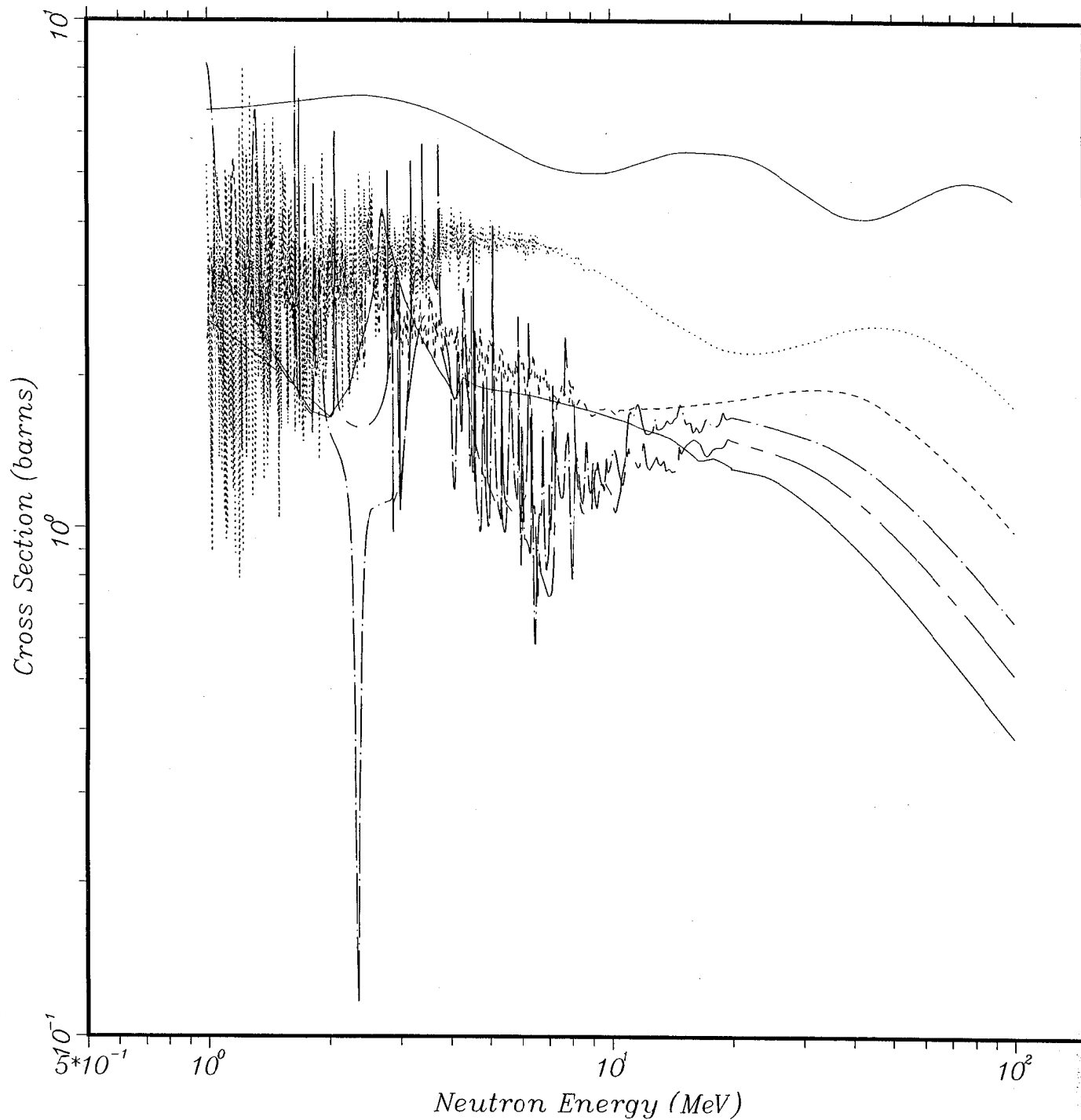
References

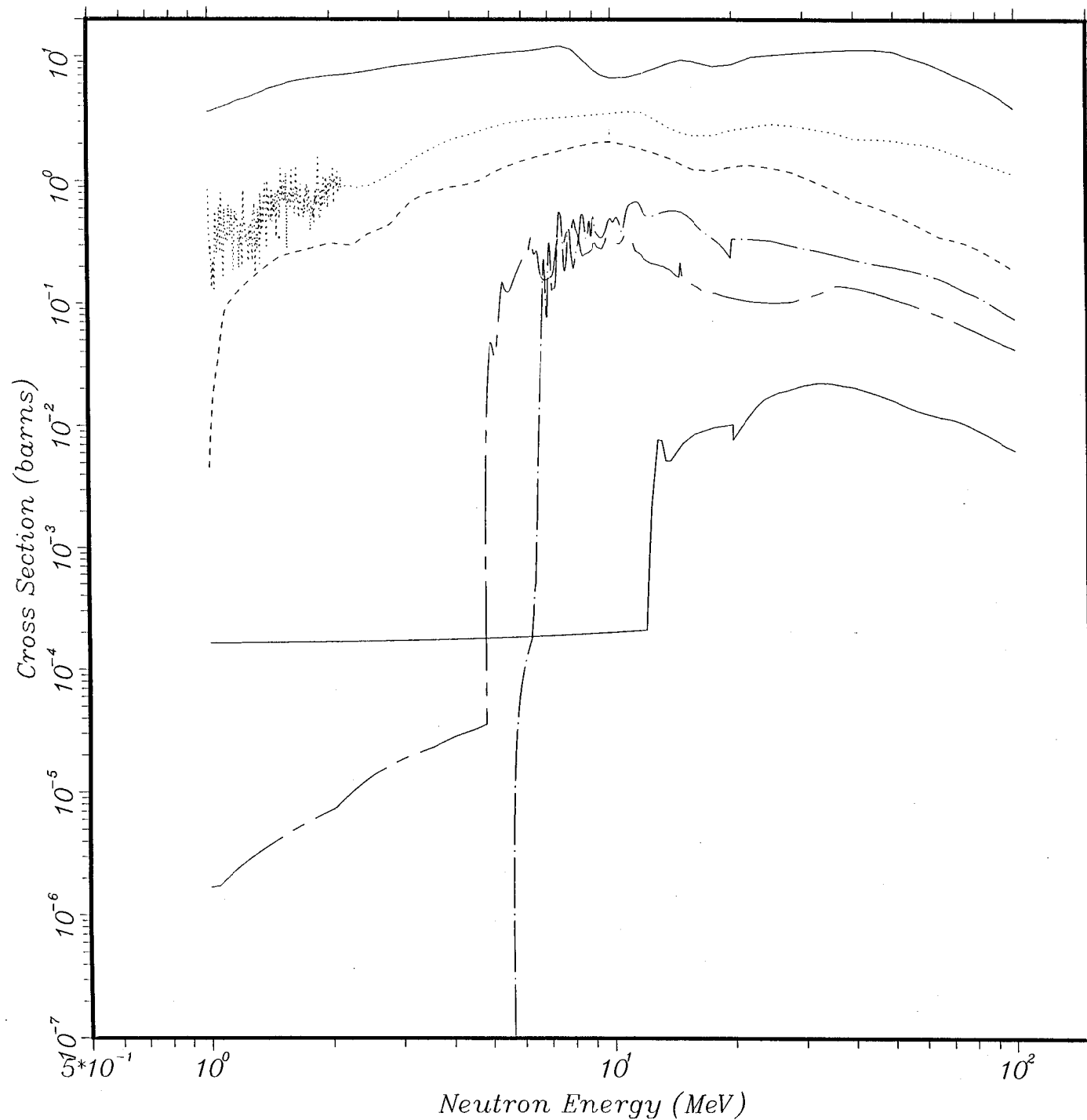
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11/25/96

From 100XS1

MT = 202

GAMMA PRODUCTION

ZAID = 4009.21C

Be - 9

ZAID = 6012.21C

C - 12

ZAID = 8016.21C

O - 16

ZAID = 13027.21C

Al - 27

ZAID = 26000.21C

Fe - NAT

ZAID = 74000.21C

W - NAT

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Los Alamos
NATIONAL LABORATORY
memorandum

*Applied Theoretical & Computational
Physics Division
Los Alamos, New Mexico 87545*

To/MS: MCNP Distribution
From/MS: R. C. Little and R. E. Seamon / MS B226
Phone/FAX:
Symbol:
Date: March 13, 1984

SUBJECT: Dosimetry/Activation Cross Sections for MCNP

The tally multiplier (FM) feature in MCNP allows users to calculate quantities of the form

$$C \int \phi(E)R(E)dE$$

where C is a constant, $\phi(E)$ is the fluence (n/cm^2), and $R(E)$ is a response function. The response function is often a microscopic cross section. With appropriate units for C (e.g., atom/b•cm), the quantity calculated becomes the total number of some type of reactions per unit volume. If the tally is made over a finite time interval, the quantity becomes a reaction rate per unit volume. The energy-dependent cross section for any reaction found on our standard ACE-format Monte Carlo libraries is available for use as a response function.

In addition to the large choice of reaction cross sections described above, cross sections for dosimetry or activation reactions may also be used as response functions.¹⁻³ These specially-formatted dosimetry cross sections may be used only for tallying, and not for neutron transport. The purpose of this memo is to document the availability of cross sections for nearly two thousand dosimetry reactions involving over four hundred target nuclei.

The available cross sections are from three sources: ENDF/B-V Dosimetry Tape 531, ENDF/B-V Activation Tape 532, and ACTL - an evaluated neutron activation cross-section library from the Lawrence Livermore National Laboratory (LLNL). One dosimetry cross-section library in MCNP-compatible format has been prepared from each of the three sources as described in the next paragraphs.

ENDF/B-V Tape 531

Cross sections for 36 dosimetry reactions involving 26 target nuclei are given on ENDF/B-V Tape 531. The 26 targets are listed in Table I along with their ENDF/B MAT identifiers. Cross sections for the eight targets with asterisks in Table I were not processed into dosimetry format for MCNP because the expensive resonance reconstruction would only provide cross sections that were duplicates of those already available on our normal cross-section libraries such as RMCCS.

The remaining targets were pre-processed using D. E. Cullen's LINEAR and RECENT codes⁴⁻⁶ with a linearization tolerance of 0.1% and a resonance reconstruction tolerance of 1.0%. The cross sections were then processed into dosimetry format appropriate for MCNP using P. D. Soran's DOSGEN code. A complete description of the available reactions, found in Table III, will be discussed later.

ENDF/B-V Tape 532

Cross sections for 84 activation reactions involving 53 target nuclei are given on ENDF/B-V Tape 532. The 53 targets are listed in Table II along with their ENDF/B MAT identifiers. For the reasons given above, cross sections for the 10 targets with asterisks in Table II were not processed into MCNP dosimetry format. The remaining targets were processed through the LINEAR-RECENT-DOSGEN series of codes. The available reactions are listed in Table IV. It is understood that there is some repetition in the reactions on Tape 531 and Tape 532.

ACTL

ACTL is an evaluated neutron activation cross-section library from LLNL. The evaluation techniques are described in Ref. 7 and the format of the library in Ref. 8. A wealth of cross-section data is available on the library. Cross sections are provided for over 1800 reactions involving 373 target nuclei, many of which are isomeric targets. A new code, READACT, has been written to process the ACTL library into MCNP dosimetry format. An index to the reactions available from ACTL is given in Table V.

Description of Tables III-V

ZAID: The ZAID is a unique MCNP identifier for a specific cross-section evaluation of a certain target isotope. For dosimetry cross sections, the ZAID is of the form ZZAAA.nnY, where ZZ is the atomic number and AAA is the mass number. The value of nn is 24 for those isotopes from Tape 531, 26 for those from Tape 532, and either 30 or 31 for those from ACTL. Two values of nn are necessary when cross sections are given for both the ground state and an excited state of the same target nuclide.

MT: The ENDF/B reaction type identifier.⁹ This value must be entered on an FM card to get a particular reaction cross section used as a response function. Livermore C values have been converted to ENDF/B MT values according to the prescription given in Table VI.

Target: The target isotope.

E-Targ: The level energy of the target isotope (in MeV).

Reaction: The type of reaction corresponding to the given MT. Abbreviations used are: N-neutron, G-gamma, P-proton, D-deuteron, T-triton, and A-alpha. In all cases, (N,G)

represents the radiative capture cross section and not the total gamma-production cross section.

Product: The product isotope from the nuclear reaction. In many cases, the product is radioactive.

E-Prod: The level energy of the product isotope (in MeV). "Ground" indicates that the product isotope is in its ground state: "All" indicates that the given cross section is the sum of the cross sections to all final states. Note that if reaction cross sections are given to both the ground state and an excited state of a product isotope, the MT of the second reaction is arbitrarily assigned a value 1000 greater than the MT of the first reaction.

EMIN: The minimum energy (in MeV) at which cross sections are given.

EMAX: The maximum energy (in MeV) at which cross sections are given.

NES: The number of energies at which cross sections are given. This provides an idea of the detail that has been put into the evaluation.

Using Dosimetry Cross Sections

Check Tables III-V for the availability of cross sections for the dosimetry reaction in which you are interested. You need the target ZAID and the reaction MT to fully identify a cross-section set. The FM feature is described completely in the MCNP manual;¹⁰ only a simple example will be given here. The following three cards in your input deck would result in a calculation of the total number of $^{58}\text{Ni}(n,p)^{58}\text{Co}$ reactions per unit volume in cell N.

F4	N		
FM4	C	12	103
M12	28058.26Y	1.0	

C would be the atom/b•cm density of ^{58}Ni .

Availability of Dosimetry Libraries

We next describe the availability of dosimetry cross-section libraries for MCNP at various installations. The discussion is appropriate for Version 3 of MCNP.

The MCNP user at Los Alamos does not need to manually retrieve any dosimetry libraries from the Common File System. All ZAID'S found in Tables III-V will also be found on the public directory file, XSDIR. MCNP will fetch any and all dosimetry libraries needed for a particular problem.

On the MFE network, all ZAI'D'S will be included on the XSDIR file. Users will have to retrieve any dosimetry libraries they need from FILEM. The libraries are named 531DOS2, 532DOS2, and LLLDOS2, and are stored under the .VER3XS subdirectory of user 5014. Procedures for accessing cross-section libraries for MCNP Version 3 at MFE are discussed in Ref. 11.

For other MCNP users, a tape with the three dosimetry libraries and a companion directory file will be sent to the Radiation Shielding Information Center (RSIC) at the Oak Ridge National Laboratory in the near future.

Summary

This memo has documented the existence of cross sections for nearly 2000 dosimetry or activation reactions involving over 400 target nuclei. These cross sections may be used with the FM feature as response functions in MCNP. Plots and tabulations of specific cross sections are available on request.

Acknowledgments

Much credit must be given to R. J. Howerton at LLNL for his evaluation work and for transmitting the ACTL library to us. P. D. Soran did the original coding allowing dosimetry cross sections to be used in MCNP.

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Table I: Nuclei Available on ENDF/B-V Dosimetry Tape 531

<u>Nuclide</u>	<u>MAT</u>	<u>Nuclide</u>	<u>MAT</u>
Na-23 *	6311	Ti-46	6427
Al-27	6313	Ti-47	6428
Mn-55	6325	Ti-48	6429
Co-59*	6327	Fe-54	6430
Np-237 *	6337	Fe-56	6431
Au-197 *	6379	Fe-58	6432
Th-232 *	6390	Ni-58	6433
U-235 *	6395	Ni-60	6434
U-238 *	6398	Cu-63	6435
Pu-239 *	6399	Cu-65	6436
Li-6	6424	In-115	6437
B-10	6425	I-127	6438
Sc-45	6426	S-32	6439

Table II: Nuclei Available on ENDF/B-V Dosimetry Tape 532

<u>Nuclide</u>	<u>MAT</u>	<u>Nuclide</u>	<u>MAT</u>
Li-6	7036	Ni-58	7288
Li-7	7037	Cu-63	7293
B-10	7050	Cu-65	7295
N-14	7074	Zr-90	7400
O-16	7086	Zr-92	7402
F-19	7099	Zr-94	7404
Na-23 *	7113	Nb-93 *	7413
Mg-24	7124	Mo-100	7420
Al-27	7137	Mo-92	7422
P-31	7151	Mo-98	7428
S-32	7162	Ag-109 *	7479
Ar-40	7180	In-115	7495
K-41	7191	Sn-120	7500
SC-45	7215	Sn-122	7502
Ti-50	7220	Sn-124	7504
Ti-46	7226	I-127	7537
Ti-47	7227	La-139	7579
Ti-48	7228	Dy-164	7664
Cr-50	7240	Ta-181 *	7731
Cr-52	7242	Au-197 *	7797
Mn-55 *	7255	Th-232 *	7902
Fe-54	7264	Pa-231	7911
Fe-56	7266	Pa-233	7913
Co-59 *	7279	U-238 *	7928
Ni-60	7280	Am-241 *	7951
Ni-62	7282		

*Available on standard Monte Carlo library from ENDF/B-V.

Table III: Index To Reaction Cross Sections Available on TAPE 531 (531DOS)

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
13027.24y	103	AL-27	GROUND	(N, P)	MG-27	ALL	1.90E+00	2.00E+01	306
	107	AL-27	GROUND	(N, A)	NA-24	ALL	3.25E+00	2.00E+01	272
25055.24y	16	MN-55	GROUND	(N, 2N)	MN-54	ALL	1.04E+01	2.00E+01	57
3006.24y	207	LI-6	GROUND	ALPHA PRODUCTION			1.00E-11	2.00E+01	365
5010.24y	207	B-10	GROUND	ALPHA PRODUCTION			1.00E-11	2.00E+01	382
21045.24y	102	SC-45	GROUND	(N, G)	SC-46	GROUND	1.00E-11	2.00E+01	10087
22046.24y	103	TI-46	GROUND	(N, P)	SC-46	GROUND	1.62E+00	2.00E+01	24
22047.24y	28	TI-47	GROUND	(N, N') P	SC-46	GROUND	1.07E+01	2.00E+01	66
	103	TI-47	GROUND	(N, P)	SC-47	GROUND	1.00E-11	2.00E+01	34
22048.24y	28	TI-48	GROUND	(N, N') P	SC-47	GROUND	1.17E+01	2.00E+01	42
	103	TI-48	GROUND	(N, P)	SC-48	GROUND	3.27E+00	2.00E+01	26
26054.24y	103	FE-54	GROUND	(N, P)	MN-54	GROUND	1.00E-11	2.00E+01	256
26056.24y	103	FE-56	GROUND	(N, P)	MN-5G	ALL	2.97E+00	2.00E+01	222
26058.24y	102	FE-5e	GROUND	(N, G)	FE-59	GROUND	1.00E-11	2.00E+01	3536
28058.24y	16	NI-58	GROUND	(N, 2N)	NI-57	ALL	1.24E+01	2.00E+01	14
	103	NI-56	GROUND	(N, P)	CO-58	ALL	1.00E-11	2.00E+01	187
28060.24y	103	NI-60	GROUND	(N, P)	CO-60	ALL	2.08E+00	2.00E+01	215
29063.24y	102	CU-63	GROUND	(N, G)	CU-64	ALL	1.00E-11	2.00E+01	1467
	107	CU-S3	GROUND	(N, A)	CO-6C	ALL	1.00E-11	2.00E+01	216
29065.24y	16	CU-65	GROUND	(N, 2N)	CU-64	ALL	1.01E+01	2.00E+01	22
49115.24y	51	IN-115	GROUND	(N, N')	IN-115	3.36E-01	3.39E-01	2.00E+01	211
	102	IN-115	GROUND	(N, G)	IN-116	1.27E-01	1.00E-11	2.00E+01	12785
53127.24y	16	I-127	GROUND	(N, 2N)	I-126	GROUND	9.23E+00	2.00E+01	55
16032.24y	103	S-32	GROUND	(N, P)	P-32	ALL	9.57E-01	2.00E+01	70

Table IV: Index To Reaction Cross Sections Available on the Tape 532 (532DOS)

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
3006.26y	105	LI-6	GROUND	(N, T)	HE-4	ALL	1.00E-11	2.00E+01	354
3007.26y	91	LX-7	GROUND	(N, N') T	HE-4	ALL	2.82E+00	2.00E+01	93
	102	LI-7	GROUND	(N, G)	LI-8	ALL	1.00E-11	2.00E+01	269
5010.26y	113	E-I0	GROUND	(N, T) 2A	HE-4	ALL	1.00E-11	2.00E+01	292
7014.26y	103	N-I4	GROUND	(N, P)	C-14	ALL	1.00E-11	2.00E+01	504
8016.26y	103	O-I6	GROUND	(N, P)	N-16	ALL	1.02E+01	2.00E+01	45
9019.26y	16	F-I9	GROUND	(N, 2N)	F-18	ALL	1.10E+01	2.00E+01	13
12024.26y	103	MG-24	GROUND	(N, P)	NA-24	ALL	4.93E+00	2.00E+01	80
13027.26y	102	AL-27	GROUND	(N, G)	AL-28	ALL	1.00E-11	2.00E+01	292
	103	AL-27	GROUND	(N, P)	MG-27	ALL	1.90E+00	2.00E+01	306
	107	AL-27	GROUND	(N, A)	TI-24	ALL	3.25E+00	2.00E+01	272
15031.26y	103	P-3I	GROUND	(N, P)	51-31	GROUND	7.33E-01	2.00E+01	30
16032.26y	103	S-.32	GROUND	(N, P)	P-32	GROUND	9.49E-01	2.00E+01	15
18040.26y	102	AR-40	GROUND	(N, G)	AR-41	ALL	1.00E-11	2.00E+01	1928
19041.26y	103	K-4I	GROUND	(N, P)	AR-41	ALL	1.75E+00	2.00E+01	14
21045.26y	16	SC-45	GROUND	(N, 2N)	SC-44	GROUND	1.16E+01	2.00E+01	14
	102	SC-45	GROUND	(N, G)	SC-46	GROUND	1.00E-11	2.00E+01	10087
22050.26y	107	TI-50	GROUND	(N, A)	CA-47	GROUND	3.51E+00	2.00E+01	14
22046.26y	103	TI-46	GROUND	(N, P)	SC-46	GROUND	1.62E+00	2.00E+01	24
22047.26y	28	TI-47	GROUND	(N, N') P	SC-46	GROUND	1.07E+01	2.00E+01	66
	103	TI-47	GROUND	(N, P)	SC-47	GROUND	1.00E-11	2.00E+01	34
22048.26y	28	TI-48	GROUND	(N, N') P	SC-47	GROUND	1.17E+01	2.00E+01	42
	103	TI-48	GROUND	(N, P)	SC-48	GROUND	3.27E+00	2.00E+01	26
	107	TI-48	GROUND	(N, A)	CA-45	GROUND	2.07E+00	2.00E+01	14
24050.26y	16	CR-50	GROUND	(N, 2N)	CR-49	ALL	1.32E+01	2.00E+01	18
	102	CR-50	GROUND	(N, G)	CR-51	ALL	1.00E-11	2.00E+01	3680
24052.26y	16	CR-52	GROUND	(N, 2N)	CR-51	ALL	1.23E+01	2.00E+01	11

Table IV cont.

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
26054.26y	102	FE-54	GROUND	(N,G)	FE-55	ALL	1.00E-11	2.00E+01	10393
	103	FE-54	GROUND	(N,P)	MN-54	GROUND	1.00E-11	2.00E+01	256
	107	FE-54	GROUND	(N,A)	CR-51	ALL	1.00E-11	2.00E+01	126
26056.26y	16	FE-56	GROUND	(N,2N)	FE-55	ALL	1.14E+01	2.00E+01	64
	103	FE-56	GROUND	(N,P)	MN-56	ALL	2.97E+00	2.00E+01	222
26058.26y	102	FE-58	GROUND	(N,G)	FE-59	GROUND	1.00E-11	2.00E+01	3546
28060.26y	16	NI-60	GROUND	(N,2N)	NI-59	GROUND	1.16E+01	2.00E+01	20
	103	NI-60	GROUND	(N,P)	CO-60	GROUND	2.08E+00	2.00E+01	215
28062.26y	102	NI-62	GROUND	(N,G)	NI-63	GROUND	1.00E-11	2.00E+01	1890
	107	NI-62	GROUND	(N,A)	FE-59	GROUND	4.42E-01	2.00E+01	29
28058.26y	16	NI-58	GROUND	(N,2N)	NI-57	GROUND	1.24E+01	2.00E+01	14
	102	NI-58	GROUND	(N,G)	NI-55	GROUND	1.00E-11	2.00E+01	1767
	103	NI-58	GROUND	(N,P)	CO-59	GROUND	1.00E-11	2.00E+01	187
	104	NI-58	GROUND	(N,O)	CO-57	GROUND	6.06E+00	2.00E+01	23
	107	NI-58	GROUND	(N,A)	FE-55	GROUND	1.00E-11	2.00E+01	38
29062.26y	102	CU-63	GROUND	(N,G)	CU-64	ALL	1.00E-11	2.00E+01	1467
	103	cu-t53	GROUND	(N,P)	NI-62	ALL	1.00E-11	2.00E+01	118
	107	CU-63	GROUND	(N,A)	CO-60	ALL	1.00E-11	2.00E+01	216
29065.26y	16	CU-65	GROUND	(N,2N)	CU-64	ALL	1.01E+01	2.00E+01	22
40090.26y	16	ZR-90	GROUND	(N,2N)	zR-e9	.CILL	1.21E+01	2.00E+01	16
40092.26y	102	ZR-92	GROUND	(N,G)	ZR-93	ALL	1.00E-11	2.00E+01	1908
40094.26y	102	ZR-94	GROUND	(N,G)	ZR-95	ALL	1.00E-11	2.00E+01	2625
42100.26y	102	MD-I00	GROUND	(N,G)	MO-101	ALL	1.00E-11	2.00E+01	2483
42092.26y	102	MO-92	GROUND	(N,G)	MO-93	GROUND	1.00E-11	2.00E+01	3905
42098.26y	102	MO-99	GROUND	(N,G)	MO-99	ALL	1.00E-11	2.00E+01	3242
49115.26y	51	IN-II5	GROUND	(N,N')	IN-115	3.36E-01	3.39E-01	2.00E+01	211
	1132	IN-II5	GROUND	(N,G)	IN-116	1.27E-01	1.00E-11	2.00E+01	12789
50120.26y	102	SN-I20	GROUND	(N,G)	SN-121	ALL	1.00E-11	2.00E+01	6438
50122.26y	102	SN-I22	GROUND	(N,G)	SN-123	ALL	1.00E-11	2.00E+01	943
50124.26y	102	SN-I24	GROUND	(N,G)	SN-125	ALL	1.00E-11	2.00E+01	844
53127.26y	16	I-I27	GROUND	(N,2N)	I-126	GROUND	9.23E+00	2.00E+01	55
	102	I-I27	GROUND	(N,G)	I-128	ALL	1.00E-11	2.00E+01	7013
57139.26y	102	LA-I39	GROUND	(N,G)	LA-140	ALL	1.00E-11	2.00E+01	7735
66164.26y	102	Dy-I64	GROUND	(N,G)	Dy-165	ALL	1.00E-11	2.00E+01	288
91231.26y	102	PA-231	GROUND	(N,G)	PA-232	GROUND	1.00E-11	2.00E+01	1428
91233.26y	16	PA-232	GROUND	(N,2N)	PA-232	ALL	6.68E+00	2.00E+01	34
92232.26y	16	U-233	GROUND	(N,2N)	U-232	ALL	5.77E+00	2.00E+01	35

**Table V: Index To Reaction Cross Sections Available
on the ACTL (LLLDOS) Library**

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
1001.30y	102	h-1	ground	(n,g)	h-2	ground	1.00E-10	2.00E+01	102
1002.30y	102	h-2	ground	(n,g)	h-3	ground	1.00E-10	2.00E+01	72
1003.30y	16	h-3	ground	(n,2n)	h-2	ground	8.36E+00	2.00E+01	11
2003.30y	103	he-3	ground	(n,p)	h-3	ground	1.00E-10	2.00E+01	102
	104	he-3	ground	(n,d)	h-2	ground	4.35E+00	2.00E+01	23
	102	he-3	ground	(n,g)	he-4	ground	1.00E-10	2.00E+01	2
3006.30y	32	li-6	ground	(n,n') t	he-4	ground	1.70E+00	2.00E+01	20
	24	li-6	ground	(n,2n) a	he-4	ground	6.61E+00	2.00E+01	5
	103	li-6	ground	(n,p)	he-6	ground	3.18E+00	2.00E+01	14
	105	li-6	ground	(n,t)	he-4	ground	1.00E-10	2.00E+01	314
	102	li-6	ground	(n,g)	li-7	ground	1.00E-10	2.00E+01	102
3007.30y	16	li-7	ground	(n,2n)	li-6	ground	8.31E+00	2.00E+01	5
	33	li-7	ground	(n,n) he3	he-4	ground	2.83E+00	2.00E+01	11
	102	li-7	ground	(n,g)	li-8	ground	1.00E-10	2.00E+01	78
4007.30y	103	be-7	ground	(n,p)	li-7	ground	1.00E-10	2.00E+01	113
	107	be-7	ground	(n,a)	he-4	ground	1.00E-10	2.00E+01	9
4009.30y	16	be-9	ground	(n,2n)	he-4	ground	1.87E+00	2.00E+01	19
	103	be-9	ground	(n,p)	li-9	ground	1.43E+01	2.00E+01	4
	104	be-9	ground	(n,d)	li-8	ground	1.63E+01	2.00E+01	9
	105	be-9	ground	(n,t)	li-7	ground	1.16E+01	2.00E+01	6
	107	be-9	ground	(n,a)	li-6	ground	6.70E-01	2.00E+01	15
	102	be-9	ground	(n,g)	be-10	ground	1.00E-10	2.00E+01	102
5010.30y	35	b-10	ground	(n,n) t2a	he-4	ground	6.70E+00	2.00E+01	5
	30	b-10	ground	(n,2n) 2a	he-4	ground	9.30E+00	2.00E+01	4
	113	b-10	ground	(n,t) 2a	he-4	ground	1.50E+00	2.00E+01	13
	107	b-10	ground	(n,a)	li-7	ground	1.00E-10	2.00E+01	102
	102	b-10	ground	(n,g)	b-11	ground	1.00E-10	2.00E+01	56
5011.30y	16	b-11	ground	(n,2n)	b-10	ground	1.25E+01	2.00E+01	3
	103	b-11	ground	(n,p)	be-11	ground	1.17E+01	2.00E+01	3
	105	b-11	ground	(n,t)	be-10	ground	1.04E+01	2.00E+01	3
	107	b-11	ground	(n,a)	li-8	ground	7.24E+00	2.00E+01	5
	102	b-11	ground	(n,g)	b-12	ground	1.00E-10	2.00E+01	35
6012.30y	23	c-12	ground	(n,n') 3a	he-4	ground	7.98E+00	2.00E+01	10
	103	c-12	ground	(n,p)	b-12	ground	1.36E+01	2.00E+01	4
	104	c-12	ground	(n,d)	b-11	ground	1.49E+01	2.00E+01	4
	107	c-12	ground	(n,a)	be-9	ground	6.17E+00	2.00E+01	9
	102	c-12	ground	(n,g)	c-13	ground	1.00E-10	2.00E+01	11
6013.30y	16	c-13	ground	(n,2n)	c-12	ground	5.33E+00	2.00E+01	14
	22	c-13	ground	(n,n') a	be-9	ground	1.15E+01	2.00E+01	5
	103	c-13	ground	(n,p)	b-13	ground	1.38E+01	2.00E+01	12
	105	c-13	ground	(n,t)	b-11	ground	1.34E+01	2.00E+01	6
	107	c-13	ground	(n,a)	be-10	ground	4.13E+00	2.00E+01	11
	102	c-13	ground	(n,g)	c-14	ground	1.00E-10	2.00E+01	179
6014.30y	16	c-14	ground	(n,2n)	c-13	ground	8.77E+00	2.00E+01	7
	17	c-14	ground	(n,3n)	c-12	ground	1.41E+01	2.00E+01	5
	22	c-14	ground	(n,n') a	be-10	ground	1.29E+01	2.00E+01	5
	107	c-14	ground	(n,a)	be-11	ground	1.23E+01	2.00E+01	6
7014.30y	16	n-14	ground	(n,2n)	n-13	ground	1.13E+01	2.00E+01	10
	28	n-14	ground	(n,n') p	c-13	ground	8.09E+00	2.00E+01	11
	103	n-14	ground	(n,p)	c-14	ground	1.00E-10	2.00E+01	102
	104	n-14	ground	(n,d)	c-13	ground	5.70E+00	2.00E+01	18
	105	n-14	ground	(n,t)	c-12	ground	4.29E+00	2.00E+01	24
	107	n-14	ground	(n,a)	b-11	ground	1.10E+00	2.00E+01	228
	102	n-14	ground	(n,g)	n-15	ground	1.00E-10	2.00E+01	50
8016.30y	16	o-16	ground	(n,2n)	o-15	ground	1.67E+01	2.00E+01	3
	28	o-16	ground	(n,n') p	n-15	ground	1.29E+01	2.00E+01	4
	22	o-16	ground	(n,n') a	c-12	ground	7.60E+00	2.00E+01	12
	103	o-16	ground	(n,p)	n-16	ground	1.02E+01	2.00E+01	13
	104	o-16	ground	(n,d)	n-15	ground	1.05E+01	2.00E+01	4

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES	
8017.30y	107	o-16	ground	(n,a)	c-13	ground	2.35E+00	2.00E+01	59	
	107	o-17	ground	(n,a)	c-14	ground	1.00E-10	2.00E+01	117	
	9019.30y	16	f-19	ground	(n,2n)	f-18	ground	1.10E+01	2.00E+01	8
	28	f-19	ground	(n,n')p	o-18	ground	8.42E+00	2.00E+01	23	
	22	f-19	ground	(n,n')a	n-15	ground	5.15E+00	2.00E+01	34	
	103	f-19	ground	(n,p)	o-19	ground	4.25E+00	2.00E+01	34	
	104	f-19	ground	(n,d)	o-18	ground	6.07E+00	2.00E+01	15	
11023.30y	105	f-19	ground	(n,t)	o-17	ground	7.96E+00	2.00E+01	9	
	107	f-19	ground	(n,a)	n-16	ground	1.60E+00	2.00E+01	17	
	102	f-19	ground	(n,g)	f-20	ground	1.00E-10	2.00E+01	102	
	16	na-23	ground	(n,2n)	na-22	ground	1.30E+01	2.00E+01	13	
	28	na-23	ground	(n,n')p	ne-22	ground	9.25E+00	2.00E+01	23	
	22	na-23	ground	(n,n')a	f-19	ground	1.10E+01	2.00E+01	13	
	103	na-23	ground	(n,p)	ne-23	ground	4.00E+00	2.00E+01	106	
12023.30y	107	na-23	ground	(n,a)	f-20	ground	4.80E+00	2.00E+01	41	
	102	na-23	ground	(n,g)	na-24	ground	1.00E-10	2.00E+01	102	
	16	mg-23	ground	(n,2n)	mg-22	ground	1.37E+01	2.00E+01	15	
	28	mg-23	ground	(n,n')p	na-22	ground	9.26E+00	2.00E+01	16	
	103	mg-23	ground	(n,p)	na-23	ground	1.00E-10	2.00E+01	14	
	107	mg-23	ground	(n,a)	ne-20	ground	1.00E-02	2.00E+01	9	
	102	mg-23	ground	(n,g)	mg-24	ground	1.00E-10	2.00E+01	102	
12024.30y	16	mg-24	ground	(n,2n)	mg-23	ground	1.72E+01	2.00E+01	5	
	28	mg-24	ground	(n,n')p	na-23	ground	1.35E+01	2.00E+01	5	
	103	mg-24	ground	(n,p)	na-24	ground	4.93E+00	2.00E+01	23	
	107	mg-24	ground	(n,a)	ne-21	ground	6.00E+00	2.00E+01	9	
	102	mg-24	ground	(n,g)	mg-25	ground	1.00E-10	2.00E+01	102	
	16	mg-25	ground	(n,2n)	mg-24	ground	7.62E+00	2.00E+01	16	
	103	mg-25	ground	(n,p)	na-25	ground	4.57E+00	2.00E+01	12	
12025.30y	107	mg-25	ground	(n,a)	ne-22	ground	3.06E+00	2.00E+01	16	
	102	mg-25	ground	(n,g)	mg-26	ground	1.00E-10	2.00E+01	102	
	16	mg-26	ground	(n,2n)	mg-25	ground	1.15E+01	2.00E+01	19	
	17	mg-26	ground	(n,3n)	mg-24	ground	1.91E+01	2.00E+01	5	
	103	mg-26	ground	(n,p)	na-26	ground	9.61E+00	2.00E+01	16	
	107	mg-26	ground	(n,a)	ne-23	ground	9.16E+00	2.00E+01	8	
	102	mg-26	ground	(n,g)	mg-27	ground	1.00E-10	2.00E+01	102	
12026.30y	16	mg-27	ground	(n,2n)	mg-26	ground	6.68E+00	2.00E+01	21	
	17	mg-27	ground	(n,3n)	mg-25	ground	1.82E+01	2.00E+01	8	
	107	mg-27	ground	(n,a)	ne-24	ground	6.60E+00	2.00E+01	15	
	102	mg-27	ground	(n,g)	mg-28	ground	1.00E-10	2.00E+01	102	
	16	al-26	ground	(n,2n)	al-25	ground	1.18E+01	2.00E+01	14	
	28	al-26	ground	(n,n')p	mg-25	ground	8.03E+00	2.00E+01	15	
	22	al-26	ground	(n,n')a	na-22	ground	1.38E+01	2.00E+01	14	
13026.30y	103	al-26	ground	(n,p)	mg-26	ground	1.00E-02	2.00E+01	7	
	104	al-26	ground	(n,d)	mg-25	ground	5.89E+00	2.00E+01	16	
	105	al-26	ground	(n,t)	mg-24	ground	7.03E+00	2.00E+01	16	
	107	al-26	ground	(n,a)	na-23	ground	8.76E-01	2.00E+01	23	
	102	al-26	ground	(n,g)	al-27	ground	1.00E-10	2.00E+01	102	
	16	al-27	ground	(n,2n)	al-26	ground	1.35E+01	2.00E+01	14	
	1016	al-27	ground	(n,2n)	al-26	2.28E-01	1.38E+01	2.00E+01	7	
13027.30y	103	al-27	ground	(n,p)	mg-27	ground	2.70E+00	2.00E+01	52	
	104	al-27	ground	(n,d)	mg-26	ground	9.00E+00	2.00E+01	18	
	105	al-27	ground	(n,t)	mg-25	ground	1.40E+01	2.00E+01	11	
	107	al-27	ground	(n,a)	na-24	ground	5.00E+00	2.00E+01	27	
	102	al-27	ground	(n,g)	al-28	ground	1.00E-10	2.00E+01	102	
	16	si-27	ground	(n,2n)	si-26	ground	1.38E+01	2.00E+01	15	
	28	si-27	ground	(n,n')p	al-26	ground	9.39E+00	2.00E+01	17	
14027.30y	22	si-27	ground	(n,n')a	mg-23	ground	1.41E+01	2.00E+01	13	
	103	si-27	ground	(n,p)	al-27	ground	1.00E-02	2.00E+01	21	
	107	si-27	ground	(n,a)	mg-24	ground	1.00E-02	2.00E+01	20	
	102	si-27	ground	(n,g)	si-28	ground	1.00E-10	2.00E+01	102	
	16	si-28	ground	(n,2n)	si-27	ground	1.78E+01	2.00E+01	7	
	28	si-28	ground	(n,n')p	al-27	ground	1.39E+01	2.00E+01	15	
	22	si-28	ground	(n,n')a	mg-24	ground	1.47E+01	2.00E+01	13	
14028.30y	103	si-28	ground	(n,p)	al-28	ground	4.91E+00	2.00E+01	17	

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
14029.30y	107	si-28	ground	(n,a)	mg-25	ground	7.07E+00	2.00E+01	22
	102	si-28	ground	(n,g)	si-29	ground	1.00E-10	2.00E+01	102
	16	si-29	ground	(n,2n)	si-28	ground	8.76E+00	2.00E+01	15
	28	si-29	ground	(n,n')p	al-28	ground	1.44E+01	2.00E+01	15
	22	si-29	ground	(n,n')a	mg-25	ground	1.58E+01	2.00E+01	11
	103	si-29	ground	(n,p)	al-29	ground	4.70E+00	2.00E+01	21
14030.30y	107	si-29	ground	(n,a)	mg-26	ground	4.32E+00	2.00E+01	21
	102	si-29	ground	(n,g)	si-30	ground	1.00E-10	2.00E+01	99
	16	si-30	ground	(n,2n)	si-29	ground	1.10E+01	2.00E+01	15
	17	si-30	ground	(n,3n)	si-28	ground	1.97E+01	2.00E+01	3
	28	si-30	ground	(n,n')p	al-29	ground	1.56E+01	2.00E+01	15
	22	si-30	ground	(n,n')a	mg-26	ground	1.53E+01	2.00E+01	11
14031.30y	103	si-30	ground	(n,p)	al-30	ground	9.71E+00	2.00E+01	19
	107	si-30	ground	(n,a)	mg-27	ground	8.59E+00	2.00E+01	21
	102	si-30	ground	(n,g)	si-31	ground	1.00E-10	2.00E+01	99
	16	si-31	ground	(n,2n)	si-30	ground	6.80E+00	2.00E+01	19
	17	si-31	ground	(n,3n)	si-29	ground	1.78E+01	2.00E+01	7
	22	si-31	ground	(n,n')a	mg-27	ground	1.54E+01	2.00E+01	11
15031.30y	107	si-31	ground	(n,a)	mg-28	ground	6.57E+00	2.00E+01	22
	102	si-31	ground	(n,g)	si-32	ground	1.00E-10	2.00E+01	99
	16	p-31	ground	(n,2n)	p-30	ground	1.27E+01	2.00E+01	8
	28	p-31	ground	(n,n')p	si-30	ground	7.54E+00	2.00E+01	4
	103	p-31	ground	(n,p)	si-31	ground	1.50E+00	2.00E+01	13
	107	p-31	ground	(n,a)	al-28	ground	3.00E+00	2.00E+01	8
16031.30y	102	p-31	ground	(n,g)	p-32	ground	1.00E-10	2.00E+01	88
	16	s-31	ground	(n,2n)	s-30	ground	1.34E+01	2.00E+01	14
	28	s-31	ground	(n,n')p	p-30	ground	8.23E+00	2.00E+01	17
	22	s-31	ground	(n,n')a	si-27	ground	1.45E+01	2.00E+01	13
	103	s-31	ground	(n,p)	p-31	ground	1.00E-10	2.00E+01	16
	107	s-31	ground	(n,a)	si-28	ground	1.00E-10	2.00E+01	22
16032.30y	102	s-31	ground	(n,g)	s-32	ground	1.00E-10	2.00E+01	102
	16	s-32	ground	(n,2n)	s-31	ground	1.56E+01	2.00E+01	7
	28	s-32	ground	(n,n')p	p-31	ground	9.14E+00	2.00E+01	3
	103	s-32	ground	(n,p)	p-32	ground	1.60E+00	2.00E+01	36
	105	s-32	ground	(n,t)	p-30	ground	1.31E+01	2.00E+01	3
	107	s-32	ground	(n,a)	si-29	ground	1.00E-10	2.00E+01	45
16033.30y	102	s-32	ground	(n,g)	s-33	ground	1.00E-10	2.00E+01	102
	16	s-33	ground	(n,2n)	s-32	ground	8.90E+00	2.00E+01	14
	28	s-33	ground	(n,n')p	p-32	ground	1.18E+01	2.00E+01	15
	22	s-33	ground	(n,n')a	si-29	ground	1.24E+01	2.00E+01	14
	103	s-33	ground	(n,p)	p-33	ground	1.00E-10	2.00E+01	22
	104	s-33	ground	(n,d)	p-32	ground	9.70E+00	2.00E+01	14
16034.30y	107	s-33	ground	(n,a)	si-30	ground	1.00E-10	2.00E+01	26
	102	s-33	ground	(n,g)	s-34	ground	1.00E-10	2.00E+01	98
	16	s-34	ground	(n,2n)	s-33	ground	1.18E+01	2.00E+01	15
	28	s-34	ground	(n,n')p	p-33	ground	1.31E+01	2.00E+01	16
	22	s-34	ground	(n,n')a	si-30	ground	1.32E+01	2.00E+01	15
	103	s-34	ground	(n,p)	p-34	ground	6.46E+00	2.00E+01	21
16035.30y	104	s-34	ground	(n,d)	p-33	ground	1.10E+01	2.00E+01	14
	107	s-34	ground	(n,a)	si-31	ground	6.38E+00	2.00E+01	25
	102	s-34	ground	(n,g)	s-35	ground	1.00E-10	2.00E+01	98
	16	s-35	ground	(n,2n)	s-34	ground	7.19E+00	2.00E+01	19
	17	s-35	ground	(n,3n)	s-33	ground	1.89E+01	2.00E+01	5
	22	s-35	ground	(n,n')a	si-31	ground	1.36E+01	2.00E+01	14
16036.30y	107	s-35	ground	(n,a)	si-32	ground	4.05E+00	2.00E+01	24
	102	s-35	ground	(n,g)	s-36	ground	1.00E-10	2.00E+01	97
	16	s-36	ground	(n,2n)	s-35	ground	1.02E+01	2.00E+01	18
	17	s-36	ground	(n,3n)	s-34	ground	1.73E+01	2.00E+01	10
	22	s-36	ground	(n,n')a	si-32	ground	1.42E+01	2.00E+01	14
	102	s-36	ground	(n,g)	s-37	ground	1.00E-10	2.00E+01	96
16037.30y	16	s-37	ground	(n,2n)	s-36	ground	4.43E+00	2.00E+01	24
	17	s-37	ground	(n,3n)	s-35	ground	1.46E+01	2.00E+01	14
	102	s-37	ground	(n,g)	s-38	ground	1.00E-10	2.00E+01	95

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
17034.30y	16	cl-34	1.45E-01	(n,2n)	cl-33	ground	1.17E+01	2.00E+01	14
	28	cl-34	1.45E-01	(n,n')p	s-33	ground	7.24E+00	2.00E+01	16
	22	cl-34	1.45E-01	(n,n')a	p-30	ground	1.22E+01	2.00E+01	14
	103	cl-34	1.45E-01	(n,p)	s-34	ground	1.00E-10	2.00E+01	24
	107	cl-34	1.45E-01	(n,a)	p-31	ground	1.00E-10	2.00E+01	25
	102	cl-34	1.45E-01	(n,g)	cl-35	ground	1.00E-10	2.50E+00	95
17035.30y	16	cl-35	ground	(n,2n)	cl-34	ground	1.30E+01	2.00E+01	6
	1016	cl-35	ground	(n,2n)	cl-34	1.45E-01	1.32E+01	2.00E+01	15
	28	cl-35	ground	(n,n')p	s-34	ground	8.63E+00	2.00E+01	14
	22	cl-35	ground	(n,n')a	p-31	ground	1.27E+01	2.00E+01	14
	103	cl-35	ground	(n,p)	s-35	ground	1.00E-10	2.00E+01	44
	107	cl-35	ground	(n,a)	p-32	ground	1.00E-10	2.00E+01	27
17036.30y	102	cl-35	ground	(n,g)	cl-36	ground	1.00E-10	2.00E+01	95
	16	cl-36	ground	(n,2n)	cl-35	ground	8.82E+00	2.00E+01	14
	28	cl-36	ground	(n,n')p	s-35	ground	1.02E+01	2.00E+01	17
	22	cl-36	ground	(n,n')a	p-32	ground	1.33E+01	2.00E+01	15
	103	cl-36	ground	(n,p)	s-36	ground	1.00E-10	2.00E+01	105
	107	cl-36	ground	(n,a)	p-33	ground	2.85E+00	2.00E+01	23
17037.30y	102	cl-36	ground	(n,g)	cl-37	ground	1.00E-10	2.00E+01	95
	16	cl-37	ground	(n,2n)	cl-36	ground	1.06E+01	2.00E+01	16
	17	cl-37	ground	(n,3n)	cl-35	ground	1.94E+01	2.00E+01	3
	28	cl-37	ground	(n,n')p	s-36	ground	1.07E+01	2.00E+01	15
	22	cl-37	ground	(n,n')a	p-33	ground	1.34E+01	2.00E+01	14
	103	cl-37	ground	(n,p)	s-37	ground	6.34E+00	2.00E+01	19
17038.30y	105	cl-37	ground	(n,t)	s-35	ground	1.23E+01	2.00E+01	13
	107	cl-37	ground	(n,a)	p-34	ground	6.66E+00	2.00E+01	23
	102	cl-37	ground	(n,g)	cl-38	ground	1.00E-10	2.00E+01	84
	16	cl-38	ground	(n,2n)	cl-37	ground	6.27E+00	2.00E+01	19
	17	cl-38	ground	(n,3n)	cl-36	ground	1.69E+01	2.00E+01	9
	28	cl-38	ground	(n,n')p	s-37	ground	1.25E+01	2.00E+01	15
18036.30y	103	cl-38	ground	(n,p)	s-38	ground	4.35E+00	2.00E+01	21
	102	cl-38	ground	(n,g)	cl-39	ground	1.00E-10	2.00E+01	94
	16	ar-36	ground	(n,2n)	ar-36	ground	1.57E+01	2.00E+01	14
	103	ar-36	ground	(n,p)	cl-36	ground	2.31E+00	2.00E+01	15
	107	ar-36	ground	(n,a)	s-33	ground	3.78E+00	2.00E+01	15
	102	ar-36	ground	(n,g)	ar-37	ground	1.00E-10	2.00E+01	102
18037.30y	16	ar-37	ground	(n,2n)	ar-36	ground	9.03E+00	2.00E+01	15
	103	ar-37	ground	(n,p)	cl-37	ground	7.23E-01	2.00E+01	15
	107	ar-37	ground	(n,a)	s-34	ground	1.04E+00	2.00E+01	15
	102	ar-37	ground	(n,g)	ar-38	ground	1.00E-10	2.00E+01	102
18038.30y	16	ar-38	ground	(n,2n)	ar-37	ground	1.22E+01	2.00E+01	15
	103	ar-38	ground	(n,p)	cl-38	ground	6.59E+00	2.00E+01	15
	107	ar-38	ground	(n,a)	s-35	ground	5.99E+00	2.00E+01	15
	102	ar-38	ground	(n,g)	ar-39	ground	1.00E-10	2.00E+01	102
18039.30y	16	ar-39	ground	(n,2n)	ar-38	ground	6.77E+00	2.00E+01	20
	17	ar-39	ground	(n,3n)	ar-37	ground	1.89E+01	2.00E+01	6
	103	ar-39	ground	(n,p)	cl-39	ground	5.06E+00	2.00E+01	15
	107	ar-39	ground	(n,a)	s-36	ground	2.58E+00	2.00E+01	15
18040.30y	102	ar-39	ground	(n,g)	ar-40	ground	1.00E-10	2.00E+01	102
	16	ar-40	ground	(n,2n)	ar-39	ground	1.01E+01	2.00E+01	21
	17	ar-40	ground	(n,3n)	ar-38	ground	1.69E+01	2.00E+01	10
	103	ar-40	ground	(n,p)	cl-40	ground	9.21E+00	2.00E+01	15
18041.30y	107	ar-40	ground	(n,a)	s-37	ground	8.24E+00	2.00E+01	15
	102	ar-40	ground	(n,g)	ar-41	ground	1.00E-10	2.00E+01	102
	16	ar-41	ground	(n,2n)	ar-40	ground	6.25E+00	2.00E+01	21
	17	ar-41	ground	(n,3n)	ar-39	ground	1.64E+01	2.00E+01	12
18042.30y	107	ar-41	ground	(n,a)	s-38	ground	6.23E+00	2.00E+01	15
	102	ar-41	ground	(n,g)	ar-42	ground	1.00E-10	2.00E+01	102
	16	ar-42	ground	(n,2n)	ar-41	ground	9.65E+00	2.00E+01	24
	17	ar-42	ground	(n,3n)	ar-40	ground	1.59E+01	2.00E+01	13
18043.30y	102	ar-42	ground	(n,g)	ar-43	ground	1.00E-10	2.00E+01	102
	16	ar-43	ground	(n,2n)	ar-42	ground	5.73E+00	2.00E+01	27
	17	ar-43	ground	(n,3n)	ar-41	ground	1.59E+01	2.00E+01	12
	102	ar-43	ground	(n,g)	ar-44	ground	1.00E-10	2.00E+01	102

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
19038.30y	16	k-38	ground	(n,2n)	k-37	ground	1.24E+01	2.00E+01	17
	103	k-38	ground	(n,p)	ar-38	ground	1.00E-10	2.00E+01	72
	107	k-38	ground	(n,a)	cl-35	ground	1.00E-10	2.00E+01	102
	102	k-38	ground	(n,g)	k-39	ground	1.00E-10	2.00E+01	102
19039.30y	16	k-39	ground	(n,2n)	k-38	ground	1.34E+01	2.00E+01	16
	1016	k-39	ground	(n,2n)	k-38	1.30E-01	1.36E+01	2.00E+01	15
	28	k-39	ground	(n,n')p	ar-38	ground	9.09E+00	2.00E+01	15
	103	k-39	ground	(n,p)	ar-39	ground	2.32E+00	2.00E+01	17
	105	k-39	ground	(n,t)	ar-37	ground	1.27E+01	2.00E+01	13
	107	k-39	ground	(n,a)	cl-36	ground	2.00E+00	2.00E+01	10
	102	k-39	ground	(n,g)	k-40	ground	1.00E-10	2.00E+01	102
	16	k-40	ground	(n,2n)	k-39	ground	8.00E+00	2.00E+01	17
19040.30y	103	k-40	ground	(n,p)	ar-40	ground	1.00E-10	2.00E+01	85
	104	k-40	ground	(n,d)	ar-39	ground	8.13E+00	2.00E+01	15
	107	k-40	ground	(n,a)	cl-37	ground	1.00E-10	2.00E+01	108
	102	k-40	ground	(n,g)	k-41	ground	1.00E-10	2.00E+01	102
19041.30y	16	k-41	ground	(n,2n)	k-40	ground	1.03E+01	2.00E+01	20
	17	k-41	ground	(n,3n)	k-39	ground	1.83E+01	2.00E+01	8
	103	k-41	ground	(n,p)	ar-41	ground	4.26E+00	2.00E+01	17
	105	k-41	ground	(n,t)	ar-39	ground	1.21E+01	2.00E+01	13
	107	k-41	ground	(n,a)	cl-38	ground	6.23E+00	2.00E+01	12
	102	k-41	ground	(n,g)	k-42	ground	1.00E-10	2.00E+01	102
19042.30y	16	k-42	ground	(n,2n)	k-41	ground	7.71E+00	2.00E+01	20
	17	k-42	ground	(n,3n)	k-40	ground	1.80E+01	2.00E+01	8
	103	k-42	ground	(n,p)	ar-42	ground	2.31E+00	2.00E+01	17
	107	k-42	ground	(n,a)	cl-39	ground	5.64E+00	2.00E+01	14
	102	k-42	ground	(n,g)	k-43	ground	1.00E-10	2.00E+01	102
19043.30y	16	k-43	ground	(n,2n)	k-42	ground	9.85E+00	2.00E+01	21
	17	k-43	ground	(n,3n)	k-41	ground	1.76E+01	2.00E+01	9
	102	k-43	ground	(n,g)	k-44	ground	1.00E-10	2.00E+01	102
19044.30y	16	k-44	ground	(n,2n)	k-43	ground	7.44E+00	2.00E+01	20
	17	k-44	ground	(n,3n)	k-42	ground	1.73E+01	2.00E+01	9
	102	k-44	ground	(n,g)	k-45	ground	1.00E-10	2.00E+01	102
19045.30y	16	k-45	ground	(n,2n)	k-44	ground	9.10E+00	2.00E+01	22
	17	k-45	ground	(n,3n)	k-43	ground	1.65E+01	2.00E+01	11
	102	k-45	ground	(n,g)	k-46	ground	1.00E-10	2.00E+01	102
19046.30y	16	k-46	ground	(n,2n)	k-45	ground	7.03E+00	2.00E+01	21
	17	k-46	ground	(n,3n)	k-44	ground	1.61E+01	2.00E+01	12
	102	k-46	ground	(n,g)	k-47	ground	1.00E-10	2.00E+01	102
20039.30y	16	ca-39	ground	(n,2n)	ca-38	ground	1.37E+01	2.00E+01	5
	28	ca-39	ground	(n,n')p	k-38	ground	5.93E+00	2.00E+01	6
	22	ca-39	ground	(n,n')a	ar-35	ground	6.83E+00	2.00E+01	6
	103	ca-39	ground	(n,p)	k-39	ground	1.00E-10	2.00E+01	67
	107	ca-39	ground	(n,a)	ar-36	ground	1.00E-10	2.00E+01	102
	102	ca-39	ground	(n,g)	ca-40	ground	1.00E-10	2.00E+01	102
20040.30y	16	ca-40	ground	(n,2n)	ca-39	ground	1.60E+01	2.00E+01	6
	28	ca-40	ground	(n,n')p	k-39	ground	8.54E+00	2.00E+01	5
	22	ca-40	ground	(n,n')a	ar-36	ground	7.22E+00	2.00E+01	4
	103	ca-40	ground	(n,p)	k-40	ground	5.40E-01	2.00E+01	10
	105	ca-40	ground	(n,t)	k-38	ground	1.33E+01	2.00E+01	3
	107	ca-40	ground	(n,a)	ar-37	ground	1.00E-10	2.00E+01	10
	102	ca-40	ground	(n,g)	ca-41	ground	1.00E-10	2.00E+01	102
20041.30y	16	ca-41	ground	(n,2n)	ca-40	ground	8.57E+00	2.00E+01	7
	28	ca-41	ground	(n,n')p	k-40	ground	9.11E+00	2.00E+01	6
	22	ca-41	ground	(n,n')a	ar-37	ground	6.77E+00	2.00E+01	6
	103	ca-41	ground	(n,p)	k-41	ground	1.50E+00	2.00E+01	10
	105	ca-41	ground	(n,t)	k-39	ground	8.41E+00	2.00E+01	4
	107	ca-41	ground	(n,a)	ar-38	ground	1.00E+00	2.00E+01	7
	102	ca-41	ground	(n,g)	ca-42	ground	1.00E-10	2.00E+01	102
20042.30y	16	ca-42	ground	(n,2n)	ca-41	ground	1.17E+01	2.00E+01	5
	28	ca-42	ground	(n,n')p	k-41	ground	1.05E+01	2.00E+01	4
	22	ca-42	ground	(n,n')a	ar-38	ground	6.40E+00	2.00E+01	5
	103	ca-42	ground	(n,p)	k-42	ground	2.80E+00	2.00E+01	8
	107	ca-42	ground	(n,a)	ar-39	ground	5.00E+00	2.00E+01	6

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
20043.30y	102	ca-42	ground	(n,g)	ca-43	ground	1.00E-10	2.00E+01	102
	16	ca-43	ground	(n,2n)	ca-42	ground	8.12E+00	2.00E+01	8
	22	ca-43	ground	(n,n') a	ar-39	ground	7.76E+00	2.00E+01	6
	103	ca-43	ground	(n,p)	k-43	ground	1.06E+00	2.00E+01	8
	104	ca-43	ground	(n,d)	k-42	ground	8.64E+00	2.00E+01	4
20044.30y	107	ca-43	ground	(n,a)	ar-40	ground	1.00E+00	2.00E+01	7
	102	ca-43	ground	(n,g)	ca-44	ground	1.00E-10	2.00E+01	102
	16	ca-44	ground	(n,2n)	ca-43	ground	1.14E+01	2.00E+01	6
	103	ca-44	ground	(n,p)	k-44	ground	5.01E+00	2.00E+01	6
	104	ca-44	ground	(n,d)	k-43	ground	1.01E+01	2.00E+01	4
20045.30y	107	ca-44	ground	(n,a)	ar-41	ground	2.81E+00	2.00E+01	6
	102	ca-44	ground	(n,g)	ca-45	ground	1.00E-10	2.00E+01	102
	16	ca-45	ground	(n,2n)	ca-44	ground	7.58E+00	2.00E+01	8
	103	ca-45	ground	(n,p)	k-45	ground	3.49E+00	2.00E+01	6
	104	ca-45	ground	(n,d)	k-44	ground	1.03E+01	2.00E+01	4
20046.30y	107	ca-45	ground	(n,a)	ar-42	ground	7.60E-01	2.00E+01	5
	102	ca-45	ground	(n,g)	ca-46	ground	1.00E-10	2.00E+01	102
	16	ca-46	ground	(n,2n)	ca-45	ground	1.06E+01	2.00E+01	8
	103	ca-46	ground	(n,p)	k-46	ground	7.08E+00	2.00E+01	5
	107	ca-46	ground	(n,a)	ar-43	ground	6.00E+00	2.00E+01	4
20047.30y	102	ca-46	ground	(n,g)	ca-47	ground	1.00E-10	2.00E+01	102
	16	ca-47	ground	(n,2n)	ca-46	ground	7.43E+00	2.00E+01	7
	103	ca-47	ground	(n,p)	k-47	ground	6.10E+00	2.00E+01	6
	102	ca-47	ground	(n,g)	ca-48	ground	1.00E-10	2.00E+01	102
	16	ca-48	ground	(n,2n)	ca-47	ground	1.02E+01	2.00E+01	7
20048.30y	103	ca-48	ground	(n,p)	k-48	ground	1.20E+01	2.00E+01	4
	102	ca-48	ground	(n,g)	ca-49	ground	1.00E-10	2.00E+01	102
	16	ca-49	ground	(n,2n)	ca-48	ground	5.25E+00	2.00E+01	8
	102	ca-49	ground	(n,g)	ca-50	ground	1.00E-10	2.00E+01	102
	21044.30y	16	sc-44	ground	(n,2n)	sc-43	ground	9.93E+00	2.00E+01
21044.31y	103	sc-44	ground	(n,p)	ca-44	ground	5.00E-02	2.00E+01	14
	107	sc-44	ground	(n,a)	k-41	ground	3.44E+00	2.00E+01	17
	102	sc-44	ground	(n,g)	sc-45	ground	1.00E-10	2.00E+01	102
	16	sc-44	2.71E-01	(n,2n)	sc-43	ground	9.65E+00	2.00E+01	15
	103	sc-44	2.71E-01	(n,p)	ca-44	ground	5.00E-02	2.00E+01	13
21045.30y	107	sc-44	2.71E-01	(n,a)	k-41	ground	3.17E+00	2.00E+01	17
	102	sc-44	2.71E-01	(n,g)	sc-45	ground	1.00E-10	2.00E+01	102
	16	sc-45	ground	(n,2n)	sc-44	ground	1.16E+01	2.00E+01	14
	1016	sc-45	ground	(n,2n)	sc-44	2.71E-01	1.18E+01	2.00E+01	15
	103	sc-45	ground	(n,p)	ca-45	ground	2.30E+00	2.00E+01	17
21046.30y	107	sc-45	ground	(n,a)	k-42	ground	7.27E+00	2.00E+01	11
	102	sc-45	ground	(n,g)	sc-46	1.42E-01	1.00E-10	2.00E+01	102
	1102	sc-45	ground	(n,g)	sc-46	ground	1.00E-10	2.00E+01	102
	16	sc-46	ground	(n,2n)	sc-45	ground	8.96E+00	2.00E+01	15
	103	sc-46	ground	(n,p)	ca-46	ground	6.12E-01	2.00E+01	19
21046.31y	107	sc-46	ground	(n,a)	k-43	ground	6.36E+00	2.00E+01	17
	102	sc-46	ground	(n,g)	sc-47	ground	1.00E-10	2.00E+01	102
	16	sc-46	1.42E-01	(n,2n)	sc-45	ground	8.82E+00	2.00E+01	15
	103	sc-46	1.42E-01	(n,p)	ca-46	ground	4.69E-01	2.00E+01	19
	107	sc-46	1.42E-01	(n,a)	k-43	ground	6.21E+00	2.00E+01	17
21047.30y	102	sc-46	1.42E-01	(n,g)	sc-47	ground	1.00E-10	2.00E+01	102
	16	sc-47	ground	(n,2n)	sc-46	ground	1.09E+01	2.00E+01	16
	17	sc-47	ground	(n,3n)	sc-45	ground	1.98E+01	2.00E+01	3
	103	sc-47	ground	(n,p)	ca-47	ground	4.05E+00	2.00E+01	17
	107	sc-47	ground	(n,a)	k-44	ground	9.76E+00	2.00E+01	17
21048.30y	102	sc-47	ground	(n,g)	sc-48	ground	1.00E-10	2.00E+01	102
	16	sc-48	ground	(n,2n)	sc-47	ground	8.41E+00	2.00E+01	17
	17	sc-48	ground	(n,3n)	sc-46	ground	1.93E+01	2.00E+01	4
	103	sc-48	ground	(n,p)	ca-48	ground	2.28E+00	2.00E+01	17
	107	sc-48	ground	(n,a)	k-45	ground	9.04E+00	2.00E+01	17
22045.30y	102	sc-48	ground	(n,g)	sc-49	ground	1.00E-10	2.00E+01	97
	16	ti-45	ground	(n,2n)	ti-44	ground	9.74E+00	2.00E+01	16
	28	ti-45	ground	(n,n') p	sc-44	ground	1.16E+01	2.00E+01	16
	1028	ti-45	ground	(n,n') p	sc-44	2.71E-01	1.18E+01	2.00E+01	16

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
22046.30y	22	ti-45	ground	(n,n')a	ca-41	ground	1.38E+01	2.00E+01	14
	103	ti-45	ground	(n,p)	sc-45	ground	1.22E-01	2.00E+01	23
	107	ti-45	ground	(n,a)	ca-42	ground	2.01E+00	2.00E+01	23
	102	ti-45	ground	(n,g)	ti-46	ground	1.00E-10	2.00E+01	102
	16	ti-46	ground	(n,2n)	ti-45	ground	1.35E+01	2.00E+01	9
	28	ti-46	ground	(n,n')p	sc-45	ground	1.34E+01	2.00E+01	15
	22	ti-46	ground	(n,n')a	ca-42	ground	1.55E+01	2.00E+01	12
	103	ti-46	ground	(n,p)	sc-46	ground	4.63E+00	2.00E+01	21
	105	ti-46	ground	(n,t)	sc-44	ground	1.37E+01	2.00E+01	7
	107	ti-46	ground	(n,a)	ca-43	ground	7.36E+00	2.00E+01	19
22047.30y	102	ti-46	ground	(n,g)	ti-47	ground	1.00E-10	2.00E+01	98
	16	ti-47	ground	(n,2n)	ti-46	ground	9.07E+00	2.00E+01	16
	28	ti-47	ground	(n,n')p	sc-46	ground	1.35E+01	2.00E+01	17
	22	ti-47	ground	(n,n')a	ca-43	ground	1.64E+01	2.00E+01	10
	103	ti-47	ground	(n,p)	sc-47	ground	1.22E+00	2.00E+01	20
	104	ti-47	ground	(n,d)	sc-46	ground	1.15E+01	2.00E+01	14
	107	ti-47	ground	(n,a)	ca-44	ground	5.01E+00	2.00E+01	18
22048.30y	102	ti-47	ground	(n,g)	ti-48	ground	1.00E-10	2.00E+01	100
	16	ti-48	ground	(n,2n)	ti-47	ground	1.19E+01	2.00E+01	16
	28	ti-48	ground	(n,n')p	sc-47	ground	1.44E+01	2.00E+01	14
	22	ti-48	ground	(n,n')a	ca-44	ground	1.76E+01	2.00E+01	9
	103	ti-48	ground	(n,p)	sc-48	ground	4.23E+00	2.00E+01	21
	104	ti-48	ground	(n,d)	sc-47	ground	1.25E+01	2.00E+01	13
	107	ti-48	ground	(n,a)	ca-45	ground	6.97E+00	2.00E+01	18
22049.30y	102	ti-48	ground	(n,g)	ti-49	ground	1.00E-10	2.00E+01	102
	16	ti-49	ground	(n,2n)	ti-48	ground	8.31E+00	2.00E+01	17
	28	ti-49	ground	(n,n')p	sc-48	ground	1.44E+01	2.00E+01	14
	22	ti-49	ground	(n,n')a	ca-45	ground	1.76E+01	2.00E+01	9
	103	ti-49	ground	(n,p)	sc-49	ground	4.21E+00	2.00E+01	21
	104	ti-49	ground	(n,d)	sc-48	ground	1.24E+01	2.00E+01	13
	107	ti-49	ground	(n,a)	ca-46	ground	6.93E+00	2.00E+01	18
22050.30y	102	ti-49	ground	(n,g)	ti-50	ground	1.00E-10	2.00E+01	98
	16	ti-50	ground	(n,2n)	ti-49	ground	1.12E+01	2.00E+01	17
	17	ti-50	ground	(n,3n)	ti-48	ground	1.95E+01	2.00E+01	4
	28	ti-50	ground	(n,n')p	sc-49	ground	1.52E+01	2.00E+01	13
	22	ti-50	ground	(n,n')a	ca-46	ground	1.81E+01	2.00E+01	9
	103	ti-50	ground	(n,p)	sc-50	ground	9.19E+00	2.00E+01	17
	107	ti-50	ground	(n,a)	ca-47	ground	1.06E+01	2.00E+01	17
22051.30y	102	ti-50	ground	(n,g)	ti-51	ground	1.00E-10	2.00E+01	81
	16	ti-51	ground	(n,2n)	ti-50	ground	6.51E+00	2.00E+01	20
	17	ti-51	ground	(n,3n)	ti-49	ground	1.77E+01	2.00E+01	8
	28	ti-51	ground	(n,n')p	sc-50	ground	1.55E+01	2.00E+01	12
	22	ti-51	ground	(n,n')a	ca-47	ground	1.72E+01	2.00E+01	10
	103	ti-51	ground	(n,p)	sc-51	ground	8.78E+00	2.00E+01	18
	107	ti-51	ground	(n,a)	ca-48	ground	6.96E+00	2.00E+01	19
23047.30y	102	ti-51	ground	(n,g)	ti-52	ground	1.00E-10	2.00E+01	93
23048.30y	102	v-47	ground	(n,g)	v-48	ground	1.00E-10	2.00E+01	102
23049.30y	16	v-48	ground	(n,2n)	v-47	ground	1.07E+01	2.00E+01	15
	28	v-48	ground	(n,n')p	ti-47	ground	9.99E+00	2.00E+01	15
	22	v-48	ground	(n,n')a	sc-44	ground	1.70E+01	2.00E+01	10
	1022	v-48	ground	(n,n')a	sc-44	2.71E-01	1.72E+01	2.00E+01	9
	103	v-48	ground	(n,p)	ti-48	ground	1.00E-10	2.00E+01	17
	107	v-48	ground	(n,a)	sc-45	ground	5.36E+00	2.00E+01	17
	102	v-48	ground	(n,g)	v-49	ground	1.00E-10	2.00E+01	102
23050.30y	16	v-49	ground	(n,2n)	v-48	ground	1.18E+01	2.00E+01	15
	28	v-49	ground	(n,n')p	ti-48	ground	9.90E+00	2.00E+01	16
	22	v-49	ground	(n,n')a	sc-45	ground	1.72E+01	2.00E+01	9
	103	v-49	ground	(n,p)	ti-49	ground	1.75E+00	2.00E+01	22
	107	v-49	ground	(n,a)	sc-46	ground	8.17E+00	2.00E+01	17
	1107	v-49	ground	(n,a)	sc-46	1.42E-01	8.31E+00	2.00E+01	16
	102	v-49	ground	(n,g)	v-50	ground	1.00E-10	2.00E+01	102
23050.30y	16	v-50	ground	(n,2n)	v-49	ground	9.52E+00	2.00E+01	15
	28	v-50	ground	(n,n')p	ti-49	ground	1.11E+01	2.00E+01	17
	22	v-50	ground	(n,n')a	sc-46	ground	1.77E+01	2.00E+01	8

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
23051.30y	1022	v-50	ground	(n,n')a	sc-46	1.42E-01	1.78E+01	2.00E+01	9
	103	v-50	ground	(n,p)	ti-50	ground	8.18E-02	2.00E+01	21
	107	v-50	ground	(n,a)	sc-47	ground	6.79E+00	2.00E+01	17
	102	v-50	ground	(n,g)	v-51	ground	1.00E-10	2.00E+01	102
	16	v-51	ground	(n,2n)	v-50	ground	1.13E+01	2.00E+01	15
	28	v-51	ground	(n,n')p	ti-50	ground	1.12E+01	2.00E+01	15
	22	v-51	ground	(n,n')a	sc-47	ground	1.30E+01	2.00E+01	6
	103	v-51	ground	(n,p)	ti-51	ground	4.83E+00	2.00E+01	22
	107	v-51	ground	(n,a)	sc-48	ground	9.00E+00	2.00E+01	6
	102	v-51	ground	(n,g)	v-52	ground	1.00E-10	2.00E+01	102
23052.30y	16	v-52	ground	(n,2n)	v-51	ground	7.45E+00	2.00E+01	17
	17	v-52	ground	(n,3n)	v-50	ground	1.87E+01	2.00E+01	6
	28	v-52	ground	(n,n')p	ti-51	ground	1.21E+01	2.00E+01	15
	22	v-52	ground	(n,n')a	sc-48	ground	1.71E+01	2.00E+01	10
	103	v-52	ground	(n,p)	ti-52	ground	4.31E+00	2.00E+01	20
	107	v-52	ground	(n,a)	sc-49	ground	6.72E+00	2.00E+01	16
	102	v-52	ground	(n,g)	v-53	ground	1.00E-10	2.00E+01	102
	16	cr-49	ground	(n,2n)	cr-48	ground	1.09E+01	2.00E+01	16
	28	cr-49	ground	(n,n')p	v-48	ground	1.16E+01	2.00E+01	10
	22	cr-49	ground	(n,n')a	ti-45	ground	1.71E+01	2.00E+01	8
24049.30y	103	cr-49	ground	(n,p)	v-49	ground	1.00E-04	2.00E+01	22
	107	cr-49	ground	(n,a)	ti-46	ground	3.58E+00	2.00E+01	18
	102	cr-49	ground	(n,g)	cr-50	ground	1.00E-10	2.00E+01	102
	16	cr-50	ground	(n,2n)	cr-49	ground	1.32E+01	2.00E+01	11
	28	cr-50	ground	(n,n')p	v-49	ground	1.30E+01	2.00E+01	11
	22	cr-50	ground	(n,n')a	ti-46	ground	1.68E+01	2.00E+01	10
	103	cr-50	ground	(n,p)	v-50	ground	3.60E+00	2.00E+01	23
	104	cr-50	ground	(n,d)	v-49	ground	1.13E+01	2.00E+01	14
	105	cr-50	ground	(n,t)	v-48	ground	1.35E+01	2.00E+01	12
	107	cr-50	ground	(n,a)	ti-47	ground	7.69E+00	2.00E+01	18
24051.30y	102	cr-50	ground	(n,g)	cr-51	ground	1.00E-10	2.00E+01	102
	16	cr-51	ground	(n,2n)	cr-50	ground	9.44E+00	2.00E+01	15
	28	cr-51	ground	(n,n')p	v-50	ground	1.29E+01	2.00E+01	9
	22	cr-51	ground	(n,n')a	ti-47	ground	1.71E+01	2.00E+01	9
	103	cr-51	ground	(n,p)	v-51	ground	1.76E+00	2.00E+01	23
	107	cr-51	ground	(n,a)	ti-48	ground	5.24E+00	2.00E+01	18
	102	cr-51	ground	(n,g)	cr-52	ground	1.00E-10	2.00E+01	102
	16	cr-52	ground	(n,2n)	cr-51	ground	1.23E+01	2.00E+01	12
	28	cr-52	ground	(n,n')p	v-50	ground	1.39E+01	2.00E+01	10
	22	cr-52	ground	(n,n')a	ti-48	ground	1.75E+01	2.00E+01	9
24052.30y	103	cr-52	ground	(n,p)	v-52	ground	4.50E+00	2.00E+01	16
	104	cr-52	ground	(n,d)	v-51	ground	1.19E+01	2.00E+01	14
	105	cr-52	ground	(n,t)	v-50	ground	1.40E+01	2.00E+01	11
	107	cr-52	ground	(n,a)	ti-49	ground	9.17E+00	2.00E+01	18
	102	cr-52	ground	(n,g)	cr-53	ground	1.00E-10	2.00E+01	102
	16	cr-53	ground	(n,2n)	cr-52	ground	8.09E+00	2.00E+01	14
	28	cr-53	ground	(n,n')p	v-52	ground	1.45E+01	2.00E+01	8
	22	cr-53	ground	(n,n')a	ti-49	ground	1.73E+01	2.00E+01	9
	103	cr-53	ground	(n,p)	v-53	ground	5.97E+00	2.00E+01	19
	104	cr-53	ground	(n,d)	v-52	ground	1.25E+01	2.00E+01	13
24053.30y	105	cr-53	ground	(n,t)	v-51	ground	1.36E+01	2.00E+01	14
	107	cr-53	ground	(n,a)	ti-50	ground	6.07E+00	2.00E+01	17
	102	cr-53	ground	(n,g)	cr-54	ground	1.00E-10	2.00E+01	102
	16	cr-54	ground	(n,2n)	cr-53	ground	9.90E+00	2.00E+01	17
	17	cr-54	ground	(n,3n)	cr-52	ground	1.80E+01	2.00E+01	8
	28	cr-54	ground	(n,n')p	v-53	ground	1.57E+01	2.00E+01	8
	22	cr-54	ground	(n,n')a	ti-50	ground	1.60E+01	2.00E+01	12
	103	cr-54	ground	(n,p)	v-54	ground	9.50E+00	2.00E+01	18
	104	cr-54	ground	(n,d)	v-53	ground	1.37E+01	2.00E+01	14
	105	cr-54	ground	(n,t)	v-52	ground	1.35E+01	2.00E+01	13
24054.30y	107	cr-54	ground	(n,a)	ti-51	ground	9.43E+00	2.00E+01	20
	102	cr-54	ground	(n,g)	cr-55	ground	1.00E-10	2.00E+01	102
	16	cr-55	ground	(n,2n)	cr-54	ground	6.37E+00	2.00E+01	22
	17	cr-55	ground	(n,3n)	cr-53	ground	1.63E+01	2.00E+01	13

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
	28	cr-55	ground	(n,n')p	v-54	ground	1.59E+01	2.00E+01	8
	22	cr-55	ground	(n,n')a	ti-51	ground	1.58E+01	2.00E+01	12
	103	cr-55	ground	(n,p)	v-55	ground	8.95E+00	2.00E+01	19
	107	cr-55	ground	(n,a)	ti-52	ground	7.83E+00	2.00E+01	19
	102	cr-55	ground	(n,g)	cr-56	ground	1.00E-10	2.00E+01	102
24056.30y	16	cr-56	ground	(n,2n)	cr-55	ground	8.37E+00	2.00E+01	25
	17	cr-56	ground	(n,3n)	cr-54	ground	1.47E+01	2.00E+01	15
	102	cr-56	ground	(n,g)	cr-57	ground	1.00E-10	2.00E+01	102
25051.30y	102	mn-51	ground	(n,g)	mn-52	3.77E-01	1.00E-10	2.00E+01	102
	1102	mn-51	ground	(n,g)	mn-52	ground	1.00E-10	2.00E+01	102
25052.30y	16	mn-52	ground	(n,2n)	mn-51	ground	1.07E+01	2.00E+01	16
	28	mn-52	ground	(n,n')p	cr-51	ground	1.00E+01	2.00E+01	15
	22	mn-52	ground	(n,n')a	v-48	ground	1.73E+01	2.00E+01	9
	103	mn-52	ground	(n,p)	cr-52	ground	1.00E-10	2.00E+01	19
	107	mn-52	ground	(n,a)	v-49	ground	5.44E+00	2.00E+01	16
	102	mn-52	ground	(n,g)	mn-53	ground	1.00E-10	2.00E+01	102
25053.30y	16	mn-53	ground	(n,2n)	mn-52	ground	1.23E+01	2.00E+01	16
	1016	mn-53	ground	(n,2n)	mn-52	3.77E-01	1.27E+01	2.00E+01	16
	28	mn-53	ground	(n,n')p	cr-52	ground	1.00E+01	2.00E+01	15
	22	mn-53	ground	(n,n')a	v-49	ground	1.77E+01	2.00E+01	9
	103	mn-53	ground	(n,p)	cr-53	ground	2.07E+00	2.00E+01	23
	107	mn-53	ground	(n,a)	v-50	ground	8.17E+00	2.00E+01	17
	102	mn-53	ground	(n,g)	mn-54	ground	1.00E-10	2.00E+01	102
25054.30y	16	mn-54	ground	(n,2n)	mn-53	ground	9.11E+00	2.00E+01	16
	28	mn-54	ground	(n,n')p	cr-53	ground	1.10E+01	2.00E+01	17
	22	mn-54	ground	(n,n')a	v-50	ground	1.73E+01	2.00E+01	9
	103	mn-54	ground	(n,p)	cr-54	ground	1.26E+00	2.00E+01	22
	107	mn-54	ground	(n,a)	v-51	ground	5.98E+00	2.00E+01	17
	102	mn-54	ground	(n,g)	mn-55	ground	1.00E-10	2.00E+01	102
25055.30y	16	mn-55	ground	(n,2n)	mn-54	ground	1.04E+01	2.00E+01	11
	28	mn-55	ground	(n,n')p	cr-54	ground	1.15E+01	2.00E+01	17
	22	mn-55	ground	(n,n')a	v-51	ground	1.64E+01	2.00E+01	10
	103	mn-55	ground	(n,p)	cr-55	ground	5.29E+00	2.00E+01	21
	104	mn-55	ground	(n,d)	cr-54	ground	9.55E+00	2.00E+01	14
	105	mn-55	ground	(n,t)	cr-53	ground	1.31E+01	2.00E+01	8
	107	mn-55	ground	(n,a)	v-52	ground	8.92E+00	2.00E+01	18
	102	mn-55	ground	(n,g)	mn-56	ground	1.00E-10	2.00E+01	102
25056.30y	16	mn-56	ground	(n,2n)	mn-55	ground	7.40E+00	2.00E+01	19
	17	mn-56	ground	(n,3n)	mn-54	ground	1.78E+01	2.00E+01	8
	28	mn-56	ground	(n,n')p	cr-55	ground	1.25E+01	2.00E+01	16
	22	mn-56	ground	(n,n')a	v-52	ground	1.63E+01	2.00E+01	11
	103	mn-56	ground	(n,p)	cr-56	ground	4.30E+00	2.00E+01	22
	107	mn-56	ground	(n,a)	v-53	ground	7.63E+00	2.00E+01	19
	102	mn-56	ground	(n,g)	mn-57	ground	1.00E-10	2.00E+01	102
25057.30y	16	mn-57	ground	(n,2n)	mn-56	ground	8.93E+00	2.00E+01	21
	17	mn-57	ground	(n,3n)	mn-55	ground	1.63E+01	2.00E+01	10
	28	mn-57	ground	(n,n')p	cr-56	ground	1.31E+01	2.00E+01	17
	22	mn-57	ground	(n,n')a	v-53	ground	1.66E+01	2.00E+01	9
	103	mn-57	ground	(n,p)	cr-57	ground	8.45E+00	2.00E+01	18
	107	mn-57	ground	(n,a)	v-54	ground	9.66E+00	2.00E+01	18
	102	mn-57	ground	(n,g)	mn-58	ground	1.00E-10	2.00E+01	102
25058.30y	16	mn-58	ground	(n,2n)	mn-57	ground	6.61E+00	2.00E+01	22
	17	mn-58	ground	(n,3n)	mn-56	ground	1.55E+01	2.00E+01	12
	102	mn-58	ground	(n,g)	mn-59	ground	1.00E-10	2.00E+01	102
26053.30y	16	fe-53	ground	(n,2n)	fe-52	ground	1.09E+01	2.00E+01	15
	28	fe-53	ground	(n,n')p	mn-52	ground	1.12E+01	2.00E+01	8
	1028	fe-53	ground	(n,n')p	mn-52	3.77E-01	1.16E+01	2.00E+01	7
	22	fe-53	ground	(n,n')a	cr-49	ground	1.70E+01	2.00E+01	9
	103	fe-53	ground	(n,p)	mn-53	ground	1.00E-04	2.00E+01	21
	107	fe-53	ground	(n,a)	cr-50	ground	3.75E+00	2.00E+01	17
	102	fe-53	ground	(n,g)	fe-54	ground	1.00E-10	2.00E+01	102
26054.30y	16	fe-54	ground	(n,2n)	fe-53	ground	1.36E+01	2.00E+01	15
	28	fe-54	ground	(n,n')p	mn-53	ground	1.25E+01	2.00E+01	8
	22	fe-54	ground	(n,n')a	cr-50	ground	1.74E+01	2.00E+01	9

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
	103	fe-54	ground	(n,p)	mn-54	ground	5.00E-02	2.00E+01	23
	104	fe-54	ground	(n,d)	mn-53	ground	1.06E+01	2.00E+01	13
	105	fe-54	ground	(n,t)	mn-52	ground	1.35E+01	2.00E+01	11
	1105	fe-54	ground	(n,t)	mn-52	3.77E-01	1.39E+01	2.00E+01	9
	107	fe-54	ground	(n,a)	cr-51	ground	5.00E-02	2.00E+01	20
	102	fe-54	ground	(n,g)	fe-55	ground	1.00E-10	2.00E+01	102
26055.30y	16	fe-55	ground	(n,2n)	fe-54	ground	9.47E+00	2.00E+01	15
	28	fe-55	ground	(n,n')p	mn-54	ground	1.28E+01	2.00E+01	8
	22	fe-55	ground	(n,n')a	cr-51	ground	1.74E+01	2.00E+01	9
	103	fe-55	ground	(n,p)	mn-55	ground	2.61E+00	2.00E+01	22
	107	fe-55	ground	(n,a)	cr-52	ground	5.08E+00	2.00E+01	18
	102	fe-55	ground	(n,g)	fe-56	ground	1.00E-10	2.00E+01	102
26056.30y	16	fe-56	ground	(n,2n)	fe-55	ground	1.14E+01	2.00E+01	15
	28	fe-56	ground	(n,n')p	mn-55	ground	1.38E+01	2.00E+01	7
	22	fe-56	ground	(n,n')a	cr-52	ground	1.65E+01	2.00E+01	9
	103	fe-56	ground	(n,p)	mn-56	ground	3.50E+00	2.00E+01	19
	104	fe-56	ground	(n,d)	mn-55	ground	1.19E+01	2.00E+01	10
	105	fe-56	ground	(n,t)	mn-54	ground	1.34E+01	2.00E+01	11
	107	fe-56	ground	(n,a)	cr-53	ground	8.36E+00	2.00E+01	18
	102	fe-56	ground	(n,g)	fe-57	ground	1.00E-10	2.00E+01	102
26057.30y	16	fe-57	ground	(n,2n)	fe-56	ground	7.78E+00	2.00E+01	17
	17	fe-57	ground	(n,3n)	fe-55	ground	1.92E+01	2.00E+01	4
	28	fe-57	ground	(n,n')p	mn-56	ground	1.42E+01	2.00E+01	10
	22	fe-57	ground	(n,n')a	cr-53	ground	1.61E+01	2.00E+01	10
	103	fe-57	ground	(n,p)	mn-57	ground	5.42E+00	2.00E+01	20
	104	fe-57	ground	(n,d)	fe-56	ground	1.23E+01	2.00E+01	13
	105	fe-57	ground	(n,t)	mn-55	ground	1.33E+01	2.00E+01	10
	107	fe-57	ground	(n,a)	cr-54	ground	6.21E+00	2.00E+01	19
	102	fe-57	ground	(n,g)	fe-58	ground	1.00E-10	2.00E+01	102
26058.30y	16	fe-58	ground	(n,2n)	fe-57	ground	1.02E+01	2.00E+01	18
	17	fe-58	ground	(n,3n)	fe-56	ground	1.80E+01	2.00E+01	9
	28	fe-58	ground	(n,n')p	mn-57	ground	1.54E+01	2.00E+01	7
	22	fe-58	ground	(n,n')a	cr-54	ground	1.64E+01	2.00E+01	11
	103	fe-58	ground	(n,p)	mn-58	ground	9.00E+00	2.00E+01	20
	105	fe-58	ground	(n,t)	mn-56	ground	1.40E+01	2.00E+01	14
	107	fe-58	ground	(n,a)	cr-55	ground	1.00E+01	2.00E+01	18
	102	fe-58	ground	(n,g)	fe-59	ground	1.00E-10	2.00E+01	102
26059.30y	16	fe-59	ground	(n,2n)	fe-58	ground	6.70E+00	2.00E+01	21
	17	fe-59	ground	(n,3n)	fe-57	ground	1.69E+01	2.00E+01	9
	28	fe-59	ground	(n,n')p	mn-58	ground	1.55E+01	2.00E+01	7
	22	fe-59	ground	(n,n')a	cr-55	ground	1.67E+01	2.00E+01	10
	103	fe-59	ground	(n,p)	mn-59	ground	8.60E+00	2.00E+01	18
	107	fe-59	ground	(n,a)	cr-56	ground	8.33E+00	2.00E+01	17
	102	fe-59	ground	(n,g)	fe-60	ground	1.00E-10	2.00E+01	102
26060.30y	16	fe-60	ground	(n,2n)	fe-59	ground	8.99E+00	2.00E+01	22
	17	fe-60	ground	(n,3n)	fe-58	ground	1.57E+01	2.00E+01	12
	102	fe-60	ground	(n,g)	fe-61	ground	1.00E-10	2.00E+01	102
27057.30y	16	co-57	ground	(n,2n)	co-56	ground	1.16E+01	2.00E+01	8
	28	co-57	ground	(n,n')p	fe-56	ground	6.84E+00	2.00E+01	13
	22	co-57	ground	(n,n')a	mn-53	ground	9.50E+00	2.00E+01	10
	103	co-57	ground	(n,p)	fe-57	ground	1.00E-10	2.00E+01	47
	107	co-57	ground	(n,a)	mn-54	ground	1.00E+00	2.00E+01	18
	102	co-57	ground	(n,g)	co-58	2.49E-02	1.00E-10	2.00E+01	102
	1102	co-57	ground	(n,g)	co-58	ground	1.00E-10	2.00E+01	102
27058.30y	4	co-58	ground	(n,n')	co-58	2.49E-02	2.53E-02	2.00E+01	30
	16	co-58	ground	(n,2n)	co-57	ground	8.72E+00	2.00E+01	11
	28	co-58	ground	(n,n')p	fe-57	ground	1.07E+01	2.00E+01	16
	22	co-58	ground	(n,n')a	mn-54	ground	1.00E+01	2.00E+01	9
	103	co-58	ground	(n,p)	fe-58	ground	6.46E-01	2.00E+01	23
	107	co-58	ground	(n,a)	mn-55	ground	1.00E-10	2.00E+01	60
	102	co-58	ground	(n,g)	co-59	ground	1.00E-10	2.00E+01	102
27058.31y	4	co-58	2.49E-02	(n,n')	co-58	ground	1.00E-10	2.00E+01	51
	16	co-58	2.49E-02	(n,2n)	co-57	ground	8.69E+00	2.00E+01	11
	28	co-58	2.49E-02	(n,n')p	fe-57	ground	1.07E+01	2.00E+01	16

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
27059.30y	22	co-58	2.49E-02	(n,n')a	mn-54	ground	9.98E+00	2.00E+01	9
	103	co-58	2.49E-02	(n,p)	fe-58	ground	6.21E-01	2.00E+01	23
	107	co-58	2.49E-02	(n,a)	mn-55	ground	1.00E-10	2.00E+01	58
	102	co-58	2.49E-02	(n,g)	co-59	ground	1.00E-10	2.00E+01	102
	16	co-59	ground	(n,2n)	co-58	ground	1.06E+01	2.00E+01	13
	1016	co-59	ground	(n,2n)	co-58	2.49E-02	1.07E+01	2.00E+01	13
	28	co-59	ground	(n,n')p	fe-58	ground	8.19E+00	2.00E+01	11
	22	co-59	ground	(n,n')a	mn-55	ground	9.37E+00	2.00E+01	10
	103	co-59	ground	(n,p)	fe-59	ground	1.29E+00	2.00E+01	19
	104	co-59	ground	(n,d)	fe-58	ground	5.23E+00	2.00E+01	4
	105	co-59	ground	(n,t)	fe-57	ground	9.08E+00	2.00E+01	4
	107	co-59	ground	(n,a)	mn-56	ground	4.90E+00	2.00E+01	30
	102	co-59	ground	(n,g)	co-60	5.86E-02	1.00E-10	2.00E+01	102
	1102	co-59	ground	(n,g)	co-60	ground	1.00E-10	2.00E+01	102
27060.30y	4	co-60	ground	(n,n')	co-60	5.86E-02	5.96E-02	2.00E+01	29
	16	co-60	ground	(n,2n)	co-59	ground	7.62E+00	2.00E+01	12
	17	co-60	ground	(n,3n)	co-58	ground	1.82E+01	2.00E+01	2
	1017	co-60	ground	(n,3n)	co-58	2.49E-02	1.83E+01	2.00E+01	2
	28	co-60	ground	(n,n')p	fe-59	ground	1.00E+01	2.00E+01	9
	22	co-60	ground	(n,n')a	mn-56	ground	1.30E+01	2.00E+01	6
	103	co-60	ground	(n,p)	fe-60	ground	5.00E-01	2.00E+01	20
	107	co-60	ground	(n,a)	mn-57	ground	2.00E+00	2.00E+01	17
	102	co-60	ground	(n,g)	co-61	ground	1.00E-10	2.00E+01	102
	4	co-60	5.86E-02	(n,n')	co-60	ground	1.00E-10	2.00E+01	58
	16	co-60	5.86E-02	(n,2n)	co-59	ground	7.56E+00	2.00E+01	12
	17	co-60	5.86E-02	(n,3n)	co-58	ground	1.82E+01	2.00E+01	2
	1017	co-60	5.86E-02	(n,3n)	co-58	2.49E-02	1.82E+01	2.00E+01	2
	28	co-60	5.86E-02	(n,n')p	fe-59	ground	9.94E+00	2.00E+01	9
27060.31y	22	co-60	5.86E-02	(n,n')a	mn-56	ground	1.29E+01	2.00E+01	6
	103	co-60	5.86E-02	(n,p)	fe-60	ground	4.40E-01	2.00E+01	23
	107	co-60	5.86E-02	(n,a)	mn-57	ground	1.94E+00	2.00E+01	17
	102	co-60	5.86E-02	(n,g)	co-61	ground	1.00E-10	2.00E+01	102
	16	co-61	ground	(n,2n)	co-60	ground	9.49E+00	2.00E+01	20
	1016	co-61	ground	(n,2n)	co-60	5.86E-02	9.55E+00	2.00E+01	20
	17	co-61	ground	(n,3n)	co-59	ground	1.71E+01	2.00E+01	10
	28	co-61	ground	(n,n')p	fe-60	ground	1.25E+01	2.00E+01	17
	103	co-61	ground	(n,p)	fe-61	ground	6.90E+00	2.00E+01	21
	102	co-61	ground	(n,g)	co-62	2.50E-02	1.00E-10	2.00E+01	102
	1102	co-61	ground	(n,g)	co-62	ground	1.00E-10	2.00E+01	102
	4	co-62	ground	(n,n')	co-62	2.50E-02	2.54E-02	2.00E+01	30
	16	co-62	ground	(n,2n)	co-61	ground	6.79E+00	2.00E+01	22
	17	co-62	ground	(n,3n)	co-60	ground	1.63E+01	2.00E+01	12
27062.30y	1017	co-62	ground	(n,3n)	co-60	5.86E-02	1.63E+01	2.00E+01	12
	28	co-62	ground	(n,n')p	fe-61	ground	1.35E+01	2.00E+01	17
	103	co-62	ground	(n,p)	fe-62	ground	5.88E+00	2.00E+01	22
	102	co-62	ground	(n,g)	co-63	ground	1.00E-10	2.00E+01	102
	4	co-62	2.50E-02	(n,n')	co-62	ground	1.00E-10	2.00E+01	58
	16	co-62	2.50E-02	(n,2n)	co-61	ground	6.76E+00	2.00E+01	22
	17	co-62	2.50E-02	(n,3n)	co-60	ground	1.63E+01	2.00E+01	12
	1017	co-62	2.50E-02	(n,3n)	co-60	5.86E-02	1.63E+01	2.00E+01	12
	28	co-62	2.50E-02	(n,n')p	fe-61	ground	1.35E+01	2.00E+01	17
	103	co-62	2.50E-02	(n,p)	fe-62	ground	5.85E+00	2.00E+01	22
	102	co-62	2.50E-02	(n,g)	co-63	ground	1.00E-10	2.00E+01	102
	16	co-63	ground	(n,2n)	co-62	ground	8.53E+00	2.00E+01	23
	1016	co-63	ground	(n,2n)	co-62	2.50E-02	8.56E+00	2.00E+01	24
	17	co-63	ground	(n,3n)	co-61	ground	1.53E+01	2.00E+01	12
27063.30y	102	co-63	ground	(n,g)	co-64	ground	1.00E-10	2.00E+01	102
	16	co-64	ground	(n,2n)	co-63	ground	6.42E+00	2.00E+01	23
	17	co-64	ground	(n,3n)	co-62	ground	1.49E+01	2.00E+01	14
	1017	co-64	ground	(n,3n)	co-62	2.50E-02	1.50E+01	2.00E+01	14
27064.30y	102	co-64	ground	(n,g)	co-65	ground	1.00E-10	2.00E+01	102
	16	ni-57	ground	(n,2n)	ni-56	ground	1.04E+01	2.00E+01	9
	28	ni-57	ground	(n,n')p	co-56	ground	7.98E+00	2.00E+01	11
	22	ni-57	ground	(n,n')a	fe-53	ground	1.35E+01	2.00E+01	6

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
28058.30y	103	ni-57	ground	(n,p)	co-57	ground	8.50E-02	2.00E+01	14
	107	ni-57	ground	(n,a)	fe-54	ground	1.00E-10	2.00E+01	58
	102	ni-57	ground	(n,g)	ni-58	ground	1.00E-10	2.00E+01	110
	16	ni-58	ground	(n,2n)	ni-57	ground	1.24E+01	2.00E+01	11
	28	ni-58	ground	(n,n')p	co-57	ground	8.82E+00	2.00E+01	10
	22	ni-58	ground	(n,n')a	fe-54	ground	9.75E+00	2.00E+01	10
	103	ni-58	ground	(n,p)	co-58	ground	6.00E-01	2.00E+01	23
	1103	ni-58	ground	(n,p)	co-58	2.49E-02	6.25E-01	2.00E+01	23
	104	ni-58	ground	(n,d)	co-57	ground	1.02E+01	2.00E+01	14
	105	ni-58	ground	(n,t)	co-56	ground	1.40E+01	2.00E+01	13
28059.30y	107	ni-58	ground	(n,a)	fe-55	ground	5.00E-01	2.00E+01	22
	102	ni-58	ground	(n,g)	ni-59	ground	1.00E-10	2.00E+01	110
	16	ni-59	ground	(n,2n)	ni-58	ground	9.15E+00	2.00E+01	10
	28	ni-59	ground	(n,n')p	co-58	ground	1.00E+01	2.00E+01	9
	1028	ni-59	ground	(n,n')p	co-58	2.49E-02	1.00E+01	2.00E+01	9
	22	ni-59	ground	(n,n')a	fe-55	ground	9.50E+00	2.00E+01	10
	103	ni-59	ground	(n,p)	co-59	ground	8.00E-04	2.00E+01	36
	107	ni-59	ground	(n,a)	fe-56	ground	1.00E-10	2.00E+01	58
	102	ni-59	ground	(n,g)	ni-60	ground	1.00E-10	2.00E+01	110
	16	ni-60	ground	(n,2n)	ni-59	ground	1.16E+01	2.00E+01	8
28060.30y	28	ni-60	ground	(n,n')p	co-59	ground	1.05E+01	2.00E+01	9
	22	ni-60	ground	(n,n')a	fe-56	ground	9.75E+00	2.00E+01	10
	103	ni-60	ground	(n,p)	co-60	ground	2.07E+00	2.00E+01	20
	1103	ni-60	ground	(n,p)	co-60	5.86E-02	2.13E+00	2.00E+01	19
	104	ni-60	ground	(n,d)	co-59	ground	1.16E+01	2.00E+01	14
	105	ni-60	ground	(n,t)	co-58	ground	1.34E+01	2.00E+01	12
	1105	ni-60	ground	(n,t)	co-58	2.49E-02	1.34E+01	2.00E+01	12
	107	ni-60	ground	(n,a)	fe-57	ground	2.50E+00	2.00E+01	17
	102	ni-60	ground	(n,g)	ni-61	ground	1.00E-10	2.00E+01	110
	16	ni-61	ground	(n,2n)	ni-60	ground	7.95E+00	2.00E+01	11
28061.30y	28	ni-61	ground	(n,n')p	co-60	ground	1.25E+01	2.00E+01	7
	1028	ni-61	ground	(n,n')p	co-60	5.86E-02	1.26E+01	2.00E+01	7
	22	ni-61	ground	(n,n')a	fe-57	ground	1.05E+01	2.00E+01	9
	103	ni-61	ground	(n,p)	co-61	ground	2.25E+00	2.00E+01	18
	104	ni-61	ground	(n,d)	co-60	ground	1.19E+01	2.00E+01	14
	1104	ni-61	ground	(n,d)	co-60	5.86E-02	1.19E+01	2.00E+01	13
	105	ni-61	ground	(n,t)	co-59	ground	1.32E+01	2.00E+01	13
	107	ni-61	ground	(n,a)	fe-58	ground	1.50E-01	2.00E+01	22
	102	ni-61	ground	(n,g)	ni-62	ground	1.00E-10	2.00E+01	110
	16	ni-62	ground	(n,2n)	ni-61	ground	1.08E+01	2.00E+01	9
28062.30y	28	ni-62	ground	(n,n')p	co-61	ground	1.30E+01	2.00E+01	6
	22	ni-62	ground	(n,n')a	fe-58	ground	1.17E+01	2.00E+01	8
	103	ni-62	ground	(n,p)	co-62	ground	6.50E+00	2.00E+01	13
	1103	ni-62	ground	(n,p)	co-62	2.50E-02	6.53E+00	2.00E+01	13
	104	ni-62	ground	(n,d)	co-61	ground	1.32E+01	2.00E+01	13
	105	ni-62	ground	(n,t)	co-60	ground	1.37E+01	2.00E+01	11
	1105	ni-62	ground	(n,t)	co-60	5.86E-02	1.39E+01	2.00E+01	12
	107	ni-62	ground	(n,a)	fe-59	ground	5.50E+00	2.00E+01	14
	102	ni-62	ground	(n,g)	ni-63	ground	1.00E-10	2.00E+01	110
	16	ni-63	ground	(n,2n)	ni-62	ground	6.95E+00	2.00E+01	12
28063.30y	17	ni-63	ground	(n,3n)	ni-61	ground	1.77E+01	2.00E+01	3
	28	ni-63	ground	(n,n')p	co-62	ground	1.45E+01	2.00E+01	5
	1028	ni-63	ground	(n,n')p	co-62	2.50E-02	1.45E+01	2.00E+01	5
	22	ni-63	ground	(n,n')a	fe-59	ground	1.25E+01	2.00E+01	7
	103	ni-63	ground	(n,p)	co-63	ground	5.00E+00	2.00E+01	14
	107	ni-63	ground	(n,a)	fe-60	ground	3.50E+00	2.00E+01	15
	102	ni-63	ground	(n,g)	ni-64	ground	1.00E-10	2.00E+01	110
	16	ni-64	ground	(n,2n)	ni-63	ground	9.81E+00	2.00E+01	10
	17	ni-64	ground	(n,3n)	ni-62	ground	1.67E+01	2.00E+01	3
	28	ni-64	ground	(n,n')p	co-63	ground	1.55E+01	2.00E+01	4
28064.30y	22	ni-64	ground	(n,n')a	fe-60	ground	1.35E+01	2.00E+01	6
	103	ni-64	ground	(n,p)	co-64	ground	8.50E+00	2.00E+01	11
	104	ni-64	ground	(n,d)	co-63	ground	1.38E+01	2.00E+01	14
	105	ni-64	ground	(n,t)	co-62	ground	1.38E+01	2.00E+01	10

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
28065.30y	107	ni-64	ground	(n,a)	fe-61	ground	7.00E+00	2.00E+01	12
	102	ni-64	ground	(n,g)	ni-65	ground	1.00E-10	2.00E+01	110
	16	ni-65	ground	(n,2n)	ni-64	ground	6.19E+00	2.00E+01	13
	17	ni-65	ground	(n,3n)	ni-63	ground	1.60E+01	2.00E+01	3
	28	ni-65	ground	(n,n')p	co-64	ground	1.70E+01	2.00E+01	3
	22	ni-65	ground	(n,n')a	fe-61	ground	1.45E+01	2.00E+01	5
29062.30y	103	ni-65	ground	(n,p)	co-65	ground	7.50E+00	2.00E+01	9
	107	ni-65	ground	(n,a)	fe-62	ground	4.50E+00	2.00E+01	15
	102	ni-65	ground	(n,g)	ni-66	ground	1.00E-10	2.00E+01	110
	16	cu-62	ground	(n,2n)	cu-61	ground	9.03E+00	2.00E+01	16
	28	cu-62	ground	(n,n')p	ni-61	ground	6.47E+00	2.00E+01	15
	22	cu-62	ground	(n,n')a	co-58	ground	7.87E+00	2.00E+01	13
29063.30y	1022	cu-62	ground	(n,n')a	co-58	2.49E-02	7.89E+00	2.00E+01	13
	103	cu-62	ground	(n,p)	ni-62	ground	8.50E-02	2.00E+01	12
	107	cu-62	ground	(n,a)	co-59	ground	1.00E-10	2.00E+01	58
	102	cu-62	ground	(n,g)	cu-63	ground	1.00E-10	2.00E+01	112
	16	cu-63	ground	(n,2n)	cu-62	ground	1.10E+01	2.00E+01	10
	28	cu-63	ground	(n,n')p	ni-62	ground	7.75E+00	2.00E+01	26
29064.30y	22	cu-63	ground	(n,n')a	co-59	ground	1.20E+01	2.00E+01	17
	103	cu-63	ground	(n,p)	ni-63	ground	1.00E+00	2.00E+01	27
	107	cu-63	ground	(n,a)	co-60	5.86E-02	5.36E+00	2.00E+01	25
	1107	cu-63	ground	(n,a)	co-60	ground	5.30E+00	2.00E+01	25
	102	cu-63	ground	(n,g)	cu-64	ground	1.00E-10	2.00E+01	112
	16	cu-64	ground	(n,2n)	cu-63	ground	8.04E+00	2.00E+01	18
29065.30y	17	cu-64	ground	(n,3n)	cu-62	ground	1.91E+01	2.00E+01	5
	28	cu-64	ground	(n,n')p	ni-63	ground	7.81E+00	2.00E+01	15
	22	cu-64	ground	(n,n')a	co-60	ground	1.15E+01	2.00E+01	10
	1022	cu-64	ground	(n,n')a	co-60	5.86E-02	1.16E+01	2.00E+01	10
	103	cu-64	ground	(n,p)	ni-64	ground	1.60E+00	2.00E+01	14
	107	cu-64	ground	(n,a)	co-61	ground	3.90E+00	2.00E+01	18
29066.30y	102	cu-64	ground	(n,g)	cu-65	ground	1.00E-10	2.00E+01	112
	16	cu-65	ground	(n,2n)	cu-64	ground	1.01E+01	2.00E+01	35
	17	cu-65	ground	(n,3n)	cu-63	ground	1.81E+01	2.00E+01	9
	28	cu-65	ground	(n,n')p	ni-64	ground	8.30E+00	2.00E+01	25
	22	cu-65	ground	(n,n')a	co-61	ground	1.32E+01	2.00E+01	15
	103	cu-65	ground	(n,p)	ni-65	ground	2.50E+00	2.00E+01	22
30064.30y	107	cu-65	ground	(n,a)	co-62	ground	6.50E+00	2.00E+01	25
	1107	cu-65	ground	(n,a)	co-62	2.50E-02	6.53E+00	2.00E+01	25
	102	cu-65	ground	(n,g)	cu-66	ground	1.00E-10	2.00E+01	109
	16	cu-66	ground	(n,2n)	cu-65	ground	7.17E+00	2.00E+01	14
	17	cu-66	ground	(n,3n)	cu-64	ground	1.72E+01	2.00E+01	4
	28	cu-66	ground	(n,n')p	ni-65	ground	1.10E+01	2.00E+01	10
30066.30y	22	cu-66	ground	(n,n')a	co-62	ground	1.45E+01	2.00E+01	7
	1022	cu-66	ground	(n,n')a	co-62	2.50E-02	1.45E+01	2.00E+01	7
	103	cu-66	ground	(n,p)	ni-66	ground	3.60E+00	2.00E+01	18
	107	cu-66	ground	(n,a)	co-63	ground	1.12E+01	2.00E+01	10
	102	cu-66	ground	(n,g)	cu-67	ground	1.00E-10	2.00E+01	112
	16	zn-64	ground	(n,2n)	zn-63	ground	1.24E+01	2.00E+01	74
30067.30y	28	zn-64	ground	(n,n')p	cu-63	ground	1.20E+01	2.00E+01	16
	22	zn-64	ground	(n,n')a	ni-60	ground	1.43E+01	2.00E+01	14
	103	zn-64	ground	(n,p)	cu-64	ground	1.00E-01	2.00E+01	35
	107	zn-64	ground	(n,a)	ni-61	ground	6.30E+00	2.00E+01	24
	102	zn-64	ground	(n,g)	zn-65	ground	1.00E-10	2.00E+01	102
	16	zn-66	ground	(n,2n)	zn-65	ground	1.12E+01	2.00E+01	91
30066.30y	28	zn-66	ground	(n,n')p	cu-65	ground	1.31E+01	2.00E+01	16
	22	zn-66	ground	(n,n')a	ni-62	ground	1.48E+01	2.00E+01	13
	103	zn-66	ground	(n,p)	cu-66	ground	4.15E+00	2.00E+01	22
	107	zn-66	ground	(n,a)	ni-63	ground	7.85E+00	2.00E+01	24
	102	zn-66	ground	(n,g)	zn-67	ground	1.00E-10	2.00E+01	102
	16	zn-67	ground	(n,2n)	zn-66	ground	7.16E+00	2.00E+01	16
30067.30y	28	zn-67	ground	(n,n')p	cu-66	ground	1.31E+01	2.00E+01	17
	22	zn-67	ground	(n,n')a	ni-63	ground	1.50E+01	2.00E+01	13
	103	zn-67	ground	(n,p)	cu-67	ground	4.03E+00	2.00E+01	22
	107	zn-67	ground	(n,a)	ni-64	ground	5.17E+00	2.00E+01	23

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
30068.30y	102	zn-67	ground	(n,g)	zn-68	ground	1.00E-10	2.00E+01	102
	16	zn-68	ground	(n,2n)	zn-67	ground	1.03E+01	2.00E+01	15
	28	zn-68	ground	(n,n')p	cu-67	ground	1.42E+01	2.00E+01	16
	22	zn-68	ground	(n,n')a	ni-64	ground	1.55E+01	2.00E+01	12
	103	zn-68	ground	(n,p)	cu-68	ground	8.12E+00	2.00E+01	22
	1103	zn-68	ground	(n,p)	cu-68	7.21E-01	8.84E+00	2.00E+01	14
	107	zn-68	ground	(n,a)	ni-65	ground	9.29E+00	2.00E+01	22
	102	zn-68	ground	(n,g)	zn-69	4.39E-01	1.00E-10	2.00E+01	102
	1102	zn-68	ground	(n,g)	zn-69	ground	1.00E-10	2.00E+01	102
	16	zn-70	ground	(n,2n)	zn-79	ground	9.35E+00	2.00E+01	15
30070.30y	1016	zn-70	ground	(n,2n)	zn-69	1.57E-01	9.50E+00	2.00E+01	15
	22	zn-70	ground	(n,n')a	ni-66	ground	1.61E+01	2.00E+01	11
	103	zn-70	ground	(n,p)	cu-70	ground	9.66E+00	2.00E+01	14
	1103	zn-70	ground	(n,p)	cu-70	1.40E-01	9.80E+00	2.00E+01	14
	107	zn-70	ground	(n,a)	ni-67	ground	1.04E+01	2.00E+01	20
	102	zn-70	ground	(n,g)	zn-71	1.57E-01	1.00E-10	2.00E+01	102
	1102	zn-70	ground	(n,g)	zn-71	ground	1.00E-10	2.00E+01	102
	16	ga-69	ground	(n,2n)	ga-68	ground	1.05E+01	2.00E+01	96
	31069.30y	16	ga-71	ground	ga-70	ground	9.70E+00	2.00E+01	104
	102	ga-71	ground	(n,g)	ga-72	ground	1.00E-10	2.00E+01	101
32070.30y	16	ge-70	ground	(n,2n)	ge-69	ground	1.17E+01	2.00E+01	14
	28	ge-70	ground	(n,n')p	ga-69	ground	1.31E+01	2.00E+01	15
	22	ge-70	ground	(n,n')a	zn-66	ground	1.49E+01	2.00E+01	14
	103	ge-70	ground	(n,p)	ga-70	ground	5.49E+00	2.00E+01	21
	107	ge-70	ground	(n,a)	zn-67	ground	7.73E+00	2.00E+01	24
	102	ge-70	ground	(n,g)	ge-71	ground	1.00E-10	2.00E+01	102
	16	ge-72	ground	(n,2n)	ge-71	ground	1.09E+01	2.00E+01	18
	17	ge-72	ground	(n,3n)	ge-70	ground	1.84E+01	2.00E+01	7
	28	ge-72	ground	(n,n')p	ga-71	ground	1.42E+01	2.00E+01	16
	22	ge-72	ground	(n,n')a	zn-68	ground	1.58E+01	2.00E+01	12
32072.30y	103	ge-72	ground	(n,p)	ga-72	ground	7.82E+00	2.00E+01	21
	107	ge-72	ground	(n,a)	zn-69	4.39E-01	9.16E+00	2.00E+01	21
	102	ge-72	ground	(n,g)	ge-73	ground	1.00E-10	2.00E+01	102
	16	ge-73	ground	(n,2n)	ge-72	ground	6.87E+00	2.00E+01	20
	17	ge-73	ground	(n,3n)	ge-71	ground	1.78E+01	2.00E+01	8
	28	ge-73	ground	(n,n')p	ga-72	ground	1.45E+01	2.00E+01	16
	22	ge-73	ground	(n,n')a	zn-69	ground	1.60E+01	2.00E+01	11
	103	ge-73	ground	(n,p)	ga-73	ground	5.34E+00	2.00E+01	22
	107	ge-73	ground	(n,a)	zn-70	ground	6.65E+00	2.00E+01	22
	102	ge-73	ground	(n,g)	ge-74	ground	1.00E-10	2.00E+01	102
32073.30y	16	ge-74	ground	(n,2n)	ge-73	ground	1.03E+01	2.00E+01	20
	17	ge-74	ground	(n,3n)	ge-72	ground	1.72E+01	2.00E+01	11
	28	ge-74	ground	(n,n')p	ga-73	ground	1.55E+01	2.00E+01	15
	22	ge-74	ground	(n,n')a	zn-70	ground	1.70E+01	2.00E+01	10
	103	ge-74	ground	(n,p)	ga-74	ground	9.22E+00	2.00E+01	20
	107	ge-74	ground	(n,a)	zn-71	ground	1.10E+01	2.00E+01	18
	102	ge-74	ground	(n,g)	ge-75	ground	1.00E-10	2.00E+01	102
	1102	ge-74	ground	(n,g)	ge-75	1.40E-01	1.00E-10	2.00E+01	102
	16	ge-76	ground	(n,2n)	ge-75	ground	9.55E+00	2.00E+01	21
	17	ge-76	ground	(n,3n)	ge-74	ground	1.69E+01	2.00E+01	11
32074.30y	28	ge-76	ground	(n,n')p	ga-75	ground	1.56E+01	2.00E+01	15
	22	ge-76	ground	(n,n')a	zn-72	ground	1.82E+01	2.00E+01	8
	103	ge-76	ground	(n,p)	ga-76	ground	1.06E+01	2.00E+01	19
	107	ge-76	ground	(n,a)	zn-73	ground	1.31E+01	2.00E+01	17
	102	ge-76	ground	(n,g)	ge-77	ground	1.00E-10	2.00E+01	102
	1102	ge-76	ground	(n,g)	ge-77	1.60E-01	1.00E-10	2.00E+01	102
	16	as-75	ground	(n,2n)	as-74	ground	1.04E+01	2.00E+01	98
	103	as-75	ground	(n,p)	ge-75	ground	4.10E+00	2.00E+01	160
	107	as-75	ground	(n,a)	ga-72	ground	7.60E+00	2.00E+01	125
	102	as-75	ground	(n,g)	as-76	ground	1.00E-10	2.00E+01	102
34074.30y	16	se-74	ground	(n,2n)	se-73	ground	1.24E+01	2.00E+01	77
34076.30y	16	se-76	ground	(n,2n)	se-75	ground	1.15E+01	2.00E+01	86
34080.30y	16	se-80	ground	(n,2n)	se-79	ground	1.01E+01	2.00E+01	100
34082.30y	16	se-82	ground	(n,2n)	se-81	ground	9.38E+00	2.00E+01	109

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
35079.30y	16	br-79	ground	(n,2n)	br-78	ground	1.09E+01	2.00E+01	92
	103	br-79	ground	(n,p)	se-79	ground	4.34E+00	2.00E+01	20
	107	br-79	ground	(n,a)	as-76	ground	9.60E+00	2.00E+01	13
35081.30y	16	br-81	ground	(n,2n)	br-80	ground	1.03E+01	2.00E+01	96
	1016	br-81	ground	(n,2n)	br-80	8.59E-02	1.09E+01	2.00E+01	92
	22	br-81	ground	(n,n')a	as-77	ground	1.10E+01	2.00E+01	16
	102	br-81	ground	(n,g)	br-82	ground	1.00E-10	2.00E+01	135
37085.30y	16	rb-85	ground	(n,2n)	rb-84	ground	1.07E+01	2.00E+01	94
37087.30y	16	rb-87	ground	(n,2n)	rb-86	ground	1.04E+01	2.00E+01	97
38084.30y	16	sr-84	ground	(n,2n)	sr-83	ground	1.22E+01	2.00E+01	79
38086.30y	22	sr-86	ground	(n,n')a	kr-82	ground	1.35E+01	2.00E+01	5
	107	sr-86	ground	(n,a)	kr-83	1.46E-02	1.10E+01	2.00E+01	7
39089.30y	16	y-89	ground	(n,2n)	y-88	ground	1.16E+01	2.00E+01	13
	28	y-89	ground	(n,n')p	sr-88	ground	1.24E+01	2.00E+01	17
	103	y-89	ground	(n,p)	sr-89	ground	6.23E+00	2.00E+01	21
	104	y-89	ground	(n,d)	sr-88	ground	1.02E+01	2.00E+01	15
	107	y-89	ground	(n,a)	rb-86	5.60E-01	1.18E+01	2.00E+01	13
	1107	y-89	ground	(n,a)	rb-86	ground	1.19E+01	2.00E+01	14
	102	y-89	ground	(n,g)	y-90	ground	1.00E-10	2.00E+01	102
	16	zr-89	ground	(n,2n)	zr-88	ground	9.40E+00	2.00E+01	15
40089.30y	103	zr-89	ground	(n,p)	y-89	ground	2.03E+00	2.00E+01	18
	107	zr-89	ground	(n,a)	sr-86	ground	7.54E+00	2.00E+01	17
	102	zr-89	ground	(n,g)	zr-90	ground	1.00E-10	2.00E+01	102
	16	zr-90	ground	(n,2n)	zr-89	ground	1.21E+01	2.00E+01	11
40090.30y	28	zr-90	ground	(n,n')p	y-89	ground	1.39E+01	2.00E+01	16
	103	zr-90	ground	(n,p)	y-90	ground	7.19E+00	2.00E+01	22
	104	zr-90	ground	(n,d)	y-89	ground	1.21E+01	2.00E+01	14
	107	zr-90	ground	(n,a)	sr-87	ground	1.11E+01	2.00E+01	15
	102	zr-90	ground	(n,g)	zr-91	ground	1.00E-10	2.00E+01	102
	16	zr-91	ground	(n,2n)	zr-90	ground	7.27E+00	2.00E+01	19
40091.30y	17	zr-91	ground	(n,3n)	zr-89	ground	1.94E+01	2.00E+01	4
	28	zr-91	ground	(n,n')p	y-90	ground	1.39E+01	2.00E+01	16
	103	zr-91	ground	(n,p)	y-91	ground	6.43E+00	2.00E+01	18
	104	zr-91	ground	(n,d)	y-90	ground	1.20E+01	2.00E+01	13
	107	zr-91	ground	(n,a)	sr-88	ground	7.11E+00	2.00E+01	17
	102	zr-91	ground	(n,g)	zr-92	ground	1.00E-10	2.00E+01	102
40092.30y	16	zr-92	ground	(n,2n)	zr-91	ground	8.73E+00	2.00E+01	26
	17	zr-92	ground	(n,3n)	zr-90	ground	1.60E+01	2.00E+01	13
	28	zr-92	ground	(n,n')p	y-91	ground	1.49E+01	2.00E+01	13
	103	zr-92	ground	(n,p)	y-92	ground	8.51E+00	2.00E+01	18
	104	zr-92	ground	(n,d)	y-91	ground	1.31E+01	2.00E+01	13
	107	zr-92	ground	(n,a)	sr-89	ground	9.37E+00	2.00E+01	16
	102	zr-92	ground	(n,g)	zr-93	ground	1.00E-10	2.00E+01	102
40093.30y	16	zr-93	ground	(n,2n)	zr-92	ground	6.83E+00	2.00E+01	24
	17	zr-93	ground	(n,3n)	zr-91	ground	1.56E+01	2.00E+01	14
	103	zr-93	ground	(n,p)	y-93	ground	7.76E+00	2.00E+01	18
	107	zr-93	ground	(n,a)	sr-90	ground	8.27E+00	2.00E+01	17
	102	zr-93	ground	(n,g)	zr-94	ground	1.00E-10	2.00E+01	102
40094.30y	16	zr-94	ground	(n,2n)	zr-93	ground	8.28E+00	2.00E+01	25
	17	zr-94	ground	(n,3n)	zr-92	ground	1.51E+01	2.00E+01	15
	22	zr-94	ground	(n,n')a	sr-90	ground	1.07E+01	2.00E+01	20
	103	zr-94	ground	(n,p)	y-94	ground	9.87E+00	2.00E+01	18
	107	zr-94	ground	(n,a)	sr-91	ground	1.06E+01	2.00E+01	16
	102	zr-94	ground	(n,g)	zr-95	ground	1.00E-10	2.00E+01	102
40095.30y	16	zr-95	ground	(n,2n)	zr-94	ground	6.54E+00	2.00E+01	27
	17	zr-95	ground	(n,3n)	zr-93	ground	1.48E+01	2.00E+01	14
	103	zr-95	ground	(n,p)	y-95	ground	9.28E+00	2.00E+01	18
	107	zr-95	ground	(n,a)	sr-92	ground	9.75E+00	2.00E+01	16
	102	zr-95	ground	(n,g)	zr-96	ground	1.00E-10	2.00E+01	102
40096.30y	16	zr-96	ground	(n,2n)	zr-95	ground	7.94E+00	2.00E+01	26
40097.30y	16	zr-97	ground	(n,2n)	zr-96	ground	5.64E+00	2.00E+01	28
	17	zr-97	ground	(n,3n)	zr-95	ground	1.36E+01	2.00E+01	15
	107	zr-97	ground	(n,a)	sr-94	ground	1.11E+01	2.00E+01	16
	102	zr-97	ground	(n,g)	zr-98	ground	1.00E-10	2.00E+01	102

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
41091.30y	16	nb-91	ground	(n,2n)	nb-90	ground	1.23E+01	2.00E+01	15
	103	nb-91	ground	(n,p)	zr-91	ground	3.77E+00	2.00E+01	18
	102	nb-91	ground	(n,g)	nb-92	1.35E-01	1.00E-10	2.00E+01	102
	1102	nb-91	ground	(n,g)	nb-92	ground	1.00E-10	2.00E+01	102
41091.31y	16	nb-91	1.05E-01	(n,2n)	nb-90	ground	1.21E+01	2.00E+01	15
	103	nb-91	1.05E-01	(n,p)	zr-91	ground	3.67E+00	2.00E+01	18
	102	nb-91	1.05E-01	(n,g)	nb-92	1.35E-01	1.00E-10	2.00E+01	102
	1102	nb-91	1.05E-01	(n,g)	nb-92	ground	1.00E-10	2.00E+01	102
41092.30y	16	nb-92	ground	(n,2n)	nb-91	1.05E-01	8.09E+00	2.00E+01	16
	103	nb-92	ground	(n,p)	zr-92	ground	3.00E+00	2.00E+01	18
	102	nb-92	ground	(n,g)	nb-93	ground	1.00E-10	2.00E+01	102
41092.31y	16	nb-92	1.35E-01	(n,2n)	nb-91	1.05E-01	7.96E+00	2.00E+01	16
	103	nb-92	1.35E-01	(n,p)	zr-92	ground	2.86E+00	2.00E+01	18
	102	nb-92	1.35E-01	(n,g)	nb-93	ground	1.00E-10	2.00E+01	102
41093.30y	4	nb-93	ground	(n,n')	nb-93	3.04E-02	3.08E-02	2.00E+01	23
	16	nb-93	ground	(n,2n)	nb-92	ground	8.93E+00	2.00E+01	21
	1016	nb-93	ground	(n,2n)	nb-92	1.35E-01	9.06E+00	2.00E+01	21
	17	nb-93	ground	(n,3n)	nb-91	ground	1.69E+01	2.00E+01	10
	1017	nb-93	ground	(n,3n)	nb-91	1.05E-01	1.70E+01	2.00E+01	11
	22	nb-93	ground	(n,n') a	y-89	9.09E-01	2.88E+00	2.00E+01	6
	103	nb-93	ground	(n,p)	zr-93	ground	5.07E+00	2.00E+01	18
	107	nb-93	ground	(n,a)	y-90	ground	4.50E+00	2.00E+01	16
	102	nb-93	ground	(n,g)	nb-94	ground	1.00E-10	2.00E+01	102
	16	nb-94	ground	(n,2n)	nb-93	ground	7.31E+00	2.00E+01	24
41094.30y	17	nb-94	ground	(n,3n)	nb-92	1.35E-01	1.64E+01	2.00E+01	13
	103	nb-94	ground	(n,p)	zr-94	ground	4.09E+00	2.00E+01	18
	102	nb-94	ground	(n,g)	nb-95	ground	1.00E-10	2.00E+01	102
	16	nb-95	ground	(n,2n)	nb-94	ground	8.58E+00	2.00E+01	25
41095.30y	17	nb-95	ground	(n,3n)	nb-93	ground	1.59E+01	2.00E+01	13
	103	nb-95	ground	(n,p)	zr-95	ground	6.12E+00	2.00E+01	18
	102	nb-95	ground	(n,g)	nb-96	ground	1.00E-10	2.00E+01	102
	16	nb-96	ground	(n,2n)	nb-95	ground	6.96E+00	2.00E+01	25
41096.30y	17	nb-96	ground	(n,3n)	nb-94	ground	1.56E+01	2.00E+01	14
	103	nb-96	ground	(n,p)	zr-96	ground	5.15E+00	2.00E+01	18
	102	nb-96	ground	(n,g)	nb-97	ground	1.00E-10	2.00E+01	102
	16	nb-97	ground	(n,2n)	nb-96	ground	8.15E+00	2.00E+01	26
41097.30y	17	nb-97	ground	(n,3n)	nb-95	ground	1.51E+01	2.00E+01	15
	103	nb-97	ground	(n,p)	zr-97	ground	7.65E+00	2.00E+01	18
	102	nb-97	ground	(n,g)	nb-98	ground	1.00E-10	2.00E+01	102
	16	nb-98	ground	(n,2n)	nb-97	ground	6.04E+00	2.00E+01	26
41098.30y	17	nb-98	ground	(n,3n)	nb-96	ground	1.42E+01	2.00E+01	16
	103	nb-98	ground	(n,p)	zr-98	ground	7.19E+00	2.00E+01	18
	102	nb-98	ground	(n,g)	nb-99	ground	1.00E-10	2.00E+01	102
	16	nb-100	ground	(n,2n)	nb-99	ground	5.15E+00	2.00E+01	28
41100.30y	17	nb-100	ground	(n,3n)	nb-98	ground	1.29E+01	2.00E+01	18
	103	nb-100	ground	(n,p)	zr-100	ground	8.52E+00	2.00E+01	18
	102	nb-100	ground	(n,g)	nb-101	ground	1.00E-10	2.00E+01	102
	16	mo-90	ground	(n,2n)	mo-89	ground	1.34E+01	2.00E+01	24
42090.30y	102	mo-90	ground	(n,g)	mo-91	ground	1.00E-10	2.00E+01	102
	16	mo-91	ground	(n,2n)	mo-90	ground	1.02E+01	2.00E+01	34
42091.30y	102	mo-91	ground	(n,g)	mo-92	ground	1.00E-10	2.00E+01	102
	16	mo-92	ground	(n,2n)	mo-91	ground	1.28E+01	2.00E+01	25
	28	mo-92	ground	(n,n') p	nb-91	ground	1.33E+01	2.00E+01	16
	1028	mo-92	ground	(n,n') p	nb-91	1.05E-01	1.33E+01	2.00E+01	16
	103	mo-92	ground	(n,p)	nb-92	1.35E-01	5.56E+00	2.00E+01	31
	1103	mo-92	ground	(n,p)	nb-92	ground	5.57E+00	2.00E+01	21
	104	mo-92	ground	(n,d)	nb-91	1.05E-01	1.16E+01	2.00E+01	15
	105	mo-92	ground	(n,t)	nb-90	ground	1.34E+01	2.00E+01	7
	107	mo-92	ground	(n,a)	zr-89	ground	9.75E+00	2.00E+01	17
	102	mo-92	ground	(n,g)	mo-93	ground	1.00E-10	2.00E+01	102
	16	mo-93	ground	(n,2n)	mo-92	ground	8.16E+00	2.00E+01	41
	103	mo-93	ground	(n,p)	nb-93	ground	4.79E+00	2.00E+01	29
	104	mo-93	ground	(n,d)	nb-92	1.35E-01	1.16E+01	2.00E+01	16
42093.30y	107	mo-93	ground	(n,a)	zr-90	ground	5.77E+00	2.00E+01	16

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
42093.31y	102	mo-93	ground	(n,g)	mo-94	ground	1.00E-10	2.00E+01	102
	16	mo-93	2.43E+00	(n,2n)	mo-92	ground	5.70E+00	2.00E+01	49
	17	mo-93	2.43E+00	(n,3n)	mo-91	ground	1.85E+01	2.00E+01	7
	103	mo-93	2.43E+00	(n,p)	nb-93	ground	2.33E+00	2.00E+01	28
	104	mo-93	2.43E+00	(n,d)	nb-92	1.35E-01	9.19E+00	2.00E+01	16
	107	mo-93	2.43E+00	(n,a)	zr-90	ground	3.31E+00	2.00E+01	16
42094.30y	102	mo-93	2.43E+00	(n,g)	mo-94	ground	1.00E-10	2.00E+01	102
	16	mo-94	ground	(n,2n)	mo-93	ground	9.77E+00	2.00E+01	10
	1016	mo-94	ground	(n,2n)	mo-93	2.43E+00	1.22E+01	2.00E+01	6
	17	mo-94	ground	(n,3n)	mo-92	ground	1.79E+01	2.00E+01	3
	28	mo-94	ground	(n,n')p	nb-93	ground	1.43E+01	2.00E+01	16
	103	mo-94	ground	(n,p)	nb-94	ground	7.24E+00	2.00E+01	22
	104	mo-94	ground	(n,d)	nb-93	ground	1.25E+01	2.00E+01	14
	105	mo-94	ground	(n,t)	nb-92	1.35E-01	1.38E+01	2.00E+01	14
	107	mo-94	ground	(n,a)	zr-91	ground	8.24E+00	2.00E+01	16
	102	mo-94	ground	(n,g)	mo-95	ground	1.00E-10	2.00E+01	102
42095.30y	16	mo-95	ground	(n,2n)	mo-94	ground	7.45E+00	2.00E+01	43
	17	mo-95	ground	(n,3n)	mo-93	ground	1.72E+01	2.00E+01	11
	28	mo-95	ground	(n,n')p	nb-94	ground	1.44E+01	2.00E+01	16
	22	mo-95	ground	(n,n')a	zr-91	ground	1.57E+01	2.00E+01	12
	103	mo-95	ground	(n,p)	nb-95	ground	6.09E+00	2.00E+01	24
	104	mo-95	ground	(n,d)	nb-94	ground	1.26E+01	2.00E+01	14
	107	mo-95	ground	(n,a)	zr-92	ground	6.93E+00	2.00E+01	23
	102	mo-95	ground	(n,g)	mo-96	ground	1.00E-10	2.00E+01	102
	16	mo-96	ground	(n,2n)	mo-95	ground	9.25E+00	2.00E+01	37
	17	mo-96	ground	(n,3n)	mo-94	ground	1.67E+01	2.00E+01	13
42096.30y	28	mo-96	ground	(n,n')p	nb-95	ground	1.51E+01	2.00E+01	14
	22	mo-96	ground	(n,n')a	zr-92	ground	1.62E+01	2.00E+01	12
	103	mo-96	ground	(n,p)	nb-96	ground	8.36E+00	2.00E+01	20
	104	mo-96	ground	(n,d)	nb-95	ground	1.33E+01	2.00E+01	14
	107	mo-96	ground	(n,a)	zr-93	ground	9.34E+00	2.00E+01	22
	102	mo-96	ground	(n,g)	mo-97	ground	1.00E-10	2.00E+01	102
	16	mo-97	ground	(n,2n)	mo-96	ground	6.89E+00	2.00E+01	45
	17	mo-97	ground	(n,3n)	mo-95	ground	1.61E+01	2.00E+01	15
	103	mo-97	ground	(n,p)	nb-97	ground	7.07E+00	2.00E+01	25
	107	mo-97	ground	(n,a)	zr-94	ground	7.89E+00	2.00E+01	16
42097.30y	102	mo-97	ground	(n,g)	mo-98	ground	1.00E-10	2.00E+01	102
	16	mo-98	ground	(n,2n)	mo-97	ground	8.73E+00	2.00E+01	39
	17	mo-98	ground	(n,3n)	mo-96	ground	1.56E+01	2.00E+01	17
	103	mo-98	ground	(n,p)	nb-98	ground	9.76E+00	2.00E+01	26
	107	mo-98	ground	(n,a)	zr-95	ground	1.01E+01	2.00E+01	16
	102	mo-98	ground	(n,g)	mo-99	ground	1.00E-10	2.00E+01	102
42098.30y	16	mo-99	ground	(n,2n)	mo-98	ground	5.98E+00	2.00E+01	48
	17	mo-99	ground	(n,3n)	mo-97	ground	1.47E+01	2.00E+01	19
	103	mo-99	ground	(n,p)	nb-99	ground	8.20E+00	2.00E+01	27
	107	mo-99	ground	(n,a)	zr-96	ground	8.07E+00	2.00E+01	16
	102	mo-99	ground	(n,g)	mo-100	ground	1.00E-10	2.00E+01	102
	16	mo-100	ground	(n,2n)	mo-99	ground	8.38E+00	2.00E+01	40
42099.30y	17	mo-100	ground	(n,3n)	mo-98	ground	1.44E+01	2.00E+01	21
	103	mo-100	ground	(n,p)	nb-100	ground	1.11E+01	2.00E+01	24
	107	mo-100	ground	(n,a)	zr-97	ground	1.08E+01	2.00E+01	16
	102	mo-100	ground	(n,g)	mo-101	ground	1.00E-10	2.00E+01	102
	16	mo-101	ground	(n,2n)	mo-100	ground	5.45E+00	2.00E+01	50
	17	mo-101	ground	(n,3n)	mo-99	ground	1.38E+01	2.00E+01	23
42100.30y	103	mo-101	ground	(n,p)	nb-101	ground	9.33E+00	2.00E+01	22
	107	mo-101	ground	(n,a)	zr-98	ground	9.72E+00	2.00E+01	16
	102	mo-101	ground	(n,g)	mo-102	ground	1.00E-10	2.00E+01	102
	16	tc-99	ground	(n,2n)	tc-98	ground	7.36E+00	2.00E+01	22
	17	tc-99	ground	(n,3n)	tc-97	ground	1.64E+01	2.00E+01	12
	28	tc-99	ground	(n,n')p	mo-98	ground	1.20E+01	2.00E+01	15
42101.30y	22	tc-99	ground	(n,n')a	nb-95	ground	1.01E+01	2.00E+01	5
	103	tc-99	ground	(n,p)	mo-99	ground	3.57E+00	2.00E+01	22
	104	tc-99	ground	(n,d)	mo-98	ground	1.02E+01	2.00E+01	15
	105	tc-99	ground	(n,t)	mo-97	ground	1.08E+01	2.00E+01	15
	102	mo-101	ground	(n,g)	mo-102	ground	1.00E-10	2.00E+01	102
	16	tc-99	ground	(n,2n)	tc-98	ground	7.36E+00	2.00E+01	22
43099.30y	17	tc-99	ground	(n,3n)	tc-97	ground	1.64E+01	2.00E+01	12
	28	tc-99	ground	(n,n')p	mo-98	ground	1.20E+01	2.00E+01	15
	22	tc-99	ground	(n,n')a	nb-95	ground	1.01E+01	2.00E+01	5
	103	tc-99	ground	(n,p)	mo-99	ground	3.57E+00	2.00E+01	22
	104	tc-99	ground	(n,d)	mo-98	ground	1.02E+01	2.00E+01	15
	105	tc-99	ground	(n,t)	mo-97	ground	1.08E+01	2.00E+01	15

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
43099.31y	107	tc-99	ground	(n,a)	nb-96	ground	7.53E+00	2.00E+01	8
	102	tc-99	ground	(n,g)	tc-100	ground	1.00E-10	2.00E+01	102
	16	tc-99	1.43E-01	(n,2n)	tc-98	ground	7.21E+00	2.00E+01	22
	17	tc-99	1.43E-01	(n,3n)	tc-97	ground	1.63E+01	2.00E+01	12
	28	tc-99	1.43E-01	(n,n')p	mo-98	ground	1.19E+01	2.00E+01	15
	22	tc-99	1.43E-01	(n,n')a	nb-95	ground	1.00E+01	2.00E+01	5
	103	tc-99	1.43E-01	(n,p)	mo-99	ground	3.42E+00	2.00E+01	22
	104	tc-99	1.43E-01	(n,d)	mo-98	ground	1.01E+01	2.00E+01	15
	105	tc-99	1.43E-01	(n,t)	mo-97	ground	1.07E+01	2.00E+01	15
	107	tc-99	1.43E-01	(n,a)	nb-96	ground	7.39E+00	2.00E+01	8
	102	tc-99	1.43E-01	(n,g)	tc-100	ground	1.00E-10	2.00E+01	102
	4	rh-103	ground	(n,n')	rh-103	3.97E-02	4.02E-02	2.00E+01	29
45103.30y	16	rh-103	ground	(n,2n)	rh-102	ground	9.70E+00	2.00E+01	104
46110.30y	102	pd-110	ground	(n,g)	pd-111	1.72E-01	1.00E-10	2.00E+01	102
	1102	pd-110	ground	(n,g)	pd-111	ground	1.00E-10	2.00E+01	102
47106.30y	16	ag-106	ground	(n,2n)	ag-105	ground	7.99E+00	2.00E+01	17
	17	ag-106	ground	(n,3n)	ag-104	ground	1.79E+01	2.00E+01	6
47106.31y	102	ag-106	ground	(n,g)	ag-107	ground	1.00E-10	2.00E+01	102
	16	ag-106	3.00E-01	(n,2n)	ag-105	ground	7.69E+00	2.00E+01	17
	17	ag-106	3.00E-01	(n,3n)	ag-104	ground	1.76E+01	2.00E+01	7
	102	ag-106	3.00E-01	(n,g)	ag-107	ground	1.00E-10	2.00E+01	102
47107.30y	16	ag-107	ground	(n,2n)	ag-106	ground	9.63E+00	2.00E+01	7
	1016	ag-107	ground	(n,2n)	ag-106	3.00E-01	9.94E+00	2.00E+01	6
	17	ag-107	ground	(n,3n)	ag-105	ground	1.76E+01	2.00E+01	2
	103	ag-107	ground	(n,p)	pd-107	ground	4.50E+00	2.00E+01	20
	107	ag-107	ground	(n,a)	rh-104	1.29E-01	5.50E+00	2.00E+01	5
	102	ag-107	ground	(n,g)	ag-108	1.10E-01	1.00E-10	2.00E+01	102
47108.30y	1102	ag-107	ground	(n,g)	ag-108	ground	1.00E-10	2.00E+01	102
	16	ag-108	ground	(n,2n)	ag-107	ground	7.34E+00	2.00E+01	20
	17	ag-108	ground	(n,3n)	ag-106	ground	1.70E+01	2.00E+01	9
	102	ag-108	ground	(n,g)	ag-109	ground	1.00E-10	2.00E+01	102
47108.31y	16	ag-108	1.10E-01	(n,2n)	ag-107	ground	7.23E+00	2.00E+01	20
	17	ag-108	1.10E-01	(n,3n)	ag-106	ground	1.69E+01	2.00E+01	9
	102	ag-108	1.10E-01	(n,g)	ag-109	ground	1.00E-10	2.00E+01	102
47109.30y	16	ag-109	ground	(n,2n)	ag-108	ground	9.27E+00	2.00E+01	21
	1016	ag-109	ground	(n,2n)	ag-108	1.10E-01	9.39E+00	2.00E+01	13
	17	ag-109	ground	(n,3n)	ag-107	ground	1.66E+01	2.00E+01	11
	103	ag-109	ground	(n,p)	pd-109	ground	3.50E+00	2.00E+01	16
	107	ag-109	ground	(n,a)	rh-106	ground	5.50E+00	2.00E+01	12
	102	ag-109	ground	(n,g)	ag-110	1.18E-01	1.00E-10	2.00E+01	102
47110.30y	1102	ag-109	ground	(n,g)	ag-110	ground	1.00E-10	2.00E+01	102
	16	ag-110	ground	(n,2n)	ag-109	ground	6.87E+00	2.00E+01	22
	17	ag-110	ground	(n,3n)	ag-108	ground	1.61E+01	2.00E+01	13
	102	ag-110	ground	(n,g)	ag-111	ground	1.00E-10	2.00E+01	97
47110.31y	16	ag-110	1.18E-01	(n,2n)	ag-109	ground	6.75E+00	2.00E+01	23
	17	ag-110	1.18E-01	(n,3n)	ag-108	ground	1.60E+01	2.00E+01	13
	102	ag-110	1.18E-01	(n,g)	ag-111	ground	1.00E-10	2.00E+01	98
	16	cd-106	ground	(n,2n)	cd-105	ground	1.15E+01	2.00E+01	86
48106.30y	4	cd-111	ground	(n,n')	cd-111	3.96E-01	4.00E-01	2.00E+01	26
48111.30y	103	cd-111	ground	(n,p)	ag-111	ground	7.30E+00	2.00E+01	128
	107	cd-112	ground	(n,a)	pd-109	ground	9.30E+00	2.00E+01	108
48112.30y	16	cd-116	ground	(n,2n)	cd-115	ground	8.80E+00	2.00E+01	113
49113.30y	16	in-113	ground	(n,2n)	in-112	ground	9.90E+00	2.00E+01	102
	28	in-113	ground	(n,n')p	cd-112	ground	1.28E+01	2.00E+01	17
	1028	in-113	ground	(n,n')p	cd-112	ground	1.30E+01	2.00E+01	17
	103	in-113	ground	(n,p)	cd-113	2.64E-01	6.71E+00	2.00E+01	22
	1103	in-113	ground	(n,p)	cd-113	ground	6.44E+00	2.00E+01	22
	107	in-113	ground	(n,a)	ag-110	1.18E-01	1.15E+01	2.00E+01	14
	1107	in-113	ground	(n,a)	ag-110	ground	1.14E+01	2.00E+01	14
	102	in-113	ground	(n,g)	in-114	1.90E-01	1.00E-10	2.00E+01	102
	1102	in-113	ground	(n,g)	in-114	ground	1.00E-10	2.00E+01	102
	16	in-115	ground	(n,2n)	in-114	ground	9.30E+00	2.00E+01	108
	1016	in-115	ground	(n,2n)	in-114	1.90E-01	9.40E+00	2.00E+01	107
	28	in-115	ground	(n,n')p	cd-114	ground	1.35E+01	2.00E+01	16

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
50112.30y	1028	in-115	ground	(n,n')p	cd-114	ground	1.36E+01	2.00E+01	16
	103	in-115	ground	(n,p)	cd-115	ground	7.54E+00	2.00E+01	23
	1103	in-115	ground	(n,p)	cd-115	1.73E-01	7.72E+00	2.00E+01	23
	107	in-115	ground	(n,a)	ag-109	ground	1.23E+01	2.00E+01	13
	102	in-115	ground	(n,g)	in-116	2.90E-01	1.00E-10	2.00E+01	102
	1102	in-115	ground	(n,g)	in-116	1.27E-01	1.00E-10	2.00E+01	102
	2102	in-115	ground	(n,g)	in-116	ground	1.00E-10	2.00E+01	102
	16	sn-112	ground	(n,2n)	sn-111	ground	1.12E+01	2.00E+01	89
	28	sn-112	ground	(n,n')p	in-111	ground	1.44E+01	2.00E+01	16
	1028	sn-112	ground	(n,n')p	in-111	ground	1.46E+01	2.00E+01	15
50114.30y	103	sn-112	ground	(n,p)	in-112	1.55E-01	7.14E+00	2.00E+01	19
	1103	sn-112	ground	(n,p)	in-112	ground	6.98E+00	2.00E+01	19
	104	sn-112	ground	(n,d)	in-111	ground	8.46E+00	2.00E+01	16
	102	sn-112	ground	(n,g)	sn-113	7.93E-02	1.00E-10	2.00E+01	102
	1102	sn-112	ground	(n,g)	sn-113	ground	1.00E-10	2.00E+01	102
	16	sn-114	ground	(n,2n)	sn-113	ground	1.04E+01	2.00E+01	15
	1016	sn-114	ground	(n,2n)	sn-113	7.93E-02	1.05E+01	2.00E+01	15
	28	sn-114	ground	(n,n')p	in-113	3.92E-01	1.53E+01	2.00E+01	15
	1028	sn-114	ground	(n,n')p	in-113	3.92E-01	1.55E+01	2.00E+01	14
	103	sn-114	ground	(n,p)	in-114	ground	8.28E+00	2.00E+01	21
50115.30y	1103	sn-114	ground	(n,p)	in-114	1.90E-01	8.47E+00	2.00E+01	21
	102	sn-114	ground	(n,g)	sn-115	ground	1.00E-10	2.00E+01	102
	16	sn-115	ground	(n,2n)	sn-114	ground	7.61E+00	2.00E+01	15
	28	sn-115	ground	(n,n')p	in-114	1.90E-01	1.56E+01	2.00E+01	13
	1028	sn-115	ground	(n,n')p	in-114	1.90E-01	1.59E+01	2.00E+01	13
	103	sn-115	ground	(n,p)	in-115	ground	6.76E+00	2.00E+01	20
	1103	sn-115	ground	(n,p)	in-115	3.36E-01	7.53E+00	2.00E+01	19
	102	sn-115	ground	(n,g)	sn-116	ground	1.00E-10	2.00E+01	102
	16	sn-116	ground	(n,2n)	sn-115	ground	9.65E+00	2.00E+01	15
	28	sn-116	ground	(n,n')p	in-115	ground	1.61E+01	2.00E+01	13
50116.30y	1028	sn-116	ground	(n,n')p	in-115	ground	1.62E+01	2.00E+01	13
	103	sn-116	ground	(n,p)	in-116	ground	9.55E+00	2.00E+01	21
	1103	sn-116	ground	(n,p)	in-116	1.27E-01	9.68E+00	2.00E+01	21
	102	sn-116	ground	(n,g)	sn-117	3.15E-01	1.00E-10	2.00E+01	102
	1102	sn-116	ground	(n,g)	sn-117	ground	1.00E-10	2.00E+01	102
	16	sn-117	ground	(n,2n)	sn-116	ground	7.00E+00	2.00E+01	16
	103	sn-117	ground	(n,p)	in-117	ground	7.70E+00	2.00E+01	15
	1103	sn-117	ground	(n,p)	in-117	3.15E-01	8.02E+00	2.00E+01	15
	102	sn-117	ground	(n,g)	sn-118	ground	1.00E-10	2.00E+01	102
	16	sn-118	ground	(n,2n)	sn-117	ground	9.41E+00	2.00E+01	15
50118.30y	1016	sn-118	ground	(n,2n)	sn-117	3.15E-01	9.72E+00	2.00E+01	15
	103	sn-118	ground	(n,p)	in-118	ground	1.05E+01	2.00E+01	14
	107	sn-118	ground	(n,a)	cd-115	ground	8.90E+00	2.00E+01	112
	102	sn-118	ground	(n,g)	sn-119	8.95E-02	1.00E-10	2.00E+01	102
	1102	sn-118	ground	(n,g)	sn-119	ground	1.00E-10	2.00E+01	102
	16	sn-119	ground	(n,2n)	sn-118	ground	6.54E+00	2.00E+01	16
	103	sn-119	ground	(n,p)	in-119	ground	8.55E+00	2.00E+01	15
	1103	sn-119	ground	(n,p)	in-119	9.00E-02	8.65E+00	2.00E+01	14
	102	sn-119	ground	(n,g)	sn-120	ground	1.00E-10	2.00E+01	102
	16	sn-120	ground	(n,2n)	sn-119	ground	9.18E+00	2.00E+01	15
50120.30y	1016	sn-120	ground	(n,2n)	sn-119	8.95E-02	9.27E+00	2.00E+01	15
	103	sn-120	ground	(n,p)	in-120	ground	1.16E+01	2.00E+01	14
	102	sn-120	ground	(n,g)	sn-121	ground	1.00E-10	2.00E+01	102
	16	sn-122	ground	(n,2n)	sn-121	ground	8.89E+00	2.00E+01	15
	103	sn-122	ground	(n,p)	in-122	ground	1.25E+01	2.00E+01	14
	102	sn-122	ground	(n,g)	sn-123	ground	1.00E-10	2.00E+01	102
	16	sn-124	ground	(n,2n)	sn-123	ground	8.56E+00	2.00E+01	16
	103	sn-124	ground	(n,p)	in-124	ground	1.33E+01	2.00E+01	14
	102	sn-124	ground	(n,g)	sn-125	2.75E-02	1.00E-10	2.00E+01	102
	1102	sn-124	ground	(n,g)	sn-125	ground	1.00E-10	2.00E+01	102
51121.30y	16	sb-121	ground	(n,2n)	sb-120	ground	9.60E+00	2.00E+01	105
	28	sb-121	ground	(n,n')p	sn-120	ground	1.27E+01	2.00E+01	17
	1028	sb-121	ground	(n,n')p	sn-120	ground	1.27E+01	2.00E+01	17
	103	sb-121	ground	(n,p)	sn-121	6.30E-03	6.72E+00	2.00E+01	24

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
51123.30y	1103	sb-121	ground	(n,p)	sn-121	ground	6.71E+00	2.00E+01	24
	102	sb-121	ground	(n,g)	sb-122	1.63E-01	1.00E-10	2.00E+01	102
	1102	sb-121	ground	(n,g)	sb-122	ground	1.00E-10	2.00E+01	102
	16	sb-123	ground	(n,2n)	sb-122	ground	9.30E+00	2.00E+01	108
	28	sb-123	ground	(n,n')p	sn-122	ground	1.34E+01	2.00E+01	16
	1028	sb-123	ground	(n,n')p	sn-122	ground	1.34E+01	2.00E+01	16
	103	sb-123	ground	(n,p)	sn-123	ground	7.70E+00	2.00E+01	22
	1103	sb-123	ground	(n,p)	sn-123	2.47E-02	7.73E+00	2.00E+01	22
	102	sb-123	ground	(n,g)	sb-124	3.50E-02	1.00E-10	2.00E+01	102
	1102	sb-123	ground	(n,g)	sb-124	1.07E-02	1.00E-10	2.00E+01	102
	2102	sb-123	ground	(n,g)	sb-124	ground	1.00E-10	2.00E+01	102
53127.30y	16	i-127	ground	(n,2n)	i-126	ground	9.30E+00	2.00E+01	108
55133.30y	16	cs-133	ground	(n,2n)	cs-132	ground	9.60E+00	2.00E+01	105
58140.30y	16	ce-140	ground	(n,2n)	ce-139	ground	9.30E+00	2.00E+01	108
	1016	ce-140	ground	(n,2n)	ce-139	7.54E-01	1.01E+01	2.00E+01	101
58142.30y	16	ce-142	ground	(n,2n)	ce-141	ground	7.21E+00	2.00E+01	130
59141.30y	16	pr-141	ground	(n,2n)	pr-140	ground	9.60E+00	2.00E+01	105
60142.30y	16	nd-142	ground	(n,2n)	nd-141	ground	1.00E+01	2.00E+01	101
60148.30y	16	nd-148	ground	(n,2n)	nd-147	ground	7.60E+00	2.00E+01	125
60150.30y	16	nd-150	ground	(n,2n)	nd-149	ground	7.41E+00	2.00E+01	127
62144.30y	16	sm-144	ground	(n,2n)	sm-143	ground	1.09E+01	2.00E+01	92
62148.30y	16	sm-148	ground	(n,2n)	sm-147	ground	8.20E+00	2.00E+01	120
62152.30y	16	sm-152	ground	(n,2n)	sm-151	ground	8.50E+00	2.00E+01	116
62154.30y	16	sm-154	ground	(n,2n)	sm-153	ground	8.05E+00	2.00E+01	121
63151.30y	16	eu-151	ground	(n,2n)	eu-150	ground	8.02E+00	2.00E+01	10
	1016	eu-151	ground	(n,2n)	eu-150	1.00E-01	8.02E+00	2.00E+01	10
	17	eu-151	ground	(n,3n)	eu-149	ground	1.45E+01	2.00E+01	6
	28	eu-151	ground	(n,n')p	sm-150	ground	8.00E+00	2.00E+01	13
	22	eu-151	ground	(n,n')a	pm-147	ground	8.00E+00	2.00E+01	12
	103	eu-151	ground	(n,p)	sm-151	ground	3.00E+00	2.00E+01	23
	104	eu-151	ground	(n,d)	sm-150	ground	2.68E+00	2.00E+01	20
	105	eu-151	ground	(n,t)	sm-149	ground	4.42E+00	2.00E+01	18
	106	eu-151	ground	(n,he-3)	pm-149	ground	5.48E+00	2.00E+01	8
	107	eu-151	ground	(n,a)	pm-148	ground	3.50E+00	2.00E+01	17
	102	eu-151	ground	(n,g)	eu-152	4.85E-02	1.00E-10	2.00E+01	102
63153.30y	1102	eu-151	ground	(n,g)	eu-152	ground	1.00E-10	2.00E+01	102
	16	eu-153	ground	(n,2n)	eu-152	ground	8.61E+00	2.00E+01	11
	1016	eu-153	ground	(n,2n)	eu-152	4.80E-02	8.61E+00	2.00E+01	11
	17	eu-153	ground	(n,3n)	eu-151	ground	1.50E+01	2.00E+01	7
	28	eu-153	ground	(n,n')p	sm-152	ground	5.93E+00	2.00E+01	23
	22	eu-153	ground	(n,n')a	pm-149	ground	6.00E+00	2.00E+01	20
	103	eu-153	ground	(n,p)	sm-153	ground	2.67E-02	2.00E+01	23
	104	eu-153	ground	(n,d)	sm-152	ground	3.69E+00	2.00E+01	20
	105	eu-153	ground	(n,t)	sm-151	ground	5.70E+00	2.00E+01	18
	106	eu-153	ground	(n,he-3)	pm-151	ground	6.88E+00	2.00E+01	7
	107	eu-153	ground	(n,a)	pm-150	ground	2.53E-08	2.00E+01	18
64150.30y	102	eu-153	ground	(n,g)	eu-154	ground	1.00E-10	2.00E+01	102
	16	gd-150	ground	(n,2n)	gd-149	ground	8.78E+00	2.00E+01	12
	102	gd-150	ground	(n,g)	gd-151	ground	1.00E-10	2.00E+01	102
64151.30y	16	gd-151	ground	(n,2n)	gd-150	ground	6.57E+00	2.00E+01	14
	102	gd-151	ground	(n,g)	gd-152	ground	1.00E-10	2.00E+01	102
67163.30y	16	ho-163	ground	(n,2n)	ho-162	1.00E-01	8.46E+00	2.00E+01	25
	17	ho-163	ground	(n,3n)	ho-161	ground	1.54E+01	2.00E+01	14
	103	ho-163	ground	(n,p)	dy-163	ground	8.18E+00	2.00E+01	13
	102	ho-163	ground	(n,g)	ho-164	1.40E-01	1.00E-10	2.00E+01	102
	1102	ho-163	ground	(n,g)	ho-164	ground	1.00E-10	2.00E+01	102
67164.30y	16	ho-164	ground	(n,2n)	ho-163	ground	6.72E+00	2.00E+01	27
	17	ho-164	ground	(n,3n)	ho-162	1.00E-01	1.52E+01	2.00E+01	13
	103	ho-164	ground	(n,p)	dy-164	ground	7.19E+00	2.00E+01	13
	102	ho-164	ground	(n,g)	ho-165	ground	1.00E-10	2.00E+01	102
67164.31y	16	ho-164	1.40E-01	(n,2n)	ho-163	ground	6.58E+00	2.00E+01	27
	17	ho-164	1.40E-01	(n,3n)	ho-162	1.00E-01	1.51E+01	2.00E+01	13
	103	ho-164	1.40E-01	(n,p)	dy-164	ground	7.05E+00	2.00E+01	13
	102	ho-164	1.40E-01	(n,g)	ho-165	ground	1.00E-10	2.00E+01	102

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
67165.30y	16	ho-165	ground	(n,2n)	ho-164	ground	8.04E+00	2.00E+01	25
	1016	ho-165	ground	(n,2n)	ho-164	1.40E-01	8.18E+00	2.00E+01	25
	17	ho-165	ground	(n,3n)	ho-163	ground	1.48E+01	2.00E+01	15
	103	ho-165	ground	(n,p)	dy-165	ground	9.46E+00	2.00E+01	13
	102	ho-165	ground	(n,g)	ho-166	9.00E-03	1.00E-10	2.00E+01	102
67166.30y	1102	ho-165	ground	(n,g)	ho-166	ground	1.00E-10	2.00E+01	102
	16	ho-166	ground	(n,2n)	ho-165	ground	6.28E+00	2.00E+01	27
	17	ho-166	ground	(n,3n)	ho-164	ground	1.44E+01	2.00E+01	16
	103	ho-166	ground	(n,p)	dy-166	ground	8.63E+00	2.00E+01	13
	102	ho-166	ground	(n,g)	ho-167	ground	1.00E-10	2.00E+01	102
67166.31y	16	ho-166	9.00E-03	(n,2n)	ho-165	ground	6.27E+00	2.00E+01	27
	17	ho-166	9.00E-03	(n,3n)	ho-164	ground	1.43E+01	2.00E+01	16
	103	ho-166	9.00E-03	(n,p)	dy-166	ground	8.62E+00	2.00E+01	13
	102	ho-166	9.00E-03	(n,g)	ho-167	ground	1.00E-10	2.00E+01	102
69169.30y	16	tm-169	ground	(n,2n)	tm-168	ground	8.11E+00	2.00E+01	120
	102	tm-169	ground	(n,g)	tm-170	ground	1.00E-10	2.00E+01	102
71173.30y	16	lu-173	ground	(n,2n)	lu-172	ground	8.23E+00	2.00E+01	26
	1016	lu-173	ground	(n,2n)	lu-172	4.20E-02	8.27E+00	2.00E+01	27
	17	lu-173	ground	(n,3n)	lu-171	ground	1.52E+01	2.00E+01	12
	1017	lu-173	ground	(n,3n)	lu-171	7.10E-02	1.53E+01	2.00E+01	12
	102	lu-173	ground	(n,g)	lu-174	1.71E-01	1.00E-10	2.00E+01	102
71174.30y	1102	lu-173	ground	(n,g)	lu-174	ground	1.00E-10	2.00E+01	102
	4	lu-174	ground	(n,n')	lu-174	1.71E-01	1.72E-01	2.00E+01	42
	16	lu-174	ground	(n,2n)	lu-173	ground	6.83E+00	2.00E+01	30
	17	lu-174	ground	(n,3n)	lu-172	ground	1.51E+01	2.00E+01	12
	1017	lu-174	ground	(n,3n)	lu-172	4.20E-02	1.51E+01	2.00E+01	12
71174.31y	102	lu-174	ground	(n,g)	lu-175	ground	1.00E-10	2.00E+01	102
	4	lu-174	1.71E-01	(n,n')	lu-174	ground	1.00E-10	2.00E+01	65
	16	lu-174	1.71E-01	(n,2n)	lu-173	ground	6.66E+00	2.00E+01	29
	17	lu-174	1.71E-01	(n,3n)	lu-172	ground	1.49E+01	2.00E+01	13
	1017	lu-174	1.71E-01	(n,3n)	lu-172	4.20E-02	1.49E+01	2.00E+01	13
71175.30y	102	lu-174	1.71E-01	(n,g)	lu-175	ground	1.00E-10	2.00E+01	102
	16	lu-175	ground	(n,2n)	lu-174	ground	7.70E+00	2.00E+01	29
	1016	lu-175	ground	(n,2n)	lu-174	1.71E-01	7.88E+00	2.00E+01	28
	17	lu-175	ground	(n,3n)	lu-173	ground	1.45E+01	2.00E+01	8
	102	lu-175	ground	(n,g)	lu-176	1.27E-01	1.00E-10	2.00E+01	102
71176.30y	1102	lu-175	ground	(n,g)	lu-176	ground	1.00E-10	2.00E+01	102
	4	lu-176	ground	(n,n')	lu-176	1.27E-01	1.28E-01	2.00E+01	29
	16	lu-176	ground	(n,2n)	lu-175	ground	6.33E+00	2.00E+01	30
	17	lu-176	ground	(n,3n)	lu-174	ground	1.40E+01	2.00E+01	18
	1017	lu-176	ground	(n,3n)	lu-174	1.71E-01	1.42E+01	2.00E+01	17
71176.31y	102	lu-176	ground	(n,g)	lu-177	9.70E-01	1.00E-10	2.00E+01	102
	1102	lu-176	ground	(n,g)	lu-177	ground	1.00E-10	2.00E+01	102
	4	lu-176	1.27E-01	(n,n')	lu-176	ground	1.50E-02	2.00E+01	37
	16	lu-176	1.27E-01	(n,2n)	lu-175	ground	6.20E+00	2.00E+01	31
	17	lu-176	1.27E-01	(n,3n)	lu-174	ground	1.39E+01	2.00E+01	17
71177.30y	1017	lu-176	1.27E-01	(n,3n)	lu-174	1.71E-01	1.41E+01	2.00E+01	17
	102	lu-176	1.27E-01	(n,g)	lu-177	9.70E-01	1.00E-10	2.00E+01	102
	1102	lu-176	1.27E-01	(n,g)	lu-177	ground	1.00E-10	2.00E+01	102
	16	lu-177	ground	(n,2n)	lu-176	ground	7.11E+00	2.00E+01	29
	1016	lu-177	ground	(n,2n)	lu-176	1.27E-01	7.24E+00	2.00E+01	27
71177.31y	17	lu-177	ground	(n,3n)	lu-175	ground	1.34E+01	2.00E+01	16
	102	lu-177	ground	(n,g)	lu-178	2.30E-01	1.00E-10	2.00E+01	102
	1102	lu-177	ground	(n,g)	lu-178	ground	1.00E-10	2.00E+01	102
	16	lu-177	9.70E-01	(n,2n)	lu-176	ground	6.13E+00	2.00E+01	29
	1016	lu-177	9.70E-01	(n,2n)	lu-176	1.27E-01	6.26E+00	2.00E+01	27
72174.30y	17	lu-177	9.70E-01	(n,3n)	lu-175	ground	1.25E+01	2.00E+01	16
	102	lu-177	9.70E-01	(n,g)	lu-178	2.30E-01	1.00E-10	2.00E+01	102
	1102	lu-177	9.70E-01	(n,g)	lu-178	ground	1.00E-10	2.00E+01	102
	16	hf-174	ground	(n,2n)	hf-173	ground	8.68E+00	2.00E+01	24
	17	hf-174	ground	(n,3n)	hf-172	ground	1.57E+01	2.00E+01	13
72175.30y	103	hf-174	ground	(n,p)	lu-174	1.71E-01	9.20E+00	2.00E+01	14
	1103	hf-174	ground	(n,p)	lu-174	ground	9.03E+00	2.00E+01	14
	16	hf-175	ground	(n,2n)	hf-174	ground	6.83E+00	2.00E+01	24

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
72176.30y	17	hf-175	ground	(n,3n)	hf-173	ground	1.55E+01	2.00E+01	15
	103	hf-175	ground	(n,p)	lu-175	ground	8.13E+00	2.00E+01	15
	16	hf-176	ground	(n,2n)	hf-175	ground	8.14E+00	2.00E+01	25
	17	hf-176	ground	(n,3n)	hf-174	ground	1.50E+01	2.00E+01	15
	103	hf-176	ground	(n,p)	lu-176	ground	9.92E+00	2.00E+01	14
72177.30y	1103	hf-176	ground	(n,p)	lu-176	1.27E-01	1.00E+01	2.00E+01	14
	16	hf-177	ground	(n,2n)	hf-176	ground	6.42E+00	2.00E+01	26
	17	hf-177	ground	(n,3n)	hf-175	ground	1.46E+01	2.00E+01	16
	103	hf-177	ground	(n,p)	lu-177	ground	9.21E+00	2.00E+01	14
	1103	hf-177	ground	(n,p)	lu-177	9.70E-01	1.02E+01	2.00E+01	14
72178.30y	16	hf-178	ground	(n,2n)	hf-177	ground	7.67E+00	2.00E+01	25
	17	hf-178	ground	(n,3n)	hf-176	ground	1.55E+01	2.00E+01	15
	103	hf-178	ground	(n,p)	lu-178	ground	1.08E+01	2.00E+01	14
	1103	hf-178	ground	(n,p)	lu-178	3.00E-01	1.11E+01	2.00E+01	14
	16	hf-179	ground	(n,2n)	hf-178	ground	6.13E+00	2.00E+01	48
72179.30y	17	hf-179	ground	(n,3n)	hf-177	ground	1.38E+01	2.00E+01	23
	103	hf-179	ground	(n,p)	lu-179	ground	1.00E+01	2.00E+01	35
	102	hf-179	ground	(n,g)	hf-180	ground	1.00E-10	2.00E+01	102
	16	hf-180	ground	(n,2n)	hf-179	ground	7.43E+00	2.00E+01	43
	17	hf-180	ground	(n,3n)	hf-178	ground	1.36E+01	2.00E+01	23
72180.30y	103	hf-180	ground	(n,p)	lu-180	ground	1.20E+01	2.00E+01	28
	102	hf-180	ground	(n,g)	hf-181	ground	1.00E-10	2.00E+01	102
	16	hf-181	ground	(n,2n)	hf-180	ground	5.72E+00	2.00E+01	49
	17	hf-181	ground	(n,3n)	hf-179	ground	1.32E+01	2.00E+01	25
	102	hf-181	ground	(n,g)	hf-182	ground	1.00E-10	2.00E+01	102
72183.30y	16	hf-183	ground	(n,2n)	hf-182	ground	5.42E+00	2.00E+01	50
	17	hf-183	ground	(n,3n)	hf-181	ground	1.21E+01	2.00E+01	28
	102	hf-183	ground	(n,g)	hf-184	ground	1.00E-10	2.00E+01	102
	16	ta-179	ground	(n,2n)	ta-178	ground	7.93E+00	2.00E+01	42
	17	ta-179	ground	(n,3n)	ta-177	ground	1.49E+01	2.00E+01	19
73179.30y	103	ta-179	ground	(n,p)	hf-179	ground	8.70E+00	2.00E+01	39
	102	ta-179	ground	(n,g)	ta-180	2.12E-01	1.00E-10	2.00E+01	102
	1102	ta-179	ground	(n,g)	ta-180	ground	1.00E-10	2.00E+01	102
	16	ta-180	ground	(n,2n)	ta-179	ground	6.62E+00	2.00E+01	46
	17	ta-180	ground	(n,3n)	ta-178	ground	1.46E+01	2.00E+01	20
73180.30y	28	ta-180	ground	(n,n')p	hf-179	1.11E+00	1.61E+01	2.00E+01	12
	103	ta-180	ground	(n,p)	hf-180	1.14E+00	9.02E+00	2.00E+01	38
	104	ta-180	ground	(n,d)	hf-179	1.11E+00	1.37E+01	2.00E+01	13
	107	ta-180	ground	(n,a)	lu-177	ground	1.11E+01	2.00E+01	16
	102	ta-180	ground	(n,g)	ta-181	ground	1.00E-10	2.00E+01	102
73180.31y	16	ta-180	2.12E-01	(n,2n)	ta-179	ground	6.40E+00	2.00E+01	47
	17	ta-180	2.12E-01	(n,3n)	ta-178	ground	1.43E+01	2.00E+01	21
	103	ta-180	2.12E-01	(n,p)	hf-180	1.14E+00	8.80E+00	2.00E+01	39
	102	ta-180	2.12E-01	(n,g)	ta-181	ground	1.00E-10	2.00E+01	102
	16	ta-181	ground	(n,2n)	ta-180	ground	7.67E+00	2.00E+01	43
73181.30y	1016	ta-181	ground	(n,2n)	ta-180	2.12E-01	7.89E+00	2.00E+01	42
	17	ta-181	ground	(n,3n)	ta-179	ground	1.43E+01	2.00E+01	21
	22	ta-181	ground	(n,n')a	lu-177	ground	1.27E+01	2.00E+01	15
	103	ta-181	ground	(n,p)	hf-181	ground	1.00E+01	2.00E+01	5
	102	ta-181	ground	(n,g)	ta-182	5.03E-01	1.00E-10	2.00E+01	75
73182.30y	1102	ta-181	ground	(n,g)	ta-182	ground	1.00E-10	2.00E+01	142
	16	ta-182	ground	(n,2n)	ta-181	ground	6.09E+00	2.00E+01	48
	17	ta-182	ground	(n,3n)	ta-180	2.12E-01	1.40E+01	2.00E+01	22
	103	ta-182	ground	(n,p)	hf-182	ground	9.28E+00	2.00E+01	37
	102	ta-182	ground	(n,g)	ta-183	ground	1.00E-10	2.00E+01	102
73182.31y	16	ta-182	5.03E-01	(n,2n)	ta-181	ground	5.59E+00	2.00E+01	50
	17	ta-182	5.03E-01	(n,3n)	ta-180	2.12E-01	1.35E+01	2.00E+01	24
	103	ta-182	5.03E-01	(n,p)	hf-182	ground	8.77E+00	2.00E+01	39
	102	ta-182	5.03E-01	(n,g)	ta-183	ground	1.00E-10	2.00E+01	102
	16	ta-183	ground	(n,2n)	ta-182	ground	6.97E+00	2.00E+01	45
73183.30y	17	ta-183	ground	(n,3n)	ta-181	ground	1.31E+01	2.00E+01	25
	103	ta-183	ground	(n,p)	hf-183	ground	1.08E+01	2.00E+01	32
	102	ta-183	ground	(n,g)	ta-184	ground	1.00E-10	2.00E+01	102
	16	ta-184	ground	(n,2n)	ta-183	ground	5.48E+00	2.00E+01	50

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
73186.30y	17	ta-184	ground	(n,3n)	ta-182	ground	1.24E+01	2.00E+01	27
	102	ta-184	ground	(n,g)	ta-185	ground	1.00E-10	2.00E+01	102
	16	ta-186	ground	(n,2n)	ta-185	ground	5.29E+00	2.00E+01	51
	17	ta-186	ground	(n,3n)	ta-184	ground	1.21E+01	2.00E+01	29
74179.30y	102	ta-186	ground	(n,g)	ta-187	ground	1.00E-10	2.00E+01	102
	16	w-179	ground	(n,2n)	w-178	ground	8.66E+00	2.00E+01	17
	17	w-179	ground	(n,3n)	w-177	ground	1.60E+01	2.00E+01	6
74180.30y	102	w-179	ground	(n,g)	w-180	ground	1.00E-10	2.00E+01	102
	16	w-180	ground	(n,2n)	w-179	ground	8.66E+00	2.00E+01	17
	17	w-180	ground	(n,3n)	w-178	ground	1.54E+01	2.00E+01	6
	28	w-180	ground	(n,n')p	ta-179	ground	1.60E+01	2.00E+01	11
74181.30y	103	w-180	ground	(n,p)	ta-180	ground	9.65E+00	2.00E+01	18
	104	w-180	ground	(n,d)	ta-179	ground	1.36E+01	2.00E+01	13
	107	w-180	ground	(n,a)	hf-177	2.74E+00	8.87E+00	2.00E+01	17
	102	w-180	ground	(n,g)	w-181	ground	1.00E-10	2.00E+01	102
	16	w-181	ground	(n,2n)	w-180	ground	6.68E+00	2.00E+01	17
	17	w-181	ground	(n,3n)	w-179	ground	1.53E+01	2.00E+01	6
	102	w-181	ground	(n,g)	w-182	ground	1.00E-10	2.00E+01	102
74182.30y	16	w-182	ground	(n,2n)	w-181	ground	8.10E+00	2.00E+01	18
	17	w-182	ground	(n,3n)	w-180	ground	1.48E+01	2.00E+01	7
	22	w-182	ground	(n,n')a	hf-178	2.45E+00	1.26E+01	2.00E+01	12
	103	w-182	ground	(n,p)	ta-182	ground	1.03E+00	2.00E+01	27
74183.30y	107	w-182	ground	(n,a)	hf-179	ground	1.10E+01	2.00E+01	29
	102	w-182	ground	(n,g)	w-183	ground	1.00E-10	2.00E+01	102
	16	w-183	ground	(n,2n)	w-182	ground	6.25E+00	2.00E+01	18
	17	w-183	ground	(n,3n)	w-181	ground	1.43E+01	2.00E+01	7
	28	w-183	ground	(n,n')p	ta-182	ground	1.17E+01	2.00E+01	30
	22	w-183	ground	(n,n')a	hf-179	1.11E+00	1.14E+01	2.00E+01	13
	103	w-183	ground	(n,p)	ta-183	ground	1.00E+01	2.00E+01	24
74184.30y	104	w-183	ground	(n,d)	ta-182	ground	1.32E+01	2.00E+01	13
	107	w-183	ground	(n,a)	hf-180	ground	1.10E+01	2.00E+01	26
	102	w-183	ground	(n,g)	w-184	ground	1.00E-10	2.00E+01	102
	16	w-184	ground	(n,2n)	w-183	ground	7.45E+00	2.00E+01	18
	17	w-184	ground	(n,3n)	w-182	ground	1.37E+01	2.00E+01	7
	28	w-184	ground	(n,n')p	ta-183	ground	1.71E+01	2.00E+01	10
	103	w-184	ground	(n,p)	ta-184	ground	2.26E+00	2.00E+01	27
74185.30y	104	w-184	ground	(n,d)	ta-183	ground	1.37E+01	2.00E+01	13
	107	w-184	ground	(n,a)	hf-181	ground	1.00E-11	2.00E+01	30
	102	w-184	ground	(n,g)	w-185	ground	1.00E-10	2.00E+01	102
	16	w-185	ground	(n,2n)	w-184	ground	5.78E+00	2.00E+01	18
	17	w-185	ground	(n,3n)	w-183	ground	1.32E+01	2.00E+01	7
	102	w-185	ground	(n,g)	w-186	ground	1.00E-10	2.00E+01	102
	16	w-186	ground	(n,2n)	w-185	ground	7.25E+00	2.00E+01	19
74186.30y	17	w-186	ground	(n,3n)	w-184	ground	1.30E+01	2.00E+01	8
	22	w-186	ground	(n,n')a	hf-182	ground	1.39E+01	2.00E+01	13
	103	w-186	ground	(n,p)	ta-186	ground	1.14E+01	2.00E+01	23
	107	w-186	ground	(n,a)	hf-183	ground	1.10E+01	2.00E+01	29
74187.30y	102	w-186	ground	(n,g)	w-187	ground	1.00E-10	2.00E+01	102
	16	w-187	ground	(n,2n)	w-186	ground	5.50E+00	2.00E+01	19
	17	w-187	ground	(n,3n)	w-185	ground	1.27E+01	2.00E+01	8
	37	w-187	ground	(n,4n)	w-183	ground	1.85E+01	2.00E+01	2
74188.30y	102	w-187	ground	(n,g)	w-188	ground	1.00E-10	2.00E+01	102
	16	w-188	ground	(n,2n)	w-187	ground	6.88E+00	2.00E+01	19
	17	w-188	ground	(n,3n)	w-186	ground	1.24E+01	2.00E+01	8
	102	w-188	ground	(n,g)	w-189	ground	1.00E-10	2.00E+01	102
75184.30y	16	re-184	ground	(n,2n)	re-183	ground	6.54E+00	2.00E+01	24
	17	re-184	ground	(n,3n)	re-182	ground	1.56E+01	2.00E+01	14
	103	re-184	ground	(n,p)	w-184	ground	7.88E+00	2.00E+01	17
	102	re-184	ground	(n,g)	re-185	ground	1.00E-10	2.00E+01	102
75184.31y	16	re-184	1.88E-01	(n,2n)	re-183	ground	6.35E+00	2.00E+01	25
	17	re-184	1.88E-01	(n,3n)	re-182	ground	1.54E+01	2.00E+01	15
	103	re-184	1.88E-01	(n,p)	w-184	ground	7.69E+00	2.00E+01	17
	102	re-184	1.88E-01	(n,g)	re-185	ground	1.00E-10	2.00E+01	102
75185.30y	16	re-185	ground	(n,2n)	re-184	ground	7.85E+00	2.00E+01	27

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES	
75186.30y	1016	re-185	ground	(n,2n)	re-184	1.88E-01	8.04E+00	2.00E+01	16	
	17	re-185	ground	(n,3n)	re-183	ground	1.45E+01	2.00E+01	15	
	103	re-185	ground	(n,p)	w-185	ground	9.43E+00	2.00E+01	16	
	102	re-185	ground	(n,g)	re-186	ground	1.00E-10	2.00E+01	102	
	16	re-186	ground	(n,2n)	re-185	ground	6.21E+00	2.00E+01	29	
	17	re-186	ground	(n,3n)	re-184	ground	1.38E+01	2.00E+01	16	
	1017	re-186	ground	(n,3n)	re-184	1.88E-01	1.38E+01	2.00E+01	16	
75187.30y	103	re-186	ground	(n,p)	w-186	ground	8.38E+00	2.00E+01	17	
	102	re-186	ground	(n,g)	re-187	ground	1.00E-10	2.00E+01	102	
	16	re-187	ground	(n,2n)	re-186	ground	7.36E+00	2.00E+01	26	
	17	re-187	ground	(n,3n)	re-185	ground	1.36E+01	2.00E+01	17	
	103	re-187	ground	(n,p)	w-187	ground	1.03E+01	2.00E+01	16	
	102	re-187	ground	(n,g)	re-188	1.72E-01	1.00E-10	2.00E+01	102	
	1102	re-187	ground	(n,g)	re-188	ground	1.00E-10	2.00E+01	102	
75188.30y	16	re-188	ground	(n,2n)	re-187	ground	5.90E+00	2.00E+01	27	
	17	re-188	ground	(n,3n)	re-186	ground	1.33E+01	2.00E+01	16	
	103	re-188	ground	(n,p)	w-188	ground	9.30E+00	2.00E+01	16	
	102	re-188	ground	(n,g)	re-189	ground	1.00E-10	2.00E+01	102	
	16	re-188	1.72E-01	(n,2n)	re-187	ground	5.73E+00	2.00E+01	28	
	17	re-188	1.72E-01	(n,3n)	re-186	ground	1.32E+01	2.00E+01	16	
	103	re-188	1.72E-01	(n,p)	w-188	ground	9.13E+00	2.00E+01	16	
75188.31y	102	re-188	1.72E-01	(n,g)	re-189	ground	1.00E-10	2.00E+01	102	
	16	ir-191	ground	(n,2n)	ir-190	ground	8.50E+00	2.00E+01	116	
	77191.30y	16	ir-193	ground	(n,2n)	ir-192	ground	8.20E+00	2.00E+01	119
	77193.30y	16	ir-194	ground	(n,2n)	ir-193	ground	6.10E+00	2.00E+01	48
	77194.30y	17	ir-194	ground	(n,3n)	ir-192	ground	1.40E+01	2.00E+01	22
	103	ir-194	ground	(n,p)	os-194	ground	9.21E+00	2.00E+01	30	
	102	ir-194	ground	(n,g)	ir-195	ground	1.00E-10	2.00E+01	102	
78190.30y	16	pt-190	ground	(n,2n)	pt-189	ground	8.87E+00	2.00E+01	24	
	17	pt-190	ground	(n,3n)	pt-188	ground	1.58E+01	2.00E+01	15	
	103	pt-190	ground	(n,p)	ir-190	1.75E-01	1.01E+01	2.00E+01	14	
	1103	pt-190	ground	(n,p)	ir-190	ground	9.93E+00	2.00E+01	14	
	78192.30y	16	pt-192	ground	(n,2n)	pt-191	ground	8.70E+00	2.00E+01	25
	17	pt-192	ground	(n,3n)	pt-190	ground	1.52E+01	2.00E+01	15	
	103	pt-192	ground	(n,p)	ir-192	ground	1.07E+01	2.00E+01	14	
78193.30y	1103	pt-192	ground	(n,p)	ir-192	1.61E-01	1.09E+01	2.00E+01	14	
	16	pt-193	ground	(n,2n)	pt-192	ground	6.28E+00	2.00E+01	26	
	17	pt-193	ground	(n,3n)	pt-191	ground	1.50E+01	2.00E+01	15	
	103	pt-193	ground	(n,p)	ir-193	ground	9.21E+00	2.00E+01	14	
	78193.31y	16	pt-193	1.50E-01	(n,2n)	pt-192	ground	6.13E+00	2.00E+01	26
	17	pt-193	1.50E-01	(n,3n)	pt-191	ground	1.48E+01	2.00E+01	15	
	103	pt-193	1.50E-01	(n,p)	ir-193	ground	9.06E+00	2.00E+01	14	
78194.30y	16	pt-194	ground	(n,2n)	pt-193	ground	8.42E+00	2.00E+01	26	
	1016	pt-194	ground	(n,2n)	pt-193	1.50E-01	8.57E+00	2.00E+01	27	
	17	pt-194	ground	(n,3n)	pt-192	ground	1.47E+01	2.00E+01	14	
	103	pt-194	ground	(n,p)	ir-194	ground	1.15E+01	2.00E+01	14	
	1103	pt-194	ground	(n,p)	ir-194	4.50E-01	1.20E+01	2.00E+01	14	
	78195.30y	16	pt-195	ground	(n,2n)	pt-194	ground	6.14E+00	2.00E+01	26
	17	pt-195	ground	(n,3n)	pt-193	ground	1.46E+01	2.00E+01	16	
78196.30y	103	pt-195	ground	(n,p)	ir-195	ground	1.04E+01	2.00E+01	14	
	1103	pt-195	ground	(n,p)	ir-195	1.20E-01	1.05E+01	2.00E+01	14	
	16	pt-196	ground	(n,2n)	pt-195	ground	7.96E+00	2.00E+01	27	
	17	pt-196	ground	(n,3n)	pt-194	ground	1.44E+01	2.00E+01	15	
	103	pt-196	ground	(n,p)	ir-196	ground	1.25E+01	2.00E+01	14	
	1103	pt-196	ground	(n,p)	ir-196	4.30E-01	1.29E+01	2.00E+01	14	
	78197.30y	16	pt-197	ground	(n,2n)	pt-196	ground	5.88E+00	2.00E+01	49
78197.31y	17	pt-197	ground	(n,3n)	pt-195	ground	1.38E+01	2.00E+01	23	
	103	pt-197	ground	(n,p)	ir-197	ground	1.12E+01	2.00E+01	31	
	102	pt-197	ground	(n,g)	pt-198	ground	1.00E-10	2.00E+01	102	
	16	pt-197	4.00E-01	(n,2n)	pt-196	ground	5.48E+00	2.00E+01	28	
	17	pt-197	4.00E-01	(n,3n)	pt-195	ground	1.34E+01	2.00E+01	16	
	103	pt-197	4.00E-01	(n,p)	ir-197	ground	1.08E+01	2.00E+01	14	
	78198.30y	16	pt-198	ground	(n,2n)	pt-197	ground	7.60E+00	2.00E+01	27
1016	pt-198	ground	(n,2n)	pt-197	4.00E-01	8.00E+00	2.00E+01	28		

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
78199.30y	17	pt-198	ground	(n,3n)	au-196	ground	1.35E+01	2.00E+01	14
	103	pt-198	ground	(n,p)	ir-198	ground	1.36E+01	2.00E+01	14
	16	pt-199	ground	(n,2n)	pt-198	ground	5.60E+00	2.00E+01	29
78199.31y	17	pt-199	ground	(n,3n)	pt-197	ground	1.32E+01	2.00E+01	16
	16	pt-199	4.24E-01	(n,2n)	pt-198	ground	5.17E+00	2.00E+01	29
79193.30y	17	pt-199	4.24E-01	(n,3n)	pt-197	ground	1.28E+01	2.00E+01	16
79194.30y	102	au-193	ground	(n,g)	au-194	ground	1.00E-10	2.00E+01	102
	16	au-194	ground	(n,2n)	au-193	ground	6.80E+00	2.00E+01	15
79195.30y	17	au-194	ground	(n,3n)	au-192	ground	1.57E+01	2.00E+01	7
	102	au-194	ground	(n,g)	au-195	ground	1.00E-10	2.00E+01	102
	16	au-195	ground	(n,2n)	au-194	ground	8.39E+00	2.00E+01	15
	17	au-195	ground	(n,3n)	au-193	ground	1.55E+01	2.00E+01	7
79196.30y	102	au-195	ground	(n,g)	au-196	ground	1.00E-10	2.00E+01	102
	16	au-196	ground	(n,2n)	au-195	ground	6.70E+00	2.00E+01	16
	17	au-196	ground	(n,3n)	au-194	ground	1.51E+01	2.00E+01	8
79196.31y	102	au-196	ground	(n,g)	au-197	ground	1.00E-10	2.00E+01	102
	16	au-196	5.96E-01	(n,2n)	au-195	ground	6.10E+00	2.00E+01	16
	17	au-196	5.96E-01	(n,3n)	au-194	ground	1.45E+01	2.00E+01	8
	102	au-196	5.96E-01	(n,g)	au-197	ground	1.00E-10	2.00E+01	102
79197.30y	16	au-197	ground	(n,2n)	au-196	ground	8.13E+00	2.00E+01	12
	1016	au-197	ground	(n,2n)	au-196	5.96E-01	8.70E+00	2.00E+01	7
	17	au-197	ground	(n,3n)	au-195	ground	1.48E+01	2.00E+01	5
	103	au-197	ground	(n,p)	pt-197	ground	1.00E+01	2.00E+01	9
	107	au-197	ground	(n,a)	ir-194	ground	7.00E+00	2.00E+01	6
79198.30y	102	au-197	ground	(n,g)	au-198	ground	1.00E-10	2.00E+01	102
	16	au-198	ground	(n,2n)	au-197	ground	6.55E+00	2.00E+01	16
	17	au-198	ground	(n,3n)	au-196	ground	1.47E+01	2.00E+01	8
	102	au-198	ground	(n,g)	au-199	ground	1.00E-10	2.00E+01	102
79199.30y	16	au-199	ground	(n,2n)	au-198	ground	7.61E+00	2.00E+01	17
	17	au-199	ground	(n,3n)	au-197	ground	1.41E+01	2.00E+01	9
	102	au-199	ground	(n,g)	au-200	ground	1.00E-10	2.00E+01	102
79200.30y	16	au-200	ground	(n,2n)	au-199	ground	6.32E+00	2.00E+01	17
80202.30y	16	hg-202	ground	(n,2n)	hg-201	ground	7.79E+00	2.00E+01	26
	17	hg-202	ground	(n,3n)	hg-200	ground	1.41E+01	2.00E+01	17
	28	hg-202	ground	(n,n')p	au-201	ground	1.81E+01	2.00E+01	8
80203.30y	103	hg-202	ground	(n,p)	au-202	ground	1.29E+01	2.00E+01	16
	107	hg-202	ground	(n,a)	pt-199	ground	1.10E+01	2.00E+01	9
	102	hg-202	ground	(n,g)	hg-203	ground	1.00E-10	2.00E+01	102
	16	hg-203	ground	(n,2n)	hg-202	ground	6.02E+00	2.00E+01	26
	17	hg-203	ground	(n,3n)	hg-201	ground	1.38E+01	2.00E+01	16
	28	hg-203	ground	(n,n')p	au-202	ground	1.86E+01	2.00E+01	7
	103	hg-203	ground	(n,p)	au-203	ground	1.17E+01	2.00E+01	16
	107	hg-203	ground	(n,a)	pt-200	ground	1.10E+01	2.00E+01	10
	102	hg-203	ground	(n,g)	hg-204	ground	1.00E-10	2.00E+01	102
	16	hg-204	ground	(n,2n)	hg-203	ground	7.53E+00	2.00E+01	27
80204.30y	17	hg-204	ground	(n,3n)	hg-202	ground	1.36E+01	2.00E+01	14
	28	hg-204	ground	(n,n')p	au-203	ground	1.89E+01	2.00E+01	6
	103	hg-204	ground	(n,p)	au-204	ground	1.39E+01	2.00E+01	14
	107	hg-204	ground	(n,a)	pt-201	ground	1.20E+01	2.00E+01	7
	102	hg-204	ground	(n,g)	hg-205	ground	1.00E-10	2.00E+01	102
	16	tl-202	ground	(n,2n)	tl-201	ground	6.91E+00	2.00E+01	26
	17	tl-202	ground	(n,3n)	tl-200	ground	1.51E+01	2.00E+01	15
	28	tl-202	ground	(n,n')p	hg-201	ground	1.56E+01	2.00E+01	15
	103	tl-202	ground	(n,p)	hg-202	ground	8.15E+00	2.00E+01	20
	102	tl-202	ground	(n,g)	tl-203	ground	1.00E-10	2.00E+01	102
81203.30y	16	tl-203	ground	(n,2n)	tl-202	ground	7.89E+00	2.00E+01	26
	17	tl-203	ground	(n,3n)	tl-201	ground	1.48E+01	2.00E+01	14
	28	tl-203	ground	(n,n')p	hg-202	ground	1.57E+01	2.00E+01	15
81204.30y	103	tl-203	ground	(n,p)	hg-203	ground	1.00E+01	2.00E+01	20
	102	tl-203	ground	(n,g)	tl-204	ground	1.00E-10	2.00E+01	102
	16	tl-204	ground	(n,2n)	tl-203	ground	6.69E+00	2.00E+01	26
	17	tl-204	ground	(n,3n)	tl-202	ground	1.46E+01	2.00E+01	15
	28	tl-204	ground	(n,n')p	hg-203	ground	1.63E+01	2.00E+01	14
	103	tl-204	ground	(n,p)	hg-204	ground	9.14E+00	2.00E+01	19

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
81205.30y	102	tl-204	ground	(n,g)	tl-205	ground	1.00E-10	2.00E+01	102
	16	tl-205	ground	(n,2n)	tl-204	ground	7.59E+00	2.00E+01	26
	17	tl-205	ground	(n,3n)	tl-203	ground	1.43E+01	2.00E+01	15
	28	tl-205	ground	(n,n')p	hg-204	ground	1.64E+01	2.00E+01	13
	103	tl-205	ground	(n,p)	hg-205	ground	1.10E+01	2.00E+01	18
82203.30y	102	tl-205	ground	(n,g)	tl-206	ground	1.00E-10	2.00E+01	102
	16	pb-203	ground	(n,2n)	pb-202	ground	6.85E+00	2.00E+01	14
	17	pb-203	ground	(n,3n)	pb-201	ground	1.56E+01	2.00E+01	6
	102	pb-203	ground	(n,g)	pb-204	ground	1.00E-10	2.00E+01	102
	4	pb-204	ground	(n,n')	pb-204	2.19E+00	2.20E+00	2.00E+01	40
82204.30y	16	pb-204	ground	(n,2n)	pb-203	ground	8.45E+00	2.00E+01	15
	17	pb-204	ground	(n,3n)	pb-202	ground	1.53E+01	2.00E+01	6
	103	pb-204	ground	(n,p)	tl-204	ground	1.04E+01	2.00E+01	17
	105	pb-204	ground	(n,t)	tl-202	ground	1.20E+01	2.00E+01	10
	102	pb-204	ground	(n,g)	pb-205	ground	1.00E-10	2.00E+01	102
82205.30y	16	pb-205	ground	(n,2n)	pb-204	ground	6.80E+00	2.00E+01	14
	17	pb-205	ground	(n,3n)	pb-203	ground	1.52E+01	2.00E+01	6
	102	pb-205	ground	(n,g)	pb-206	ground	1.00E-10	2.00E+01	102
	16	pb-206	ground	(n,2n)	pb-205	ground	8.12E+00	2.00E+01	15
	17	pb-206	ground	(n,3n)	pb-204	ground	1.49E+01	2.00E+01	6
82206.30y	103	pb-206	ground	(n,p)	tl-206	ground	1.11E+01	2.00E+01	16
	105	pb-206	ground	(n,t)	tl-204	ground	1.20E+01	2.00E+01	10
	107	pb-206	ground	(n,a)	hg-203	ground	1.19E+01	2.00E+01	12
	102	pb-206	ground	(n,g)	pb-207	ground	1.00E-10	2.00E+01	102
	16	pb-207	ground	(n,2n)	pb-206	ground	6.78E+00	2.00E+01	14
82207.30y	17	pb-207	ground	(n,3n)	pb-205	ground	1.49E+01	2.00E+01	6
	22	pb-207	ground	(n,n')a	hg-203	ground	1.20E+01	2.00E+01	4
	103	pb-207	ground	(n,p)	tl-207	ground	1.10E+01	2.00E+01	15
	107	pb-207	ground	(n,a)	pb-204	ground	1.11E+01	2.00E+01	13
	102	pb-207	ground	(n,g)	pb-208	ground	1.00E-10	2.00E+01	102
82208.30y	16	pb-208	ground	(n,2n)	pb-207	ground	7.41E+00	2.00E+01	16
	17	pb-208	ground	(n,3n)	pb-206	ground	1.42E+01	2.00E+01	7
	102	pb-208	ground	(n,g)	pb-209	ground	1.00E-10	2.00E+01	102
	16	pb-209	ground	(n,2n)	pb-208	ground	3.96E+00	2.00E+01	18
	17	pb-209	ground	(n,3n)	pb-207	ground	1.14E+01	2.00E+01	9
82209.30y	37	pb-209	ground	(n,4n)	pb-206	ground	1.81E+01	2.00E+01	2
	102	pb-209	ground	(n,g)	pb-210	ground	1.00E-10	2.00E+01	102
	16	pb-210	ground	(n,2n)	pb-209	ground	5.21E+00	2.00E+01	27
	17	pb-210	ground	(n,3n)	pb-208	ground	9.16E+00	2.00E+01	15
	28	pb-210	ground	(n,n')p	tl-209	ground	1.84E+01	2.00E+01	7
83208.30y	103	pb-210	ground	(n,p)	tl-210	ground	1.51E+01	2.00E+01	14
	102	pb-210	ground	(n,g)	pb-211	ground	1.00E-10	2.00E+01	102
	16	bi-208	ground	(n,2n)	bi-207	ground	6.93E+00	2.00E+01	24
	17	bi-208	ground	(n,3n)	bi-206	ground	1.51E+01	2.00E+01	14
	28	bi-208	ground	(n,n')p	pb-207	ground	1.38E+01	2.00E+01	15
83209.30y	22	bi-208	ground	(n,n')a	tl-204	ground	1.92E+01	2.00E+01	3
	103	bi-208	ground	(n,p)	pb-208	ground	6.80E+00	2.00E+01	19
	107	bi-208	ground	(n,a)	tl-205	ground	1.16E+01	2.00E+01	13
	102	bi-208	ground	(n,g)	bi-209	ground	1.00E-10	2.00E+01	102
	16	bi-209	ground	(n,2n)	bi-208	ground	7.44E+00	2.00E+01	23
83210.30y	17	bi-209	ground	(n,3n)	bi-207	ground	1.43E+01	2.00E+01	9
	22	bi-209	ground	(n,n')a	tl-205	ground	1.00E-11	2.00E+01	7
	103	bi-209	ground	(n,p)	pb-209	ground	9.90E+00	2.00E+01	12
	107	bi-209	ground	(n,a)	tl-206	ground	1.29E+01	2.00E+01	6
	102	bi-209	ground	(n,g)	bi-210	ground	1.00E-10	2.00E+01	102
83210.30y	1102	bi-209	ground	(n,g)	bi-210	2.71E-01	1.00E-10	2.00E+01	102
	16	bi-210	ground	(n,2n)	bi-209	ground	4.63E+00	2.00E+01	27
	17	bi-210	ground	(n,3n)	bi-208	ground	1.21E+01	2.00E+01	16
	28	bi-210	ground	(n,n')p	pb-209	ground	1.46E+01	2.00E+01	14
	22	bi-210	ground	(n,n')a	tl-206	ground	1.72E+01	2.00E+01	9
83210.31y	103	bi-210	ground	(n,p)	pb-210	ground	9.73E+00	2.00E+01	20
	107	bi-210	ground	(n,a)	tl-207	ground	1.03E+01	2.00E+01	17
	102	bi-210	ground	(n,g)	bi-211	ground	1.00E-10	2.00E+01	93
	16	bi-210	2.71E-01	(n,2n)	bi-209	ground	4.35E+00	2.00E+01	27

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
	17	bi-210	2.71E-01	(n,3n)	bi-208	ground	1.19E+01	2.00E+01	16
	28	bi-210	2.71E-01	(n,n')p	pb-209	ground	1.40E+01	2.00E+01	14
	22	bi-210	2.71E-01	(n,n')a	tl-206	ground	1.70E+01	2.00E+01	9
	103	bi-210	2.71E-01	(n,p)	pb-210	ground	9.80E+00	2.00E+01	20
	107	bi-210	2.71E-01	(n,a)	tl-207	ground	1.00E+01	2.00E+01	17
	102	bi-210	2.71E-01	(n,g)	bi-211	ground	1.00E-10	2.00E+01	93
84210.30y	16	po-210	ground	(n,2n)	po-209	ground	7.70E+00	2.00E+01	26
	17	po-210	ground	(n,3n)	po-208	ground	1.47E+01	2.00E+01	16
	28	po-210	ground	(n,n')p	bi-209	ground	1.52E+01	2.00E+01	13
	22	po-210	ground	(n,n')a	pb-206	ground	1.71E+01	2.00E+01	11
	103	po-210	ground	(n,p)	bi-210	ground	1.10E+01	2.00E+01	20
	107	po-210	ground	(n,a)	pb-207	ground	1.03E+01	2.00E+01	18
	102	po-210	ground	(n,g)	po-211	ground	1.00E-10	2.00E+01	102
90230.30y	102	th-230	ground	(n,g)	th-231	ground	1.00E-10	2.00E+01	102
90231.30y	16	th-231	ground	(n,2n)	th-230	ground	5.15E+00	2.00E+01	19
	17	th-231	ground	(n,3n)	th-229	ground	1.20E+01	2.00E+01	11
	37	th-231	ground	(n,4n)	th-228	ground	1.72E+01	2.00E+01	5
	18	th-231	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	152
	102	th-231	ground	(n,g)	th-232	ground	1.00E-10	2.00E+01	102
90232.30y	16	th-232	ground	(n,2n)	th-231	ground	6.50E+00	2.00E+01	17
	17	th-232	ground	(n,3n)	th-230	ground	1.16E+01	2.00E+01	8
	37	th-232	ground	(n,4n)	th-229	ground	1.85E+01	2.00E+01	2
	18	th-232	ground	(n,f)	f-prod	ground	1.00E+00	2.00E+01	34
	102	th-232	ground	(n,g)	th-233	ground	1.00E-10	2.00E+01	102
90233.30y	16	th-233	ground	(n,2n)	th-232	ground	4.81E+00	2.00E+01	19
	17	th-233	ground	(n,3n)	th-231	ground	1.13E+01	2.00E+01	14
	37	th-233	ground	(n,4n)	th-230	ground	1.63E+01	2.00E+01	5
	18	th-233	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	130
	102	th-233	ground	(n,g)	th-234	ground	1.00E-10	2.00E+01	102
90234.30y	16	th-234	ground	(n,2n)	th-233	ground	6.20E+00	2.00E+01	16
91233.30y	16	pa-233	ground	(n,2n)	pa-232	ground	6.68E+00	2.00E+01	15
	17	pa-233	ground	(n,3n)	pa-230	ground	1.22E+01	2.00E+01	13
	18	pa-233	ground	(n,f)	f-prod	ground	4.80E-01	2.00E+01	42
	102	pa-233	ground	(n,g)	pa-234	ground	1.00E-10	2.00E+01	102
92233.30y	16	u-233	ground	(n,2n)	u-232	ground	5.76E+00	2.00E+01	12
	17	u-233	ground	(n,3n)	u-231	ground	1.31E+01	2.00E+01	6
	18	u-233	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	u-233	ground	(n,g)	u-234	ground	1.00E-10	2.00E+01	102
92234.30y	16	u-234	ground	(n,2n)	u-233	ground	6.87E+00	2.00E+01	13
	17	u-234	ground	(n,3n)	u-232	ground	1.26E+01	2.00E+01	7
	18	u-234	ground	(n,f)	f-prod	ground	1.00E-01	2.00E+01	40
	102	u-234	ground	(n,g)	u-235	ground	1.00E-10	2.00E+01	128
92235.30y	16	u-235	ground	(n,2n)	u-234	ground	5.32E+00	2.00E+01	16
	17	u-235	ground	(n,3n)	u-233	ground	1.25E+01	2.00E+01	7
	37	u-235	ground	(n,4n)	u-232	ground	1.80E+01	2.00E+01	3
	18	u-235	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	1511
	102	u-235	ground	(n,g)	u-236	ground	1.00E-10	2.00E+01	767
92236.30y	16	u-236	ground	(n,2n)	u-235	ground	6.58E+00	2.00E+01	14
	17	u-236	ground	(n,3n)	u-234	ground	1.19E+01	2.00E+01	8
	18	u-236	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	40
	102	u-236	ground	(n,g)	u-237	ground	1.00E-10	2.00E+01	127
92237.30y	16	u-237	ground	(n,2n)	u-236	ground	5.14E+00	2.00E+01	11
	17	u-237	ground	(n,3n)	u-235	ground	1.17E+01	2.00E+01	7
	18	u-237	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	104
	102	u-237	ground	(n,g)	u-238	ground	1.00E-10	2.00E+01	174
92238.30y	16	u-238	ground	(n,2n)	u-237	ground	6.18E+00	2.00E+01	15
	17	u-238	ground	(n,3n)	u-236	ground	1.15E+01	2.00E+01	9
	37	u-238	ground	(n,4n)	u-235	ground	1.80E+01	2.00E+01	3
	18	u-238	ground	(n,f)	f-prod	ground	4.50E-01	2.00E+01	41
	102	u-238	ground	(n,g)	u-239	ground	1.00E-10	2.00E+01	1473
92239.30y	16	u-239	ground	(n,2n)	u-238	ground	4.82E+00	2.00E+01	14
	17	u-239	ground	(n,3n)	u-237	ground	1.10E+01	2.00E+01	12
	37	u-239	ground	(n,4n)	u-236	ground	1.61E+01	2.00E+01	5
	18	u-239	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	160

ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
92240.30y	102	u-239	ground	(n, g)	u-240	ground	1.00E-10	2.00E+01	211
	16	u-240	ground	(n, 2n)	u-239	ground	5.96E+00	2.00E+01	13
	17	u-240	ground	(n, 3n)	u-238	ground	1.08E+01	2.00E+01	10
	37	u-240	ground	(n, 4n)	u-237	ground	1.70E+01	2.00E+01	5
	18	u-240	ground	(n, f)	f-prod	ground	1.00E+00	2.00E+01	17
93237.30y	102	u-240	ground	(n, g)	u-241	ground	1.00E-10	2.00E+01	139
	16	np-237	ground	(n, 2n)	np-236	ground	6.57E+00	2.00E+01	21
	17	np-237	ground	(n, 3n)	np-235	ground	1.24E+01	2.00E+01	9
	18	np-237	ground	(n, f)	f-prod	ground	1.00E-10	2.00E+01	174
	102	np-237	ground	(n, g)	np-238	ground	1.00E-10	2.00E+01	102
94237.30y	16	pu-237	ground	(n, 2n)	pu-236	ground	5.90E+00	2.00E+01	22
	17	pu-237	ground	(n, 3n)	pu-235	ground	1.32E+01	2.00E+01	9
	18	pu-237	ground	(n, f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	pu-237	ground	(n, g)	pu-238	ground	1.00E-10	2.00E+01	102
	16	pu-238	ground	(n, 2n)	pu-237	ground	7.05E+00	2.00E+01	12
94238.30y	17	pu-238	ground	(n, 3n)	pu-236	ground	1.29E+01	2.00E+01	5
	18	pu-238	ground	(n, f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	pu-238	ground	(n, g)	pu-239	ground	1.00E-10	2.00E+01	102
	16	pu-239	ground	(n, 2n)	pu-238	ground	5.64E+00	2.00E+01	22
	17	pu-239	ground	(n, 3n)	pu-237	ground	1.26E+01	2.00E+01	10
94239.30y	37	pu-239	ground	(n, 4n)	pu-236	ground	1.86E+01	2.00E+01	2
	18	pu-239	ground	(n, f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	pu-239	ground	(n, g)	pu-240	ground	1.00E-10	2.00E+01	102
	16	pu-240	ground	(n, 2n)	pu-239	ground	6.60E+00	2.00E+01	17
	17	pu-240	ground	(n, 3n)	pu-238	ground	1.22E+01	2.00E+01	10
94240.30y	18	pu-240	ground	(n, f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	pu-240	ground	(n, g)	pu-241	ground	1.00E-10	2.00E+01	102
	16	pu-241	ground	(n, 2n)	pu-240	ground	5.30E+00	2.00E+01	13
	17	pu-241	ground	(n, 3n)	pu-239	ground	1.19E+01	2.00E+01	8
	37	pu-241	ground	(n, 4n)	pu-238	ground	1.76E+01	2.00E+01	4
94241.30y	18	pu-241	ground	(n, f)	f-prod	ground	1.00E-10	2.00E+01	124
	102	pu-241	ground	(n, g)	pu-242	ground	1.00E-10	2.00E+01	120
	16	pu-242	ground	(n, 2n)	pu-241	ground	6.33E+00	2.00E+01	23
	17	pu-242	ground	(n, 3n)	pu-240	ground	1.16E+01	2.00E+01	13
	37	pu-242	ground	(n, 4n)	pu-239	ground	1.81E+01	2.00E+01	2
94242.30y	18	pu-242	ground	(n, f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	pu-242	ground	(n, g)	pu-243	ground	1.00E-10	2.00E+01	102
	16	pu-243	ground	(n, 2n)	pu-242	ground	5.06E+00	2.00E+01	20
	17	pu-243	ground	(n, 3n)	pu-241	ground	1.14E+01	2.00E+01	14
	37	pu-243	ground	(n, 4n)	pu-240	ground	1.66E+01	2.00E+01	7
94243.30y	18	pu-243	ground	(n, f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	pu-243	ground	(n, g)	pu-244	ground	1.00E-10	2.00E+01	102
	16	am-241	ground	(n, 2n)	am-240	ground	5.95E+00	2.00E+01	15
	17	am-241	ground	(n, 3n)	am-239	ground	1.26E+01	2.00E+01	5
	18	am-241	ground	(n, f)	f-prod	ground	1.00E-10	2.00E+01	102
95241.30y	102	am-241	ground	(n, g)	am-242	4.82E-02	1.00E-10	2.00E+01	102
	1102	am-241	ground	(n, g)	am-242	ground	1.00E-10	2.00E+01	102
	16	am-242	4.82E-02	(n, 2n)	am-241	ground	5.60E+00	2.00E+01	11
	17	am-242	4.82E-02	(n, 3n)	am-240	ground	1.13E+01	2.00E+01	8
	37	am-242	4.82E-02	(n, 4n)	am-239	ground	1.82E+01	2.00E+01	3
95242.30y	18	am-242	4.82E-02	(n, f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	am-242	4.82E-02	(n, g)	am-243	ground	1.00E-10	2.00E+01	102
	16	am-243	ground	(n, 2n)	am-242	ground	6.41E+00	2.00E+01	18
	17	am-243	ground	(n, 3n)	am-241	ground	1.19E+01	2.00E+01	10
	37	am-243	ground	(n, 4n)	am-240	ground	1.85E+01	2.00E+01	3
95243.30y	18	am-243	ground	(n, f)	f-prod	ground	1.00E-10	2.00E+01	72
	102	am-243	ground	(n, g)	am-244	ground	1.00E-10	2.00E+01	102
	16	cm-242	ground	(n, 2n)	cm-241	ground	7.00E+00	2.00E+01	14
	17	cm-242	ground	(n, 3n)	cm-240	ground	1.31E+01	2.00E+01	7
	18	cm-242	ground	(n, f)	f-prod	ground	1.00E-10	2.00E+01	102
96242.30y	102	cm-242	ground	(n, g)	cm-243	ground	1.00E-10	2.00E+01	102
	16	cm-243	ground	(n, 2n)	cm-242	ground	5.72E+00	2.00E+01	10
	17	cm-243	ground	(n, 3n)	cm-241	ground	1.27E+01	2.00E+01	6
	37	cm-243	ground	(n, 4n)	cm-240	ground	1.88E+01	2.00E+01	2

Table V cont.

For ZAID= 36078.50m there are 21 edits available

1 2 16 51 52 53 91 101 102 103 104 105 106 107 203 204 205 206 207 301
401

For ZAID= 36000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 36080.50m there are 24 edits available

1 2 16 51 52 53 54 55 56 91 101 102 103 104 105 106 107 203 204 205
206 207 301 401

For ZAID= 36082.50m there are 25 edits available

1 2 16 51 52 53 54 55 56 57 58 59 91 101 102 103 104 105 107 203
204 205 207 301 401

For ZAID= 36083.50m there are 24 edits available

1 2 16 51 52 53 54 55 56 91 101 102 103 104 105 106 107 203 204 205
206 207 301 401

For ZAID= 36084.50m there are 29 edits available

1 2 16 51 52 53 54 55 56 57 58 59 60 61 62 63 91 101 102 103
104 105 107 203 204 205 207 301 401

For ZAID= 36086.50m there are 28 edits available

1 2 16 51 52 53 54 55 56 57 58 59 60 61 62 63 64 91 101 102
103 104 105 203 204 205 301 401

For ZAID= 40000.50m there are 31 edits available

1 2 16 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67
68 69 91 101 102 103 107 203 207 301 401

For ZAID= 40000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 41093.50m there are 27 edits available

1 2 16 17 22 51 52 53 54 55 56 57 58 59 60 61 62 91 101 102
103 107 203 207 301 302 401

For ZAID= 41000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 42000.50m there are 10 edits available

1 2 16 17 91 101 102 301 302 401

For ZAID= 42000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 45103.50m there are 22 edits available

1 2 16 51 52 53 54 55 56 57 58 59 60 61 62 63 64 91 101 102
301 401

For ZAID= 45000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 45117.90m there are 8 edits available

1 2 16 17 91 101 102 302

For ZAID= 46119.90m there are 8 edits available

1 2 16 17 91 101 102 302

Table V cont.

For ZAID= 46000.01g there are 6 edits available

1 101 502 504 516 602

For ZAID= 47000.55m there are 28 edits available

1 2 16 51 52 53 54 55 56 57 58 59 60 61 91 101 102 103 104 105
107 203 204 205 207 301 302 401

For ZAID= 47000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 47107.50m there are 22 edits available

1 2 16 51 52 53 54 55 56 91 101 102 103 104 105 107 203 204 205 207
301 401

For ZAID= 47109.50m there are 17 edits available

1 2 16 51 52 53 54 55 91 101 102 103 107 203 207 301 401

For ZAID= 48000.50m there are 16 edits available

1 2 16 51 52 53 54 91 101 102 103 107 203 207 301 401

For ZAID= 48000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 50120.35m there are 8 edits available

1 2 16 17 91 101 102 302

For ZAID= 50000.01g there are 6 edits available

1 101 502 504 516 602

For ZAID= 50998.99m there are 3 edits available

1 2 91

For ZAID= 50999.99m there are 4 edits available

1 2 16 91

For ZAID= 54000.35m there are 8 edits available

1 2 16 17 91 101 102 302

For ZAID= 54000.01g there are 6 edits available

1 101 502 504 516 602

For ZAID= 56138.50m there are 14 edits available

1 2 16 17 91 101 102 103 107 203 207 301 302 401

For ZAID= 56000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 63000.35m there are 8 edits available

1 2 16 17 91 101 102 302

For ZAID= 63000.01g there are 6 edits available

1 101 502 504 516 602

For ZAID= 63151.55m there are 37 edits available

1 2 16 17 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66
67 91 101 102 103 104 105 106 107 203 204 205 206 207 301 302 401

For ZAID= 63153.55m there are 30 edits available

1 2 16 17 51 52 53 54 55 56 57 58 59 60 91 101 102 103 104 105
106 107 203 204 205 206 207 301 302 401

Table V cont.

For ZAID= 64000.35m there are 8 edits available

1 2 16 17 91 101 102 302

For ZAID= 64000.01g there are 6 edits available

1 101 502 504 516 602

For ZAID= 67165.55m there are 23 edits available

1 2 16 17 51 52 53 54 55 56 57 58 59 60 61 62 63 91 101 102
301 302 401

For ZAID= 67000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 73181.50m there are 22 edits available

1 2 16 17 51 52 53 54 55 56 57 58 59 60 91 101 102 103 203 301
302 401

For ZAID= 73000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 74000.55m there are 53 edits available

1 2 16 17 28 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85
86 87 88 89 90 91 101 102 103 107 203 207 302

For ZAID= 74000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 74182.55m there are 34 edits available

1 2 16 17 28 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
66 67 68 69 91 101 102 103 107 203 207 301 302 401

For ZAID= 74183.55m there are 29 edits available

1 2 16 17 28 51 52 53 54 55 56 57 58 59 60 61 62 63 64 91
101 102 103 107 203 207 301 302 401

For ZAID= 74184.55m there are 33 edits available

1 2 16 17 28 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
66 67 68 91 101 102 103 107 203 207 301 302 401

For ZAID= 74186.55m there are 33 edits available

1 2 16 17 28 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
66 67 68 91 101 102 103 107 203 207 301 302 401

For ZAID= 75185.50m there are 17 edits available

1 2 16 17 51 52 53 54 55 56 57 58 91 101 102 301 401

For ZAID= 75000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 75187.50m there are 20 edits available

1 2 16 17 51 52 53 54 55 56 57 58 59 60 61 91 101 102 301 401

For ZAID= 78000.35m there are 8 edits available

1 2 16 17 91 101 102 302

For ZAID= 78000.01g there are 6 edits available

1 101 502 504 516 602

Table V cont.

For ZAID= 79197.56m there are 27 edits available
 1 2 16 17 51 52 53 54 55 56 57 58 59 60 61 62 63 91 101 102
 103 107 203 207 301 302 401

For ZAID= 79000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

For ZAID= 82000.50m there are 45 edits available
 1 2 16 17 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66
 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 91
 101 102 301 302 401

For ZAID= 82000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

For ZAID= 83209.50m there are 21 edits available
 1 2 16 17 22 51 52 53 54 55 56 91 101 102 103 107 203 207 301 302
 401

For ZAID= 83000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

For ZAID= 90232.50m there are 29 edits available
 1 2 16 17 18 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
 91 101 102 301 302 318 401 452 456

For ZAID= 90000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

For ZAID= 91233.50m there are 17 edits available
 1 2 16 17 18 51 52 53 54 55 91 101 102 301 318 401 452

For ZAID= 91000.01g there are 0 edits available

For ZAID= 92233.50m there are 17 edits available
 1 2 16 17 18 51 52 53 54 91 101 102 301 318 401 452 456

For ZAID= 92000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

For ZAID= 92234.50m there are 21 edits available
 1 2 16 17 18 19 20 21 51 52 53 54 55 56 91 101 102 301 318 401
 452

For ZAID= 92235.50m there are 33 edits available
 1 2 16 17 18 19 20 21 51 52 53 54 55 56 57 58 59 60 61 62
 63 64 65 66 91 101 102 301 302 318 401 452 456

For ZAID= 92236.50m there are 21 edits available
 1 2 16 17 18 19 20 21 51 52 53 54 55 56 91 101 102 301 318 401
 452

For ZAID= 92237.50m there are 13 edits available
 1 2 16 17 18 91 101 102 301 302 318 401 452

For ZAID= 92238.50m there are 44 edits available
 1 2 16 17 18 19 20 21 51 52 53 54 55 56 57 58 59 60 61 62
 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 91 101 102 301 302
 318 401 452 456

Table V cont.

For ZAID= 92239.35m there are 12 edits available

1 2 16 17 18 37 91 101 102 302 318 456

For ZAID= 93237.55m there are 41 edits available

1 2 16 17 18 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
66 67 68 69 70 71 72 73 74 75 76 77 78 91 101 102 301 318 401 452
456

For ZAID= 93000.01g there are 0 edits available

For ZAID= 94238.50m there are 29 edits available

1 2 16 17 18 19 20 51 52 53 54 55 56 57 58 59 60 61 62 63
64 65 91 101 102 301 318 401 452

For ZAID= 94000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 94239.55m there are 35 edits available

1 2 16 17 18 19 20 21 51 52 53 54 55 56 57 58 59 60 61 62
63 64 65 66 67 68 91 101 102 301 302 318 401 452 456

For ZAID= 94240.50m there are 29 edits available

1 2 16 17 18 19 20 21 51 52 53 54 55 56 57 58 59 60 61 62
91 101 102 301 302 318 401 452 456

For ZAID= 94241.50m there are 29 edits available

1 2 16 17 18 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
91 101 102 301 302 318 401 452 456

For ZAID= 94242.50m there are 33 edits available

1 2 16 17 18 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
66 67 68 69 91 101 102 301 302 318 401 452 456

For ZAID= 95241.50m there are 32 edits available

1 2 16 17 18 19 20 51 52 53 54 55 56 57 58 59 60 61 62 63
64 65 66 67 68 91 101 102 301 318 401 452

For ZAID= 95242.50m there are 27 edits available

1 2 16 17 18 19 20 51 52 53 54 55 56 57 58 59 60 61 62 63
91 101 102 301 318 401 452

For ZAID= 95243.50m there are 31 edits available

1 2 16 17 18 19 20 51 52 53 54 55 56 57 58 59 60 61 62 63
64 65 66 67 91 101 102 301 318 401 452

For ZAID= 96242.50m there are 17 edits available

1 2 16 17 18 19 20 51 52 53 91 101 102 301 318 401 452

For ZAID= 96244.50m there are 17 edits available

1 2 16 17 18 19 20 51 52 53 91 101 102 301 318 401 452

Table VI: Definition of ENDF/B-V Reaction Types

1	Total cross section
2	Elastic scattering cross section
6	(n, 2n) cross section for first excited state (describes first neutron)
7	(n,2n) cross section for second excited state (describes first neutron)
8	(n,2n) cross section for third excited state (describes first neutron)
9	(n,2n) cross section for fourth excited state (describes first neutron)
16	direct (n,2n) cross section
17	(n,3n) cross section
18	Total fission cross section
19	(n,f) cross section (first chance fission)
20	(n,n' f) cross section (second chance fission)
21	(n,2nf) cross section (third chance fission)
22	(n,n' α) cross section
24	(n,2n α) cross section
25	(n,3n α) cross section
28	(n,n' p) cross section
37	(n,4n) cross section
46	cross section for describing the second neutron from (n,2n) reaction for first excited state
47	cross section for describing the second neutron from (n,2n) reaction for second excited state
48	cross section for describing the second neutron from (n,2n) reaction for third excited state
49	cross section for describing the second neutron from (n,2n) reaction for fourth excited state

Table VI cont.

51	(n,n') to the first excited state
52	(n,n') to the second excited state
.	.
.	.
90	(n,n') to the 40th excited state
91	(n,n') to the continuum
101	neutron disappearance (Sum of all cross sections in which a neutron is not in the exit channel, MT=102-117); also called total capture or absorption.
102	(n,γ) radiative capture cross section
103	(n,p) cross section
104	(n,d) cross section
105	(n,t) cross section
106	$(n,^3\text{He})$ cross section
107	(n,α) cross section
108	$(n,2\alpha)$ cross section
111	$(n,2p)$ cross section
112	$(n,p\alpha)$ cross section
113	$(n,t2\alpha)$ cross section
203	Total hydrogen production
204	Total deuterium production
205	Total tritium production
206	Total ^3He production
207	Total α production

Table VI cont.

452	Total nubar
456	Prompt nubar
502	Photon coherent scattering
504	Photon incoherent scattering
516	Pair production, nuclear and electron field (i.e., pair plus triplet production)
602	Photoelectric (MT=522 in ENDF/B-VI)

Table VII: Special Reaction Numbers

302	σ_{pp} - total photon-production cross section (MT=202 in ENDF/B-VI)
301	H - average heating number (MeV/collision)
318	Q_f - fission Q (MeV)
401	$H\sigma_t$ - heating number times the total cross section (MeV • barns)

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ZAID	MT	TARGET	E-TARG	REACTION	PRODUCT	E-PROD	EMIN	EMAX	NES
96244.30y	18	cm-243	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	cm-243	ground	(n,g)	cm-244	ground	1.00E-10	2.00E+01	102
	16	cm-244	ground	(n,2n)	cm-243	ground	6.83E+00	2.00E+01	15
	17	cm-244	ground	(n,3n)	cm-242	ground	1.26E+01	2.00E+01	10
	37	cm-244	ground	(n,4n)	cm-241	ground	1.95E+01	2.00E+01	2
96245.30y	18	cm-244	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	cm-244	ground	(n,g)	cm-245	ground	1.00E-10	2.00E+01	102
	16	cm-245	ground	(n,2n)	cm-244	ground	5.60E+00	2.00E+01	10
	17	cm-245	ground	(n,3n)	cm-243	ground	1.24E+01	2.00E+01	6
	37	cm-245	ground	(n,4n)	cm-242	ground	1.82E+01	2.00E+01	2
96246.30y	18	cm-245	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	cm-245	ground	(n,g)	cm-246	ground	1.00E-10	2.00E+01	102
	16	cm-246	ground	(n,2n)	cm-245	ground	6.48E+00	2.00E+01	17
	17	cm-246	ground	(n,3n)	cm-244	ground	1.20E+01	2.00E+01	11
	37	cm-246	ground	(n,4n)	cm-243	ground	1.88E+01	2.00E+01	3
96247.30y	18	cm-246	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	cm-246	ground	(n,g)	cm-247	ground	1.00E-10	2.00E+01	102
	16	cm-247	ground	(n,2n)	cm-246	ground	5.18E+00	2.00E+01	17
	17	cm-247	ground	(n,3n)	cm-245	ground	1.16E+01	2.00E+01	10
	37	cm-247	ground	(n,4n)	cm-244	ground	1.72E+01	2.00E+01	4
96248.30y	18	cm-247	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	cm-247	ground	(n,g)	cm-248	ground	1.00E-10	2.00E+01	102
	16	cm-248	ground	(n,2n)	cm-247	ground	6.23E+00	2.00E+01	15
	17	cm-248	ground	(n,3n)	cm-246	ground	1.14E+01	2.00E+01	13
	37	cm-248	ground	(n,4n)	cm-245	ground	1.79E+01	2.00E+01	5
97249.30y	18	cm-248	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	cm-248	ground	(n,g)	cm-249	ground	1.00E-10	2.00E+01	102
	16	bk-249	ground	(n,2n)	bk-248	ground	6.23E+00	2.00E+01	18
	17	bk-249	ground	(n,3n)	bk-247	ground	1.18E+01	2.00E+01	12
	37	bk-249	ground	(n,4n)	bk-246	ground	1.79E+01	2.00E+01	5
98249.30y	18	bk-249	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	125
	102	bk-249	ground	(n,g)	bk-250	ground	1.00E-10	2.00E+01	102
	16	cf-249	ground	(n,2n)	cf-248	ground	5.62E+00	2.00E+01	16
	17	cf-249	ground	(n,3n)	cf-247	ground	1.18E+01	2.00E+01	11
	37	cf-249	ground	(n,4n)	cf-246	ground	1.87E+01	2.00E+01	4
98250.30y	18	cf-249	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	cf-249	ground	(n,g)	cf-250	ground	1.00E-10	2.00E+01	102
	16	cf-250	ground	(n,2n)	cf-249	ground	6.64E+00	2.00E+01	18
	17	cf-250	ground	(n,3n)	cf-248	ground	1.23E+01	2.00E+01	11
	37	cf-250	ground	(n,4n)	cf-247	ground	1.87E+01	2.00E+01	4
98251.30y	18	cf-250	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	22
	102	cf-250	ground	(n,g)	cf-251	ground	1.00E-10	2.00E+01	102
	16	cf-251	ground	(n,2n)	cf-250	ground	5.13E+00	2.00E+01	15
	17	cf-251	ground	(n,3n)	cf-249	ground	1.18E+01	2.00E+01	8
	37	cf-251	ground	(n,4n)	cf-248	ground	1.74E+01	2.00E+01	5
98252.30y	18	cf-251	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	cf-251	ground	(n,g)	cf-252	ground	1.00E-10	2.00E+01	102
	16	cf-252	ground	(n,2n)	cf-251	ground	6.25E+00	2.00E+01	9
	17	cf-252	ground	(n,3n)	cf-250	ground	1.14E+01	2.00E+01	8
	37	cf-252	ground	(n,4n)	cf-249	ground	1.80E+01	2.00E+01	2
	18	cf-252	ground	(n,f)	f-prod	ground	1.00E-10	2.00E+01	102
	102	cf-252	ground	(n,g)	cf-253	ground	1.00E-10	2.00E+01	102

Los Alamos
NATIONAL LABORATORY
memorandum

*Applied Theoretical & Computational
Physics Division
Los Alamos, New Mexico 87545*

To/MS: MCNP Distribution
From/MS: R. C. Little / MS B226
Phone/FAX:
Symbol: X-6: RCL-87-225
Date: April 21, 1987

SUBJECT: Neutron and Photon Multigroup Data Tables for MCNP3B

Version 3B will be our first production version of MCNP to provide the capability of using multigroup cross-section sets. The format of multigroup cross-section tables for MCNP is documented in Ref. 1. This memo describes the neutron and photon multigroup libraries that we plan to provide with MCNP3B.

We will use the transport cross sections contained on the public libraries MENDF5 and MENDF5G as the starting point for the MCNP libraries. MENDF5 and MENDF5G are used by several deterministic production codes at Los Alamos. The latest versions of the libraries are described in Ref. 2. The multigroup data on MENDF5 and MENDF5G were originally generated from pointwise cross sections using the NJOY³ and CLYDE⁴ processing codes.

Multigroup neutron cross sections will be available for 95 isotopes or elements as listed in Table I. ZAID identifiers, evaluated data sources, and references are given for each of the 95 materials. Finally, Table I indicates whether photon-production data are included for each cross-section set.

Neutron cross sections are given in 30 groups. The boundaries are listed in Table II. The spectrum used for collapse is the standard TD-Division weight function described in Ref. 5. Neutron cross sections have generally been processed at room temperature (300°K) with the exception of the ENDL85 and PERMFILE sets which have been processed at 0°K. No self-shielding has been incorporated; all cross sections are infinitely dilute. No upscatter groups are provided for.

For fissionable isotopes, the data tables will contain either prompt nubar, total nubar, or both prompt and total nubar. The type of nubar available for a particular isotope will be the same as what is available for its continuous-energy counterpart. This information is documented in Appendix G of the MCNP manual.⁶ Presently, the fission chi is independent of incident neutron energy and is generally the prompt chi. As is the case with all of our continuous-energy neutron tables, secondary photon-production data include prompt photons only.

Multigroup photon cross sections will be available for 55 elements as listed in Table III. All photon transport cross sections have been obtained from R. E. MacFarlane's file /LIB12/DTF12⁷ and are based on the DLC-7E⁸ library.

Photon cross sections are given in 12 groups. The boundaries are listed in Table IV. The weight function used for collapse of photon cross sections is described in Ref. 9.

Angular distributions of scattered particles are accounted for on MENDF5 and MENDF5G by PO through P4 Legendre scattering tables (no transport correction). MCNP requires angular distributions for each group-to-group scattering in terms of either discrete cosines or equally-probable cosine bins. We have chosen to process the data contained in the PO-P4 tables into equally-probable cosine bins. We are able to

calculate three equally-probable cosine bins for each combination of incident and exiting group using methods described in Ref. 10.

To use multigroup cross sections in MCNP3B, it is necessary that the cross-section directory contain information about the multigroup libraries. When MCNP3B becomes the production version, the public file XSDIR will be expanded to include the necessary information. Until that time, the user must fetch /X6MGXS/CTSS/DIR2NP from CFS. Then, switch DIR2NP to XSDIR, or set XSDIR=DIR2NP on the MCNP execute line. The MGOPT card¹¹ in your input deck will signal MCNP that you want a multigroup calculation. When using the cross-section tables described in this memo, the second entry on the MGOPT card (IGM) should be 30 for MODE N, 12 for MODE P, and 42 for MODE N P. Users will have the same flexibility in specifying ZAIDs on the MCNP material cards that exists now for continuous-energy calculations. For example, the following three Mn cards are all equivalent:

```
Mn 92235 1
Mn 92235.50 1
Mn 92235.50M 1
```

In a MODE N P problem, any of the above material cards would force the code to access both the 92235.50M and the 92000.01G tables.

Very complete edit cross sections for use with the FM feature of MCNP are available on these libraries. The edits have been obtained from TRANSX¹² and CLYDE⁴ calculations. In general, any cross section available to MCNP as a response function from a continuous-energy table (including any of the special negative FM reaction numbers) will be available to MCNP from the corresponding multigroup table. The edit reactions available for each multigroup table are listed in Table V. Most of the edit numbers identify reactions as defined in the ENDF/B system for evaluated data.¹³ Table VI is a list of many of these reactions. Users should be aware of exceptions for several light isotopes (D, ⁶Li, ⁷Li, ¹⁰B, C, ¹²C, ¹⁴N, ¹⁶O, and ²⁷Al). For these isotopes, reactions identified in Table VI as (n,n') inelastic scattering may actually be (n,n'X) inelastic scattering, where X indicates some combination of particles. The exceptions are described in Refs. 14 and 15. We have also defined several additional reaction numbers. These special reaction numbers are listed in Table VII.

We will conclude this memo by acknowledging our understanding that one multigroup library cannot be adequate for all problems. There will, therefore, be available a code that will read from several standard multigroup formats (e.g., DTF, FIDO), process the data, and produce multigroup tables in a format ready for use in MCNP. This code will be described in a subsequent memo.

Table 1: Neutron Multigroup Cross-Section Tables for MCNP3B

Isotope	ZAID	Source	Reference	Photon Production
H	1001.50m	ENDF/B-V	16	Yes
D	1002.55m	Group T-2	17	Yes
T	1003.50m	ENDF/B-V	16	No
He-3	2003.50m	ENDF/B-V	16	No
He-4	2004.50m	ENDF/B-V	16	No
Li-6	3006.50m	ENDF/B-V	16	Yes
Li-7	3007.55m	Group T-2	18	Yes
Be-7	4007.35m	ENDL85	19	No
Be-9	4009.50m	ENDF/B-V	16	Yes
B-10	5010.50m	ENDF/B-V	16	Yes
B-11	5011.56m	Group T-2	20	Yes
C	6000.50m	ENDF/B-V	16	Yes
C-12*	6012.50m	ENDF/B-V	16	Yes
N-14	7014.50m	ENDF/B-V	16	Yes
N-15	7015.55m	Group T-2	21	Yes
O-16	8016.50m	ENDF/B-V	16	Yes
F-19	9019.50m	ENDF/B-V	16	Yes
Na-23	11023.50m	ENDF/B-V	16	Yes
Mg	12000.50m	ENDF/B-V	16	Yes
Al-27	13027.50m	ENDF/B-V	16	Yes
Si	14000.50m	ENDF/B-V	16	Yes
P-31	15031.50m	ENDF/B-V	16	Yes
S-32	16032.50m	ENDF/B-V	16	Yes
Cl	17000.50m	ENDF/B-V	16	Yes
Ar	18000.35m	ENDL85	19	Yes
K	19000.50m	ENDF/B-V	16	Yes
Ca	20000.50m	ENDF/B-V	16	Yes
Ti	22000.50m	ENDF/B-V	16	Yes
V	23000.50m	ENDF/B-V	16	Yes
Cr	24000.50m	ENDF/B-V	16	Yes
Mn-55	25055.50m	ENDF/B-V	16	Yes
Fe	26000.55m	Group T-2	22	Yes
Co-59	27059.50m	ENDF/B-V	16	Yes
Ni	28000.50m	ENDF/B-V	16	Yes
Cu	29000.50m	ENDF/B-V	16	Yes

Table 1 cont.

Isotope	ZAID	Source	Reference	Photon Production
Ga	31000.50m	ENDF/B-V	16	Yes
As-75	33075.35m	ENDL85	19	Yes
Kr-78	36078.50m	ENDF/B-V	16	No
Kr-80	36080.50m	ENDF/B-V	16	No
Kr-82	36082.50m	ENDF/B-V	16	No
Kr-83	36083.50m	ENDF/B-V	16	No
Kr-84	36084.50m	ENDF/B-V	16	No
Kr-86	36086.50m	ENDF/B-V	16	No
Zr	40000.50m	ENDF/B-V	16	No
Nb-93	41093.50m	ENDF/B-V	16	Yes
Mo	42000.50m	ENDF/B-V	16	Yes
Rh-103	45103.50m	ENDF/B-V	16	No
U235 FP**	45117.90m	Group T-2	23	Yes
Pu239 FP**	46119.90m	Group T-2	23	Yes
Ag	47000.55m	Group T-2	24	Yes
Ag-107	47107.50m	ENDF/B-V	16	No
Ag-109	47109.50m	ENDF/B-V	16	No
Cd	48000.50m	ENDF/B-V	16	No
Ave FP**	50120.35m	ENDL85	19	Yes
FPP**	50998.99m	PERMFILE	25	No
FPA**	50999.99m	PERMFILE	25	No
Xe	54000.35m	ENDL85	19	Yes
Ba-138	56138.50m	ENDF/B-V	16	Yes
Eu	63000.35m	ENDL85	19	Yes
Eu-151	63151.55m	Group T-2	26	Yes
Eu-153	63153.55m	Group T-2	26	Yes
Gd	64000.35m	ENDL85	19	Yes
Ho-165	67165.55m	Group T-2	26	Yes
Ta-181	73181.50m	ENDF/B-V	16	Yes
W	74000.55m	Group T-2	27	Yes
W-182	74182.55m	Group T-2	28	Yes
W-183	74183.55m	Group T-2	28	Yes
w-184	74184.55m	Group T-2	28	Yes
W-186	74186.55m	Group T-2	28	Yes
Re-185	75185.50m	ENDF/B-V	16	No

Table 1 cont.

Isotope	ZAID	Source	Reference	Photon Production
Re-187	75187.50m	ENDF/B-V	16	No
Pt	78000.35m	ENDL85	19	Yes
Au-197	79197.56m	Group T-2	29	Yes
Pb	82000.50m	ENDF/B-V	16	Yes
Bi-209	83209.50m	ENDF/B-V	16	Yes
Th-232	90232.50m	ENDF/B-V	16	Yes
Pa-233	91233.50m	ENDF/B-V	16	No
U-233	92233.50m	ENDF/B-V	16	No
U-234	92234.50m	ENDF/B-V	16	No
U-235	92235.50m	ENDF/B-V	16	Yes
U-236	92236.50m	ENDF/B-V	16	No
U-237	92237.50m	ENDF/B-V	16	Yes
U-238	92238.50m	ENDF/B-V	16	Yes
U-239	92239.35m	ENDL85	19	Yes
Np-237	93237.55m	Group T-2	30	No
Pu-238	94238.50m	ENDF/B-V	16	No
Pu-239	94239.55m	Group T-2	31	Yes
Pu-240	94240.50m	ENDF/B-V	16	Yes
Pu-241	94241.50m	ENDF/B-V	16	Yes
Pu-242	94242.50m	ENDF/B-V	16	Yes
Am-241	95241.50m	ENDF/B-V	16	No
Am-242m	95242.50m	ENDF/B-V	16	No
Am-243	95243.50m	ENDF/B-V	16	No
Cm-242	96242.50m	ENDF/B-V	16	No
Cm-244	96244.50m	ENDF/B-V	16	No

* - Data for C-12 are identical to those for natural carbon.

** - Several sets of fission-product cross sections are included. ZAID=50120.35M is the cross section for an average fission product from ENDL85. ZAID=45117.90M and ZAID=46119.90M are cross sections from Group T-2 for prompt fission products from ²³⁵U fission and ²³⁹Pu fission, respectively. ZAID=50998.99M and ZAID=50999.99M are older cross section sets intended for fission-product pairs. The data have been used often over the years at Los Alamos and are included here for completeness.

Table II: Boundaries of Neutron Energy Groups

<u>Group</u>	<u>Upper Energy (MeV)</u>	<u>Group</u>	<u>Upper Energy (MeV)</u>
1	17.000	16	0.184
2	15.000	17	0.0676
3	13.500	18	0.0248
4	12.000	19	0.00912
5	10.000	20	0.00335
6	7.790	21	0.001235
7	6.070	22	4.54E-04
8	3.680	23	1.67E-04
9	2.865	24	6.14E-05
10	2.232	25	2.26E-05
11	1.738	26	8.32E-06
12	1.353	27	3.06E-06
13	0.823	28	1.13E-06
14	0.500	29	4.14E-07
15	0.303	30	1.52E-07
		lower energy of Group 30 is equal to 1.39E-10 MeV	

Table III: Photon Multigroup Cross-Section Tables for MCNP3B

<u>Element</u>	<u>ZAID</u>	<u>Element</u>	<u>ZAID</u>
H	1000.01g	Zr	40000.01g
He	2000.01g	Nb	41000.01g
Li	3000.01g	Mo	42000.01g
Be	4000.01g	Rh	45000.01g
B	5000.01g	Pd	46000.01g
C	6000.01g	Ag	47000.01g
N	7000.01g	Cd	48000.01g
O	8000.01g	Sn	50000.01g
F	9000.01g	Xe	54000.01g
Na	11000.01g	Ba	56000.01g
Mg	12000.01g	Eu	63000.01g
Al	13000.01g	Gd	64000.01g
Si	14000.01g	Ho	67000.01g
P	15000.01g	Ta	73000.01g
S	16000.01g	w	74000.01g
Cl	17000.01g	Re	75000.01g
Ar	18000.01g	Pt	78000.01g
K	19000.01g	Au	79000.01g
Ca	20000.01g	Pb	82000.01g
Ti	22000.01g	Bi	83000.01g
V	23000.01g	Th	90000.01g
Cr	24000.01g	Pa*	91000.01g
Mn	25000.01g	U	92000.01g
Fe	26000.01g	Np*	93000.01g
Co	27000.01g	Pu	94000.01g
Ni	28000.01g		
Cu	29000.01g		
Ga	31000.01g		
As	33000.01g		
Kr	36000.01g		

* - Data for Z=91 and Z=93 are not available on DLC-7E. The values here have been arrived at for Z=91 by averaging data for Z=90 and Z=92, and for Z=93 by averaging data for Z=92 and Z=94.

Table IV: Boundaries of Photon Energy Groups

<u>Group</u>	<u>Upper Energy (MeV)</u>
1	20.0
2	9.0
3	8.0
4	7.0
5	6.0
6	5.0
7	4.0
8	3.0
9	2.0
10	1.0
11	0.5
12	0.1

lower energy of Group 12 is equal to 0.01 MeV

Table V: Edit Reactions for Each Multigroup Table

For ZAID= 1001.50m there are 8 edits available
 1 2 101 102 204 301 302 401

For ZAID= 1000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

For ZAID= 1002.55m there are 39 edits available
 1 2 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68
 69 70 71 72 73 74 75 76 77 78 79 80 101 102 203 205 301 302 401

For ZAID= 1003.50m there are 6 edits available
 1 2 16 204 301 401

For ZAID= 2003.50m there are 10 edits available
 1 2 101 103 104 203 204 205 301 401

For ZAID= 2000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

For ZAID= 2004.50m there are 4 edits available
 1 2 301 401

For ZAID= 3006.50m there are 43 edits available
 1 2 24 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67
 68 69 70 71 72 73 74 75 76 77 78 79 101 102 103 105 203 204 205 207
 301 302 401

For ZAID= 3000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

For ZAID= 3007.55m there are 42 edits available
 1 2 16 24 25 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
 66 67 68 69 70 71 72 73 74 75 76 77 101 102 104 203 204 205 207 301
 302 401

For ZAID= 4007.35m there are 7 edits available
 1 2 101 103 107 203 207

For ZAID= 4000.01g there are 6 edits available
 1 101 502 504 516 602

For ZAID= 4009.50m there are 23 edits available
 1 2 6 7 8 9 46 47 48 49 101 102 103 104 105 107 203 204 205 207
 301 302 401

For ZAID= 5010.50m there are 44 edits available
 1 2 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68
 69 70 71 72 73 74 75 76 77 78 79 101 102 103 104 107 113 203 204 205
 207 301 302 401

For ZAID= 5000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

For ZAID= 5011.56m there are 26 edits available
 1 2 16 22 28 51 52 53 54 55 56 57 58 59 60 101 102 103 105 107
 203 205 207 301 302 401

Table V cont.

For ZAID= 6000.50m there are 28 edits available
 1 2 51 52 53 54 55 56 57 58 59 60 61 62 63 64 91 101 102 103
 104 107 203 204 207 301 302 401

For ZAID= 6000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

For ZAID= 6012.50m there are 28 edits available
 1 2 51 52 53 54 55 56 57 58 59 60 61 62 63 64 91 101 102 103
 104 107 203 204 207 301 302 401

For ZAID= 7014.50m there are 44 edits available
 1 2 16 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67
 68 69 70 71 72 73 74 75 76 77 101 102 103 104 105 107 108 203 204 205
 207 301 302 401

For ZAID= 7000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

For ZAID= 7015.55m there are 26 edits available
 1 2 16 22 28 51 52 53 54 55 56 57 91 101 102 103 104 105 107 203
 204 205 207 301 302 401

For ZAID= 8016.50m there are 43 edits available
 1 2 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68
 69 70 71 72 73 74 75 76 77 78 79 80 101 102 103 104 107 203 204 207
 301 302 401

For ZAID= 8000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

For ZAID= 9019.50m there are 40 edits available
 1 2 16 22 28 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
 66 67 68 69 70 71 91 101 102 103 104 105 107 203 204 205 207 301 302 401

For ZAID= 9000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

For ZAID= 11023.50m there are 31 edits available
 1 2 16 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67
 68 91 101 102 103 107 203 207 301 302 401

For ZAID= 11000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

For ZAID= 12000.50m there are 55 edits available
 1 2 16 22 28 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85
 86 87 88 89 90 91 101 102 103 107 203 207 301 302 401

For ZAID= 12000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

For ZAID= 13027.50m there are 52 edits available
 1 2 16 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67
 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 101
 102 103 104 105 107 203 204 205 207 301 302 401

For ZAID= 13000.01g there are 8 edits available
 1 101 301 401 502 504 516 602

Table V cont.

For ZAID= 14000.50m there are 39 edits available

1 2 16 22 28 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
66 67 68 69 70 71 72 91 101 102 103 104 107 203 204 207 301 302 401

For ZAID= 14000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 15031.50m there are 14 edits available

1 2 16 28 91 101 102 103 107 203 207 301 302 401

For ZAID= 15000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 16032.50m there are 16 edits available

1 2 16 28 91 101 102 103 105 107 203 205 207 301 302 401

For ZAID= 16000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 17000.50m there are 28 edits available

1 2 16 22 28 51 52 53 54 55 56 57 58 59 60 61 62 63 91 101
102 103 107 203 207 301 302 401

For ZAID= 17000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 18000.35m there are 11 edits available

1 2 16 91 101 102 103 107 203 207 302

For ZAID= 18000.01g there are 6 edits available

1 101 502 504 516 602

For ZAID= 19000.50m there are 32 edits available

1 2 16 22 28 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
66 67 91 101 102 103 107 203 207 301 302 401

For ZAID= 19000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 20000.50m there are 47 edits available

1 2 16 22 28 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
66 67 68 69 70 71 72 73 91 101 102 103 104 105 106 107 108 111 112 203
204 205 206 207 301 302 401

For ZAID= 20000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 22000.50m there are 34 edits available

1 2 16 22 28 51 52 53 54 55 56 57 58 59 60 61 62 91 101 102
103 104 105 106 107 111 203 204 205 206 207 301 302 401

For ZAID= 22000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 23000.50m there are 26 edits available

1 2 16 22 28 51 52 53 54 55 56 57 91 101 102 103 104 105 107 203
204 205 207 301 302 401

For ZAID= 23000.01g there are 8 edits available

1 101 301 401 502 504 516 602

Table V cont.

For ZAID= 24000.50m there are 61 edits available

1 2 16 22 28 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85
86 87 88 89 90 91 101 102 103 104 105 106 107 203 204 205 206 207 301 302
401

For ZAID= 24000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 25055.50m there are 31 edits available

1 2 16 22 28 51 52 53 54 55 56 57 58 59 60 61 62 91 101 102
103 104 106 107 203 204 206 207 301 302 401

For ZAID= 25000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 26000.55m there are 61 edits available

1 2 16 22 28 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85
86 87 88 89 90 91 101 102 103 104 105 106 107 203 204 205 206 207 301 302
401

For ZAID= 26000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 27059.50m there are 32 edits available

1 2 16 22 28 51 52 53 54 55 56 57 58 59 60 61 91 101 102 103
104 105 106 107 203 204 205 206 207 301 302 401

For ZAID= 27000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 28000.50m there are 44 edits available

1 2 16 22 28 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65
66 67 68 69 70 71 72 73 74 75 76 91 101 102 103 104 107 111 203 204
207 301 302 401

For ZAID= 28000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 29000.50m there are 30 edits available

1 2 16 22 28 51 52 53 54 55 56 57 58 59 60 61 91 101 102 103
104 106 107 203 204 206 207 301 302 401

For ZAID= 29000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 31000.50m there are 13 edits available

1 2 16 91 101 102 103 107 203 207 301 302 401

For ZAID= 31000.01g there are 8 edits available

1 101 301 401 502 504 516 602

For ZAID= 33075.35m there are 11 edits available

1 2 16 91 101 102 103 107 203 207 302

For ZAID= 33000.01g there are 6 edits available

1 101 502 504 516 602

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ENDF/B-VI Neutron Library for MCNP with Probability Tables

Robert C. Little
X-CI; Code Integration Group

Robert E. MacFarlane
T-2; Applied Theoretical and Nuclear Physics

Executive Summary

A neutron cross-section library that includes probability tables to represent unresolved resonances is available for use in MCNP. The name of the library is URES. Room-temperature data for 27 isotopes are provided, based on ENDF/B-VI release 4. URES is compatible with versions of MCNP beginning with 4XS. This Research Note documents the creation, verification, and utilization of the URES library.

Introduction

Versions of MCNPTM up through and including Version 4B¹ have treated neutron cross sections in the unresolved energy range as if they were infinitely dilute. The energy-dependent cross sections in this energy range, therefore, have been relatively smooth, without any substantial fluctuations or resonance structure. The result has been that neutron self-shielding effects in the unresolved energy region have not been accurately modeled with MCNP.

Figure 1 illustrates the ENDF/B-V radiative capture cross section for ²³⁸U as a function of neutron energy. The energy region between the vertical dashed lines (from 4 keV to 149 keV) is the unresolved resonance region as defined by this particular evaluation. As can be seen, there appears to be an abrupt transition at 4 keV from detailed resonance behavior to a smooth cross section.

In reality, the lack of structure exhibited in Figure 1 in the unresolved energy region is not physical. In fact, resonance behavior *does* extend to higher energies. However, resolved resonance structure is generally not extended to these higher energies in most nuclear data evaluations. The major reason for this omission is that the resolution of available experimental data is such that individual resonances can no longer be well defined at these energies. In addition, the practicality of including such explicit data eventually becomes prohibitive.

Often times, nuclear data evaluations *do* include information describing the distribution of resonance behavior in the unresolved region. For example, the ENDF format² allows evaluators to provide average level spacings and average neutron, radiative, fission, and competitive widths (all as a function of neutron energy) over the unresolved region. It has generally been the *average* (also referred to as the smooth, or unshielded, or infinitely dilute) cross sections determined from these parameters that MCNP has used.

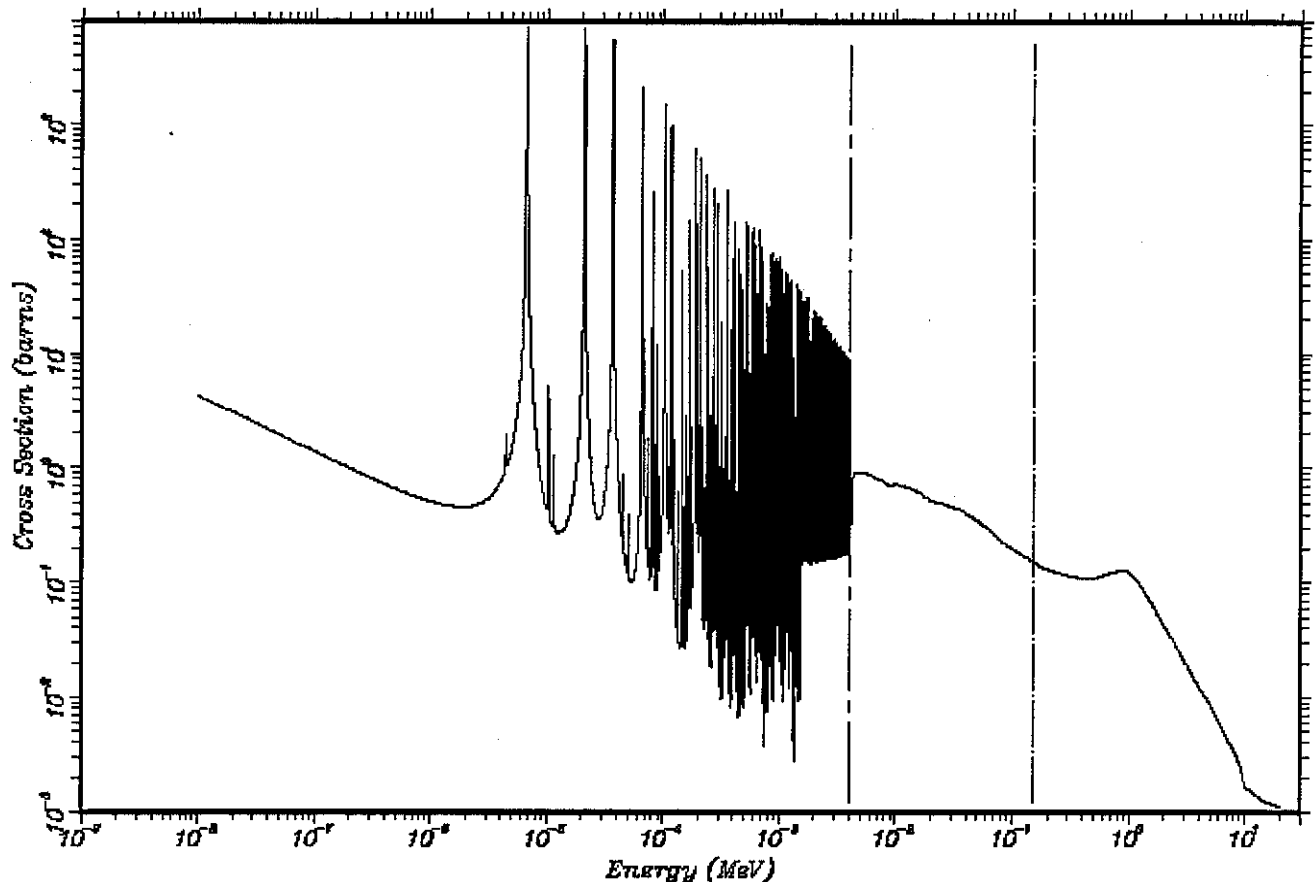
One approach that has been implemented in continuous-energy Monte Carlo neutron transport codes to account statistically for the average resonance parameters specified in the unresolved resonance range is the probability table (PT) method.³ In this method, the average parameters are used to generate ladders of representative resonances. Cross sections from these ladders are then used to form cross-section probability distribution functions, from which a table of cross sections (total, elastic, fission, and radiative capture) as a function of probability is prepared. When transporting neutrons in the unresolved energy range of a particular nuclide, the Monte Carlo code samples the total and reaction cross sections at the energy of the neutron, rather than simply using single average values at this energy. The PT method has been available in the VIM Monte Carlo code for some time.⁴ More recently, it has also been implemented in the RACER⁵ and TRIPOLI⁶ codes.

MCNP has also been upgraded to incorporate the PT method. The implementation is described in Ref. 7. Specifics concerning the first version of MCNP to be released with the PT method (Version 4XS) are provided in Ref. 8.

In parallel with the efforts to upgrade MCNP, a new MCNP neutron cross-section library (named URES) has also been developed. The library provides data for 27 isotopes based on ENDF/B-VI Release 4.⁹ This is the first time that PT data have been incorporated in an

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Fig. 1: U-238 Unresolved Region (ENDF/B-V capture)



MCNP library. Data on the library have been processed by NJOY.¹⁰ All cross sections on the library have been processed at room temperature (300 degrees Kelvin). The MCNP ZAIDs of all cross-section tables on URES end in .49c.

The remainder of this Research Note documents the creation, verification, and utilization of the URES library, including a concern about heating numbers in the probability tables. The following sections describe the ENDF/B evaluations used, the NJOY processing techniques, and the verification efforts. Then, the actual MCNP library is discussed, along with how to use it in MCNP. Relationships to data on other MCNP libraries will be described. Finally, selected calculational results will be presented.

Evaluation Sources

This work commenced by processing 29 ENDF/B-VI evaluations. The 29 materials selected for processing represent a subset of the ENDF/B-VI evaluations that include unresolved resonance data. The specific materials for this work were selected based upon their perceived importance in nuclear criticality safety applications. Note that MCNP data tables from two evaluations, Mo and ¹⁸¹Ta, were ultimately discarded. See the section below on verification for details on why this was the case.

Table I gives information about the source of each evaluation used in this work. In particular, the ENDF/B-VI Release used as the evaluation source is specified for each of the 29 materials processed.

Table I
Materials and ENDF/B-VI Releases of Evaluations Used

Material	ENDF/B-VI Release	Material	ENDF/B-VI Release	Material	ENDF/B-VI Release
Mo (Discarded)	0	W-182	0	U-235	4
Sm-149	0	W-183	0	U-236	0
Sm-150	2	W-184	0	U-238	2 (Modified)
Sm-152	2	W-186	0	Pu-238	0
Eu-151	0	Ir-191	4	Pu-239	2
Eu-152	0	Ir-193	4	Pu-240	2
Eu-153	0	Th-232	0	Pu-241	3
Eu-154	0	U-232	0	Pu-242	0
Ta-181 (Discarded)	0	U-233	0	Cm-244	0
Ta-182	0	U-234	0		

As noted in Table I, one evaluation was modified. The ^{238}U ENDF/B-VI Release 2 evaluation specifies an upper energy for the unresolved region of 300 keV. Unresolved data, however, are only provided up to 149.029 keV. To resolve this inconsistency, we changed the specified upper energy boundary of the unresolved region to 149.029 keV. (To be completely honest, this change was actually effected by changes made by hand to the processed NJOY file, not to the evaluation itself). In the subsequently available ENDF/B-VI Release 5 evaluation of ^{238}U , the upper energy of the unresolved region has indeed been changed to 149.029 keV, consistent with our modification here. This is not, however, the only change that has been made to the Release 5 evaluation.

NJOY processing

NJOY97 is a recent release of the NJOY nuclear data processing system. The code has undergone substantial modification in order to make it compatible with a much wider

variety of compilers and computer hardware, both 32-bit and 64-bit systems. All processing for this work was performed with versions of NJOY97.

For each material, two major NJOY calculations are performed. Most of the work occurs during the first calculation, which we commonly refer to as the "PENDF run." This sequence utilizes the following modules of NJOY: RECONR, BROADR, UNRESR, HEATR, PURR, THERMR, and GASPR. This sequence begins by operating on the original ENDF/B-VI evaluation. RECONR reconstructs pointwise cross sections from resonance parameters and linearizes all cross sections on a unionized energy grid. BROADR generates Doppler-broadened cross sections using the kernel broadening method. UNRESR produces effective self-shielded cross sections for resonance reactions in the unresolved energy region. The HEATR module generates various quantities related to heat production and energy deposition. The PURR module is especially important for this work – it is in this module that the unresolved probability table data are generated. THERMR generates scattering cross sections in the thermal energy range. Finally, the GASPR module calculates production cross sections of hydrogen and helium isotopes.

The NJOY input file for the PENDF run for ^{235}U is reproduced in Appendix A. All materials were processed through NJOY using input files of this form and content. The parameters of particular importance to note are the 0.001 values for fractional tolerances in both RECONR and BROADR. Also, note that the PURR input specifies 20 probability bins for each table, with each table to be generated based upon 32 unique resonance ladders. The final comment to make about the NJOY input file shown in Appendix A is that it creates as its output a file that is very general in nature – much more general, in fact, than required specifically for this work. For example, the multiple temperatures and sigma zero values are not required in this work, and the UNRESR and THERMR modules could have been omitted entirely if the only objective was to generate a file for the MCNP URES library.

The second NJOY calculation is what we commonly refer to as the "ACER run." The major task accomplished during this sequence is to reformat the previously generated data into a form that is directly useful to MCNP. Error checking and consistency checks are performed during this sequence. Interpreted listings and plots are also generated. The NJOY input file for the ACER run for ^{235}U is reproduced in Appendix B. All cross sections on the MCNP library described in this Research Note are at 300 degrees Kelvin. As in the example in Appendix B, no thinning of the energy grid was performed during the ACER runs.

Table II provides a detailed list of exactly which version of NJOY97 was used in the PENDF run and ACER run for each of the 29 materials processed. The table also includes the date of the various calculations. All PENDF runs were performed on a Sun workstation whereas all ACER runs were performed on a Cray.

Table II
Versions and Dates of NJOY Processing

Material	NJOY Version for PENDF Run	Date of PENDF Run	NJOY Version for ACER Run	Date of ACER Run
Mo (Discarded)	97.15	8/18/98	97.9	8/18/98
Sm-149	97.13	8/14/98	97.9	8/15/98
Sm-150	97.13	8/14/98	97.9	8/15/98
Sm-152	97.13	8/14/98	97.9	8/15/98
Eu-151	97.13	8/14/98	97.9	8/15/98
Eu-152	97.13	8/14/98	97.9	8/15/98
Eu-153	97.13	8/15/98	97.9	8/15/98
Eu-154	97.13	8/15/98	97.9	8/15/98
Ta-181 (Discarded)	97.13	8/18/98	97.9	8/18/98
Ta-182	97.13	8/18/98	97.9	8/18/98
W-182	97.13	8/14/98	97.9	8/15/98
W-183	97.13	8/14/98	97.9	8/15/98
W-184	97.13	8/14/98	97.9	8/15/98
W-186	97.13	8/14/98	97.9	8/15/98
Ir-191	97.13	8/18/98	97.9	8/18/98
Ir-193	97.13	8/18/98	97.9	8/18/98
Th-232	97.15	8/16/98	97.9	8/16/98
U-232	97.17	8/27/98	97.9	8/27/98
U-233	97.3	5/28/98	97.9	8/16/98
U-234	97.13	8/15/98	97.9	8/15/98
U-235	97.13	8/16/98	97.14	8/28/98
U-236	97.13	8/15/98	97.9	8/16/98
U-238	97.13	8/16/98	97.14	8/28/98
Pu-238	97.10	7/4/98	97.6	7/6/98
Pu-239	97.13	8/15/98	97.9	8/17/98
Pu-240	97.11	8/13/98	97.9	8/16/98
Pu-241	97.11	8/14/98	97.9	8/16/98
Pu-242	97.11	8/14/98	97.9	8/16/98
Cm-244	97.15	8/17/98	97.9	8/17/98

Verification

URES contains nearly 5 million words of data. The prospect of verifying every piece of data, on a word-by-word basis, is not practical. Instead, we rely on a number of methods we have developed in the past to verify continuous-energy MCNP data tables and one new code written expressly for verifying the new PT data.

We will first list the traditional verification techniques employed in this work (see Ref. 11 for more details):

- NJOY performs several consistency checks during the second execution of the ACER module shown in Appendix B. The code checks several attributes of reaction thresholds, energy grids, and secondary energy and angular distributions. It flags anything known to be wrong or considered likely to be unreasonable. All 29 materials passed these internal consistency checks. In addition, we examined all NJOY output files for "messages" generated by the code during the execution of the various modules. All such messages were understood and were considered to be of minor consequence.
- Plots of each individual cross section for every isotope on URES were generated with the XSPLIT code (an updated version of XDATAP¹²). Each of the 764 plots was displayed and examined for signs of possible processing errors. None were identified, although energy-balance problems in the original evaluations led to negative heating numbers at certain neutron energies for ¹⁵¹Eu, ^{183,184}W, ^{191,193}Ir, and ²³²Th. The plots have been archived as a postscript file.
- The WHALE code¹³ was used to calculate average cross sections in several energy ranges for each reaction from both the original ENDF/B-VI evaluation and from the final MCNP data table as processed by NJOY. Comparisons were made. All differences reported by WHALE (above a minimal tolerance level) were easily understood and deemed reasonable.
- Several small, special-purpose checking codes were executed. These codes are described in more detail in Ref. 11. No problems were identified.

The incorporation of probability tables is a major addition to MCNP data sets and a relatively untested feature of NJOY. As such, we wrote a new code specifically to perform a variety of consistency tests on the PT data. The code is named check_ures.

Check_ures does the following for each PT on an MCNP library:

- Ensures that any competition reaction specified (see the format description in Appendix C) actually exists.
- Checks the infinitely-dilute total cross section at the energy of the PT against the sum of the infinitely-dilute elastic, fission, capture, and competition cross sections. Fractional differences greater than 1.e-06 are noted.
- Checks to make sure that the cumulative probabilities as given in the PT are in ascending order, and that the distribution is correctly normalized to 1.

- Checks for negative values of the total, elastic, fission, and capture cross sections in each row of the PT. In addition, checks for zero cross sections for total, elastic, and capture.
- Checks for zero or negative values of the heating numbers in each row of the PT. Also warns if the value of a particular heating number is greater than some reasonable upper limit (chosen to be 210 MeV/collision).
- For each row, compares the total cross section from the PT with the sum of the elastic, fission, capture, and competition cross sections. Differences greater than 1% are noted.
- For each cross section (total, elastic, fission, capture, and heating), compares the infinitely dilute value at that neutron energy with the apparent average value from the PT. The infinitely dilute value is obtained from the traditional portion of the data table. The average value from the PT is obtained by summing the product of the differential probability and the cross section over each row. Fractional differences greater than 1.e-04 are noted.

One class of problem identified by check_ures led to the decision to exclude Mo and ¹⁸¹Ta from the MCNP library. For both materials, a large number of negative cross sections were found in the PT data. Upon further review, the problems were traced to negative background cross sections in the ENDF/B-VI evaluations. These negative backgrounds are not an issue when only calculating infinitely dilute cross sections. They do, however, pose an unrealistic challenge to a processing code attempting to generate probability tables. Thus, we decided to drop plans to include data for Mo and ¹⁸¹Ta on this library. This is the reason that, although we processed 29 materials, only 27 survived on URES.

Other problems identified by check_ures reflected upon the processing. Consequently, several iterations of NJOY processing resulted. In fact, the "final" NJOY processing as summarized in the previous section still led to some complaints from check_ures. They were:

- ¹⁵²Sm had small differences between the infinitely-dilute total and the sum of the infinitely-dilute partials. This issue resulted from the fact that NJOY97.9 neglected the competition cross sections from MT=22 and MT=107. These cross sections, while small, are non-zero over portions of the unresolved energy range. Instead of re-running NJOY, we directly modified the file produced to include MT=22 as an inelastic competition reaction and MT=107 as an absorption competition reaction. In addition, NJOY was updated so that it would include these competition reactions in the future.
- Two materials (²³²Th and ²⁴¹Pu) each had one row in one PT with a zero differential probability. The same row also included non positive cross sections. Although this is not a desirable situation, it is not capable of damaging an MCNP calculation (since the row in doubt will never be sampled) and effectively means only that these two probability tables have 19 rows instead of 20. No changes were made.
- Several cases were identified for rows in which the total did not equal the sum of the partials. This check was made at the 1% level; it was repeated at a 5% level and no such disagreements were found. We believe that the apparent discrepancies result from the fact that NJOY normalizes each partial cross section and the total cross section (within a particular PT) independently from one another. As implemented in MCNP, the total cross section within the unresolved range is always calculated as the

sum of the partials (i.e., the total cross section in the PT is never used directly). Therefore, we decided that no changes were necessary in this matter.

- Non-physical values of heating numbers were found in several cases for ^{232,233}U and ^{238,241}Pu. Values greater than 2900 MeV/collision were found. It should be emphasized that the average heating number from each PT was consistent with the infinitely-dilute value. However, one of us (RCL) believes that distributions of heating numbers that include such extremely large values are indicative that further work is required in this area. No changes were made at this time, however. *We warn users to be somewhat careful about MCNP F6 tallies calculated using this library.*

Finally, we are in the process of collaborating with KAPL and ANL to further verify the data in our probability tables. Both Laboratories have generated PT data from the same underlying evaluations, but by methods independent of NJOY. Thus, similarities between PT data from other Laboratories would lend additional confidence to our PT data; differences would require further investigation. This collaboration is only in its infancy; however, preliminary comparisons with KAPL PT data have been very encouraging. Results of this study will be reported at a later date.

MCNP Library

Data for the 27 isotopes were combined on to one library, named URES. Some characteristics of each data table are listed in Table III. All data have been processed at 300 degrees Kelvin. All ZAIDs on URES end in .49c.

Versions of the URES library (and associated directory files) for MCNP have been stored on CFS at Los Alamos. All files are under the /x6data/ce/special/probtable sub-directory. The relevant files are listed in Table IV. All of the files listed in Table IV were written to CFS on 10/05/98. All of the directory files listed in Table IV are "partial" directory files. That is, they contain entries for each of the 27 data tables, but nothing else. These directories need to be merged in to existing "full" directory files. Finally, all major files used in the creation of URES have been archived on CDROM.

The format of the PT data as contained in MCNP data tables has been previously documented by Lee Carter and John Hendricks (see Ref. 14). For completeness, we reproduce here in Appendix C the PT format description taken almost identically from Ref. 14.

Relationship to Other MCNP Libraries

Currently, the only generally-distributed MCNP library based upon ENDF/B-VI is ENDF60.¹⁵ ENDF60 was distributed in 1994 and contains data for 122 nuclides based on ENDF/B-VI Release 2. All of the ZAIDs on ENDF60 end in .60c. Of the 27 isotopes on URES, seven were not available on ENDF60. Thus, URES represents our first ENDF/B-VI release for ^{149,150,152}Sm, ^{152,154}Eu, and ^{191,193}Ir.

Table III
Characteristics of Data Tables on the URES Library

Material	ZAID	Length of Data Table	Number of Energies	Photon Production Data	Minimum Probability Table Energy (keV)	Maximum Probability Table Energy (keV)
Sm-149	62149.49c	57,787	7,392	No	0.1	10.
Sm-150	62150.49c	60,992	8,183	No	1.6	100.
Sm-152	62152.49c	203,407	19,737	No	5.025	100.
Eu-151	63151.49c	147,572	10,471	Yes	.09881	1.
Eu-152	63152.49c	81,509	6,540	No	.0615	3.
Eu-153	63153.49c	129,446	8,784	Yes	.09722	1.
Eu-154	63154.49c	72,804	6,627	No	.063	10.
Ta-182	73182.49c	20,850	2,463	No	.035	10.
W-182	74182.49c	150,072	16,495	Yes	4.5	100.
W-183	74183.49c	119,637	12,616	Yes	0.765	45.
W-184	74184.49c	97,118	9,794	Yes	2.65	100.
W-186	74186.49c	102,199	10,485	Yes	3.2	100.
Ir-191	77191.49c	83,955	8,976	Yes	0.16	10.
Ir-193	77193.49c	82,966	8,943	Yes	0.3	10.
Th-232	90232.49c	305,942	41,414	Yes	4.	50.
U-232	92232.49c	21,813	2,820	No	0.053	1.
U-233	92233.49c	47,100	4,601	Yes	0.06	10.
U-234	92234.49c	161,296	22,539	No	1.5	100.
U-235	92235.49c	647,347	72,649	Yes	2.25	25.
U-236	92236.49c	159,074	20,865	No	1.5	100.
U-238	92238.49c	705,623	85,021	Yes	10.	149.03
Pu-238	94238.49c	41,814	5,337	No	0.2	10.
Pu-239	94239.49c	595,005	64,841	Yes	2.5	30.
Pu-240	94240.49c	341,542	41,596	Yes	5.7	40.
Pu-241	94241.49c	155,886	17,753	Yes	0.3	40.2
Pu-242	94242.49c	130,202	14,922	Yes	0.986	10.
Cm-244	96244.49c	97,975	11,389	Yes	0.525	10.

Table IV
Versions of URES and Directories Under /x6data/ce/special/probtable CFS Node

File Name	Description	Length (MB)
URES1	Type-1 version of URES	97.6
URES1.XS	Directory file for URES1	
URES2.UCOS	Type-2 version of URES for UNICOS	38.7
URES2.XS.UCOS	Directory file for URES2.UCOS	
URES2.UNIX	Type-2 version of URES for UNIX (32-bit)	19.4
URES2.XS.UNIX	Directory file for URES2.UNIX	

The other 20 isotopes on URES have MCNP cross-section tables on both ENDF60 (ZAIDs ending in .60c) and on URES (ZAIDs ending in .49c). In no case are the equivalent pairs of data tables identical (even accounting for the obvious addition of PT data on URES relative to ENDF60). One reason is that, for a couple of isotopes (namely ^{235}U and ^{241}Pu), the URES table is based on a later ENDF/B-VI release than the corresponding ENDF60 table. The more pervasive reasons for differences between URES and ENDF60 (even when the original evaluation source was identical) are that different versions of NJOY were used and that different parameters were used in the NJOY processing. For example, the 0.001 fractional tolerances used in the current NJOY processing are, in general, tighter than the tolerances previously used to generate the data on ENDF60. One manifestation of this difference in processing strategy is that the number of energy points used to represent the ^{239}Pu cross sections is 64,841 on URES, compared to 26,847 on ENDF60. Another major difference between URES and ENDF60 is that gas-production cross sections (MTs 203-207) and damage cross sections (MT=444) exist on URES; they do not on ENDF60. In summary, the MCNP data tables on URES are much more detailed in *many* respects than their counterparts on ENDF60, not just in the addition of PT data.

Los Alamos plans to release a major new MCNP neutron cross section library in 1999. The library is tentatively named ENDF65, which is to indicate that it will be based upon ENDF/B-VI Release 5 evaluations. ENDF65 will be a much more complete library than URES, providing cross sections for well over 100 materials. Data tables on ENDF65 for all materials whose evaluations provide unresolved resonance data will contain probability tables. Thus, the number of PT materials on ENDF65 will be larger than the limited set of 27 provided on URES.

Users should not anticipate that data for those materials common to URES and ENDF65 will be identical. One reason is that URES is based upon ENDF/B-VI Release 4 whereas ENDF65 will be based upon ENDF/B-VI Release 5. Some of the evaluations for materials on URES (for example, ^{235}U and ^{238}U) will be updated for Release 5. In addition, a later version of NJOY than used in this work will be used for the entire ENDF65 processing, and a slightly different processing strategy will likely be followed. Even for those materials

without updated evaluations, we will assign new ZAIID identifiers for each table on ENDF65. Because of this, we consider URES to be an interim library that will be superseded when ENDF65 is released.

Utilization in MCNP

Probability-table data will be used for neutron transport in MCNP only in Versions 4XS and later. Version 4XS is an intermediate version between MCNP4B and MCNP4C and is documented in Ref. 8.

With respect to PT data, the default in MCNP4XS is to use such data whenever they are available. The presence and use of PT data can be confirmed in print table 100 of the MCNP "outp" file by a statement associated with each relevant data table indicating "probability tables used from 2.25e-03 to 2.5e-02 MeV" (or whatever the minimum and maximum energies of the PT data are).

Users are allowed to turn off PT data in MCNP4XS by entering a non zero value for the third variable on the PHYS:N card. The code will then print a warning message indicating that "unresolved resonance probability tables turned off." When PT data are disabled by the user, MCNP reverts to its historical treatment of cross sections in the unresolved energy range. Users are encouraged to *not* routinely disable PT data; the predominant reason for doing so should be to determine the sensitivity of a particular calculated result to the unresolved data.

Data tables such as those found on URES may be used with versions of MCNP that predate the inclusion of a PT treatment (e.g., MCNP4B). However, the PT data will be ignored by the code in such cases, except that a slight increase in memory will be required.

As noted in Ref. 8, the format of xsdir entries for tables with PT data has been changed to include 'ptable' after the existing temperature entry. Users converting a type-1 representation of URES to a type-2 binary format *must* do so with a version of the MAKXSF auxiliary code that has been updated to recognize and include this new entry.¹⁶ Otherwise, the resulting xsdir will not include the required 'ptable' entries.

Testing in MCNP

The majority of calculations reported to date using the PT method in MCNP have been performed with a version of MCNP preliminary to MCNP4XS and with a version of a cross-section library preliminary to URES. Nevertheless, these results may be of interest; we will therefore summarize them here.

Ref. 7 includes results of sample problems that were calculated with and without PT data. The problems were somewhat contrived, in that they were designed to emphasize unresolved region effects. Two shielding problems and one eigenvalue problem were described. The conclusion from the shielding problems was that neutron transmission through deep-penetration problems could be significantly increased in the unresolved region when PT data were used. The eigenvalue problem was designed to emphasize

neutrons in the ^{238}U unresolved energy range; by doing this, a relatively significant increase in the eigenvalue was obtained as a result of the decrease in ^{238}U capture effected by using PT data. Although these calculations were based upon ENDF/B-V PT data and an interim version of MCNP, the general nature of the conclusions is not expected to change.

Refs. 17 and 18 include calculated results for several uranium and plutonium benchmark critical assemblies. Again, although these results were generated with preliminary versions of the code and data, we are confident that the conclusions based upon the results are valid. The major conclusion, in terms of impact of PT data, was that the method could produce substantial increases in reactivity for systems that include large amounts of ^{238}U and have high fluxes within the unresolved resonance region. Therefore, these results indicated that calculations for such systems would produce significantly non conservative results unless the PT method is employed. Unfortunately, agreement between calculations and experiments was actually worse when PT data were used. This disparity suggests that the unresolved resonance data for ^{238}U may need to be revised.

Finally, we have repeated calculations for three of the critical assemblies studied in Refs. 17 and 18 using URES and MCNP4XS. The assemblies are BIG TEN, ZEBRA-8B, and HISS/HPG (see Ref. 17 or 18 for details on these assemblies). Results from current calculations are compared with results reported in Ref. 18 in Table V (uncertainties shown are one standard deviation). Current calculations were performed on a Sun Ultra workstation. All current results are within two standard deviations of the results reported in Ref. 18.

Table V
Results for Selected Critical Assembly Benchmarks

Benchmark Name	Benchmark k_{eff}	Calculation Description	Calculated k_{eff} with PT data	Calculated k_{eff} without PT data	Δk_{eff} (PT - no PT)
BIG TEN	0.9960 (.0030)	Ref. 18	1.0112 (.0005)	1.0069 (.0005)	0.0043 (.0007)
		MCNP4XS and URES	1.0125 (.0005)	1.0077 (.0005)	0.0048 (.0007)
HISS/HPG	1.0000 (.0100)	Ref. 18	1.0115 (.0005)	1.0132 (.0006)	-0.0017 (.0008)
		MCNP4XS and URES	1.0119 (.0005)	1.0126 (.0006)	-0.0007 (.0008)
ZEBRA-8B	1.0010 (.0023)	Ref. 18	1.0177 (.0004)	1.0045 (.0005)	0.0132 (.0006)
		MCNP4XS and URES	1.0169 (.0004)	1.0056 (.0004)	0.0113 (.0006)

Availability

MCNP4XS and URES are *not* generally available to the worldwide MCNP user community. The reason is straightforward. The PT data on URES and functionality in MCNP4XS were developed largely under the support of the USDOE Nuclear Criticality Safety Program (NCSP), in particular through funding from DOE-EM. Another major MCNP4XS feature, enhanced criticality perturbation, was also developed under this support. Therefore, the distribution of MCNP4XS and URES has been put on a "fast track" primarily for the DOE nuclear criticality safety community.

Appropriate users from outside Los Alamos should contact the Radiation Shielding Information Computational Center (RSICC) at Oak Ridge to obtain MCNP4XS and URES. To contact RSICC, send e-mail to pdg@ornl.gov or telephone 423-574-6176.

All internal Los Alamos users may obtain access to both MCNP4XS and URES. For access to URES, send e-mail to nucldata@lanl.gov.

Other users will be able to access the capabilities embodied in MCNP4XS and URES when MCNP4C is made available for general distribution via RSICC. This distribution is currently anticipated in mid-late 1999.

Summary

This Research Note describes the first formal neutron library for MCNP that includes probability-table data to allow modeling of unresolved resonance self-shielding. The library, URES, was generated by the NJOY processing code based upon ENDF/B-VI Release 4. Data for 27 materials are included on URES, which is compatible with MCNP4XS and subsequent versions. The library has undergone significant verification. Only a modest amount of code / library validation has been presented here. The distribution policy for URES is that it is immediately available to the DOE nuclear criticality safety community (via RSICC) and to all users at Los Alamos.

Acknowledgments

We wish to acknowledge that this work was sponsored as part of the DOE Nuclear Criticality Safety Program (NCSP). In particular, the funding was made possible by the Office of Environmental Management (EM) at DOE and by the Program Manager, Dennis Cabrilla.

Lee Carter and John Hendricks are responsible for the code development that led to the probability-table method as included in MCNP4XS. In addition, Lee did much of the initial testing of cross-section tables from NJOY. Russ Mosteller has performed most of the more recent calculations.

We appreciate Bernadette Kirk and RSICC for agreeing to distribute MCNP4XS and URES to the nuclear criticality safety community.

Feedback from Stephanie Frankle and Pat Mendius on earlier drafts of this report have improved the final version.

Finally, the developing collaboration with KAPL and ANL on probability-table data can only make all of our capabilities stronger in the future. We wish to acknowledge the roles of Tom Sutton at KAPL and Roger Blomquist at ANL in this collaboration.

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Distribution

Internal LANL Distribution:

X-DO File, MS B218

R. Krajcik / P. Soran / W. Krauser, X-DO, MS B218

X-CI File, MS F663

D. Shirk, X-CI, MS F663

A. Heath, X-CI, MS F663

R. MacFarlane, T-2, MS B243

M. Chadwick, T-2, MS B243

D. Madland, T-2, MS B243

P. Young, T-2, MS B243

S. Frankle, X-CI, MS F663

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G. Estes, X-TM, MS D409
R. Roberts, X-TM, MS D409
W. Urban, X-TM, MS D409
S. Becker, X-TA, MS B220
C. Lebeda, X-TA, MS B220
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D. Hayes, ESH-6, MS F691
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S. Vessard, ESH-6, MS F691
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T. Sutton, KAPL

Appendix A - NJOY INPUT File for "PENDF Run" for ^{235}U

```
moder
20 -21
reconr
-21 -22
'pendf tape for endf/b-vi.4 u-235'/
9228 9/
.001/
'92-u-235 from endf/b-vi.4'/
'processed with njoy 97.9'/
'the following reaction types are added'/
' mt152 bondarenko unresolved'/
' mt153 unresolved probability tables'/
' mt20x gas production'/
' mt221 free thermal scattering'/
' mt301 total heating kerma factor'/
' mt443 kinematic kerma'/
0/
broadr
-21 -22 -23
9228 9 0 1/
.001/
300 400 600 800 1000 1200 1600 2000 3000/
0/
unresr
-21 -23 -24
9228 9 7 1
300 400 600 800 1000 1200 1600 2000 3000/
```

1e10 1e4 1e3 300 100 30 10/

0/

heatr

-21 -24 -25/

9228 4/

302 318 402 443/

heatr

-21 -24 -22/

9228 6 0 1 0 2/

302 303 304 318 402 443

purr

-21 -25 -26

9228 9 7 20 32/

300 400 600 800 1000 1200 1600 2000 3000/

1e10 1e4 1e3 300 100 30 10/

0/

thermr

0 -26 -27

0 9228 8 9 1 0 1 221 0/

300 400 600 800 1000 1200 1600 2000 3000/

.005 4.5/

gaspr

-21 -27 -28

moder

-28 29

stop

APPENDIX B - NJOY INPUT File for "ACER Run" for ²³⁵U

acer

20 21 0 31 32

1 0 1/

'92-u-235 from endf-vi.4 njoy97.9/

9228 300./

.01/

/

acer

0 31 33 34 35

7 1 2/

'92-u-235 from endf-vi.4 njoy97.9/

viewr

33 36/

stop

APPENDIX C – Format Description for MCNP Probability-Table Data

This format description is intended to be included in Appendix F of the MCNP manual (Ref. 1).

If probability-table data exist; JXS(23) LUNR = location of probability tables

UNR Block — contains the unresolved resonance range probability tables. The UNR block exists if JXS(23) > 0.

Format of UNR Block:

Location in XSS	Parameter	Description
JXS(23)	N	number of incident energies where there is a probability table.
JXS(23)+1	M	length of table; i.e., number of probabilities, typically 20.
JXS(23)+2	INT	interpolation parameter between tables; =2 lin-lin; =5 log-log
JXS(23)+3	ILF	inelastic competition flag (see below)
JXS(23)+4	IOA	other absorption flag (see below)
JXS(23)+5	IFF	factors flag (see below)
JXS(23)+6	E(I), I=1,N	incident energies
JXS(23)+6+N	P(I,J,K)	probability tables (see below)

ILF is the inelastic competition flag. If this flag is less than zero, the inelastic cross section is zero within the entire unresolved energy range. If ILF is greater than zero, then it is a special MT number whose tabulation is the sum of the inelastic competition reactions. An exception to this scheme is typically made when there is only one inelastic reaction within the unresolved energy range, since the flag can then just be set to the MT number of the reaction and a special tabulation is not needed. The flag can also be set to zero, which means that the sum of the contribution of the inelastic competition reactions will be made "on-the-fly" by MCNP using a balance relationship involving the smooth cross sections.

IOA is the other absorption flag for determining the contribution of "other absorptions" (no neutron out or destruction reactions). The "other" in "other absorptions" refers to absorption reactions other than radiative capture. If this flag is less than zero, the "other absorption" cross section is zero within the entire unresolved energy range. If IOA is greater than zero, then its value is a special MT number whose tabulation is the sum of the "other absorption" reactions. An exception to this scheme is typically made when there is only one "other absorption" reaction within the unresolved energy range, since the flag can then just be set to the MT number of the reaction and a special tabulation is not needed. The flag can also be set to zero, which means that the sum of the contribution of the "other absorption" reactions will be made "on-the-fly" by MCNP using a balance relationship involving the smooth cross sections.

IFF is the factors flag. If this flag is zero, then the tabulations in the probability tables are cross sections. If the flag is one, the tabulations in the probability tables are factors that must be multiplied by the corresponding "smooth" cross sections to obtain the actual cross sections.

$P(I,J,K)$, where $I=1,N$, $J=1,6$, and $K=1,M$, are the tables at N incident energies for M cumulative probabilities. For each of these probabilities the J values are:

- 1 cumulative probability
- 2 total cross section or total factor
- 3 elastic cross section or elastic factor
- 4 fission cross section or fission factor
- 5 (n,γ) cross section or (n,γ) factor
- 6 heating number or heating factor

The ordering of the probability-table entries is as follows:

- M cumulative probabilities for energy $I=1$ ($K=1$ through $K=M$)
- M total cross sections (or factors) for energy $I=1$ ($K=1$ through $K=M$)
- ...
- M cumulative probabilities for energy $I=2$ ($K=1$ through $K=M$)
- ...
- M heating numbers (or factors) for energy $I=N$ ($K=1$ through $K=M$)

Notes: The cumulative probabilities are monotonically increasing from an implied lower value of zero to the upper value of $P(I,1,K=M) = 1.0$. The total cross section, $P(I,2,J)$, is not used in MCNP; the total is recalculated from sampled partials to avoid round-off error. The (n,γ) cross section is radiative capture only; it is not the usual MCNP "capture" cross section which is really absorption or destruction with other no-neutron-out reactions included.

Los Alamos

NATIONAL LABORATORY

memorandum

*Applied Theoretical & Computational
Physics Division
Transport Methods Group, XTM
Los Alamos, New Mexico 87545*

To/MS: Distribution
From/MS: S. Frankle, XTM / MS B226
Phone/FAX: 5-6461 / 5-5538
Symbol: XTM:96-153
Date: April 11, 1996

SUBJECT: ENDF62MT, A Multi-temperature Neutron Library for MCNP (Rev.0)

A multi-temperature neutron data library has been produced for use with MCNP. Seven temperatures (77, 400, 500, 600, 800, 900, 1200 °K) for three nuclides, ²³⁵U, ²³⁸U, and ²³⁹Pu are available on the library as a companion to ENDF60 (300°K). The entries for this library are a part of the latest Table 1 for Appendix G available on the WWW or in hardcopy, and are listed below. We have used the same source evaluations as were used for the ENDF60 library, Release 2 of each nuclide. However, a tighter fractional tolerance on thinning was used for ENDF62MT, so the data files are larger in size than the corresponding 300° data files in the ENDF60 library. Sample input decks for the NJOY runs are included with this memorandum.

The standard QA procedures were used for each data file in this library:

- * the NJOY interpretative output files was examined
- * all cross-section data and heating numbers were examined graphically
- * a number of codes were run to check the threshold energies, secondary energy distributions for both neutrons and photons, etc.
- * and sample MCNP problems were run

Figures 1-6 show example plots of the total neutron cross section for ²³⁵U, ²³⁸U, and ²³⁹Pu over two energy regions, 10⁻⁷-10⁻⁵ and 1x10⁻⁵-2x10⁻⁵ MeV, at 300, 600, 900, and 1200 °K. One can see the most notable difference due to thinning in the ENDF60 data library for ²³⁸U in Figure 4.

Currently, this data library is available on CFS the secure network, but is not publicly available on the open. The library should only be distributed to those with support contracts with XTM. The library is stored on CFS under /x6data/ce/special/multitemp , and will become available in the standard XSDIR file during the next update this summer. This library may undergo changes, such as the addition of other temperatures and nuclides, prior to its public release.

Appendix G Information for the ENDF62MT Neutron Data Library (Rev. 0)

ZAID	Atomic Wt. Ratio	Library Name	Source	Date of Eval.	Temp. (°K)	Length (words)	Num. of Energies	E _{max} (MeV)	GPD	Nubar
92235.11c	233.0250	endf62mt	ENDF/B-VI.2	1989	77	696,398	78,912	20	yes	both
92235.12c	233.0250	endf62mt	ENDF/B-VI.2	1989	400	411,854	43,344	20	yes	both
92235.13c	233.0250	endf62mt	ENDF/B-VI.2	1989	500	379,726	39,328	20	yes	both
92235.14c	233.0250	endf62mt	ENDF/B-VI.2	1989	600	353,678	36,072	20	yes	both
92235.15c	233.0250	endf62mt	ENDF/B-VI.2	1989	800	316,622	31,440	20	yes	both
92235.16c	233.0250	endf62mt	ENDF/B-VI.2	1989	900	300,278	29,397	20	yes	both
92235.17c	233.0250	endf62mt	ENDF/B-VI.2	1989	1200	269,062	25,495	20	yes	both
92238.11c	236.0060	endf62mt	ENDF/B-VI.2	1993	77	621,385	74,481	20	yes	both
92238.12c	236.0060	endf62mt	ENDF/B-VI.2	1993	400	456,593	53,882	20	yes	both
92238.13c	236.0060	endf62mt	ENDF/B-VI.2	1993	500	433,681	51,018	20	yes	both
92238.14c	236.0060	endf62mt	ENDF/B-VI.2	1993	600	414,185	48,581	20	yes	both
92238.15c	236.0060	endf62mt	ENDF/B-VI.2	1993	800	386,305	45,096	20	yes	both
92238.16c	236.0060	endf62mt	ENDF/B-VI.2	1993	900	372,625	43,386	20	yes	both
92238.17c	236.0060	endf62mt	ENDF/B-VI.2	1993	1200	348,137	40,325	20	yes	both
94239.11c	236.9986	endf62mt	ENDF/B-VI.2	1993	77	568,756	62,522	20	yes	both
94239.12c	236.9986	endf62mt	ENDF/B-VI.2	1993	400	418,556	43,747	20	yes	both
94239.13c	236.9986	endf62mt	ENDF/B-VI.2	1993	500	395,964	40,923	20	yes	both
94239.14c	236.9986	endf62mt	ENDF/B-VI.2	1993	600	377,116	38,567	20	yes	both
94239.15c	236.9986	endf62mt	ENDF/B-VI.2	1993	800	350,292	35,214	20	yes	both
94239.16c	236.9986	endf62mt	ENDF/B-VI.2	1993	900	338,236	33,707	20	yes	both
94239.17c	236.9986	endf62mt	ENDF/B-VI.2	1993	1200	312,572	30,499	20	yes	both

Corresponding entries for the ENDF60 Neutron Data Library

92235.60c	233.0250	endf60	ENDF/B-VI.2	1989	294	289,975	28,110	20	yes	both
92238.60c	236.0060	endf60	ENDF/B-VI.2	1993	294	206,322	22,600	20	yes	both
94239.60c	236.9986	endf60	ENDF/B-VI.2	1993	294	283,354	26,847	20	yes	both

Input deck for creating pendf tape with NJOY based on that used for the ENDF60 library
(²³⁹Pu).

```
0
6
moder
20 -21
reconr
-21 -22
*pendf tape for endf/b-vi.2 pu-239b*/
9437 7 0 /
.002 0. 7 /
*94-pu-239b from endf/b-vi.2 tape 117 (young, lanl) */
*processed with the njoy nuclear data processing system*/
*see original endf/b-vi tape for details of evaluation*/
*the following reaction types are added*/
*   mt221   free thermal scattering*/
*   mt301   total heating kerma factor*/
*   mt443   kinematic kerma*/
0/
broadr
-22 -23
9437 9 0 1/
.002/
0 77 300 400 500 600 800 900 1200 /
0/
unresr
-21 -23 -24
9437 9 7 1
0 77 300 400 500 600 800 900 1200 /
1e10 1e4 1e3 300 100 30 10
0/
heatr
-21 -24 -25/
9437 1/
443/
stop
```

Input deck for creating ACE data file and interpretative output file with NJOY based on that used for the ENDF60 library at 1200 °K (^{239}Pu).

```
0
6
moder
20 -21
acer
-21 -25 0 31 32
1 0 1 .17/
*94-pu-239b from endf/b-vi.2*/
9437 1200.0/
.01/
/
acer
0 31 33 34 35
7 1 2 .17/
*94-pu-239b from endf/b-vi.2*/
stop
```

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memorandum

*Applied Theoretical & Computational
Physics Division
Code Integration Group, XCI
Los Alamos, New Mexico 87545*

To/MS: Distribution
From/MS: C. Werner, XCI / MS F663
Phone/FAX: 7-6892 / 5-5553
Symbol: XCI:CJW-99-25(U)
Date: April 28, 1999

SUBJECT: New Data Library for MCNP Delayed Neutron Capability

A new MCNP data library has been created, appending ENDF/B-VI delayed neutron data to the existing ENDF60¹ data tables. This library is intended for use with a newly modified version of MCNP4B (to be documented in a later report) that more accurately represents the secondary neutrons produced from a fission event. This is accomplished by incorporating the capability of calculating the probability of delayed neutrons and, if sampled, their energy and time of creation after the occurrence of a fission event.

The status of delayed neutron data as well as the total and prompt information available from ENDF/B-VI is summarized in Table 1 for each fissionable nuclide. From Table 1 it can be seen that there are 23 isotopes in ENDF/B-VI that contain complete delayed neutron data. The ENDF/B-VI data libraries for each of these 23 isotopes were obtained from the T-2 web-site <http://T2.lanl.gov/cgi-bin/nuclides/endind>, and are listed in Table 2. The delayed data obtained were from ENDF/B-VI evaluations consistent with those originally used for ENDF60; the ENDF/B-VI release number is also listed for each isotope on Table 2.

These delayed neutron data were next merged with the existing MCNP data table, from ENDF60, using the stand alone utility code MERGE.F, reported in XCI:CJW-98-123(U) and available on CFS in the directory /x6data/working/delayed/. This code reads in the ENDF formatted isotopic data file, along with the MCNP data library. It then adds to the neutron production section of the data library file the ENDF/B-VI delayed neutron data for that isotope and corrects the JXS and NXS arrays. The modified version of MCNP4B requires the delayed neutron data in the format specified in XCI:CJW-98-121(U), which MERGE.F creates. Since ²³⁷U, ²³⁸Np, and ²⁴²Am are not available on ENDF60, no modified files were created for these isotopes. A new library, called ENDF6DN, has now been constructed consisting of the 20 isotopes listed in Table 2 and having ZAIDs with the extension .61c. The only difference between the corresponding files with ZAID extensions of .60c and .61c is the addition of the delayed neutron data, and therefore, these files are identical in all other places.

This MCNP data library is now ready to be used with the modified MCNP4B version. Type 1 and type 2 libraries (and directories) can be found on CFS at /x6data/ce/special/delayed/.

Table 1 – Total, prompt and delayed fission data available from ENDF/B-VI.

Element	Isotope	Total Nubar (Format)	Delayed Nubar (Format)	Prompt Nubar (Format)	Total/Prompt Energy Distribution	Delayed Energy Distribution
Th	230	Polynomial	-----	-----	7	-----
	232	Table	Table	Table	11	5
Pa	231	Table	Table	Table	11	5
	233	Polynomial	-----	-----	7	-----
U	232	Table	Table	Table	7	5
	233	Table	Table	Table	11	5
	234	Table	Table	Table	7	5
	235	Table	Table	Table	Table	5
	236	Table	Table	Table	7	5
	237	Table	Table	Table	7	5
	238	Table	Table	Table	11	5
Np	237	Table	Table	Table	Table	5
	238	Table	Table	Table	7	5
Pu	239	Polynomial	-----	-----	7	-----
	236	Polynomial	-----	-----	7	-----
	237	Polynomial	-----	-----	7	-----
	238	Table	Table	Table	7	5
	239	Table	Table	Table	Table	5
	240	Table	Table	Table	7	5
	241	Table	Table	Table	7	5
	242	Table	Table	Table	7	5
	243	Table	-----	-----	7	-----
	244	Polynomial	-----	-----	7	-----
	241	Table	Table	Table	7	5
	242	Table	-----	-----	7	-----
Am	242ma	Table	Table	Table	7	5
	243	Table	Table	Table	7	5
Cm	241	Polynomial	-----	-----	7	-----
	242	Table	Table	Table	7	5
	243	Polynomial	-----	-----	7	-----
	244	Polynomial	-----	-----	7	-----
	245	Table	Table	Table	7	5
	246	Polynomial	-----	-----	7	-----
	247	Table	-----	-----	7	-----
	248	Polynomial	-----	-----	7	-----
Bk	249	Table	Table	Table	7	-----
Cf	249	Table	Table	Table	7	5
	250	Table	-----	-----	7	-----
	251	Table	Table	Table	7	5
	252	Table	-----	-----	7	-----
	253	Polynomial	-----	-----	7	-----

LF:11 = Watt Spectrum

LF:7 = Maxwellian Spectrum

LF:5 = General Evaporation Spectrum

Table 2 – Material, ZAID from ENDF60 library, new ZAID for ENDF6DN, the evaluation and version of delayed neutron data placed in the ENDF6DN data files.

Material	ZAID from ENDF60	New ZAID for ENDF6DN	Evaluation	Release
²³² Th	90232.60c	90232.61c	BNL,ANL+	-
²³¹ Pa	91231.60c	91231.61c	HEDL	-
²³² U	92232.60c	92232.61c	HEDL	-
²³³ U	92233.60c	92233.61c	LANL,ORNL	-
²³⁴ U	92234.60c	92234.61c	BNL,GGA+	-
²³⁵ U	92235.60c	92235.61c	KAPL,ORNL,LANL+	r2
²³⁶ U	92236.60c	92236.61c	HEDL	-
²³⁸ U	92238.60c	92238.61c	ORNL,LANL+	r2
²³⁷ Np	93237.60c	93237.61c	LANL,HEDL	r1
²³⁸ Pu	94238.60c	94238.61c	HEDL,AI+	-
²³⁹ Pu	94239.60c	94239.61c	LANL,ORNL+	r2
²⁴⁰ Pu	94240.60c	94240.61c	ORNL,LANL	r2
²⁴¹ Pu	94241.60c	94241.61c	ORNL	r1
²⁴² Pu	94242.60c	94242.61c	HEDL,SRL+	-
²⁴¹ Am	95241.60c	95241.61c	CNDC	r2
²⁴³ Am	95243.60c	95243.61c	LANL,ORNL	-
²⁴² Cm	96242.60c	96242.61c	HEDL,SRL+	-
²⁴⁵ Cm	96245.60c	96245.61c	SRL,LLNL	r2
²⁴⁹ Cf	98249.60c	98249.61c	CNDC	-
²⁵¹ Cf	98251.60c	98251.61c	BNL,SRL+	r2

- Release 0 of ENDF/B-VI

Distribution:

XCI files

Comly J. C.	XCI	MS F663
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Werner Christopher J.	XCI	MS F663
MacFarlane, R. E.	T-2	MS B243

ReadURES.txt
January 31, 2001
Revision to URESA

The library URES has been distributed as part of the
DLC-200/MCNPDATA and DLC-205/MCNPXDATA packages.
URES is a neutron cross-section library for MCNP and MCNPX
that includes probability tables to represent unresolved
resonances. (Note that MCNPX does not currently use the
probability tables). Room-temperature data for 27 isotopes
were provided on URES, based on ENDF/B-VI release 4. ZAIDs
for the 27 isotopes ended in .49c.

Processing problems have been found for 6 of the isotopes
on URES: W-182, W-183, W-184, W-186, Th-232, and U-238.
The problems would impact coupled neutron - photon
calculations; neutron only problems would not be affected.
An updated version of URES, named URESA, has been prepared.
For the 6 isotopes listed above, the photon-production data
have been deleted, and the ZAID's have been changed from .49c
to .48c. Users should note that the heating (energy-
deposition) data are now inconsistent for these isotopes
and should not be used. The other 21 isotopes have simply
been copied over intact from URES to URESA and their ZAIDs
remain .49c. All of the data on URESA will be superseded by
data on the ENDF66 library, a major new release planned for
later in 2001.

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memorandum

Applied Physics Division

Diagnostics Applications Group, X-5

PHONE: (505)667-8747— FAX:(505)665-3046

To/MS: Distribution

From/MS: Kenneth J. Adams/F663

Phone/Email: 5-7000/adamsk

Symbol: X-5-RN(U)-00-14

Date: 5/25/00

SUBJECT: ELECTRON UPGRADE FOR MCNP4B

Abstract:

The focus of this effort was the integration into MCNP^{*} (ref. 1) of the ITS3.0(ref. 2) radiative and collisional stopping power (refs. 2 and 3) and bremsstrahlung production model (refs. 2, 4, 17, and 18). The integration was achieved with the development of a patch to MCNP4B and a new electron physics library database, EL03 (like EL1). SQA of the patch and database are still ongoing. However, improvements in the stopping power model and bremsstrahlung model are presented. Moreover, in the process of successfully achieving this goal several other related (bremsstrahlung angular distribution models) or otherwise unforeseen problems (electron step-size artifacts or thick target bremsstrahlung production problems) were identified and unfortunately not solved yet since they involve much more work. Several bugs have been found and eradicated.

Overview:

The memo is organized as follow:

- 0) Background
- 1) EL03 database. the new database library
- 2) code modifications for new algorithms to use the new database and other enhancements
 - 2a) radiative stopping power
 - 2b) bremsstrahlung production
 - 2c) density effect
 - 2d) relaxation
 - 2e) variance reduction
 - 2f) bug fixes

^{*}MCNP is a trademark of the Regents of the University of California, Los Alamos National Laboratory

- 3) verification
 - 3a) ITS stopping powers
 - 3b) ITS cross section comparison
 - 3c) Implications
- 4) preliminary validation based upon Gierga and Adams
 - 4a) comparison with Faddegon for bremsstrahlung production
 - 4b) depth dose in water
- 5) future work

Background:

In order to perform radiography analysis, accurate transport models are required to define the bremsstrahlung source and contamination in the object. Several authors (refs. 7 - 10, this is list is not meant to be exhaustive nor definitive) have examined the accuracy of various bremsstrahlung models in transport codes (MCNP¹, ITS² and EGS⁵). From these studies, possible deficiencies in MCNP were identified and sought to be corrected. As an example of the results found, figure1 dis-

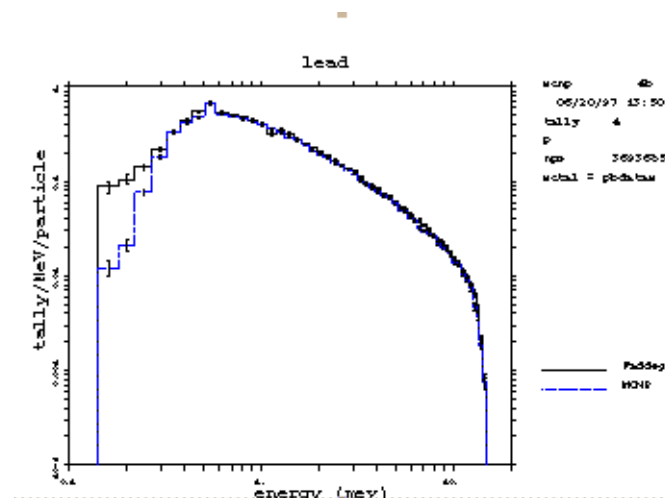


Figure 1: Bremsstrahlung Spectra for 15 MeV electrons on Lead.
Calculation (MCNP) and experiment (Faddegon)

plays the bremsstrahlung spectrum for 15 MeV electrons on lead calculated with MCNP4B and measured by Faddegon where spectral variations at low energies are present. Moreover, the electron physics package in MCNP is based mostly on ITS1.0 (ref. 6) physics with an upgrade to ITS3.0 physics for the calculation of energy loss straggling and calculation of the ionization potential. A more complete upgrade would be desired to include more of the ITS3.0 package and/or the database from ref. 11.

EL03 database

The previous electron evaluation database was stored in library file, EL1. Essentially, this file was a partial rewriting of the ITS1.0 xdata file into a format suitable for reading by MCNP (ACE format.) The initial few words of each data block for a particular Z were peculiar to MCNP but otherwise it just reformatted the data to be read by subroutine getxst for use in the subroutine xsgen with appropriate offsets. As far as I was able to determine there was no processing of the data only rewriting.

Following this philosophy, a code was written to extract and reformat the desired portions of xdata from ITS3.0 and generate the new library file EL03. The code for doing this reprocessing is presented in Appendix A and is available on cfs. The code represents an evolution of what the database needed from its progenitor and could still further evolve.

The following tables are the list of data pointers used in EL03 and EL1 and some explanation of their use. Table 1 lists the nxs array entries. These are mostly lengths of blocks of data. However, nxs(16) now has special meaning and is a flag for using the new its3.0 data and algorithms. Table 2 lists the jxs array. Finally, the crux of the puzzle is the contents of the exs array which is presented in Table 3.

Table 1: nxs array

nxs	exposition EL03	EL1
1	length of xss data set	length of xss data set
2	z	z
3	number of radiation stopping power interpolation points	number of radiation stopping power interpolation points
4	number of mott scattering cross section corrections	number of mott scattering cross section corrections
5	number of electron energy points for bremsstrahlung interpolation	number of bremsstrahlung interpolation points
6	number of photon ratio point for bremsstrahlung interpolation	number of bremsstrahlung interpolation points
7	unused	unused
8	unused	unused

Table 1: nxs array

nxs	exposition EL03	EL1
9	number of interpolation points in the bremsstrahlung database for bremsstrahlung spectrum calculation	unused
10	number of interpolation points in the bremsstrahlung database for angular/energy calculation	unused
11	number of oscillator points for density effect calculation	unused
12	unused	unused
13	unused	unused
14	unused	unused
15	unused	unused
16	3	unused

Table 2: jxs array exposition

jxs	EL03	EL1
1	1 (offset for line data)	same
2	offset for radiative stopping power numbers	same
3	offset for Mott cross section corrections	same
4	offset for Riley cross section evaluation	same
5	offset for ITS3.0 bremsstrahlung production database	offset for ITS1.0 bremsstrahlung production database
6	unused	bremsstrahlung production database
7	offset for internally calculated Riley cross sections	same
8	offset for internally calculated functions of Z	same
9	offset for photon energy ratios	unused
10	offset for photon energy ratio for angular distribution	unused
11	offset for oscillator descriptions for density effect calculation	unused

Table 3: exs array exposition

exs	EL03	EL1
jxs(1)	K edge below which no electron induced relaxation will occur (renamed for persistence edg)	K edge below which no electron induced relaxation will occur (renamed for persistence edg)
jxs(1)+1	Auger electron emission energy (renamed for persistence eek) $=\bar{E}_K - 2\bar{E}_L$	K-x-ray or Auger electron emission energy (renamed for persistence eek) $=\bar{E}_K$
jxs(1)+2	unused	transition point for bremsstrahlung extrapolation model
jxs(1)+3	unused	transition point for bremsstrahlung extrapolation model
jxs(2)	energy points for radiation stopping power interpolation	energy points for radiation stopping power interpolation
jxs(2)+nxs(3)	corresponding normalized stopping powers	corresponding normalized stopping powers
jxs(2)+2*nxs(3)	corresponding electron-electron bremsstrahlung correction (η)	unused
jxs(3)	energy of Mott scattering correction points	
jxs(3)+nxs(4)	$h(\theta)$ from eq. (A35) of ref 1 at $\theta=0$	
jxs(3)+2*nxs(4)	$h(\theta)$ from eq. (A35) of ref 1 at $\theta=\pi/4$	
jxs(3)+3*nxs(4)	$h(\theta)$ from eq. (A35) of ref 1 at $\theta=\pi/2$	
jxs(3)+4*nxs(4)	$h(\theta)$ from eq. (A35) of ref 1 at $\theta=3\pi/4$	
jxs(3)+5*nxs(4)	$h(\theta)$ from eq. (A35) of ref 1 at $\theta=\pi$	
jxs(4)	energy of riley set	
jxs(4)+1 to jxs(4)+14	riley scattering cross section parameters at that energy	
repeated 8 more times		

Table 3: exs array exposition

exs	EL03	EL1
jxs(5)	electron energy values for interpolation for bremsstrahlung cross section nxs(5) values	energy values for bremsstrahlung interpolation
jxs(5)+nxs(5)	photon energy ratio values for interpolation for bremsstrahlung cross section nxs(6) values	cross section values for interpolation
jxs(5)+nxs(5)+nxs(6)	values for interpolation for bremsstrahlung cross section nxs(5)*nxs(6) values	unused
jxs(6)	unused	energy values for bremsstrahlung interpolation cross section values for interpolation
jxs(7)	used for riley cross section evaluation (2160 float words long) ^a	
jxs(8)	various powers and functions of Z and A (10 floats long, though only three are used as below) ^a	
jxs(8)	$Z^{1/3}$	
jxs(8)+1	$\log(Z)$	
jxs(8)+2	$zf*(1./(1.+zf)+.20206-zf*(.0369-zf*(.0083-zf*.002)))$ where $zf=(Z*\alpha)**2$	
jxs(9)	ratio values of photon energy over electron energy for bremsstrahlung spectrum calculation (renamed for persistence rkt)	
jxs(10)	ratio values of photon energy over electron energy for bremsstrahlung angular distribution calculation (renamed for persistence rka)	

Table 3: exs array exposition

exs	EL03	EL1
jxs(11)	number of electrons in each level (occupation number) (< 0 signifying a conductor) used as $f_i = n_i/Z$.	
jxs(11)+nxs(11)	oscillator strength or binding energy of the level, ν_i .	

a. This explains the 2170 (2160+10) in getxst for the electron offset.

Unfortunately, the electron side of MCNP has its own peculiar way of dealing with its cross section data and storing it. That is, for neutrals, all the transport data are stored in the xss arrays and very few arrays are “hardwired” to reflect the physics manifest in the evaluation. (An example may be the rtc array which is hardwired to store only ten different cross section data.) Moreover, the xss data persists after it has undergone the ordeal of expunging; the exs data are overwritten since the relevant physics data has been processed and then stored in various arrays. As such, we have found that even though the electron portion of MCNP seems like an ideal candidate to become a “module”, it has interspersed its memory management throughout the code.

The problem is simple, the physics data are stored in the database as a reduced set of data over a range where interpolation is valid while it is used in transport on a much finer or at least different scale. The original code had the size of the expanded or internally calculated arrays parameterized and the dynamic allocation of memory was done based on that those parameters. For the new database, the parameterization has been retained but in such a way as to allow for regression compatibility. It is expected that this way of dealing with data structures will be amended in the future, so that the code can readily accept new data evaluations without being constrained to parameterization based on different evaluations.

The parameters used in the electron databases are listed and explained in the following table.

parameter	exposition on parameter	EL03	EL1
mtop*	number of ratio values of photon energy to electron energy for brems production	89	49
mwng*	number of ratio values of photon energy to electron energy for brems detailed brems angular distribution	(mtop+1)/ 2 actually 34 is used; to be replaced by nxs(10)	(mtop+1)/ 2 actually 25
mstp*	coarsening factor for electron energy grid for detailed brems angular distribution	4	8
mpng	number of photon angular bins	21	21
mbng	number of possible photon to electron ratio values from the its3.0	51	unused

* These parameters have corresponding variables, ntop, nwng, nstp respectively, to toggle between databases.

Code and Algorithm Modifications

The patch to achieve these changes is presented in Appendix B.

2a) Radiative stopping power.

The ITS3.0 database has a more recent evaluation of the radiative stopping power. To use the evaluation in the code required a few small changes since the effect of electron bremsstrahlung has been included in the database. The stopping power is

$$\frac{dE}{dx} = (Z + \eta)\Phi_{rad}$$

The database originally contained a table of energy and Φ_{rad} with the assumed value of unity for the electron bremsstrahlung contribution, η . The new evaluation includes a newer table of energy and Φ_{rad} as well as η .

2b) Bremsstrahlung production

The brems production database has been extended to cover a larger range of photon energies. The photon energy from a bremsstrahlung event is specified in terms of a ratio of the incident electron; the range of ratio values is from one to 10^{-6} for the EL03 database while the EL1 range was from 0.9999 to 10^{-3} . The following table gives the ratio values for both databases.

EL1:

rkt=.001,.00125,.0015,.00175,.002,.0025,.003,.0035,.004,
.0045,.005,.0055,.006,.007,.008,.009,.01,.0125,.015,.0175,.02,
.025,.03,.035,.04,.045,.05,.055,.06,.07,.08,.09,.1,.125,.15,.175,
.2,.25,.3,.35,.4,.45,.5,.55,.6,.7,.8,.9,.9999

EL03:

rkt=0.000001,0.0000015,0.000002,0.000003,0.000004,0.000005,
0.000006,0.000008,0.00001,0.000015,0.00002,0.00003,0.00004,
0.00005,0.00006,0.00008,0.0001,0.00015,0.0002,0.0003,0.0004,
0.0005,0.0006,0.0008,0.001,0.00125,0.0015,0.00175,0.002,0.0025,
0.003,0.0035,0.004,0.0045,0.005,0.0055,0.006,0.007,0.008,
0.009,0.01,0.0125,0.015,0.0175,0.02,0.025,0.03,0.035,0.04,
0.045,0.05,0.055,0.06,0.07,0.08,0.09,0.1,0.125,0.15,0.175,0.2,
0.225,0.25,0.275,0.3,0.325,0.35,0.375,0.4,0.425,0.45,0.475,
0.5,0.525,0.55,0.575,0.6,0.65,0.7,0.75,0.8,0.85,0.9,0.95,0.97,
0.99,0.995,0.9995,1.0

The production of bremsstrahlung from an electron of energy, E , is determined by numerical or analytic integration of the cross section over the above specified ranges of photon energy ratios. Choosing the best cross section for all energies is non-trivial and has evolved over the years^{*}. For the EL1 database, most of the production cross sections are derived from Koch and Motz (ref. 15 and referred to as K&M hereafter) with some exceptions near the “tip” (the “tip” is the photon phase space near where the photon is extremely forward peaked and has nearly all the incident electron energy.) Berger and Seltzer (ref. 16) describe their best evaluation of which cross sections should be used over various energy ranges. The EL1 database follows that evaluation except for not using the Schiff formula (see ref. 15 for formula).

In the EL03 evaluation or database, the production cross section for bremsstrahlung photons and energy spectra are from the evaluation by Seltzer and Berger (refs. 2, 17, 18 and referred to as S&B collectively hereafter). We summarize the salient features of the evaluation below; more details can be found in the evaluators’ documentation. The evaluation uses detailed calculations of the electron-nucleus bremsstrahlung cross section for electrons with energies below 2 MeV and above 50 MeV. The evaluation below 2 MeV uses the results of Pratt, Tseng, and collaborators, based on numerical phase-shift calculations (refs. 22, 19, and 20). For 50 MeV and above, the analytical theory of Davies, Bethe, Maximom, and Olsen, (DBMO ref. 23) is used and is supplemented by the Elwert Coulomb (ref. 26) correction factor and the theory of the high-frequency limit or tip region given by Jabbur and Pratt (ref. 27). Screening effects are accounted for by the use of Hartree-Fock atomic form factors (ref. 25). The values between these firmly grounded theoretical limits are found by a cubic-spline interpolation as described in refs. 64 and S2. Seltzer reports good agreement between interpolated values and those calculated by Tseng and Pratt (ref. 21) for 5 and 10 MeV electrons in aluminum and uranium. Electron-electron bremsstrahlung is also included in the cross section evaluation based on the theory of Haug (ref. 24) with screening corrections derived from Hartree-Fock incoherent scattering factors (ref. 25). The energy spectra for the bremsstrahlung photons are provided in the evaluation. Finally, the comparisons with experiments presented in S&B indicate the validity of their approach^{**}.

In summary, then, the calculation of the production cross section for bremsstrahlung is given by numerical integration of various formula which reflect the best evaluation at the time of the bremsstrahlung process. For EL1, this integration is done in the code and most of the formula are related back to the review of K&M, while for EL03, the integration and any corrections are done in the evaluation. Consequently, all of subroutine *brem* had to be rewritten for the evaluation of the bremsstrahlung cross section, *pbr*, spectra, *eba* for EL03. The new algorithms for the EL03 database are derived from the ITS code which presumably reflect those

^{*}We quote from Seltzer and Berger, “Many theories of the bremsstrahlung process have been developed, each with its own approximations, limitations and region of applicability. To obtain accurate cross sections over a wide range of conditions, it is necessary to combine, and to interpolate between, the results of several theories.”

^{**}The validation problems discussed later will eventually be augmented to include this set.

authors' recommended interpolation schemes and have been incorporated in brem since the pbr and eba arrays are used for the calculation of thick target brems.

The detailed angular distribution calculation has been modified only in so far as the resolution of the energy grids has been more refined. For EL1, the angular distribution was

calculated for electrons with energies $E_n = \frac{E_{max}}{2^n}$ with n being a positive integer. (Recall that

electron transport parameters are calculated on an energy grid with $E_n = \frac{E_{max}}{2^{n/8}}$.) For EL03,

$E_n = \frac{E_{max}}{2^{n/2}}$. Moreover, for EL1 the photon ratios used every other ratio value in the

production evaluation. For EL03, the ratios are read from the database and are fewer than every other. (Presumably every other could be used for EL03 or even every one. However, this is what ITS does.) The angular distribution is determined by numerical integration of several formulas from K&M using these electron energy values and photon energy ratios. Consequently, the modifications to subroutine brang have been minimal. As a passing note, it is not apparent that 2BS is used in the angular distribution calculation in brang though one may have expected it to be used from ref. 16. It might be speculated that this may account for some of the differences seen between EGS and MCNP.

In MCNP, brems production is sampled along an electron substep. The probability of an event(s) occurring in a distance d is $1 - e^{-\sum_{brem} \delta}$, with the number of events being sampled from a Poisson distribution. If the probability of an event occurring is small, then only the first non-zero term in the Taylor expansion of the probability is kept and used to sample if one and only one event has occurred. This is the procedure used in MCNP for EL1. For EL03, the sampling has been modified to sample from the Poisson distribution so that zero, one or more events could occur, though two or more are extremely infrequent. The electron's energy is decremented after each event and used to scale the energy of the photon from the next event. This procedure is consistent with the method employed by ITS3.0.

2c) Density Effect Correction

The stopping power has a correction term to account for the polarization of the media through which the charged particle passes. Sternheimer, Berger and Seltzer (ref. 14 and referred to as SBS) have given a method to calculate the density correction in ref. sbs in terms of the plasma

frequency of the media, the oscillator strengths and occupation numbers given by Carlsson (ref. 28), and the ionization potential of the media. The method finds an adjustment factor, r , which gives the ionization potential for the media at the specified density from their eq. (10) where I is

$$\ln I = \sum_{i=1}^{n-1} f_i \ln \left[(h\nu_i \rho) + \frac{2}{3} f_i (h\nu_p)^2 \right]^{1/2} + f_n \ln (h\nu_p f_n^{1/2})$$

the ionization potential of the medium (consistent with the prescriptions of ref. 10), h is Planck's constant, f_i is the fraction of electrons at that level, ν_i is the oscillator frequency of that level, ν_p is

the plasma frequency of the medium ($\nu_p = \frac{\omega_{pe}^2}{2\pi}$) and n is the number density of electrons in

the medium). The adjustment factor is found in subroutine den1 using a newton search method; for high density materials the newton search method may fail and the code will issue a warning and nullify the density effect correction; this is consistent with the fundamental theory of the density effect. Once the parameter has been found, it is used in subroutine den2 to calculate the density effect correction as specified by Eq. (5) in sbs

$$\delta(\beta) = \sum_{i=1}^n f_i \ln \left[\frac{l_i^2 + l^2}{l_i^2} \right] - l^2 (1 - \beta^2)$$

where $l_i^2 = \frac{2}{3} f_i + \frac{1}{2} \frac{\nu_i \rho}{\nu_p} \frac{\omega_{pe}^2}{\nu_i^2}$ and for conductors the n th level has, $l_n^2 = f_n$ and the parameter, l ,

is the solution of eq. (6) of sbs.

$$\frac{1}{\beta^2} - 1 = \sum_{i=1}^n \frac{f_i}{l^2 + \{(\nu_i \rho) / \nu_p\}^2}$$

Previously, MCNP used the published empirical fit formula of Sternheimer and Peierls (ref. 15). The method described in sbs is an improvement upon the sp fit formula as discussed in sbs.

For the sbs treatment, the conduction state of the material is needed which is specified with cond=0 for a non-conductor, and cond=1 for a conductor; any non-conducting isotopes in a conductor are handled as non-conductors. The user can specify this on the m card with the keyword cond. The code will provide defaults which the user can override for the specification of a non-conductor with cond=-1 on the m card. The scheme is:

initially - cond=0 (non-conductor)

in range -

check if all isotopes are conductors and set cond=1 if cond=0

check that there is at least one conducting isotope is present if cond=1, if not set cond=0

if cond=-1 override any prior decision on the conduction state and make non-conductor cond=0.

The density effect calculation has a problem for materials of a single element or Z but specified with multiple isotopes or ZAIDs. This is traced to the implementation of Eq. (10) of sbs and the ionization sum rule of ICRU 37 (ref. 10). ITS has changed that implementation slightly. Making it consistent with the above rules and such that the same answer is obtained regardless of isotopic specification worsens the verification discussed later. We note this as a problem.

2d) Relaxation

Relaxation has been a problem in MCNP. There are two algorithmic branches for an atom to relax after undergoing an ionizing event depending upon the source of the ionizing event. If the source of the ionization was a photon the atom's relaxation will be described using the model of Carter and Cashwell (ref. 29) where the major K and L line photons may be emitted or an Auger electron with energy $\text{exs}(\text{jxs}(1))$ which has been stored in eek for that material. If the source of ionization was an electron then if the atom fluoresces the photon will have only an average K energy given by eek; Auger transitions are treated the same. Under many circumstances, these approximations would not be too conflicting. However, there is an inconsistency, that fluorescent radiation will be emitted with different lines. To remedy the inconsistency, relaxation from an ionizing event has been combined into a single subroutine, flaug, called by either colidp or kxray as needed. Most of flaug is similar to the relaxation code in colidp except for mostly bookkeeping changes needed to accommodate calls from kxray. The only other physics change needed in flaug for a call from kxray is that the atom is required to relax either through an Auger transition or fluorescence; for a photoionization the atom is not required to Auger or fluor (presumably it relaxes through lower energy transitions neglected in our models.) The arguments of flaug are the energy of the ionizing quanta, the photon energy for a photoionization or max K shell energy, $\text{edg} * 1.0001$ (1.0001 prevents any round-off problems), the isotope undergoing the interaction (the highest Z isotope for kxray), the material identifier, and a flag for electron or photon ionization event.

Since, eek is not used any longer as the energy given to either a kxray fluor photon or Auger electron, its value has been lowered to the more probable value (used in the POEM code of

ref. 30 and derived by inspection from ref. 31) of $E_A = E_{\bar{K}} - 2E_{\bar{L}}$ as compared to the previous value of $E_A = E_{\bar{K}}$ and is used only to give the value of Auger transition electron.

This model is still quite crude and approximate though no longer inconsistent. Further improvements will be discussed later.

2e) Variance reduction

The variance reduction discussed in this section pertains to brems production only. There are two or techniques for brems production:

1) bbrem

Bias the brems photons produced toward higher energies.

2) bnum

Produce bnum more photons.

For EL03, the format of the bbrem card has not been changed to allow for backwards compatibility. The entries on the bbrem card are mtop_EL1=49 biasing numbers for the production of photons followed by cell numbers where the biasing will be applied. For EL03, the number of relative photon energy bins is larger so to allow compatibility with the previous bbrem card, the biasing probabilities over the low energy photon bins from the card are linearly interpolated over the low energy bins in EL03.

In MCNP, when bnum was invoked on the phys:e card, bnum identical photons were generated from a brems event with wgt=wgt/bnum. For photon thin problems, such a biasing scheme is not effective and for thick problems, it may be of only marginal help. The more common method (refs. 32 and 2) is to sample the production distribution bnum times using the same reduced weight for each photon. The sampling is done in a do loop bnum times over the full Poisson sampling of events. Note for bnum<1, the sampling of energy loss is done once but photons may or may not be produced based on bnum+rang() being zero or one. The sampling of several different photons with reduced weights has been shown to be more effective in photon thin problems (Gierga and Adams). There is a question as to how to deal with the energy loss from the more numerous and different pseudo events. For bnum>0, the first sampling of the photon(s) energy is used to calculate the energy loss. For bnum<0, the average of all photons produced is used to calculate the energy loss. The first method does not conserve energy, but more properly gives the straggling contribution of radiative energy loss. The second method more nearly

conserves energy and gives the CSDA energy loss more closely at the expense of the straggling. The latter method was the one used by Gierga and Adams in their report.

Finally, for any bnum scheme to be effective the number of secondary electrons generated must be controlled; secondary electrons take up most of the time in transport calculations. Controlling the number of secondary electrons generated is done with the enum entry on the phys:e card. The default has been one regardless of the value on the bnum card. This has been changed to scale the production down by the inverse of the number of photons produced; $enum=1/bnum$. The user can override this default setting.

Another vr scheme suggested by R. A. Forster is to have a brems occur on each substep. This scheme is done when the tenth entry, numb, on the phys:e card is non-zero. On each substep only one photon is produced with weight equal to the probability of an interaction on that substep. If this was a real interaction, then the energy loss is calculated and applied to the electron. The subsequent photons generated in the Poisson sampling are given the nominal, unreduced weight. In this scheme, enum is arbitrarily set to 1%.

2f) Bug fixes

The primary bug has to do with how the electron (charged particle) algorithms handle materials described with isotopic decomposition. The ionization potential has a factor of 1.13 applied to it when the material is a mixture of different elements. However, a pure material, i.e. composed of one type of atom, may be described with its isotopic decomposition and erroneously have the factor of 1.13 applied to the potential. To correct this each isotope is checked to see if its Z is different, if not the factor of 1.13 is not applied. Clearly, more work needs to be done here, since trace elements may be included which should not effect the ionization potential.

This isotopic decomposition bug also appears in the new density effect calculation. The fit parameter calculated from Eq. (10) of sbs is different for single ZAID and composite ZAID materials. This is because when Eq. (10) is calculated in den1 it uses a different elemental weighting scheme than used for the calculation of the ionization potential. Since we obtain excellent agreement with ITS on materials, we note this discrepancy; it is usually an error of 0.1%, though can be larger.

Another related bug is in the determination of the kxray production probability. The atom fraction of the first highest Z isotope is used. If several isotopes of the same Z were present, then the probability is too low. This has been corrected to use the sum of the atom fractions of all the highest Z isotopes

Turning off kxray production was not done when xnum=0 on the phys:e card. The test has been changed from “.gt.” to “.ge.” in subroutine xsген. Moreover, there appears to be a special treatment for $xnum < 0$ where the probability, d, of kxray production for an electron is calculated for the electron till it has gone below cutoff. This would be the probability of kxray production over the full electron range. For $xnum < 0$ the probability is scaled by $|xnum|/d$. This treatment of biasing raises a point of discussion related back to the bnum treatment.

3) Verification

There exist no analytic problems against which to verify the correctness of the physics or variance reduction models implemented in the code. However, since the database for stopping powers and bremsstrahlung cross sections has been adapted from an existing code system, ITS3.0, verifying the correctness of the implementation will be obtained by comparing to that code system.

3a) Stopping powers

Stopping powers are calculated and printed in the MCNP print table 85 and in the output from ITS code xgenp. Relative differences in ionizing and radiative stopping powers normalized to the ITS values for $Z=1$ to 92 over the energy range of 1 keV to 1 GeV are presented

in Fig. 2 (ionizing), Fig. 3 (density effect), Fig. 4(radiative). Generally, the agreement is to four

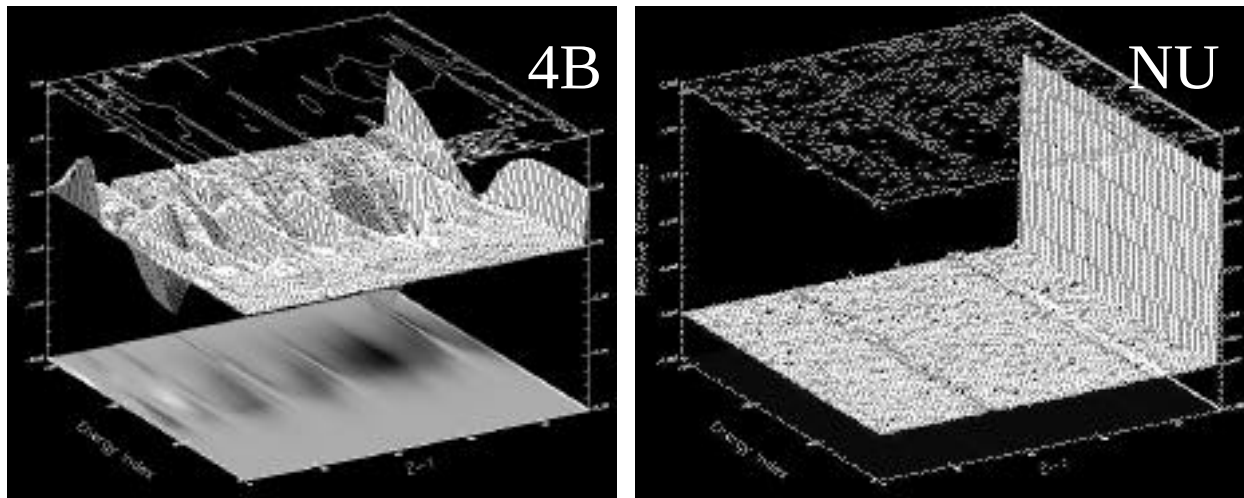


Figure 2: Relative differences between ionizing stopping powers from ITS3.0 and MCNP4B (4B) and ITS3.0 and MCNP4BNU (NU) all normalized to ITS3.0 for Z=1 to 92 from 1 keV to 1 GeV.

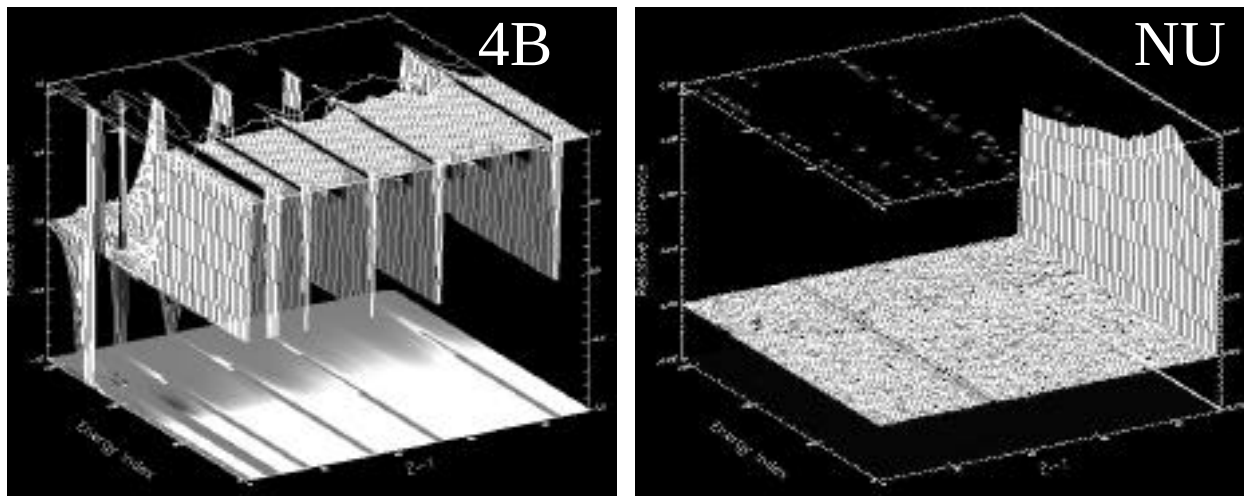


Figure 3: Relative differences between density effect corrections from ITS3.0 and MCNP4B (4B) and ITS3.0 and MCNP4BNU (NU) all normalized to ITS3.0 for Z=1 to 92 from 1 keV to 1 GeV.

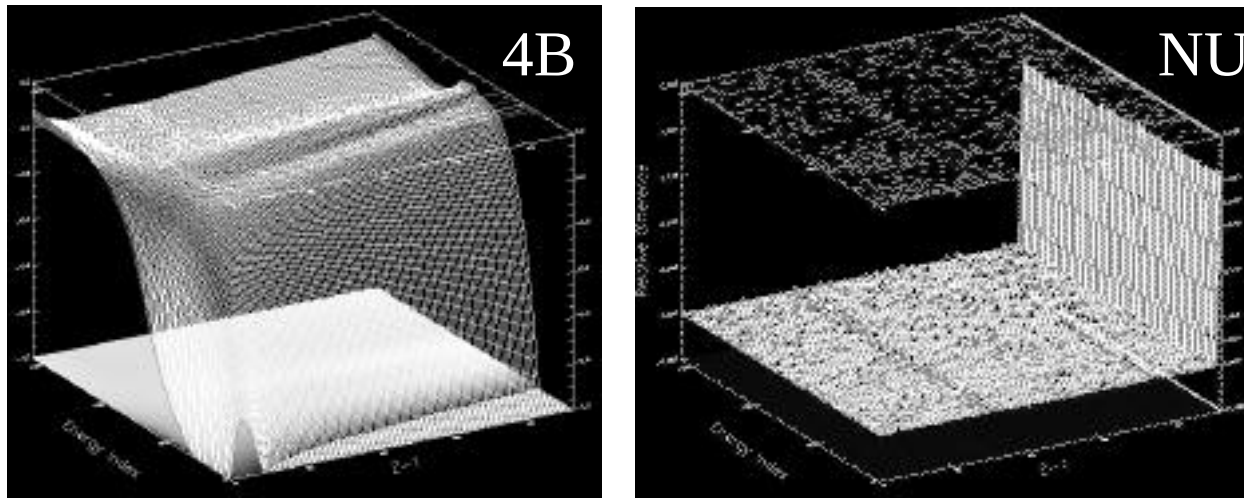


Figure 4: Relative differences between radiative stopping powers from ITS3.0 and MCNP4B (4B) and ITS3.0 and MCNP4BNU (NU) all normalized to ITS3.0 for Z=1 to 92 from 1 keV to 1 GeV.

significant figures and this level of agreement is satisfactory since the fundamental constants and atomic weights in ITS are slightly different from those in MCNP. The only problem material is Pu where the ITS atomic weight is significantly different than MCNP (and most standard tables.) Changing the atomic weight and modifying the constants in MCNP cause the stopping powers to be in agreement to four significant figures.

The stopping powers for materials was also verified. Using the materials of ICRU 37 and a few extra to span composites of a wide range of Z's, the same comparisons were done.

Figure 5 shows the comparison of the collisional or ionizing stopping powers. Figure 6 shows the

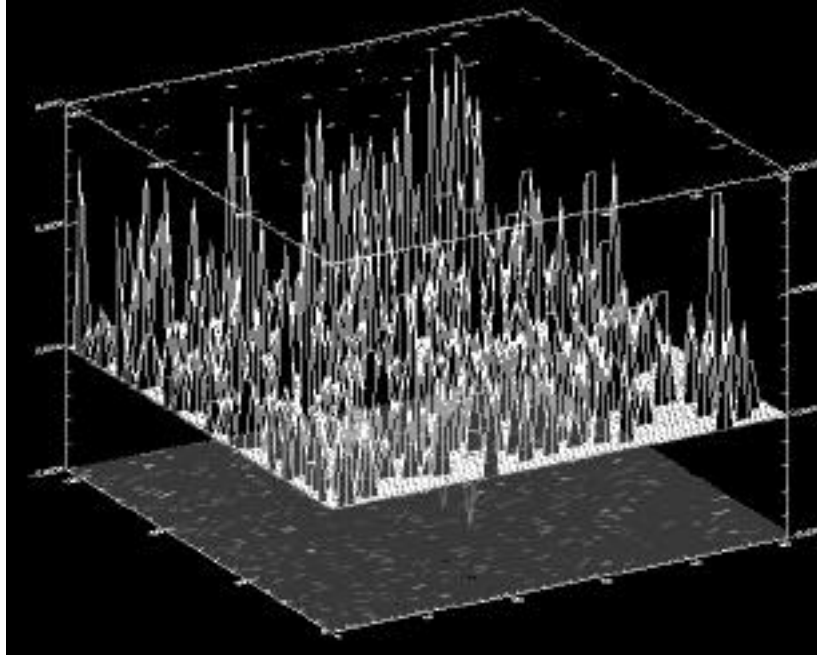


Figure 5: Relative differences between ionizing stopping powers from ITS3.0 and MCNP4BNU all normalized to ITS3.0 for a wide range of materials from 1 keV to 1 GeV.

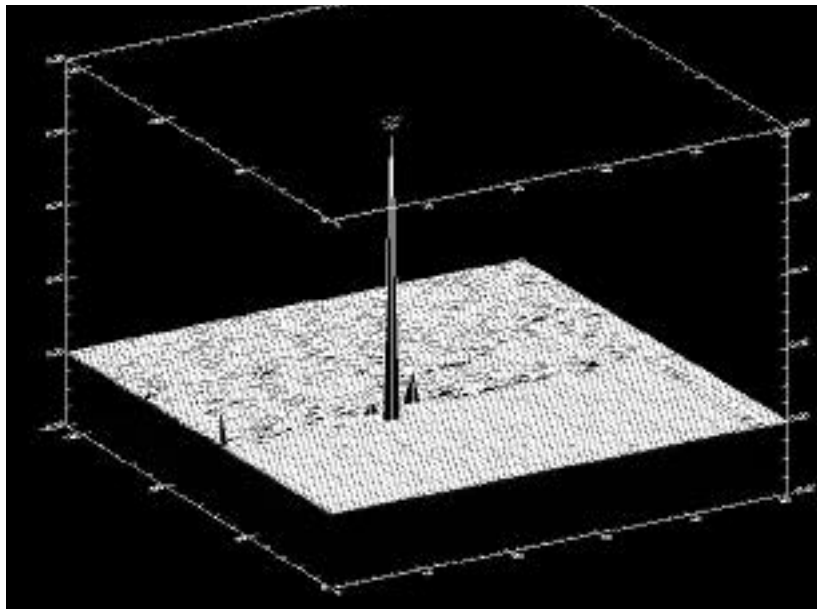


Figure 6: Relative differences between density effect from ITS3.0 and MCNP4BNU all normalized to ITS3.0 for a wide range of materials from 1 keV to 1 GeV.

density effect correction comparison and Figure 7 shows the radiative stopping power

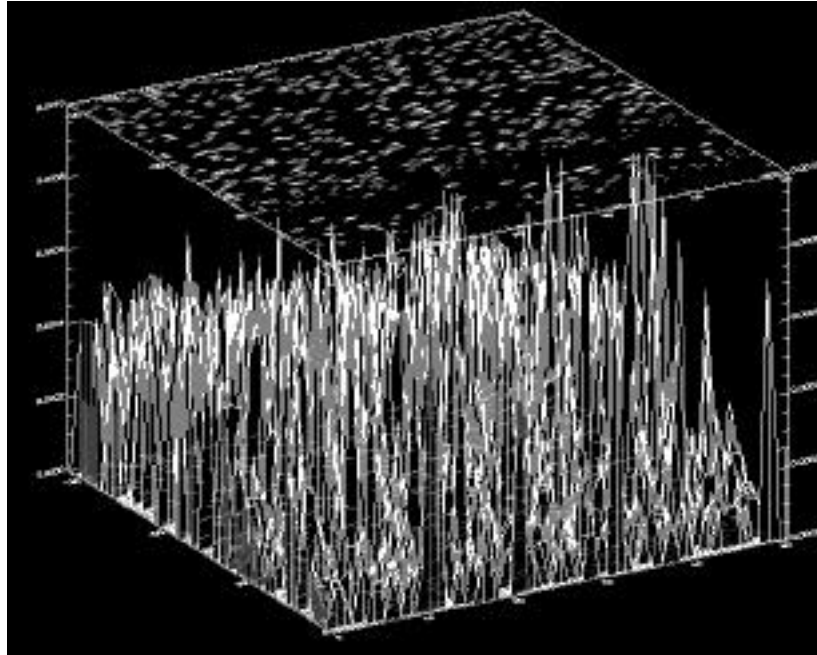


Figure 7: Relative differences between radiative stopping powers from ITS3.0 and MCNP4BNU all normalized to ITS3.0 for a wide range of materials from 1 keV to 1 GeV.

comparison. Once again the results are all within four significant figures except for a spike in the density effect calculation. This spike was for a very small correction which was near zero for both codes and so does not represent any significant problem.

In the patch, the bremsstrahlung production cross section array, pbr, the energy spectrum array, eba, and detailed angular distribution array, ech, are printed to separate files for debugging and verification purposes. H. G. Hughes maintains a version of the ITS3.0 code system where these arrays are printed to output files from the ITS3.0 cross section generating code xgenp. For all the elements for $Z=1$ to 92 and over the energy range of 1 keV to 1 GeV, the average relative error in the “pbr” array is 10^{-6} , in the spectra, 10^{-6} , and in the angular distribution array, 10^{-6} . The maximum for pbr is 10^{-5} . For the “eba” array, the maximum relative difference is 10^{-4} and is in the first bin which is typically a small number. For ech the maximum relative difference is 10^{-2} again for the same reason as for eba. Since the ITS3.0 code has changed the angular distribution calculation only cosmetically, we have not changed our routine yet. Though it is recommended that these changes be made when the angular distributions are revised since the coding is more readable and easily related back to fundamental equations. Such revisions would make the inclusion of 2BS in the sampling scheme more straightforward.

The patch also uses an adapted cubic spline fitting routine in correspondence with ITS3.0. The sampling algorithms are in agreement except for the sampling of the angular distribution. The patch includes what is needed to obtain exact ITS agreement though either method is valid.

The EL03 bremsstrahlung cross sections stored in the pbr array are displayed in figure 8. The same array for EL1 is also shown in figure 8. The curves are significantly different. Part of

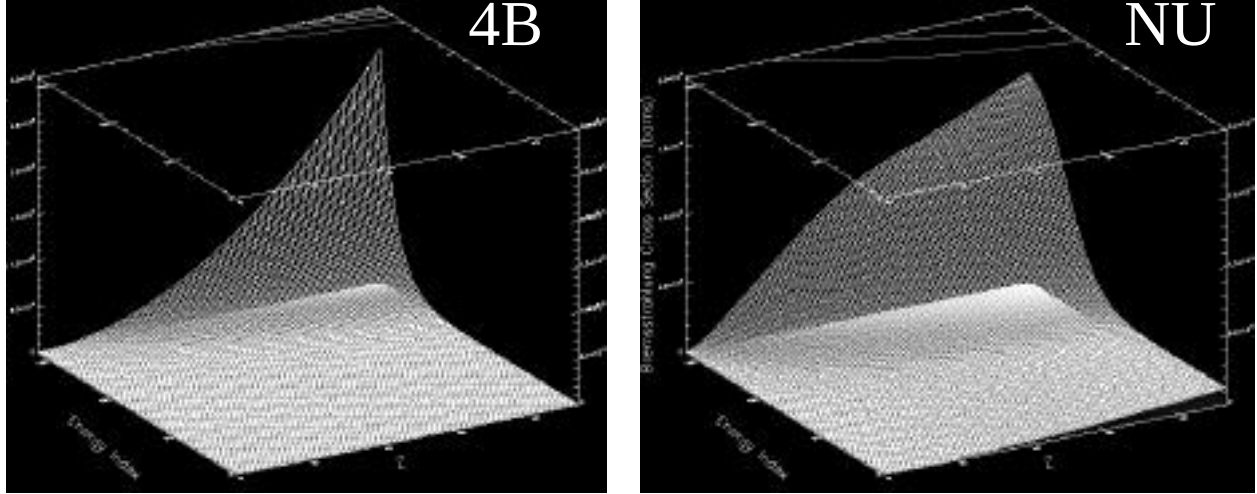


Figure 8: Bremsstrahlung production cross sections (pbr array) from MCNP4B (4B) and MCNP4BNU (NU) for Z=1 to 92 from 1 keV to 1 GeV.

the difference, as noted above, is that the EL03 database and the EL1 database have different lower limits of the photon energy for which the cross section was calculated. Symbolically, pbr for an electron of kinetic energy, T, is given by the following equation where the minimum photon

$$pbr(T) = \int_{k_{min}}^T \frac{d\sigma}{dk} dk$$

energy is kmin. Moreover, knowing that the eba array is the cumulative probability of producing a photon with energy between k/T and kmin/T, a scaling relation between the pbr arrays of the two databases would be

$$pbr_{el1}(T) = pbr_{el3w}(T)(1 - eba(0.001))$$

where $\text{eba}(0.001)$ is the value of the eba at that ratio. Thus a fairer comparison is between a scaled pbr for EL03 and EL1. Figure 9 shows the relative difference between the scaled EL03 and EL1

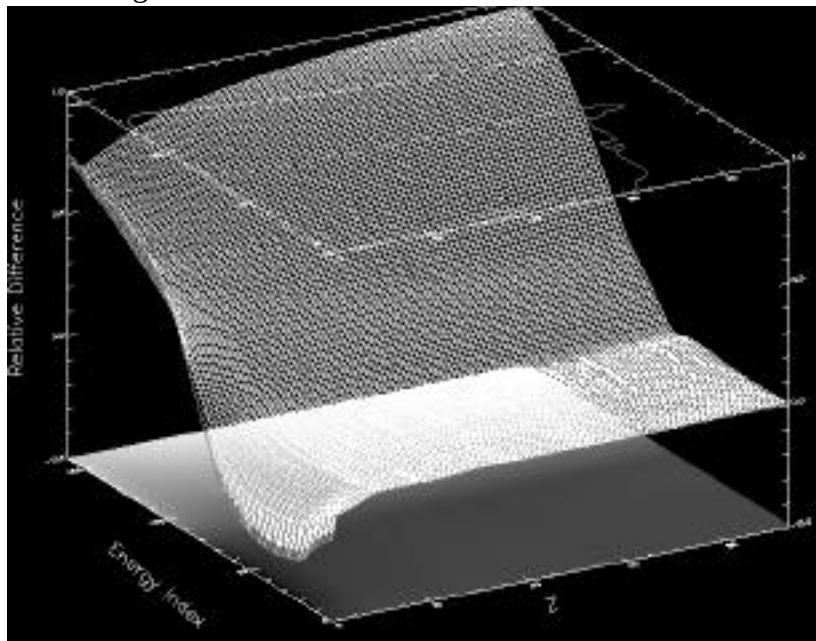


Figure 9: Scaled Bremsstrahlung production cross section relative differences between MCNP4B and MCNP4BNU for $Z=1$ to 92 from 1 keV to 1 GeV normalized to MCNP4B. For low energies the differences are quite large and show that EL1 had a very large low energy electron bremsstrahlung cross section.

At this level the algorithms for stopping power calculation and the bremsstrahlung production cross sections have been verified for all the elements and many representative materials over the energy range of 1 keV to 1 GeV. This level of verification meets or exceeds the verification standard of EL1 and MCNP4A.

Implications:

Several problems have been suggested that show that the EL1 model needed to be updated. Some of the problems still persist, however, improvements have been made. We have considered three energy ranges of problems for geometries similar to nominal converter foils.

The first is a low energy problem suggested by Newton Scientific. 20 keV electrons are normally incident on a 0.001cm W foil. The bremsstrahlung spectrum is tallied. Figure 10

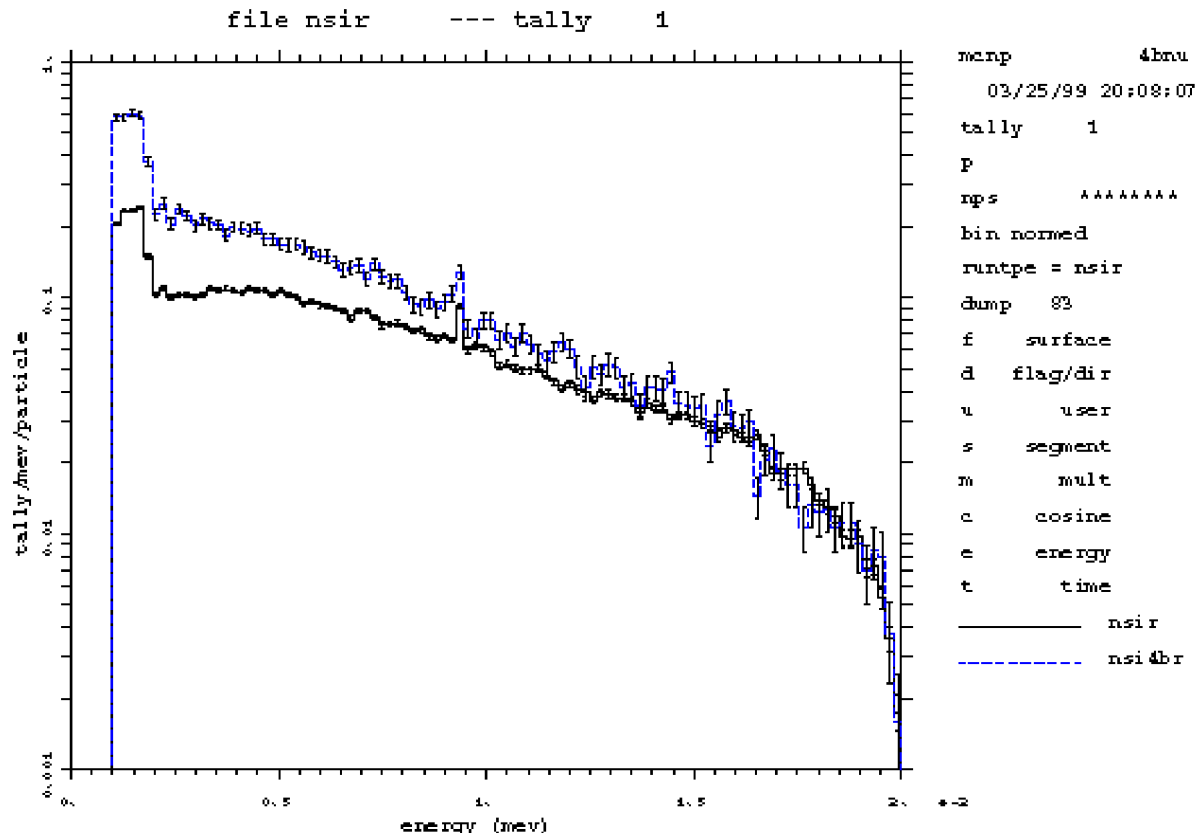


Figure 10: Bremsstrahlung spectra for 20 keV electron incident on 0.001 cm of W calculated with MCNP4B (nsi4br) and MCNP4BNU (nsir).

shows the spectrum calculated with 4bnu (the patched 4b which passed all the MCNP test problems) for the EL1(4b in the plots) and EL03 (4bnu in the plots) databases. The differences are quite large. In fact the 4b appears to have the M lines which of course it cannot. The 4bnu result is much less intense and more consistent with the NSI ITS calculation.

Higher in energy J. V. Siebers proposed a simple machine head model of a W/Cu slab. 6 MeV electrons are normally incident. He has noted that the bremsstrahlung yield downstream depends on the maximum energy of the spectrum. We have not addressed that problem yet but note that it also apparent in the ITS codes. Table JV shows the results of a calculation for that problem with two choices of EMAX (the first entry on the phys:e card). The total bremsstrahlung generated from the summary tables shows a 0.12% difference which is beyond the statistical bounds one would expect. The tally which scores the forward yield about a 1m from the converter

packs show a much larger variation. Even 4bnu has not answered this problem, however we do see that the yield is more forward directed by about 1%. Another implication apparent in this problem is inconsistency of the relaxation spectra as seen in Figure 11 where 4bnu has removed

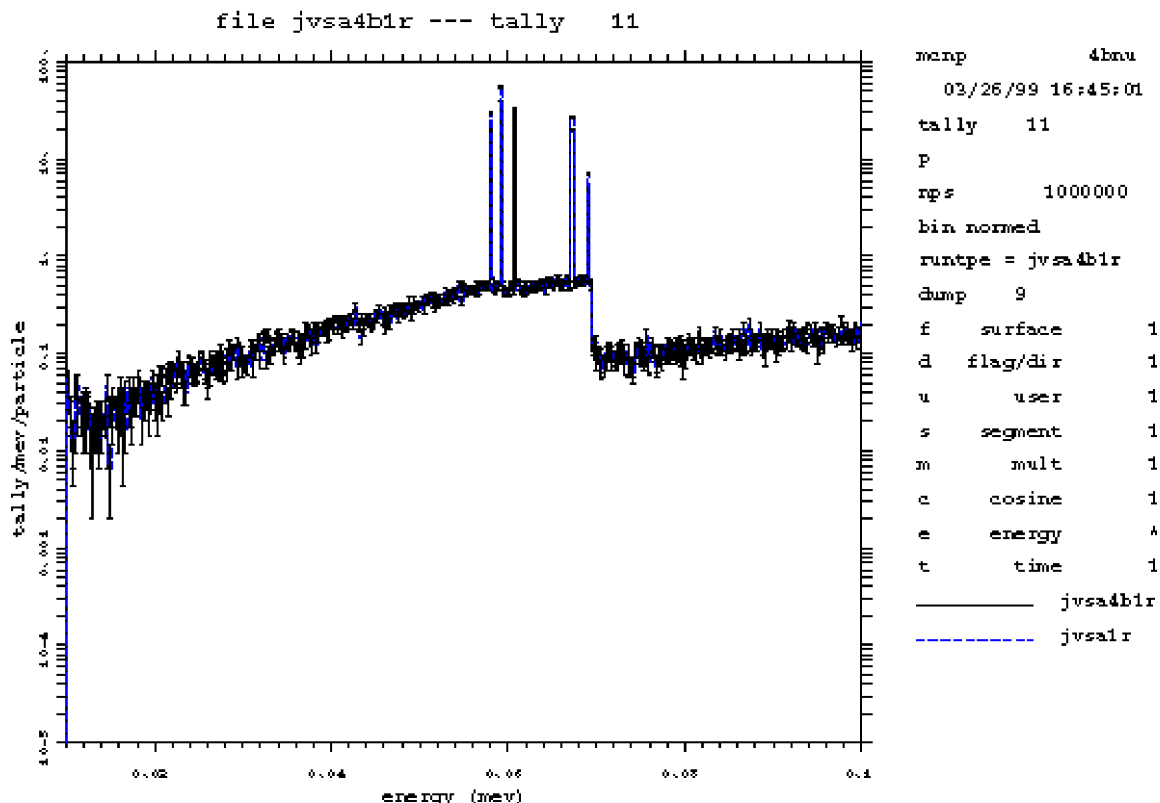


Figure 11: Bremsstrahlung and line spectra calculated with MCNP4B (jvsa4b1r) and MCNP4BNU (jvsa1r). The anomalous line has been removed in the EL03/4BNU calculation.

the anomalous line at 60.69 keV.

code	EMAX=100		EMAX=6	
mcnp4b				
bremsstrahlung	41622998	4.1623E+00	41551441	4.1551E+00
tally in forward cone	5.29151E-02	0.0014	5.15433E-02	0.0014
mcnp4bnu				
bremsstrahlung	41362833	4.1363E+00	41283862	4.1284E+00
tally in forward cone	5.35427E-02	0.0014	5.21917E-02	0.0014

Finally, at high energies, A. E. Schah von Wittenau suggested a typical converter foil 0.1 cm of W with 85.0 MeV electrons normally incident. The spectra calculated with the two different databases are presented in Fig. 12. The EL03 (not labeled asvw4br) results are all

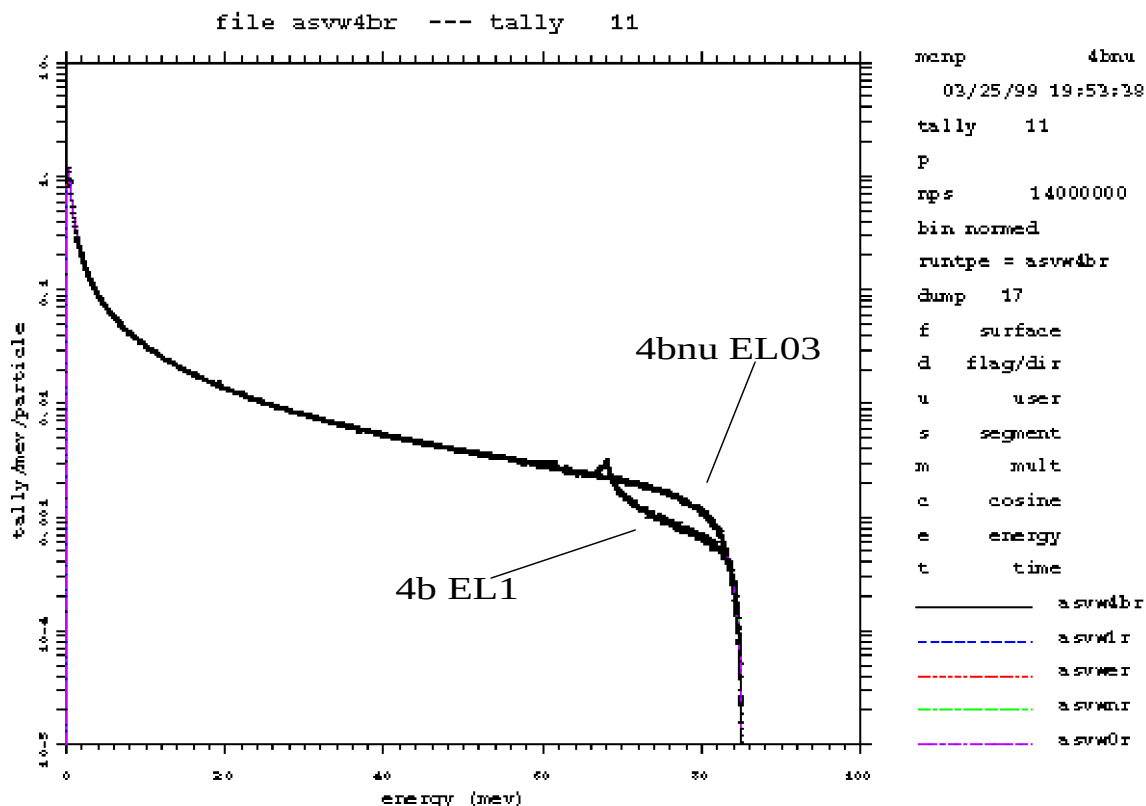


Figure 12: Bremsstrahlung spectra calculated with MCNP4B (asvw4br) and MCNP4BNU (others) for 85 MeV electrons incident on 0.1 cm W. Various variance reduction methods were tested giving nominally the same results.

smoother than the EL1. The notches in the 4b calculation are probably due to non-smooth model transitions; 4bnu by definition is immune to such artifacts. All these problems except for asvw0 were run with some variance reduction. asvw1 has the new bnum biasing, asvw1e is the same except W is made of its isotopic constituents, asvw1n uses the numb biasing, asvw4b uses the old

bnum scheme. Fig 13 shows the spectra calculated with bbrem being used for asvw1 (labeled

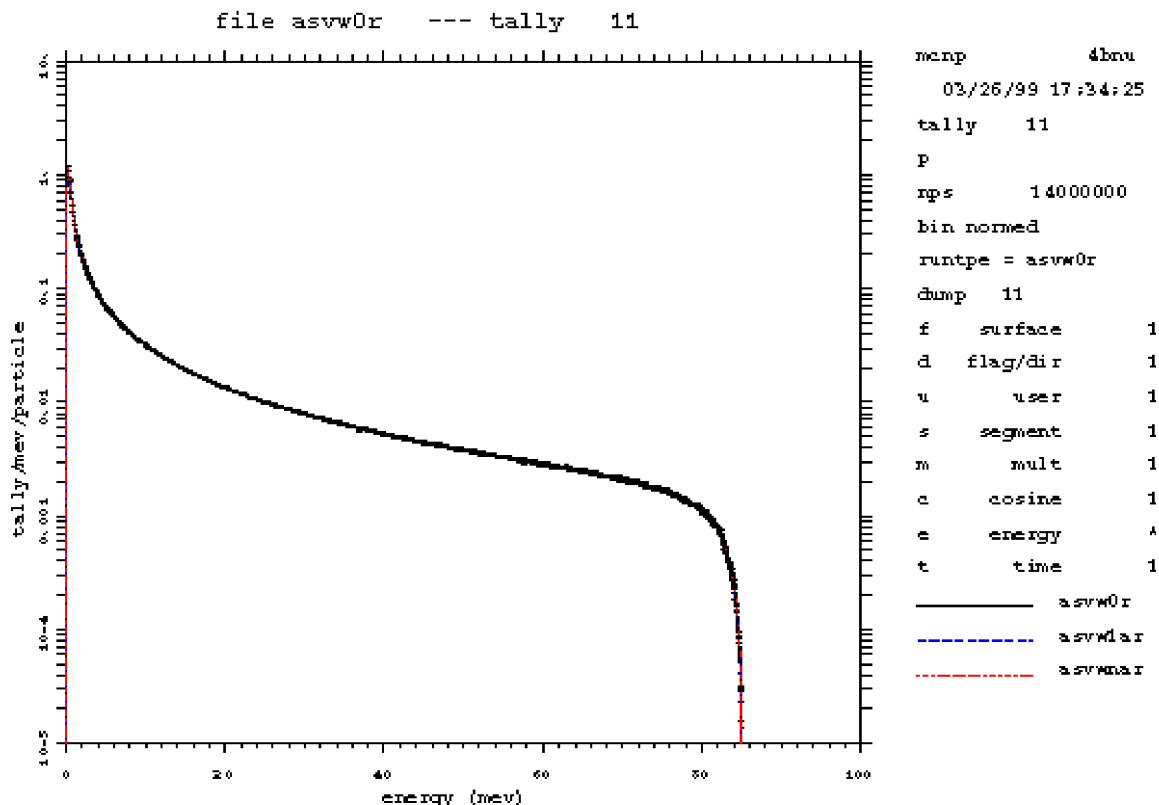


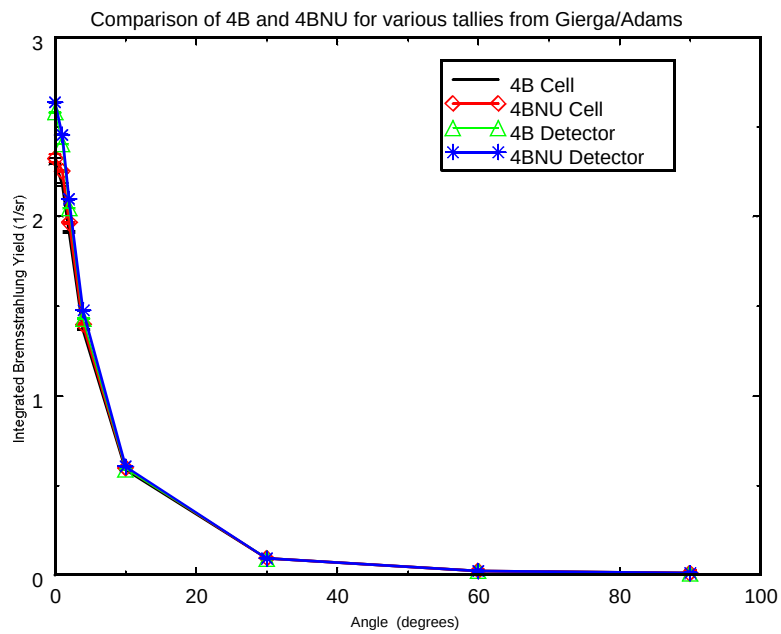
Figure 13: Bremsstrahlung spectra calculated with MCNP4BNU for 85 MeV electrons incident on 0.1 cm W with no variance reduction (asvw0r) and bbrem used in conjunction with new bnum biasins (asvw1ar) and numb biasing (asvwnar).

asvw1a) and asvwn (labeled asvwna) and asvw0. The impact of bbrem biasing has introduced no artifacts when used in conjunction with other biasing. It is amusing to note on this particularly thin problem, the largest fom was for no biasing.

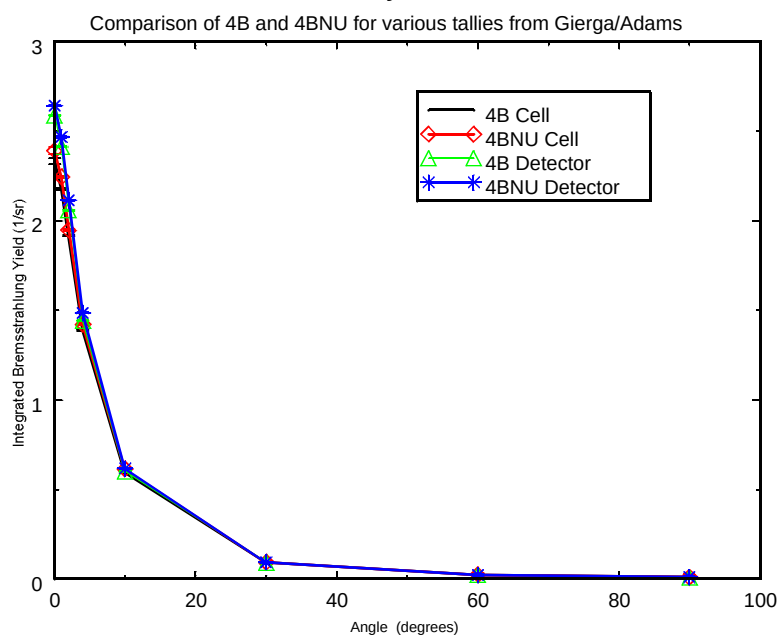
Validation:

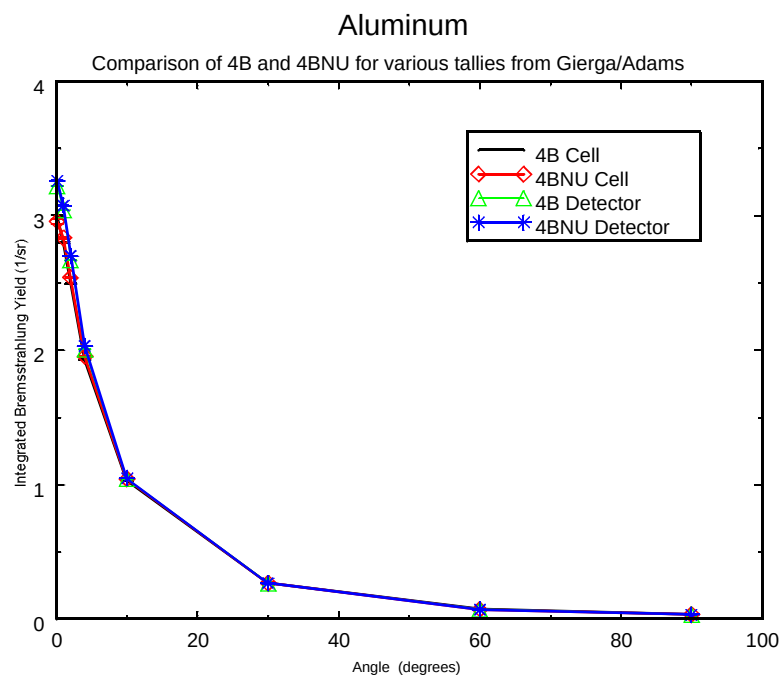
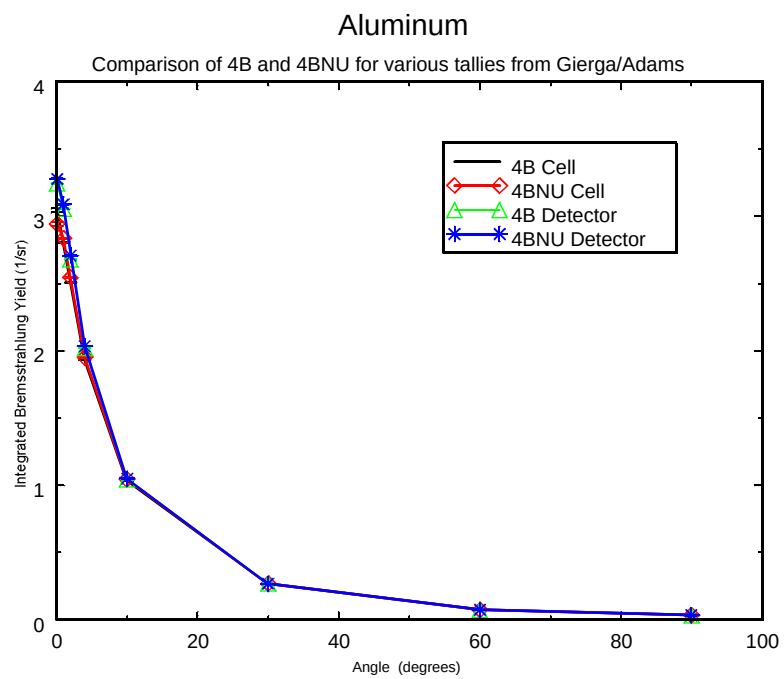
Gierga/Adams provided a very large suite of validation calculation templates. Due to budgetary limitations this set was not expanded. Nor have all the templates been run, only a few of the bremsstrahlung sensitive ones have been exercised. The results follow. Appendix C presents tables of some of the plots for more detailed comparisons.

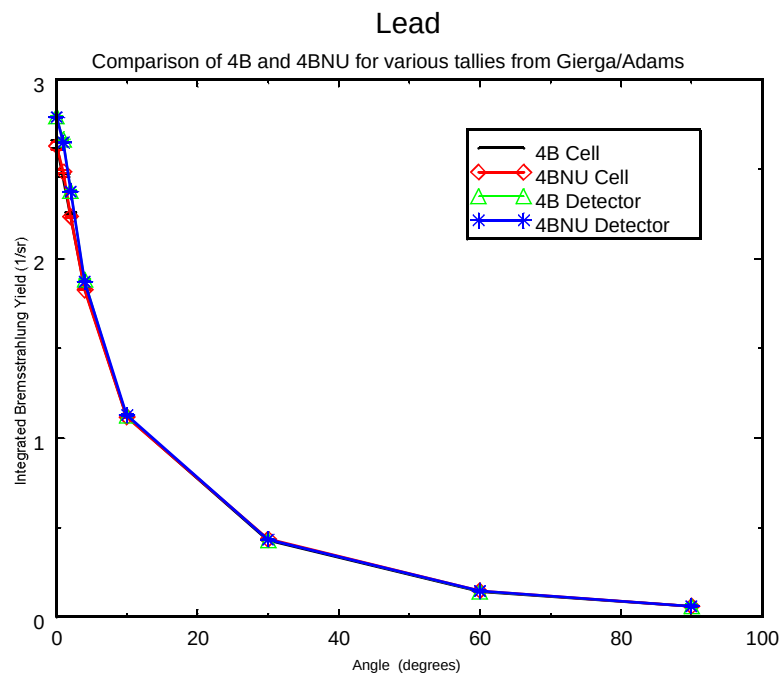
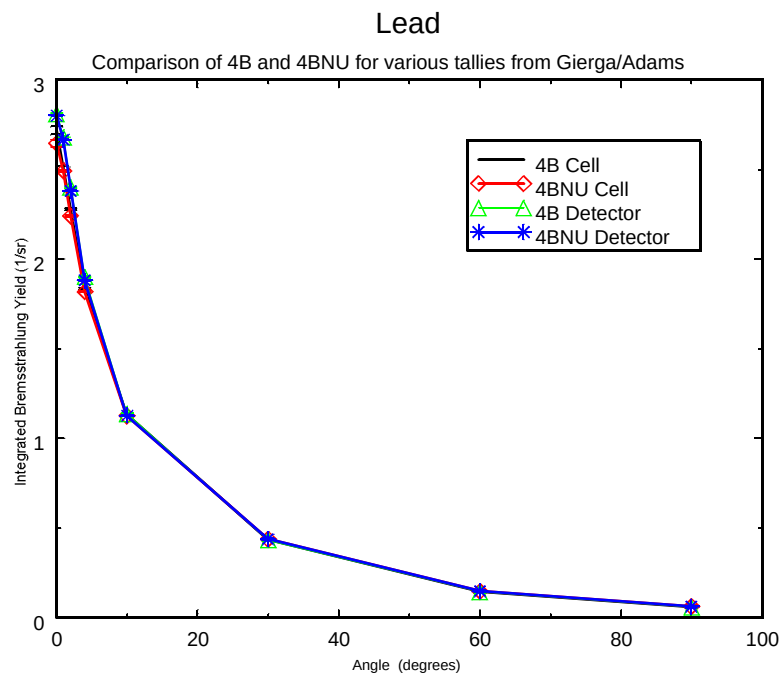
Beryllium



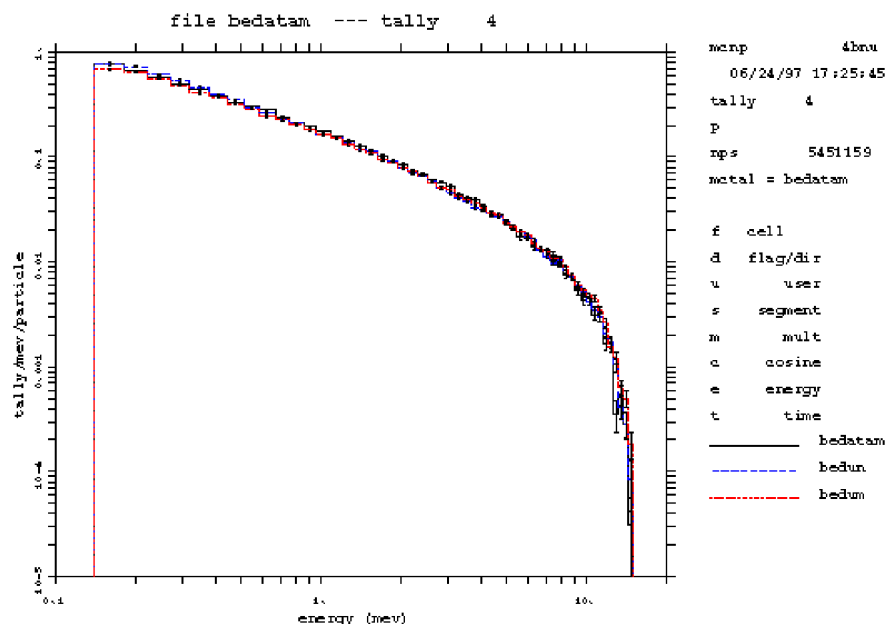
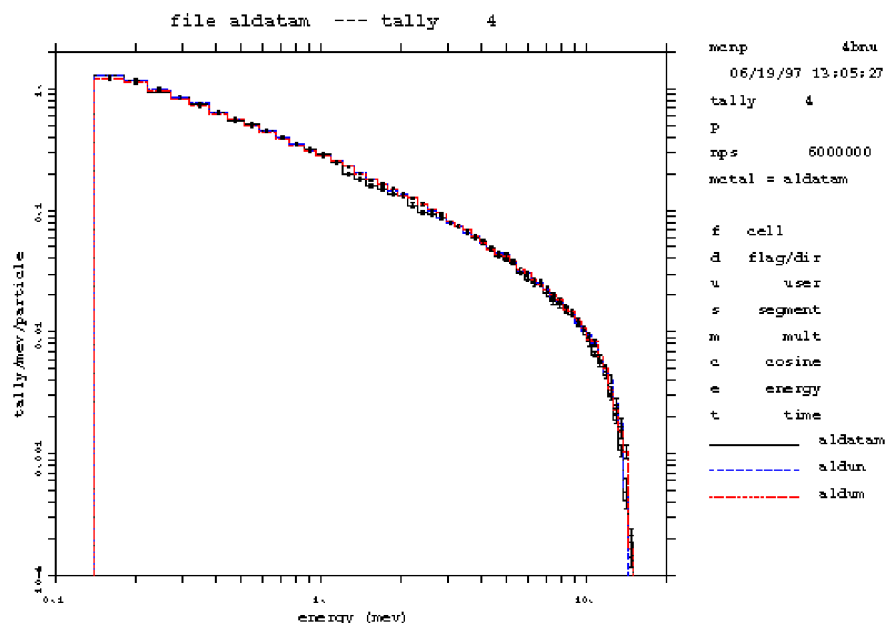
Beryllium

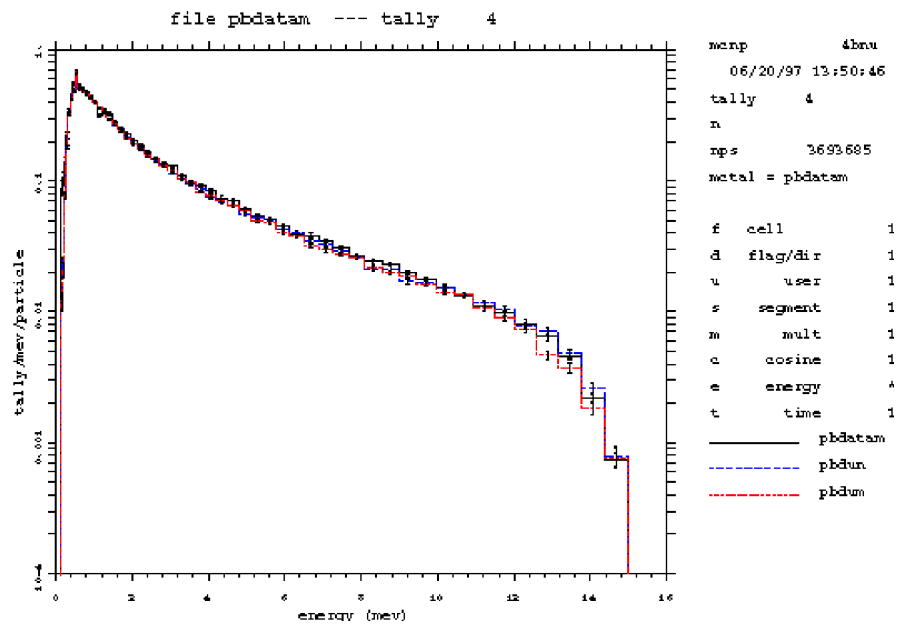




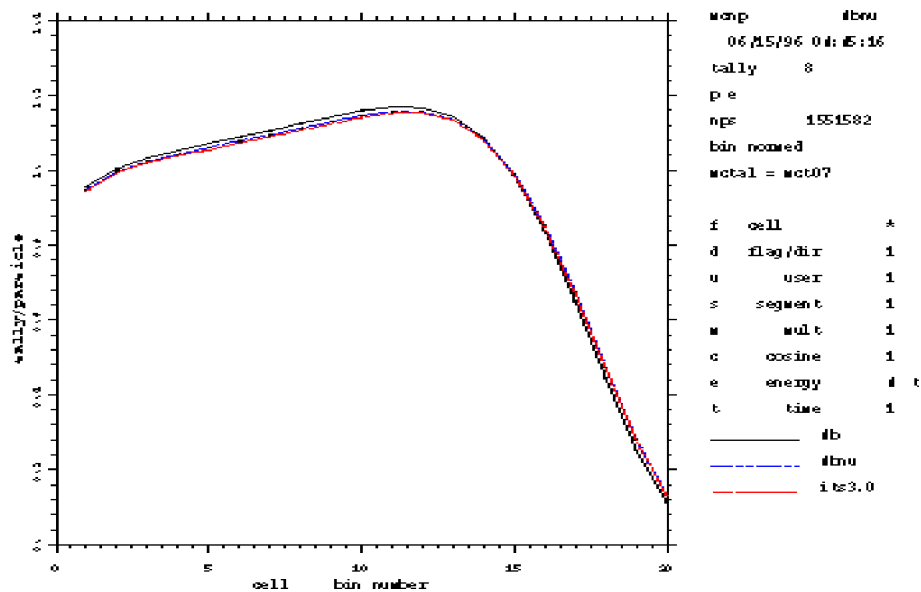


Selected spectra comparisons with 4BNU labeled with the suffix “n” and 4B “m”





20 mev electrons in water



Testing:

To facilitate merging this patch with the evolving MCNP4C, several test problems have been generated. They fall into two classes. One set are the ones derived from the previous 4B test suite which were selected if they used the EL1 database. This set consists of problems 04, 08, 10, 11, 20, 21, 22, 23, and 29. For problems 23 and 29, longer runs were done to assure that the test problems were giving reasonably assured results. A variant of 08 is included to assure that the code crashes and fails to run for mixed databases; this should not be allowed to be overcome with a fatal option since the results will probably be gibberish.

The second set focus on the new features mentioned above and are derived from the relatively straightforward “implication” validation set. inp30 and inp31 are derived from the high energy set. inp30 assures that the new bnum sampling is being done as expected with the bbrem card. inp31 checks the numb option for brems vr. inp32 is derived from the low energy problem with just fewer histories being run. It assures that the low energy portions of the database are being used as expected.

These test problems have exercised by D. Court and myself. Needless to say there are platform dependencies, however the test problems are easily checked with the converged results generated on Openblue.

Future work and conclusions:

The patch and database provide an improvement in the physics modeling of electron transport in MCNP. Comparisons with the progenitor code shows excellent agreement with calculated or derived quantities. Improvements in performance were illustrated with three sample problems. Comparisons with validation runs show improvement, albeit on the 1% level (as one might say the original developers did very good work so it is hard to improve it.) Several lessons have been learned from this effort on how new databases should be merged with the code. As such the original electron patch was a bandage and the current patch retains several of the “bandage” features.

There are still many remaining issues to be addressed for the electron/photon transport capabilities of MCNP.

- 1) The relaxation models are out of date and need to be improved (ref. 30 and 33).
- 2) Other bremsstrahlung and electron interaction databases (ref. 11) and the bremsstrahlung angular distribution calculation still is open
- 3) Emax and step size problems
- 4) Cleaning up the code and allow different databases to be used like the neutron side.
- 5) No-scatter calculation which is highly platform dependent.
- 6) Goudsmit-Saunderson treatment and general questions about multiple scattering and transport corrections such as TLC (transverse longitudinal correction).

7) Thick target bremsstrahlung.

Acknowledgements:

J. V. Siebers for helping debug the SGI version of the patch and suggesting the mid range problem. A. Schah von Wittenau. for suggesting the high energy problem and illuminating discussions on the bnum biasing schemes.

Appendix A

```
implicit real*8 (a-h,o-z)
character line*80,xline*80,hd*10,rline(15)*3
dimension nxs(16),jxs(32),nxs3(16),jxs3(32)
dimension xss(100000),exs(100000),etb(57),phi(57),eta(57)
dimension emott(100),xmott(100,5),xsc(100000),nxc(16),jxc(32)
dimension xsp(10000),nxp(16),jxp(32)
dimension tb(100),x(100),xi(100),xsec(100,100),xsic(100,100),
  1 r(100),awit(100),ek(100),el(100,15),ea(100),
  1 rtwo(9),rfac(9),riley(9,12),rkac(51)
  data kmax/89/r/0.000001d0,0.0000015d0,0.000002d0,0.000003d0,
  1 0.000004d0,0.000005d0,0.000006d0,0.000008d0,0.00001d0,
  2 0.000015d0,0.00002d0,0.00003d0,0.00004d0,0.00005d0,0.00006d0,
  3 0.00008d0,0.0001d0,0.00015d0,0.0002d0,0.0003d0,0.0004d0,
  4 0.0005d0,0.0006d0,0.0008d0,0.001d0,0.00125d0,0.0015d0,
  5 0.00175d0,0.002d0,0.0025d0,0.003d0,0.0035d0,0.004d0,0.0045d0,
  5 0.005d0,0.0055d0,0.006d0,0.007d0,0.008d0,0.009d0,0.01d0,
  6 0.0125d0,0.015d0,0.0175d0,0.02d0,0.025d0,0.03d0,0.035d0,
  7 0.04d0,0.045d0,0.05d0,0.055d0,0.06d0,0.07d0,0.08d0,0.09d0,
  8 0.1d0,0.125d0,0.15d0,0.175d0,0.20d0,0.225d0,0.25d0,0.275d0,
  9 0.3d0,0.325d0,0.35d0,0.375d0,0.4d0,0.425d0,0.45d0,0.475d0,0.5d0,
  9 0.525d0,0.55d0,0.575d0,0.6d0,0.65d0,0.7d0,0.75d0,0.8d0,0.85d0,
  9 0.9d0,0.95d0,0.97d0,0.99d0,0.995d0,0.9995d0,1.0d0,11*0.d0/,
  9 aneut/1.008664967d0/
  data rkac/.9999d0,0.8d0,0.6d0,0.5d0,0.4d0,0.3d0,0.2d0,0.15d0,
  1 0.1d0,0.08d0,0.06d0,0.05d0,0.04d0,0.03d0,0.02d0,0.015d0,0.01d0,
  2 0.008d0,0.006d0,0.005d0,0.004d0,0.003d0,0.002d0,0.0015d0,0.001d0,
  3 0.0005d0,0.0002d0,0.0001d0,0.00005d0,0.00002d0,0.00001d0,
  4 0.000005d0,0.000002d0,0.000001d0,17*0.0d0/
iel1=7
imcplib=iel1+1
ixdata=imcplib+1
ixold=ixdata+1
ixnew=ixold+1
ius=ixnew+1
ius1=ius+1
ius2=ius1+1
ius3=ius2+1
iel30=ius3+1
  open(iel1,file='el1')
open(imcplib,file='mcplib02')
open(ixdata,file='xdata.30')
open(ixold,file='xsdir1')
open(ixnew,file='xsdir3w')
open(ius,status='scratch')
open(ius1,status='scratch')
open(ius2,status='scratch')
open(ius3,status='scratch')
  open(iel30,file='el03')
c
c      read atomic weights and edge data from xdata.
read(ixdata,'(a80)')line
do 1 iz=1,100
read(ixdata,*)kzit,awit(iz),n,nn
```

```

awit(iz)=awit(iz)/aneut
cwrite(6,*)awit(iz)
ek(iz)=0.
ea(iz)=0.
if(n.eq.0)go to 1
      read(ixdata,'(5(a3,f9.1,i4))')(rline(il),el(iz,il),nm,il=1,n)
ek(iz)=el(iz,1)*1d-3
ea(iz)=ek(iz)
if(n.lt.2)go to 1
ae=0.0
nl=0
do 2 il=2,n
if(rline(il)(2:2).ne.'L')go to 2
ae=ae+el(iz,il)*1d-3
nl=nl+1
2continue
ea(iz)=ea(iz)-ae/nl
1continue
c
c      start reading and processing in "jxs" order.
c
c      read and process the radiative stopping powers.
10read(ixdata,'(a80)')line
if(line(1:11).ne.'PHIRADB.DAT')go to 10
read(ixdata,*)lmax,lrax,mrax
read(ixdata,*)(etb(lmax-i+1),i=1,lmax)
do 60 iz=1,100
read(ixdata,*)(phi(lmax-i+1),i=1,lmax),(eta(lmax-i+1),i=1,lmax)
if(iz.le.94)then
read(iel1,'(f9.2,1he,2f12.6,1x,a10)')zaid,awit(iz),tz,hd
hd='6/6/98'
write(ius,'(f9.2,1he,2f12.6,1x,a10)')zaid,awit(iz),tz,hd
call readl(5,iel1,ius)
read(iel1,'(8i9)')(nxs(i),i=1,16),(jxs(i),i=1,32)
read(iel1,'(1p4e20.12)')(xss(i),i=1,nxs(1))
call xs23(nxs,jxs,nxs3,jxs3)
else
call zeroxs(nxs,jxs)
call zeroxs(nxs3,jxs3)
zaid=1000*iz+.01
tz=0.0
hd='6/6/98'
write(ius,'(f9.2,1he,2f12.6,1x,a10)')zaid,awit(iz),tz,hd
line=' '
do 15 i=1,5
15write(ius,'(a80)')line
endif
jxs(1)=1
nxs3(1)=2+3*lmax
nxs3(2)=iz
nxs3(3)=lmax
jxs3(1)=1
jxs3(2)=3
jxs3(3)=jxs3(2)+3*lmax
c
c      flag the new treatment.

```



```

        nxs3(16)=3
    if(iz.le.94)then
    call readl(6,imcplib,0)
    read(imcplib,'(8i9)')(nxp(i),i=1,16),(jxp(i),i=1,32)
        read(imcplib,'(1p4e20.12)')(xsp(i),i=1,nxp(1))
    do 30 i=1,jxs3(2)-1
    30exs(i)=xss(i)
    c fix the kxray line and auger emission value at least for
    c consistency with fluorescence model.
    c exs(jxs3(1))=kxray edge threshold.
    c exs(jxs3(1)+1)=auger emission energy
        if(nxp(4).ne.0)then
    exs(jxs3(1))=xsp(jxp(4)+nxp(4)-1)*1d3+1d-6
    exs(jxs3(1)+1)=exs(jxs3(1))
    if(nxp(4).gt.2)exs(jxs3(1)+1)=(xsp(jxp(4)+nxp(4)-1)-
        1 2*xsp(jxp(4)+1))*1d3
    if(exs(jxs3(1)+1).lt.0.)exs(jxs3(1)+1)=
        1 min(abs(exs(jxs3(1)+1)),
        1 xss(jxs(1)+1))
    endif
    else
    exs(jxs3(1))=ek(iz)
    exs(jxs3(1)+1)=ea(iz)
    endif
        do 40 i=1,nxs3(3)
    exs(jxs3(2)+i-1)=etb(i)
    exs(jxs3(2)+i-1+nxs3(3))=phi(i)
    40exs(jxs3(2)+i-1+2*nxs3(3))=eta(i)
    write(ius,'(8i9)')(nx3(i),i=1,16),(jxs3(i),i=1,32)
    60write(ius,'(1p4e20.12)')(exs(i),i=1,nxs3(1))
        rewind ixdata
        rewind ius
    c
    c      read the old ratios for comparison sake.
        rewind iel1
    c
    c      read mott cross section ratios.
    61  read(ixdata,'(a80)')line
    if(line(1:9).ne.'MOTTE.DAT')go to 61
    read(ixdata,*)imott,ixmott
    read(ixdata,*)(emott(i),i=1,imott)
    read(ixdata,'(a80)')line
    do 63 iz=1,100
    read(ixdata,*)((xmott(i,j),i=1,imott),j=1,ixmott)
    call readl(6,ius,ius1)
    read(ius,'(8i9)')(nxs(i),i=1,16),(jxs(i),i=1,32)
    read(ius,'(1p4e20.12)')(xss(i),i=1,nxs(1))
    call upto(exs,xss,jxs(3))
        call xs23(nxs,jxs,nxs3,jxs3)
    nxs3(4)=imott
    nxs3(1)=nxs(1)+(ixmott+1)*nxs3(4)
    jxs3(4)=jxs(3)+(ixmott+1)*nxs3(4)
    do 62 i=1,imott
    62exs(jxs3(3)+i-1)=emott(i)
    do 602 j=1,ixmott
    jd=j*imott

```

```
do 602 i=1,imott
602exs(jxs3(3)+i-1+jd)=xmott(i,j)
c
c      skip over positron data
read(ixdata,*)((xmott(i,j),i=1,imott),j=1,ixmott)
write(ius1,'(8i9)')(nxs3(i),i=1,16),(jxs3(i),i=1,32)
63write(ius1,'(1p4e20.12)')(exs(i),i=1,nxs3(1))
rewind ius
rewind ius1
rewind ixdata
c
c      read riley cross sections.
cRILEY.DAT
64  read(ixdata,'(a80)')line
if(line(1:9).ne.'RILEY.DAT')go to 64
do 67 iz=1,100
do 65 ir=1,9
read(ixdata,*)kz,rtwo(ir),rfac(ir)
65read(ixdata,'(5e15.8)')(riley(ir,ik),ik=1,12)
call readl(6,ius1,ius)
read(ius1,'(8i9)')(nxs(i),i=1,16),(jxs(i),i=1,32)
read(ius1,'(1p4e20.12)')(xss(i),i=1,nxs(1))
call upto(exs,xss,jxs(4))
      call xs23(nxs,jxs,nxs3,jxs3)
jxs3(5)=jxs3(4)+9*14
nxs3(1)=nxs(1)+9*14
io=-1
do 66 ir=1,9
io=io+1
exs(jxs3(4)+io)=rtwo(ir)
io=io+1
exs(jxs3(4)+io)=rfac(ir)
do 66 ik=1,12
io=io+1
66exs(jxs3(4)+io)=riley(ir,ik)
write(ius,'(8i9)')(nxs3(i),i=1,16),(jxs3(i),i=1,32)
67write(ius,'(1p4e20.12)')(exs(i),i=1,nxs3(1))
rewind ius
rewind ius1
rewind ixdata
c
c      now do the bremsstrahlung production
70  read(ixdata,'(a80)')line
if(line(1:9).ne.'BREME.DAT')go to 70
      read(ixdata,'(2i6)') nmix,kmix
      read(ixdata,'(6f12.5)') (tb(n),n=1,nmix)
      read(ixdata,'(6f12.5)') (x(k),k=1,kmix)
read(ixdata,'(a80)')line
do 120 iz=1,100
call readl(6,ius,ius2)
read(ius,'(8i9)')(nxs(i),i=1,16),(jxs(i),i=1,32)
read(ius,'(1p4e20.12)')(xss(i),i=1,nxs(1))
call upto(exs,xss,jxs(5))
      call xs23(nxs,jxs,nxs3,jxs3)
nxs3(5)=nmix
nxs3(6)=kmix
```

```

jxs3(6)=0
nxs3(1)=nxs(1)+nmix+kmix+nmix*kmix
read (ixdata,'(6f9.5)') ((xsic(n,k),n=1,nmix),k=1,kmix)
    do 90 n=1,nmix
90exs(jxs3(5)+n-1)=tb(n)
do 100 k=1,kmix
exs(jxs3(5)+k-1+nmix)=x(k)
c
c      1d-3 converts from mb to b.
do 100 n=1,nmix
100exs(jxs3(5)+nmix+kmix+nmix*(k-1)+n-1)=xsic(n,k)*1d-3
c
c      write out the normalized photon grid numbers.
c      these are rkt and rka but to allow for flexibility, they
c      will now be read from el30.
c      the jxs indices are bumped up since jxs(7) and jxs(8) are
c      used already. Then for consistency nxs is bumped up to have
c      the same indices.
nxs3(9)=kmax
jxs3(9)=nxs3(1)+1
nxs3(1)=nxs3(1)+nxs3(9)
nxs3(10)=34
jxs3(10)=nxs3(1)+1
nxs3(1)=nxs3(1)+nxs3(10)
do 110 i=1,nxs3(9)
110exs(jxs3(9)+i-1)=r(i)
do 115 i=1,nxs3(10)
115exs(jxs3(10)+i-1)=rkac(i)
write(ius2,'(8i9)')(nxs3(i),i=1,16),(jxs3(i),i=1,32)
120write(ius2,'(1p4e20.12)')(exs(i),i=1,nxs3(1))
c
c      binding energies and occupation numbers of Carlson for density
c      effect corrections.
    rewind ixdata
    rewind ius2
140  read(ixdata,'(a80)')line
if(line(1:12).ne.'BINDENGY.DAT')go to 140
    do 160 iz=1,100
read(ixdata,*)izx,isets
    read(ixdata,*)(x(i),xi(i),i=1,isets)
    call readl(6,ius2,ius1)
read(ius2,'(8i9)')(nxs(i),i=1,16),(jxs(i),i=1,32)
read(ius2,'(1p4e20.12)')(exs(i),i=1,nxs(1))
    call xs23(nxs,jxs,nxs3,jxs3)
nxs3(11)=isets
nxs3(1)=nxs(1)+2*isets
jxs3(11)=nxs(1)+1
do 150 ij=1,isets
exs(jxs3(11)+ij-1)=x(ij)
150exs(jxs3(11)+ij+isets-1)=xi(ij)
write(ius1,'(8i9)')(nxs3(i),i=1,16),(jxs3(i),i=1,32)
160write(ius1,'(1p4e20.12)')(exs(i),i=1,nxs3(1))
    rewind ius1
c
c      all done now write the new data file, el30,
c      directory file, xsdir30.

```

```

nline=1
170read(ixold, '(a80)', end=220, err=220)xline
    iyes=0
line=xline
do 180 j=1,5
    jj=j+19
180if(line(jj:jj+2).eq.'el1')iyes=jj
    if(iyes.eq.0)then
        write(ixnew, '(a80)')xline
        go to 170
    endif
do 200 i=1,6
    read(ius1, '(a80)')line
    if(i.eq.1)then
        do 190 j=1,3
            jj=j+7
190    if(line(jj:jj).eq.'1')line(jj:jj)='3'
            write(6, '(a80)')line
        endif
200write(iel30, '(a80)')line
        read(ius1, '(8i9)')(nxs3(i), i=1, 16), (jxs3(i), i=1, 32)
        read(ius1, '(1p4e20.12)')(exs(i), i=1, nxs3(1))
        write(iel30, '(8i9)')(nxs3(i), i=1, 16), (jxs3(i), i=1, 32)
        write(iel30, '(1p4e20.12)')(exs(i), i=1, nxs3(1))
        line=xline
        line(7:8)='03'
        line(iyes+2:iyes+4)=' '
        line(iyes+2:iyes+4)='3x '
        n0=0
        n1=1
        t=0.0
        write(ixnew, 210)line(1:27), n0, n1, nline, nxs3(1),
            1 n0, n0, t
210format(a26, 1x, 2i2, 2i8, 2i2, f4.1)
        noff=13
        if(mod(nxs3(1), 4).eq.0)noff=12
            nline=nline+nx3(1)/4+noff
        write(ixnew, '(a80)')xline
        go to 170
    c
    c      add in the final zaids to 100.
220    continue
        do 250 iz=95,100
do 240 i=1,6
    read(ius1, '(a80)')line
    if(i.eq.1)then
        do 230 j=1,3
            jj=j+7
230    if(line(jj:jj).eq.'1')line(jj:jj)='3'
            write(6, '(a80)')line
        xline=line
        if(line(1:1).eq.' ')xline(1:27)=line(2:28)
        endif
240write(iel30, '(a80)')line
            read(ius1, '(8i9)')(nxs3(i), i=1, 16), (jxs3(i), i=1, 32)
            read(ius1, '(1p4e20.12)')(exs(i), i=1, nxs3(1))

```

```
write(iel30,'(8i9)')(nxs3(i),i=1,16),(jxs3(i),i=1,32)
write(iel30,'(1p4e20.12)')(exs(i),i=1,nxs3(1))
n0=0
n1=1
t=0.0
xline(22:27)=' el3x '
write(ixnew,210)xline(1:27),n0,n1,nline,nxs3(1),
      1 n0,n0,t
noff=13
if(mod(nxs3(1),4).eq.0)noff=12
250  nline=nline+nxs3(1)/4+noff
stop
      end
subroutine readl(n,iin,iout)
character line*80
do 10 i=1,n
      read(iin,'(a80)',end=999)line
cif(i.eq.1)write(6,'(a80)')line
10if(iout.gt.0)write(iout,'(a80)')line
      return
999write(*,*)' problem reading unit ',iin
stop
end
subroutine xs23(nxs,jxs,nxs3,jxs3)
dimension nxs(16),jxs(32),nxs3(16),jxs3(32)
do 10 i=1,16
nxs3(i)=nxs(i)
jxs3(i)=jxs(i)
10  jxs3(i+16)=jxs(i+16)
return
end
subroutine zeroxs(nxs,jxs)
dimension nxs(16),jxs(32)
do 10 i=1,16
nxs(i)=0
jxs(i)=0
10  jxs(i+16)=0
return
end
subroutine upto(exs,xss,jxs)
implicit real*8 (a-h,o-z)
dimension exs(1),xss(1)
do 10 i=1,jxs
10exs(i)=xss(i)
return
end
```

Distribution
X-5-RN(U)-00-14

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Appendix B

Intentionally omitted.

Appendix C

Tables of values calculated using David Gierga's input files. Each suite of runs is presented. The final three tables compare to data showing only a marginal change in the results due to the upgrade.

TABLE IV: Beryllium cell flux tallies without stainless steel

angle	Flux(4b)	Rel. Err.	Flux(nu)	Rel. Err.	rel. diff.
0	2.31E+00	0.0181	2.33E+00	0.0212	-0.0072
1	2.18E+00	0.0066	2.25E+00	0.0076	-0.0334
2	1.91E+00	0.005	1.97E+00	0.0058	-0.0275
4	1.37E+00	0.0041	1.40E+00	0.0048	-0.0223
10	5.90E-01	0.004	5.98E-01	0.0047	-0.0142
30	9.37E-02	0.0059	9.42E-02	0.007	-0.0060
60	2.29E-02	0.0091	2.34E-02	0.0106	-0.0222
90	1.08E-02	0.0123	1.11E-02	0.0144	-0.0281

TABLE V: Beryllium detector flux tallies without stainless steel

angle	Flux(4b)	Rel. Err.	Flux(nu)	Rel. Err.	rel. diff.
0	2.58E+00	0.0025	2.64E+00	0.003	-0.0215
1	2.40E+00	0.0024	2.46E+00	0.0028	-0.0218
2	2.05E+00	0.0024	2.10E+00	0.0027	-0.0246
4	1.43E+00	0.0024	1.48E+00	0.0027	-0.0328
10	5.90E-01	0.0028	6.06E-01	0.0033	-0.0267
30	9.20E-02	0.0034	9.27E-02	0.004	-0.0077
60	2.26E-02	0.003	2.28E-02	0.0034	-0.0063
90	1.05E-02	0.0027	1.06E-02	0.0038	-0.0114

TABLE VI: Beryllium cell flux tallies with stainless steel

angle	Flux(4b)	Rel. Err.	Flux(nu)	Rel. Err.	rel. diff.
0	2.33E+00	0.0181	2.39E+00	0.0209	-0.0249
1	2.18E+00	0.0066	2.25E+00	0.0076	-0.0319
2	1.92E+00	0.005	1.95E+00	0.0058	-0.0158
4	1.39E+00	0.0041	1.42E+00	0.0048	-0.0240
10	5.98E-01	0.004	6.15E-01	0.0046	-0.0294
30	9.34E-02	0.006	9.53E-02	0.0069	-0.0207
60	2.37E-02	0.009	2.38E-02	0.0105	-0.0044
90	1.10E-02	0.0124	1.10E-02	0.0144	-0.0055

TABLE VII: Beryllium detector flux tallies with stainless steel

angle	Flux(4b)	Rel. Err.	Flux(nu)	Rel. Err.	rel. diff.
0	2.59E+00	0.0026	2.64E+00	0.003	-0.0211
1	2.41E+00	0.0025	2.47E+00	0.0028	-0.0228
2	2.06E+00	0.0024	2.11E+00	0.0027	-0.0269
4	1.44E+00	0.0024	1.49E+00	0.0027	-0.0288
10	6.03E-01	0.0028	6.17E-01	0.0032	-0.0235
30	9.34E-02	0.0034	9.49E-02	0.004	-0.0158
60	2.34E-02	0.0038	2.33E-02	0.0037	0.0014
90	1.08E-02	0.0027	1.09E-02	0.004	-0.0176

TABLE VIII: Aliminum cell flux tallies without stainless steel

angle	Flux(4b)	Rel. Err.	Flux(nu)	Rel. Err.	rel. diff.
0.1	3.04E+00	0.0151	2.94E+00	0.0168	0.0325
1	2.81E+00	0.0056	2.84E+00	0.0061	-0.0100
2	2.51E+00	0.0042	2.55E+00	0.0045	-0.0149
4	1.94E+00	0.0034	1.95E+00	0.0037	-0.0081
10	1.03E+00	0.0029	1.04E+00	0.0032	-0.0089
30	2.66E-01	0.0034	2.68E-01	0.0037	-0.0064
60	7.28E-02	0.0049	7.14E-02	0.0054	0.0199
90	3.29E-02	0.0068	3.28E-02	0.0075	0.0039

TABLE IX: Aliminum detector flux tallies without stainless steel

angle	Flux(4b)	Rel. Err.	Flux(nu)	Rel. Err.	rel. diff.
0.1	3.24E+00	0.0022	3.28E+00	0.0023	-0.0107
1	3.06E+00	0.0021	3.09E+00	0.0023	-0.0103
2	2.68E+00	0.002	2.71E+00	0.0022	-0.0101
4	2.02E+00	0.002	2.04E+00	0.0022	-0.0071
10	1.04E+00	0.0024	1.05E+00	0.0026	-0.0053
30	2.67E-01	0.0027	2.66E-01	0.003	0.0055
60	7.28E-02	0.0028	7.25E-02	0.0029	0.0031
90	3.28E-02	0.0027	3.26E-02	0.003	0.0056

TABLE X: Aliminum cell flux tallies with stainless steel

angle	Flux(4b)	Rel. Err.	Flux(nu)	Rel. Err.	rel. diff.
0.1	2.99E+00	0.0153	2.96E+00	0.0188	0.0110
1	2.80E+00	0.0056	2.84E+00	0.0068	-0.0125
2	2.49E+00	0.0042	2.54E+00	0.0051	-0.0190
4	1.93E+00	0.0034	1.96E+00	0.0041	-0.0132
10	1.03E+00	0.0029	1.04E+00	0.0035	-0.0117
30	2.69E-01	0.0034	2.68E-01	0.0041	0.0040
60	7.32E-02	0.0049	7.26E-02	0.006	0.0081
90	3.33E-02	0.0068	3.32E-02	0.0083	0.0004

TABLE XI: Aliminum detector flux tallies with stainless steel

angle	Flux(4b)	Rel. Err.	Flux(nu)	Rel. Err.	rel. diff.
0.1	3.22E+00	0.0022	3.26E+00	0.0026	-0.0117
1	3.04E+00	0.0021	3.08E+00	0.0025	-0.0106
2	2.67E+00	0.002	2.70E+00	0.0025	-0.0123
4	2.01E+00	0.002	2.03E+00	0.0025	-0.0117
10	1.04E+00	0.0023	1.05E+00	0.0028	-0.0060
30	2.67E-01	0.0027	2.67E-01	0.0033	0.0011
60	7.29E-02	0.0027	7.21E-02	0.0032	0.0105
90	3.31E-02	0.0027	3.29E-02	0.0033	0.0072

TABLE XII: Lead cell flux tallies without stainless steel

angle	Flux(4b)	Rel. Err.	Flux(nu)	Rel. Err.	rel. diff.
0	2.72E+00	0.0222	2.65E+00	0.0199	0.0260
1	2.51E+00	0.0082	2.49E+00	0.0072	0.0060
2	2.28E+00	0.006	2.24E+00	0.0054	0.0168
4	1.84E+00	0.0048	1.82E+00	0.0042	0.0090
10	1.13E+00	0.0038	1.13E+00	0.0034	0.0063
30	4.34E-01	0.0037	4.37E-01	0.0032	-0.0079
60	1.44E-01	0.0048	1.48E-01	0.0042	-0.0231
90	6.03E-02	0.007	6.23E-02	0.0061	-0.0335

TABLE XIII: Lead detector flux tallies without stainless steel

angle	Flux(4b)	Rel. Err.	Flux(nu)	Rel. Err.	rel. diff.
0	2.81E+00	0.0032	2.80E+00	0.0028	0.0032
1	2.68E+00	0.0031	2.67E+00	0.0027	0.0054
2	2.40E+00	0.003	2.38E+00	0.0027	0.0080
4	1.90E+00	0.0031	1.88E+00	0.0027	0.0104
10	1.14E+00	0.0036	1.13E+00	0.0032	0.0083
30	4.31E-01	0.0045	4.37E-01	0.004	-0.0142
60	1.43E-01	0.0053	1.47E-01	0.0045	-0.0233
90	5.86E-02	0.0065	6.13E-02	0.0055	-0.0464

TABLE XIV: Lead cell flux tallies with stainless steel

angle	Flux(4b)	Rel. Err.	Flux(nu)	Rel. Err.	rel. diff.
0	2.64E+00	0.0207	2.63E+00	0.0199	0.0045
1	2.46E+00	0.0076	2.48E+00	0.0073	-0.0086
2	2.25E+00	0.0056	2.24E+00	0.0054	0.0086
4	1.83E+00	0.0044	1.83E+00	0.0042	0.0049
10	1.12E+00	0.0036	1.12E+00	0.0034	0.0041
30	4.30E-01	0.0034	4.34E-01	0.0032	-0.0098
60	1.43E-01	0.0045	1.47E-01	0.0042	-0.0324
90	6.03E-02	0.0064	6.21E-02	0.0061	-0.0296

TABLE XV: Lead detector flux tallies with stainless steel

angle	Flux(4b)	Rel. Err.	Flux(nu)	Rel. Err.	rel. diff.
0	2.80E+00	0.0029	2.79E+00	0.0028	0.0028
1	2.67E+00	0.0028	2.65E+00	0.0027	0.0070
2	2.38E+00	0.0028	2.37E+00	0.0027	0.0022
4	1.88E+00	0.0028	1.87E+00	0.0027	0.0061
10	1.13E+00	0.0032	1.13E+00	0.0032	-0.0018
30	4.31E-01	0.0042	4.33E-01	0.004	-0.0057
60	1.42E-01	0.0046	1.46E-01	0.0045	-0.0278
90	6.00E-02	0.0117	6.14E-02	0.0058	-0.0226

TABLE XVI: Data comparison with Faddegon for Lead

	4bnu cell		4bnu detect		Faddegon
0	2.63E+00	0.0199	2.79E+00	0.0028	2.92E+00(5.0)
1	2.48E+00	0.0073	2.65E+00	0.0027	2.80E+00(5.0)
2	2.24E+00	0.0054	2.37E+00	0.0027	2.48E+00(5.0)
4	1.83E+00	0.0042	1.87E+00	0.0027	1.99E+00(5.0)
10	1.12E+00	0.0034	1.13E+00	0.0032	1.2E+00(5.0)
30	4.37E-01	0.0032	4.37E-01	0.004	4.47E-01(5.0)
60	1.48E-01	0.0042	1.47E-01	0.0045	1.29E-01(5.0)
90	6.23E-02	0.0061	6.13E-02	0.0055	5.19E-02(7.0)

TABLE XVII: Data comparison with Faddegon for Beryllium

	4bnu cell		4bnu detect		Faddegon
0	2.39E+00	0.0209	2.64E+00	0.003	2.73E+00(5.1)
1	2.25E+00	0.0076	2.47E+00	0.0028	2.57E+00(5.1)
2	1.95E+00	0.0058	2.11E+00	0.0027	2.14E+00(5.1)
4	1.42E+00	0.0048	1.49E+00	0.0027	1.54E+00(5.0)
10	6.15E-01	0.0046	6.17E-01	0.0032	6.30E-01(5.1)
30	9.42E-02	0.007	9.27E-02	0.004	9.49E-02(5.1)
60	2.34E-02	0.0106	2.28E-02	0.0034	2.38E-02(5.9)
90	1.11E-02	0.0144	1.06E-02	0.0038	1.06E-02(7.0)

TABLE XVIII: Data comparison with Faddegon for Aluminum

	4bnu cell		4bnu detect		Faddegon
0	2.96E+00	0.0188	3.26E+00	0.0026	3.42E+00(5.0)
1	2.84E+00	0.0068	3.08E+00	0.0025	3.21E+00(5.0)
2	2.54E+00	0.0051	2.70E+00	0.0025	2.78E+00(5.0)
4	1.96E+00	0.0041	2.03E+00	0.0025	2.14E+00(5.0)
10	1.04E+00	0.0035	1.05E+00	0.0028	1.06E+00(5.0)
30	2.68E-01	0.0037	2.66E-01	0.003	2.65E-01(5.0)
60	7.14E-02	0.0054	7.25E-02	0.0029	6.66E-02(6.0)
90	3.28E-02	0.0075	3.26E-02	0.003	2.87E-02(6.0)

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